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Assessing the role of external photoevaporation of protoplanetary discs in nearby star-forming regions

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*A mamma e papà,
che spesso ci hanno creduto più di me.
Prometto che vi farò un riassunto in italiano!*

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A special thanks to Michele, who kindly agreed to process an image of Orion taken 10 years ago.

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Thesis overview

This thesis consists of an introductory chapter, three parts including three chapters each, and a chapter containing the conclusions and future perspectives. In the first part, I present the theoretical framework on which the thesis is built and the characteristics of the star-forming regions on which the thesis is mainly focused. The second part addresses the role of external photoevaporation on protoplanetary disc masses and radii in weakly and moderately irradiated environments, focusing on the disc population in the Upper Scorpius and Lupus star-forming regions. This part includes a summary of the first results of the AGE-PRO ALMA Large Program, to which I have actively collaborated during my PhD. In the third part, I present a novel method to evaluate the far-ultraviolet flux and its application to the nearby star-forming regions. Throughout this thesis, I outline the current state of knowledge regarding the evidence for the effect of external photoevaporation on nearby protoplanetary discs.

I summarise the content of the chapters as follows.

- **Chapter 1. Planet formation shaped by diverse stellar environments**
In this chapter, I provide an introduction to the research field of exoplanets, protoplanetary disc observations, and the role of the stellar formation environment. I present here the main motivation of the thesis and the scientific questions I aim to address.
- **Chapter 2. Dynamics of viscous discs**
This chapter presents the classic picture of star and protoplanetary disc formation, the dynamics of the gas disc, following the viscous model, and the main processes affecting the dust disc dynamics.
- **Chapter 3. Externally UV-driven disc winds**
In this chapter, I discuss a geometric model for the photoevaporative disc wind, distinguishing between EUV and FUV dominated flow. Moreover, I present the coupling between disc viscous evolution and external photoevaporation.
- **Chapter 4. Rising under the spotlight: nearby OB associations**
I illustrate, in this chapter, the main properties of the nearby OB associations, focusing on the Scorpius-Centaurus and Orion OB1 associations. Moreover, I present a study of the UV radiation field in the solar neighbourhood.
- **Chapter 5. External photoevaporation shaping disc masses and radii in Upper Scorpius**

This chapter summarises the first results of the AGE-PRO ALMA Large Program and focuses on the findings presented in the publication that I lead as first author. This study compares disc evolution models with and without the inclusion of external photoevaporation with the disc masses and radii retrieved from AGE-PRO observations in the Upper Scorpius star-forming region.

- **Chapter 6. The extended Upper Scorpius disc sample**

This chapter complements the previous one, investigating the properties of the largest disc sample available in Upper Scorpius, particularly focusing on gas and dust disc sizes and comparing with disc evolution models. The results collected here are part of three co-authored publications.

- **Chapter 7. Substructures and weak FUV flux: application to Lupus**

This chapter investigates whether including dust traps and external photoevaporation into disc evolution models can resolve the discrepancy between predicted and observed gas-to-dust size ratios in the Lupus star-forming region.

- **Chapter 8. A novel method for evaluating the FUV flux**

In this chapter, I present a new method for evaluating the FUV flux accounting for the uncertainty in the separation from massive stars in 3D space. I apply this method to a large sample of ClassII objects in the nearby star-forming regions, providing a publicly available catalogue.

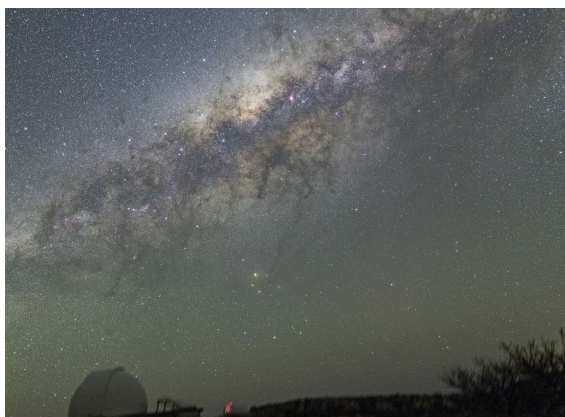
- **Chapter 9. Mapping Orion in FUV flux**

In this chapter, I use the method presented in the previous chapter to provide a map of the FUV radiation field at the stellar population of the Orion complex, and I discuss the distribution of the FUV flux across the region.

- **Chapter 10. FUV flux and disc properties**

This chapter addresses the relation between disc properties and FUV flux, investigating the distribution of dust disc masses in the nearby star-forming regions and the accretion luminosity in the Orion region. I aim to outline the current picture of the indirect evidence of external photoevaporation on discs in the solar neighbourhood.

Planet formation shaped by diverse stellar environments



Our imagination is stretched to the utmost, not, as in fiction, to imagine things which are not really there, but just to comprehend those things which are there.
[Richard Feynman] - The Milky Way as seen from the Hakos Astro Farm in Namibia.

There has always been a sense of fascination and curiosity in looking at the night sky and interpreting the nature and origin of stars, planets, and celestial phenomena. From the earliest records of eclipses and planetary motions, to the philosophical speculations of Greek natural philosophers, and later to the development of modern physics, to the present day, scientists continue pursuing a common goal of providing scientific explanation of what we see and what is out there. The sky has become a laboratory in which we probe the fundamental processes of physics: gravitation, hydrodynamics, thermodynamics, and chemistry. Unfortunately, the sky to which we have access everyday with naked eyes is far less rich than the sky visible to our ancestors due to light pollution and atmospheric contamination. To recover a less-contaminated view of the sky, astronomy relies on remote and uncontaminated places. These sites, often situated at high altitudes and in dry environments, host some of the main astronomical facilities, such as the Atacama Large Millimeter/submillimeter Array (ALMA) and the Very Large Telescope (VLT) in the Chilean desert, and the Keck Observatory on Mauna Kea in Hawaii. At the same time, space-based telescopes have revolutionized our view of the Universe by complementing and expanding ground-based observations. From the Hubble Space Telescope (HST), which has provided high-resolution imaging across optical and near-infrared wavelengths, to missions such as Spitzer and Herschel, which opened new windows in the infrared, and more recently the James Webb Space Telescope (JWST), which

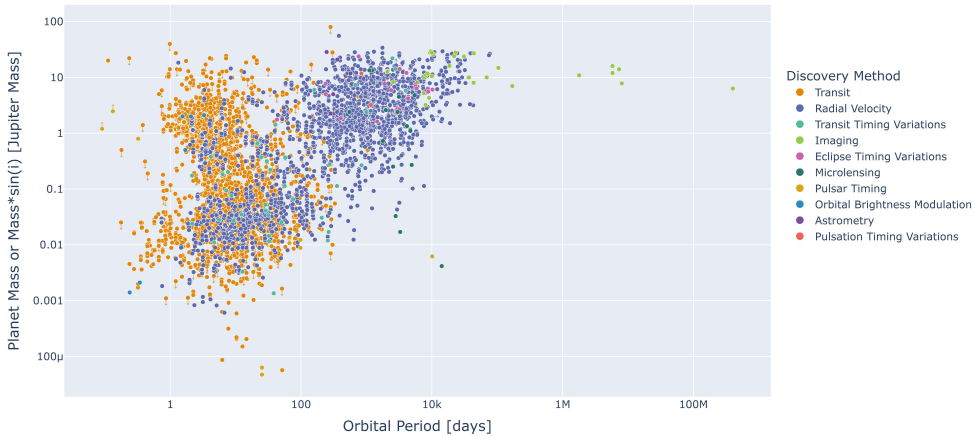


Figure 1.1: Mass and orbital period of the exoplanets discovered using the methods listed in the right box. From exoplanets.eu.

is probing the formation of galaxies, stars, and planetary systems with unprecedented sensitivity, we now have access to a vast amount of data, which are fundamental to build accurate theoretical models.

Although the interpretation of the sky has evolved dramatically with time, some of the most fundamental questions remain timeless: Are we alone in the universe? Are there other planets like Earth? What is the origin of stars, planets, and the structures that host them? These questions lie at the heart of modern astrophysics and are particularly central to the study of stars, protoplanetary discs, and planet formation.

The study of planet formation outside the Solar System has increased enormously after the discovery of the first exoplanet orbiting a Solar-like star, made in 1995 by Mayor & Queloz (1995). This planet, 51 Pegasi b, has a mass similar to Jupiter but is located at a smaller distance than Earth from the central star. Currently, 5983 exoplanets have been confirmed¹. Of all those planets, only 84 have been directly imaged to date by telescopes, because seeing planets directly in optical light is not straightforward and requires extremely deep sensitivity and high resolution. Figure 1.1 shows the mass of the detected exoplanets against their orbital period, colour-coded according to their discovery method. There are many exoplanets with masses similar to Jupiter and at smaller distances to the central star. These objects are called *Hot Jupiters*. There is also a large sample of planets located at sub-au distances from the central star and more massive than Earth, which are known as *Super Earths*. Despite the strong influence of observational biases and detection limits on current exoplanet observations (Morbidelli & Raymond 2016), the diversity revealed within the current exoplanet population raises fundamental questions about the physical processes that govern planet formation and evolution, as well as the origins of the differences among planetary systems and our Solar System. Currently, there are several theories explaining the process of planet formation, which happens in disc-like structures of gas and dust around young stars. Historically, planet formation has been explained by the *core accretion* model (e.g., Pollack et al. 1996; Armitage 2013), which involves the coagulation of dust grains in a protoplanetary disc to form gravita-

¹See the archive of exoplanets <https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=PS>.

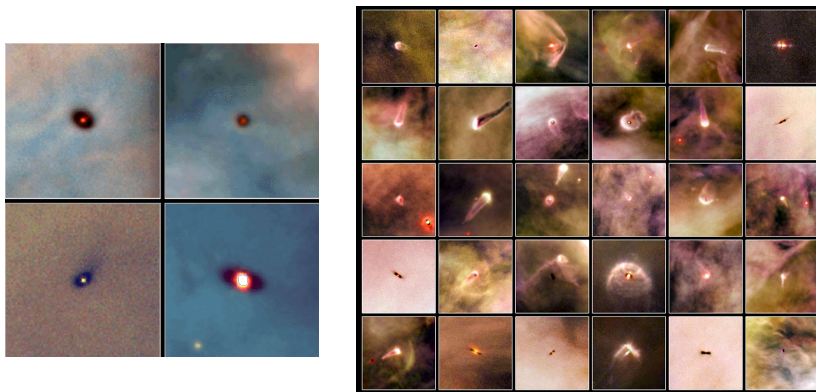


Figure 1.2: Protoplanetary discs in the Orion Nebula, as seen in visible light by the HST. Left: one of the first images of proplyds, from O'Dell et al. (1993). Right: atlas of HST proplyds by Ricci et al. (2008).

tionally bound planetesimals, and then rocky cores, which would eventually accrete gas in an external envelope. Alternatively, Boss (1997) proposed gas fragmentation in protoplanetary discs, caused by gravitational instability, as an alternative paradigm for the formation of giant planets. However, it is evident that, in order to understand the properties and origin of the exoplanet population, it is essential to investigate the physics and composition of their birthplaces, which are protoplanetary discs, and connect them with the natal stellar formation environment.

Tracers of protoplanetary discs are gas and dust. Although dust is the less abundant component, it dominates the opacity and is therefore easier to observe. In particular, the thermal emission from dusty discs around young stars produces an excess in the total flux, at infrared (IR) wavelengths, above the emission of the star. The first discs around young stars discovered via IR emission were those around Vega (Aumann et al. 1984) and β Pictoris (Smith & Terrile 1984). The first surveys detecting IR excess for populations of young stars are attributed to the Infrared Astronomical Satellite (IRAS, e.g., Strom et al. 1989), which contributed to characterise the spectral energy distribution (SED) of these objects. The study of IR excess was later reinforced and expanded by Spitzer (e.g., Evans et al. 2009) and WISE (e.g., Koenig & Leisawitz 2014). A comprehensive picture of dust distribution in protoplanetary discs can be achieved by combining observations across multiple wavelengths, since the thermal emission at each wavelength primarily traces dust grains with sizes comparable to the emission peak. Micron-sized grains, which dominate the disc surface layers, are seen in scattered light by facilities such as SPHERE (Spectro-Polarimetric High-contrast Exoplanet REsearch) on the VLT. Larger grains, with sizes up to millimetres, settle toward the disc mid-plane and are best traced at (sub-)millimetre wavelengths, accessible with ALMA at $\sim 0.35 - 3$ mm. The first protoplanetary disc detected at mm-continuum wavelengths was HL Tau (Lay et al. 1994). Beyond ALMA wavelengths, the VLA (Very Large Array) probes the emission from cm-sized pebbles in discs. For what concern

The first observations of protoplanetary discs in the early 1990s refer to continuum

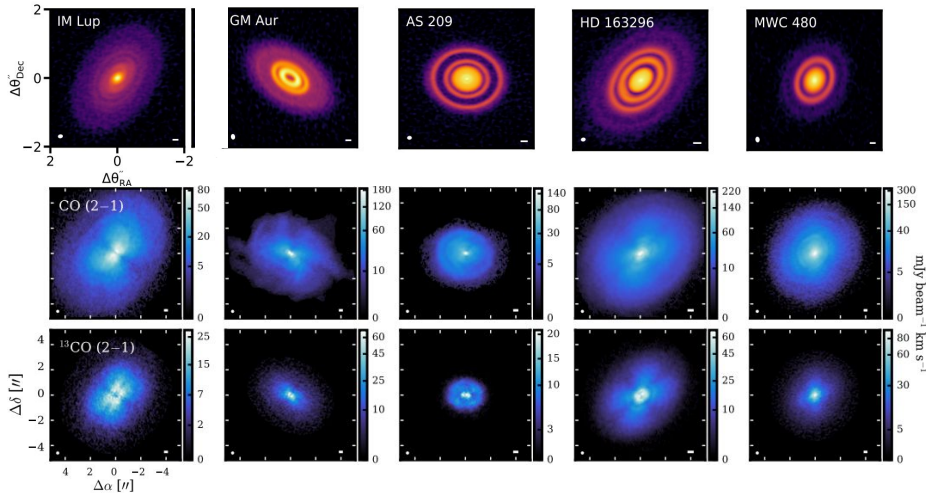


Figure 1.3: First row: 220 GHz continuum images from MAPS displaying the diversity of continuum structure present at the same spatial resolutions as probed by the MAPS molecular lines. The synthesized beams are indicated in the bottom-left corner and a 20 au scale bar in the bottom-right corner of each panel. Second and third row: ^{12}CO 2-1 and ^{13}CO 2-1 integrated intensity images, respectively.

emission of small particles at (sub-)millimetre wavelengths (Beckwith et al. 1990). Subsequently, O’Dell et al. (1993) presented images of the iconic *proplyds* in the Orion Nebula (see left panel of Fig. 1.2) seen with HST. These objects are discs around young stars whose elongated shapes are sculpted by the intense ultraviolet (UV) radiation from nearby massive stars. In the right panel of Fig. 1.2, we show part of the atlas of HST *proplyds* in the Orion Nebula published by Ricci et al. (2008).

A new era for the study of protoplanetary disc evolution and planet formation started with the advent of ALMA, which is currently the largest radio telescope in the world. The breathtaking images of the protoplanetary disc around HL Tau (ALMA Partnership et al. 2015) provided the first direct view of multiple substructures in a young dust disc, revealing a completely different morphology compared to the HST *proplyds* in Orion, and marking a turning point in the field. This discovery triggered a theoretical and observational effort to interpret disc substructures in the context of planet formation. Since then, ALMA continuum observations have revealed that substructures are a common feature of discs, manifesting as rings, gaps, spirals, and arcs.

For what concerns the gas component in protoplanetary discs, investigating gas kinematics is possible through observations of spectral line emission, particularly CO (e.g., Sargent & Beckwith 1987; Simon et al. 2000). Observing spectral line emission with ALMA has enabled unprecedented insights into disc kinematics and the dynamical processes that regulate disc evolution and planet formation. Figure 1.3 shows a gallery of the five discs investigated by the Molecules with ALMA at Planet-forming Scales (MAPS) Large Program (Öberg et al. 2021), as seen in dust continuum emission (first row) and in CO and ^{13}CO line emission (second and third row). Surveys of protoplanetary discs such as DSHARP (Andrews et al. 2018a), the surveys in the Lupus, Taurus, and Upper Scorpion regions (Ansdell et al. 2016; Long et al. 2018; Carpenter et al. 2014, 2025), and the recent AGE-PRO (Zhang et al. 2025) and exoALMA (Teague et al. 2025), shifted the

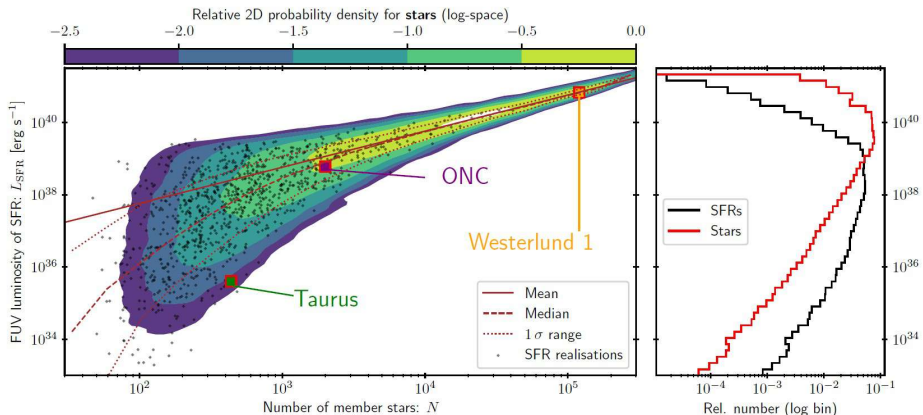


Figure 1.4: Left: FUV luminosities at an age of 1 Myr. The number of members is taken from Fatuzzo & Adams (2008). The solid brown line follows the mean luminosity as a function of the number of members of the clusters, the dashed brown line is the median, while the dotted lines bound the 14th and 86th percentile. The colour bar shows an estimate for the relative density of stars in logarithmic space. Right: the corresponding distribution in the number of stars (red) and SFRs (black) per logarithmic bin in total FUV luminosity space. From Winter & Haworth (2022).

focus from individual systems to populations of discs. These surveys provided us with a statistically large sample of data that revealed correlations between disc and stellar properties and that allowed us to test disc evolution and planet formation models (Manara et al. 2023a).

However, to gain a comprehensive understanding of how discs typically evolve, how planets form, and to connect with the observed exoplanet population, it is essential to include an additional ingredient to the picture presented above: the role of the stellar formation environment. Disc evolution and ultimately planet formation do not happen in isolation. Protostars and their discs form and evolve in crowded star-forming regions, where the disc properties are directly linked to the angular momentum content of the molecular cloud core in which they form. Moreover, the properties and the dynamical history of the star-forming environment affect the future of discs and planets. Dynamic encounters can change the morphology of discs, producing features such as spirals and truncating discs (e.g., Cuello et al. 2019; Rosotti et al. 2014). Streamers of infalling material on the disc can provide an additional contribution to the total mass reservoir to form planets. Evidence of the interaction between discs and the interstellar medium is seen in SPHERE images in the Taurus region (e.g., Garufi et al. 2024), in the spatial correlation between stellar accretion and gas density in the Lupus clouds (Winter et al. 2024a), and in isolated systems presenting streamers (e.g., Gupta et al. 2024). Moreover, proximity to massive stars that emit intense UV radiation influences disc morphology, physical, and chemical properties through the process known as external photoevaporation (Hollenbach et al. 1994; Johnstone et al. 1998), as directly seen in the Orion Nebula (Fig. 1.2). This process is responsible for the outside-in depletion of disc material and drastically reduces the possibility of forming planets in the outer disc regions (e.g., Winter & Haworth 2022).

The reason why external photoevaporation has often been neglected in planet formation models (until recently) lies primarily in the fact that the available sample of observed

discs is strongly biased toward nearby, low-mass star-forming regions. Such regions were naturally prioritised in surveys, given their proximity and relative abundance, but they host only a few B-type stars and lack the extreme conditions where proplyds are detected. Nevertheless, there are two main motivations for investigating the role of external radiation on discs. First, the UV flux experienced at the outer disc boundary depends strongly on the distance from the nearest massive stars, such that even late-type B or early-type A stars in close proximity can induce considerable photoevaporation. Second, it is widely established that the majority of young stars form in massive cluster that host numerous OBA-type stars (Lada & Lada 2003; Fatuzzo & Adams 2008), making low-mass star-forming regions unrepresentative of the typical birth environments of stars and their planetary systems. As shown in Fig. 1.4, regions like Taurus (~ 140 pc from the Sun) are common but present $\sim 1/100$ of the number of stars in clusters such as Westerlund 1 (~ 4 kpc from the Sun). The most common type of birth environment for stars lies between the Orion Nebula Cluster (ONC, ~ 400 pc) and Westerlund 1, where the FUV luminosity is $\sim 10^{40}$ erg s $^{-1}$ at an age of 1 Myr. The corresponding average typical FUV flux (F_{FUV}) is in the range $10^2 - 10^4$ G_0 , where G_0 is the Habing unit (Habing 1968) and describes the flux normalised to the average value in the solar neighbourhood, and integrated in the far-ultraviolet (FUV) range of wavelengths

$$\left(\frac{F_{\text{FUV}}}{G_0}\right) = \int_{912 \text{ \AA}}^{2400 \text{ \AA}} \frac{F_{\lambda} d\lambda}{1.6 \times 10^{-3} \text{ erg s}^{-1} \text{ cm}^{-2}}. \quad (1.1)$$

Therefore, in order to investigate how planet formation typically happens in our Galaxy, we need to characterise the impact of the external environment on protoplanetary disc evolution. The sole direct observational evidence of the effect of external photoevaporation on protoplanetary discs is represented by proplyds (Fig. 1.2), which are illuminated by extreme UV radiation ($\gtrsim 3 \times 10^3$ G_0 , Kim et al. 2016) and are not representative of the typical conditions for star formation. Indirect evidence demonstrates the influence of external photoevaporation on disc properties, such as the inverse correlation between the number of massive stars and the disc fraction observed in Cygnus (Guarcello et al. 2023), and the disc dust mass and FUV flux in the L1641 and L1647 regions of Orion (van Terwisga & Hacar 2023) as well as in the σ -Orionis cluster (Ansdell et al. 2017a, Maucó et al. 2023a) and in the ONC (Eisner et al. 2018a, Otter et al. 2021).

Within this background, fundamental questions remain open:

- (1) To what degree is external photoevaporation affecting protoplanetary disc dynamics when proplyds are not detected?
- (2) Which disc observables better reveal the influence of the external radiation on protoplanetary discs?
- (3) How can we characterise the FUV radiation field experienced by young stars in the solar neighbourhood?
- (4) How does external photoevaporation impacts disc demographics?

This Thesis will explore the above questions considering nearby star-forming regions ($\lesssim 500$ pc). Specifically, Part I provides an introduction to the formation of protoplanetary discs, their evolution in a classical viscous framework, and a description of photoevaporative winds and how to couple external photoevaporation to disc viscous evolution. Part II addresses questions (1) and (2) by using disc evolution models in which the effect of external photoevaporation is included to investigate the evolution of disc sizes and masses. The results of the models are compared with recent observations of protoplanetary discs in the Upper Scorpius region. Part III focuses on question (3) by

addressing the effect of geometrical projection in the calculation of the FUV radiation field experienced by young stars, presenting a new method and a catalogue of the FUV flux for young stars. Moreover, Part III explores question (4) by comparing disc dust masses and accretion luminosities with the FUV flux experienced by young stars, and highlighting future observational and theoretical steps to help fill the current gap in understanding the role of the environment on disc dynamics.

Part I

Dynamics of viscous and externally irradiated accretion discs

Dynamics of viscous discs



In myth, Jupiter used to attract countless lovers, as the planet does with its 95 moons. He was born from Cronus and Rhea, or, in the less exciting version, from a disc of gas and dust. [Osservatorio Astronomico G. V. Schiaparelli, Varese]

The content of this Chapter is primarily based on Hartmann (2008), Lodato (2008), Armitage (2013), Clarke & Carswell (2014).

Planets typically form in disc-like structures of gas and dust around young stars, known as protoplanetary discs. The typical gas-to-dust mass ratio in protoplanetary discs is inferred from the composition of the interstellar medium (Draine 2011), and it is often assumed to be ~ 100 . Although the gas controls the dynamics of the system, due to its abundance, the dust is responsible for radiative transport due to its opacity. The dust absorbs and reprocesses stellar radiation, emitting at wavelengths which are proportional to the grain size. Gas can be treated as a viscous fluid, which is subject to pressure forces, while dust is considered pressure-less and inviscid. In this Chapter, we briefly describe the formation and classification of young stellar objects (YSOs). Subsequently, we describe the dynamics of gas and dust in a protoplanetary disc subject to viscous transport of angular momentum.

2.1 Star and protoplanetary disc formation

Stars form from the collapse of a rotating Molecular Cloud Core (MCC), that is a substructure of a Giant Molecular Cloud (GMC) made of gas (mainly hydrogen in molecular form H_2) and dust. GMCs have typical masses of the order of $10^4 - 10^6 M_\odot$, typical radii between 10 and 100 pc, and contain filamentary substructures. The hydrostatic equilibrium of a GMC depends on the balance between gravitational attraction, thermal pressure, magnetic field forces, and turbulent motions, that can lead the clouds to collapse or expand. The thermal energy of a GMC is orders of magnitude smaller than its gravitational, kinetic, and magnetic energies. Therefore, GMCs are not supported by thermal pressure, but rather by turbulent motions and magnetic fields. Numerical hydrodynamic and magneto-hydrodynamic (MHD) simulations of GMCs show that that turbulent internal motions and magnetic fields of sufficient strength can mitigate or even impede cloud collapse (e.g., Gammie & Ostriker 1996; Mac Low et al. 1998). Internal random motions in molecular clouds are often referred to as *turbulence*, although their nature remains unclear. Turbulence is represented by a cascade of energy which is injected at large scales and dissipates at small scales with an energy spectrum, $E(k)$, which is the kinetic energy per unit mass per unit wave number, linked to the power spectrum in a three-dimensional (3D) space, $P(k)$, as:

$$E(k) \sim P(k)k^2 \sim \frac{\sigma_{v,k}^2}{k} \propto k^{-\beta} \propto \lambda^\beta, \quad (2.1)$$

where k is the wave-number, $\sigma_{v,k}$ is the velocity dispersion related to k , and λ is the wave-length. Models and observations of turbulent molecular clouds support a power-law velocity dispersion-size relation, with β assuming different values. Larson (1981), retrieved the relation $\sigma_v \propto R^{0.38}$, with R being the length scale of the cluster (which corresponds to $\beta = 1.76$) from observations of molecular lines available for several earlier studies of molecular clouds. This relation is known as the first Larson law. The same work (Larson 1981) showed that this relation remains valid considering subregions of the molecular clouds, with differences in amplitude depending on the region. Supersonic and rapidly cooling turbulent molecular clouds, dominated by shocks, are predicted to show $\beta = 2$ (Burgers 1948). The Kolmogorov (1941) model for incompressible turbulence predicts $\beta = 5/3$. Observations in the Taurus star-forming region support the double power-law behaviour of the velocity dispersion (Qian et al. 2018), showing $\beta = 2$ at smaller scales, between 3 and 1 pc, which suggests a compressive fluid behaviour, and the Kolmogorov relation with $\beta = 5/3$ at larger scales, in the range 5-10 pc.

Although turbulence can balance gravity on global scales, it causes density enhancement at local scales, where thermal pressure becomes the most important force resisting gravity. This happens at a typical scale, called Jeans length, which is the maximum wavelength of small perturbations below which the cloud is stable

$$\lambda_J = \frac{\sqrt{\pi} c_s}{\sqrt{G\rho}}, \quad (2.2)$$

where c_s is the sound speed in the cloud, G is the gravitational constant, and ρ is the medium density. Typical Jeans lengths for clouds with typical temperature of $T \sim 10$ K and number density $\sim 10^5 \text{ cm}^{-3}$ are on the order of $\lambda_J \sim 0.1 - 1$ pc. This length scale sets the minimum mass that a cloud core must have to collapse and form stars, $\propto \lambda_J^3$, which is known as the Jeans mass and can be written as

$$M_J = 0.2 \left(\frac{T}{10 \text{ K}} \right)^{3/2} \left(\frac{n}{10^5 \text{ cm}^{-3}} \right)^{-1/2} M_\odot, \quad (2.3)$$

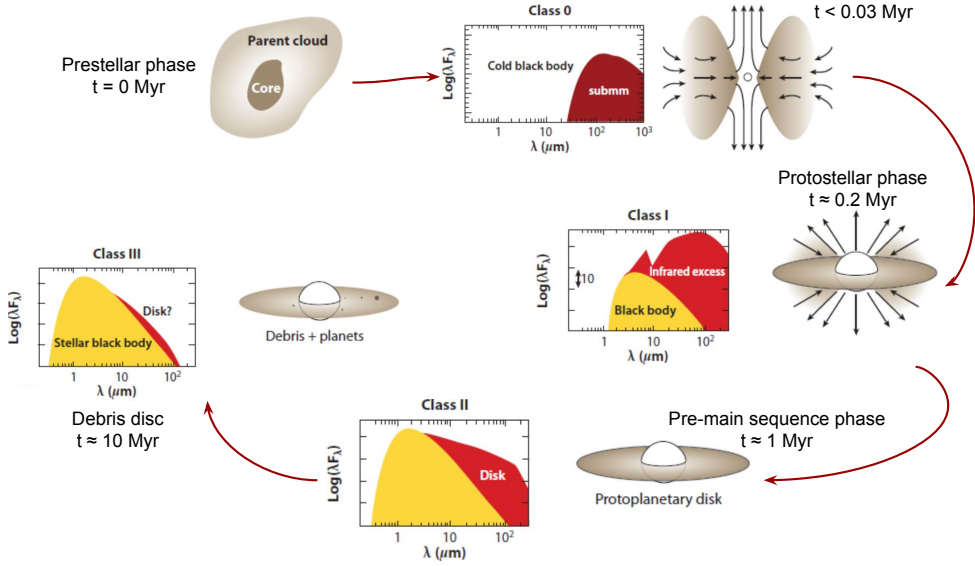


Figure 2.1: Schematic view of the classification of YSOs. A star forms from the collapse of a MCC. Between 0.03 and 0.2 Myr the envelope of gas and dust becomes flat and forms a disc-like structure around the protostar. Between ~ 1 and ~ 10 Myr the protoplanetary disc evolves, accretes onto the star, dust grains coagulate to form planetesimals, and the remnant is a debris disc with little of no gas.

where n is the number density of hydrogen, which is normalised to the typical value of the densest regions in GMCs, 10^5 cm^{-3} . Since MCCs are characterized by rotation, the centrifugal force associated with their angular momentum can become important and hold the contraction, leading to the formation of a binary or multiple system of stars. In order to conserve the total angular momentum, the reduction in the core radius (collapse) causes an increase in the angular velocity of the MCC, resulting in a strong centrifugal force that balances the horizontal component of gravity. The vertical component of the gravitational force is not balanced by any force, leading to flattening of the material and the formation of a disc around the protostar, called protostellar disc, with an extension of $R_{\text{disc}} \sim 10^2 - 10^4 \text{ au}$. Protostellar discs inherit their initial mass, size, and chemical composition from the broader star formation environment and their evolution is influenced by many processes (e.g., infalling material, perturbations from external bodies). After a few 10^5 years, which is similar to the free-fall time of dense MCCs, most of the disc material accretes onto the central protostar and the remaining accretion disc is known as protoplanetary disc, being the birthplace of planets.

2.1.1 Classification of Young Stellar Objects

A system composed of a young star and an envelope of gas and dust is usually referred to as Young Stellar Object (YSO). Conventionally, YSOs are classified based on the slope of their Spectral Energy Distribution (SED) in the infrared region of the spectrum (from $\sim 2 \mu\text{m}$ to $\sim 100 \mu\text{m}$), which is defined by the spectral index

$$\alpha_{\text{IR}} = \frac{d \log(\lambda F_{\lambda})}{d \log \lambda}, \quad (2.4)$$

where F_λ is the emitted flux of the object at wavelength λ . As shown in Fig. 2.1, different classes are interpreted as different evolutionary stages of the system (Lada 1987; Adams et al. 1988; Andre et al. 1993; Andre & Montmerle 1994; André 2002). Figure 2.1 shows that each of the main phases of the star and disc formation process is dominated by a different component, and each component gives a different energetic contribution to the SED. The three components of the main infall phase are: (i) the envelope, which emits at sub-mm wavelength, through the reprocessing of energy; (ii) the protostellar disc, which emits via viscous accretion or reprocessing the radiation from the protostar; and (iii) the protostar, which emits at in the range from UV to optical wavelengths. Based on the value of α_{IR} , YSOs are classified into four classes as follows:

- Class 0: in the first phases of cloud collapse, no SED is detected in the near infrared, while there is a peak in the far infrared. Each protostar in an MCC is surrounded by an optical thick envelope of gas and dust. This makes the stellar radiation invisible in the near infrared.
- Class I: $\alpha_{\text{IR}} > 0$, the SED is almost flat in the near and middle infrared. The young and cold objects begin to reveal a protoplanetary disc. The dominant component is still the envelope that absorbs and processes stellar and disc radiation, re-emitting at long wavelength ($\sim 100 \mu\text{m}$).
- Class II: $-\frac{4}{3} < \alpha_{\text{IR}} < 0$, the SED peaks in the middle-near infrared. In this phase, most of the envelope is accreted. An excess at infrared (IR) wavelengths, which is due to the disc, is detected in addition to the emission from the stellar photosphere.
- Class III: $\alpha_{\text{IR}} < -\frac{4}{3}$, the SED peaks in the infrared part of the spectrum. The infrared excess from the disc is negligible with respect to the black body emission from the stellar photosphere.

Since the infrared excess is considered to be an indicator of the presence of a protostellar disc around a protostar, ideally it can be used to give a first estimate of the disc lifetime. Indeed, the fraction of stars in which an infrared excess is detected, in populations of young stars of different ages, can be computed to provide an estimate of the disc lifetime, which results in ~ 10 Myr (Mamajek 2009; Williams & Cieza 2011). This timescale must be sufficient to allow planet formation. It is important to highlight that Class IIIs are not necessarily disc-less stars. Although at the current sensitivity and resolution of the telescopes Class IIIs show little or no excess from warm dust in the inner disc regions, this does not necessarily imply the absence of millimetre emission from cooler dust in the outer disc. Indeed, several Class IIIs show outer discs and even CO line flux emission as indicative of gas still orbiting the star, and be regarded as young debris discs or planetesimal belts (e.g., Lovell et al. 2021; Matr  et al. 2025). However, the dust mass detected in Class IIIs and the gas content are orders of magnitude lower than observed in Class IIs, suggesting that gas-giant planets must form before the Class III phase.

2.2 Gas evolution

Protoplanetary discs are often assumed to be geometrically thin and axisymmetric, as a first approximation. In this case, it is useful to carry out the study using a system of cylindrical coordinates (R, ϕ, z) centred on a star of mass M_* . In the geometrically thin approximation, the disc height H is assumed to be much smaller than the radial extend R , which translates in:

$$\frac{H}{R} \ll 1, \quad (2.5)$$

where H/R is known as the aspect ratio of the disc. This condition is marginally satisfied in protoplanetary discs, where typical aspect ratios are on the order of $\sim 10^{-1}$ (e.g., Sturm et al. 2023). Considering a Keplerian disc in hydrostatic equilibrium, in Sec. 2.2.2 we will see that the disc height can be written in terms of the sound speed, c_s and the Keplerian angular velocity $\Omega_K = \sqrt{GM_\star/R^3}$, and therefore the thin approximation implies:

$$\frac{H}{R} = \frac{c_s}{\Omega_K R} = \frac{c_s}{v_K} \ll 1, \quad (2.6)$$

where $v_K = \sqrt{GM/R}$ is the Keplerian velocity (i.e., in the azimuthal direction, using cylindrical coordinates). The previous expression means that the disc rotation is highly supersonic. Moreover, as will be demonstrated in Sec. 2.2.2, if the accretion takes place on a longer timescale than the dynamical timescale, the radial velocity v_R will be smaller than both the sound speed and the rotational speed:

$$v_r \ll c_s \ll v_K. \quad (2.7)$$

Following the geometrically thin approximation, it is convenient to use vertically integrated quantities:

$$\Sigma(R, t) = \int_{-\infty}^{+\infty} \rho(R, z, t) dz, \quad (2.8)$$

where Σ and ρ are the surface and volume density of the disc, respectively.

2.2.1 Evidence of accretion

Protoplanetary discs are effectively accretion discs, and this is motivated by observational evidence of young stars showing signs of accretion. Evidence of ongoing accretion include:

- A flux excess at UV and blue optical wavelengths above the stellar photosphere. This is caused by material that falls on the stellar surface, following magnetic field lines, at nearly free-fall velocity and produces accretion shocks that radiate in UV/optical (e.g. Calvet & Gullbring 1998; Herczeg & Hillenbrand 2008).
- Strong emission lines, especially H_α , which trace gas infalling onto the stellar surface at velocities of ~ 200 – 400 km/s, consistent with what predicted by magnetospheric accretion models (e.g., Fang et al. 2009; Rigliaco et al. 2012).

Based on the H_α equivalent width, YSOs are classified as Classical T Tauri Stars (CTTSs, H_α equivalent width > 10 Å) or Weak-line T Tauri Stars (WTTSs, H_α equivalent width < 10 Å). The X-Shooter instrument on the ESO Very Large Telescope (VLT) has advanced the field thanks to the possibility of accurately evaluating the accretion luminosity from the UV-excess in the Balmer continuum region (e.g., Manara et al. 2013; Alcalá et al. 2014), and from the line luminosity of several additional emission lines, which are more reliable tracers of accretion compared to measurements of the width of the H_α line. The excess luminosity above the stellar photosphere can be directly converted into accretion luminosity, from which we can estimate the stellar accretion rate. In particular, given an estimate of stellar mass and radius, and the measured accretion luminosity, the accretion rate can be computed as

$$\dot{M}_{\text{acc}} = \frac{L_{\text{acc}} R_\star}{GM_\star} \left(1 - \frac{R_\star}{R_{\text{in}}} \right)^{-1}, \quad (2.9)$$

where R_{in} is the location of the inner disc (i.e., a measure of the point from where the material is accreting). Typical accretion rates are on the order of $\sim 10^{-8} - 10^{-11} \text{ M}_{\odot}/\text{yr}$ (for a review, see Manara et al. 2023a).

Since we have evidence of accretion, this means that material from the disc has to be able to flow inward and be accreted onto the star. In the most simplistic case of a geometrically thin disc in Keplerian orbit around a star of mass $M_{\star}$, the specific angular momentum (i.e., the angular momentum per unit mass) is:

$$l = R v_K = \sqrt{GM_{\star}R}, \quad (2.10)$$

where $v_K = \sqrt{GM_{\star}/R}$ is the Keplerian velocity. Since l increases with distance from the star R , this means that the gas needs to lose angular momentum to move inward. Two main frameworks are currently competing to describe disc evolution and accretion. The viscous model predicts that angular momentum is transported outward allowing accretion onto the star (Lynden-Bell & Pringle 1974). Instead, in the magneto-hydrodynamic (MHD) winds scenario, disc material and angular momentum are removed from the disc surface following the magnetic field lines (Lesur et al. 2014; Suzuki et al. 2016). Additionally, in both scenarios material can be removed from the disc due to thermal wind from the central star and external processes due to the environment (e.g., dynamical encounters, external photoevaporation). This Chapter focuses on the pure viscous model and addresses the main equations governing the physics of gas and dust in protoplanetary discs.

2.2.2 Fluid equations

The dynamics of protoplanetary discs is described by the two fundamental fluid equations: the continuity equation, which describes the conservation of mass, and the Navier-Stokes equation, describing the conservation of angular momentum. The continuity equations, integrated in the vertical direction employing the geometrically thin approximation, reads

$$\frac{\partial \Sigma(R, t)}{\partial t} + \frac{1}{R} \frac{\partial}{\partial R} (R \Sigma(R, t) v_R) = 0, \quad (2.11)$$

while the Navier-Stokes equation is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} (\nabla P - \nabla \cdot \sigma) - \nabla \Phi, \quad (2.12)$$

where $\mathbf{v}$ is the velocity vector, P is the pressure, σ is the stress tensor, and Φ is the gravitational potential. Assuming that only the central star contributes to the gravitational potential, we can write Φ as

$$\Phi = -\frac{GM_{\star}}{r}, \quad (2.13)$$

where G is the gravitational constant, $M_{\star}$ is the stellar mass, and r is the distance from the central star. The stress tensor σ can be written in terms of its components as

$$\sigma_{ij} = \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3} (\nabla_k v_k) \delta_{ij} \right) + \zeta (\nabla_k v_k) \delta_{ij}, \quad (2.14)$$

where η and ζ are the coefficients of shear viscosity and bulk viscosity, and δ_{ij} is the Kronecker delta. In the case of a Keplerian disc, the only non-vanishing component of σ

is

$$\sigma_{R\Phi} = \eta R \frac{d\Omega}{dR}, \quad (2.15)$$

where Ω is the disc angular velocity. If we describe the disc as a set of infinitesimal annuli, since $\Omega = \Omega_K \propto R^{-3/2}$, each annulus will rotate at different speed, with the inner ones rotating faster. This results in shear in the radial direction and therefore the presence of a shear viscosity. The viscous stress opposes to the velocity gradient, slowing down the inner annulus and transferring angular momentum from the inner to the outer part of the disc. This redistribution of angular momentum allows for the accretion process, where there is more material moving inward than outward.

The analysis of each of the three components of the Navier-Stokes equation reveals important properties of the accretion mechanism.

Vertical component: hydrodynamic equilibrium

The projection in the z -component of the Navier-Stokes equation 2.12 describes the vertical structure of the disc. Under the assumption of a geometrically thin disc, v_z can be neglected and the left-hand component of Eq. (2.12) goes to zero. The viscous term has no component in this direction, so Eq. (2.12) reduces to:

$$\frac{1}{\rho} \frac{\partial P}{\partial z} = -\frac{\partial \Phi}{\partial z}, \quad (2.16)$$

which describes the hydrostatic equilibrium in the vertical direction. Considering $z \ll R$, the vertical component of the gravitational force is

$$\frac{\partial \Phi}{\partial z} = \frac{GM_\star}{R^2 + z^2} \sin \theta \sim \frac{GM_\star}{R^3} z = \Omega_K^2 z, \quad (2.17)$$

with $\theta \ll 1$. Moreover, approximating the disc as vertically isothermal, the pressure in terms of density is $P = c_s \rho$, where the sound speed c_s is constant. Therefore, Eq. (2.16) can be written as:

$$\frac{c_s^2}{\rho} \frac{\partial \rho}{\partial z} = -\Omega_K^2 z. \quad (2.18)$$

To solve this equation, we define the dimensionless variable $\xi = z/H$, where H is the typical scale height of the problem, defined as

$$H \equiv \frac{c_s}{\Omega}. \quad (2.19)$$

We solve Eq. (2.18) for ρ , finding the density vertical profile:

$$\rho(R, z) = \rho_0(R) \exp \left[-\frac{z^2}{2H^2} \right], \quad (2.20)$$

which is a Gaussian in the z direction where $\rho_0(R)$ is the density at the disc mid-plane (i.e., $z = 0$).

Radial component: centrifugal equilibrium

The radial projection of the Navier-Stokes equation (2.12) is:

$$v_\phi^2 = \frac{R}{\rho} \frac{dP}{dR} - R \frac{\partial \Phi}{\partial R}, \quad (2.21)$$

where v_ϕ is the angular velocity and this equation describes the centrifugal balance. The term dv_R/dt is not present in Eq. (2.21) because it is negligible compared to the centrifugal term v_ϕ^2/R . Considering the disc as a barotropic fluid, where pressure depends only on density, the fluid sound speed satisfies the relation:

$$c_s^2 = \frac{dP}{d\rho}, \quad (2.22)$$

and therefore the pressure term in Eq. (2.21) can be written as:

$$\frac{R}{\rho} \frac{dP}{dR} = \frac{c_s^2 R}{\rho} \frac{d\rho}{dR} = v_K^2 \left(\frac{H}{R} \right)^2 \frac{d \log \rho}{d \log R}. \quad (2.23)$$

The second term on the right-hand side of Eq. (2.21) is:

$$R \frac{\partial \Phi}{\partial R} = R \frac{GM_\star R}{(R^2 + z^2)^{3/2}} = v_K \left(1 + \frac{z^2}{R^2} \right)^{-3/2}. \quad (2.24)$$

To a first order approximation, we can assume that the density scales with distance from the star as $\rho \propto R^{-n}$ with $n > 0$ and that $z \ll R$, obtaining:

$$v_\phi = v_K \left[1 - n \left(\frac{H}{R} \right)^2 \right]^{1/2}, \quad (2.25)$$

which means that the disc rotational velocity is sub-Keplerian. The term containing H/R gives a small contribution to the total azimuthal velocity, however we will see in Sec. 2.5 that this term is significant when including the dynamics of the dust.

Azimuthal component: angular momentum conservation

We can now consider the azimuthal projection of the Navier-Stokes equation 2.12, assuming an axisymmetric disc and integrating in the vertical direction

$$\Sigma \left(\frac{\partial v_\phi}{\partial t} + \frac{v_R v_\phi}{R} + v_R \frac{\partial v_\phi}{\partial R} \right) = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 T_{R\phi}), \quad (2.26)$$

where there is no contribution from pressure or gravity in this direction, and $T_{R\phi}$ is the vertical integral of the non-vanishing component of the stress tensor (2.15)

$$T_{R\phi} = \nu \Sigma R \frac{\partial \Omega}{\partial R}, \quad (2.27)$$

where ν is the vertical averaged kinematic viscosity. Combining Eq. 2.26 with the continuity equation (2.11), we obtain:

$$\frac{\partial}{\partial t} (\Sigma R v_\phi) + \frac{1}{R} \frac{\partial}{\partial R} (R^3 v_R \Sigma \Omega) = \frac{1}{R} \frac{\partial}{\partial R} (\nu \Sigma R^3 \Omega'), \quad (2.28)$$

where $\Sigma R v_\phi$ is the angular momentum per unit mass. Equation (2.28) describes the redistribution of the angular momentum across the disc. Indeed, the left-hand term of Eq. (2.28) is the Lagrangian derivative of the total angular momentum, while the right-hand term is the variation in the viscous torque exerted on a disc annulus by stresses at the inner and outer edges:

$$\frac{1}{R} \frac{\partial}{\partial R} (\nu \Sigma R^3 \Omega') = \frac{1}{2\pi} \frac{\partial G}{\partial R}, \quad (2.29)$$

where the torque G indicates the flux of angular momentum across an annulus at distance R :

$$G(R) = (2\pi R) \nu \Sigma R \frac{d\Omega}{dR} R. \quad (2.30)$$

Using Eq. (2.28) and (2.11) and considering $\partial_t v_\phi = \partial_t \Omega = 0$, we obtain an expression for the radial velocity:

$$v_R = \frac{1}{R \Sigma \frac{\partial}{\partial R} (R v_\phi)} \frac{\partial}{\partial R} (\nu \Sigma R^3 \Omega') = -\frac{3}{\Sigma R^{1/2}} \frac{\partial}{\partial R} (\nu \Sigma R^{1/2}), \quad (2.31)$$

where we used the assumption of a Keplerian disc. Finally, using v_R in the continuity equation (2.11) we obtain the equation for the surface density of the disc:

$$\frac{\partial \Sigma}{\partial t} = \frac{3}{R} \frac{\partial}{\partial R} \left[R^{1/2} \frac{\partial}{\partial R} (\nu \Sigma R^{1/2}) \right], \quad (2.32)$$

which states that accretion is a diffusion process regulated by viscosity. Equation (2.32) can be written in terms of the mass accretion rate onto the central star:

$$\frac{\partial \Sigma}{\partial t} = \frac{1}{2\pi R} \frac{\partial \dot{M}}{\partial R}, \quad (2.33)$$

where the mass accretion rate is

$$\dot{M} = 6\pi R^{1/2} \frac{\partial}{\partial R} (\nu \Sigma R^{1/2}) = -2\pi R \Sigma v_R, \quad (2.34)$$

where v_R is the radial velocity defined by Eq. (2.31).

2.2.3 Solutions to the disc surface density diffusion equation

The diffusion equation describing the evolution of the disc surface density can be analytically solved in three illustrative cases: in the steady state, the spreading ring, and the self-similar scenarios.

Steady state solution

The stationary scenario, where $\partial \Sigma / \partial t = 0$, the continuity equation implies that the mass accretion rate in Eq. (2.34) is constant, and is positive when material accrete onto the star (negative radial velocity), and negative otherwise. The conservation of angular momentum (Eq. 2.28) in the stationary scenario leads to the definition of the constant quantity

$$C = R^3 v_R \Sigma \Omega - \nu \Sigma R^3 \Omega' \quad (2.35)$$

The angular velocity at the stellar surface must remain well below the Keplerian frequency of the disc, $\Omega_\star \ll \Omega_k \propto R^{-3/2}$, to prevent centrifugal disruption of the star. Therefore, there must be a boundary layer $R_\star + b$ between the stellar surface and the disc inner boundary where $\Omega' = 0$, and hence the viscous torque vanishes, $\pi \nu \Sigma R^3 \Omega' = 0$. Assuming $R_\star \gg b$ and combining Eq. (2.35) and Eq. (2.28), we can write

$$C = \frac{\dot{M}}{2\pi} \sqrt{GM_\star R_{\text{in}}}. \quad (2.36)$$

We highlight that in protoplanetary discs we do not expect a boundary layer but rather magnetospheric accretion. However, for this simple derivation, we can use the *no-torque*

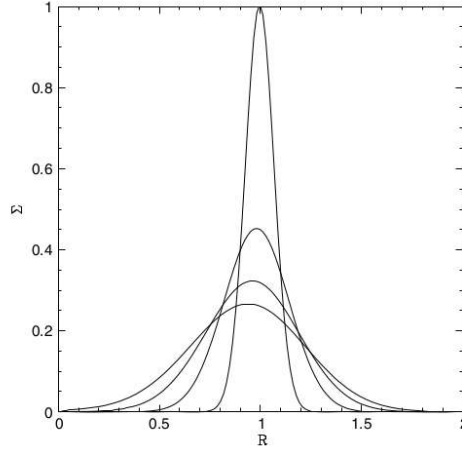


Figure 2.2: Evolution of the disc surface density predicted by the spreading ring solution (Eq. 2.39) considering $\tau = [0.01, 0.05, 0.1, 0.1]$. From Lodato (2008).

boundary condition. Using $\dot{M} = -2\pi\nu_R\Sigma$ in Eq. (2.36), we obtain the steady state solution for the surface density of the disc

$$3\pi\nu\Sigma = \dot{M} \left(1 - \sqrt{\frac{R_{\text{in}}}{R}}\right). \quad (2.37)$$

If $R \gg R_*$ we can write the relation $3\pi\nu\Sigma = \dot{M}$ and, since $\dot{M}$ is constant, $\nu\Sigma$ is also constant. We notice that, in this last case, an high value of disc density corresponds to a small viscosity. Although the steady state solution does not provide an accurate description, because in reality discs evolve with time, it is useful to derive basic properties of protoplanetary discs. For example, it has been used to provide an estimate of the viscosity given the observed mass accretion rate (e.g., Andrews et al. 2009).

Spreading ring solution

Considering a time-dependent disc surface density, the diffusion equation can be analytically solved in the simplest case by assuming that all the disc mass is concentrated in an infinitesimal annulus of mass M_d at the position R_0 :

$$\Sigma(R, t = 0) = \frac{M_d}{2\pi R_0} \delta(R - R_0), \quad (2.38)$$

where $\delta(R - R_0)$ is the Dirac function. We can consider that the viscosity is a function of the radius $\nu \propto r^\gamma$ and, for simplicity, that ν is a constant function of the radius. The solution of the diffusion equation (2.32) under this initial condition is (Lynden-Bell & Pringle 1974):

$$\Sigma(x, \tau) = \frac{M_d}{\pi R_0^2} \frac{x^{1/4}}{\tau} \exp\left(-\frac{1+x^2}{\tau}\right) I_{1/4}\left(\frac{2x}{\tau}\right), \quad (2.39)$$

where $x \equiv R/R_0$, $\tau \equiv 12\nu t R_0^{-2}$ and $I_{1/4}$ is a modified Bessel function of the first kind. This solution is known as *spreading ring solution* because the initial mass contained in an infinitesimal ring spreads inward and outward due to the viscous forces. In order to

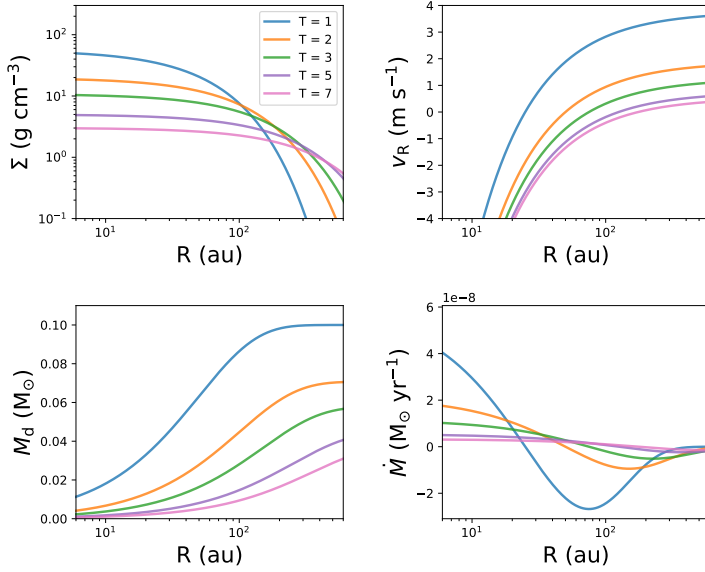


Figure 2.3: Time evolution of disc surface density (Σ), radial velocity (v_R), disc mass (M_d), and accretion rate ($\dot{M}$) following the self similar solution. The results refer to an initial disc mass of $M_0 = 0.1 M_\odot$, characteristic radius $R_c = 50$ au, and $\gamma = 1$.

allow accretion and conservation of the total angular momentum, a small amount of material must be transported outward. The radius where the flow reverts its direction, the transition radius, is the radius R_t satisfying $2(R_t/R_0) \sim t/t_\nu$, where $t_\nu = R_0^2/12\nu$ is the viscous timescale, so that $\tau = t/t_\nu$ is an increasing function of time. Figure 2.2 shows the evolution of the disc surface density as described by the spreading ring solution, with time and across the disc radial extent.

Self-similar solution

Another interesting solution of the diffusion equation can be obtained assuming that the viscosity is a power-law of radius, $\nu \propto R^\gamma$. The initial profile of the surface density is often set to a power-law exponentially truncated at a critical radius R_c :

$$\Sigma(R, t = 0) = \frac{C}{3\pi\nu(R)} \exp \left[- \left(\frac{R}{R_c} \right)^{2-\gamma} \right], \quad (2.40)$$

where C is a normalization constant. The viscosity can be written in terms of the critical radius as

$$\nu = \nu_c \left(\frac{R}{R_c} \right)^\gamma, \quad (2.41)$$

where $\nu_c = \nu(R_c)$. Given these initial conditions, the solution to the diffusion equation is known as *self-similar* (Lynden-Bell & Pringle 1974) and is

$$\Sigma(R, T) = \frac{C}{3\pi\nu(R)} T^{-\frac{5/2-\gamma}{2-\gamma}} \exp \left[- \frac{1}{T} \left(\frac{R}{R_c} \right)^{2-\gamma} \right], \quad (2.42)$$

where $T = 1 + \tau/\tau_\nu$, and τ_ν is the typical timescale of the problem, known as the viscous timescale and defined as a function of the critical radius by the relation

$$\tau_\nu = \frac{1}{3(2-\gamma)^2} \frac{R_c^2}{\nu_c}. \quad (2.43)$$

This means that while the total disc mass decreases, because material is accreted onto the star, the disc undergoes viscous spreading and the critical radius moves outward. The name self-similar is due to the similarity in the shape of the solution as time passes and the disc expands. The disc mass enclosed in a radius R varies with time following

$$M_d(R, T) = \frac{C2R_c^2}{3\nu_c(2-\gamma)} T^{-\frac{1}{2(2-\gamma)}} \left\{ 1 - \exp \left[-\frac{2}{T} \left(\frac{R}{R_c} \right)^{2-\gamma} \right] \right\}. \quad (2.44)$$

The total disc mass can be computed by integrating the surface density from 0 to infinity, obtaining

$$M_d(T) = 2\pi \int_0^{+\infty} \Sigma(R, T) dR = \frac{C2R_c^2}{3\nu_c(2-\gamma)} T^{-\frac{1}{2(2-\gamma)}}. \quad (2.45)$$

Defining the initial disc mass as $M_0 = C2R_c^2/3\nu_c(2-\gamma)$, the disc mass can be written as a power-law of time

$$M_d(T) = M_0 T^{-\frac{1}{2(2-\gamma)}}, \quad (2.46)$$

and using this expression in the self-similar solution we obtain

$$\Sigma(R, T) = \frac{M_{d,0}(2-\gamma)}{2\pi R_c^2} T^{-\frac{5/2-\gamma}{2-\gamma}} \exp \left[-\frac{1}{T} \left(\frac{R}{R_c} \right)^{2-\gamma} \right], \quad (2.47)$$

which is the product of the surface density of a disc with mass $M_{d,0}$ enclosed in a radius R_c and a time dependent part. For a better visualisation of the motion of the disc material, we can write the radial velocity using Eq. (2.31), which reads

$$v_R(R, T) = -\frac{3\pi\nu_c}{R} \left[1 - \frac{2}{T} \left(\frac{R}{R_c} \right)^{2-\gamma} \right], \quad (2.48)$$

where the inner disc regions present positive velocity, inward motion, meaning that they are undergoing accretion, while disc material in the outermost regions show negative velocities, moving outward. The transition radius between inward and outward moving flow is $R_{tr} = R_c/2^{2-\gamma}$ and increases with time. Therefore, the accretion mechanism in a viscous disc requires that a small fraction of the material is transported outwards, redistributing angular momentum and leading to an overall disc spreading. The accretion rate associated with a self-similar surface density is

$$\dot{M} = \frac{3M_{d,0}\nu_c(2-\gamma)}{2R_c^2} \exp \left[-\frac{1}{T} \left(\frac{R}{R_c} \right)^{2-\gamma} \right] \left[1 - \frac{2}{T} \left(\frac{R}{R_c} \right)^{2-\gamma} \right] T^{-\frac{5/2-\gamma}{2-\gamma}}, \quad (2.49)$$

which is positive at smaller radii and negative at larger radii, indication of accretion and outward migration. Figure 2.3 shows the evolution of the disc surface density, the radial velocity, the disc mass, and the accretion rate with time and across the radial extent of the disc, as obtained by following the self similar solution.

2.3 Active and passive discs

Protoplanetary discs can be distinguished into two categories based on the dominant heating mechanism. In active discs, most of the disc luminosity is caused by accretion, while passive discs are mainly heated by radiation from the central star. The total luminosity in the case of an active disc is called accretion luminosity

$$L_{\text{acc}} = \frac{GM_{\star}\dot{M}}{2R_{\star}}, \quad (2.50)$$

which is caused by material flowing from larger to smaller radii. The luminosity caused by stellar irradiation is

$$L_{\text{irr}} = \frac{L_{\star}}{4}. \quad (2.51)$$

Depending on the mass accretion rate, discs are more prone to being active or passive. In particular, an active disc satisfies

$$\dot{M} \gtrsim \frac{L_{\star}R_{\star}}{2GM_{\star}} \sim 10^{-8} \left(\frac{L_{\star}}{L_{\odot}} \right) \left(\frac{R_{\star}}{R_{\odot}} \right) \left(\frac{M_{\odot}}{M_{\star}} \right) M_{\odot} \text{ yr}^{-1}. \quad (2.52)$$

Since accretion rates of T Tauri stars typically cover the range $\sim 10^{-10} - 10^{-6} M_{\odot} \text{ yr}^{-1}$ (e.g., Manara et al. 2023a), discs are neither active nor passive, and both internal and external heating contribute.

2.3.1 Temperature profile

During the redistribution of angular momentum across the disc, a certain amount of gravitational potential energy is dissipated by viscous torques. The energy dissipated per unit time and surface by both sides of the disc is

$$D(R) = 2 \frac{GM\dot{M}}{4\pi R^3} = \nu \Sigma (R\Omega')^2 = \frac{9}{4} \nu \Sigma \Omega^2, \quad (2.53)$$

which is written in the specific case of a Keplerian disc. Assuming a stationary disc (Eq. 2.37) extended to infinity, the total luminosity of the disc is

$$L_{\text{tot}} = \int_{R_{\text{in}}}^{+\infty} 2\pi R D(R) dR = \int_{R_{\text{in}}}^{+\infty} \frac{3GM\dot{M}}{2R^2} \left(1 - \sqrt{\frac{R_{\text{in}}}{R}} \right) dR = \frac{1}{2} \frac{GM\dot{M}}{R_{\text{in}}}, \quad (2.54)$$

which is only half of the potential energy dissipated by accretion. The other half remains as kinetic energy to support the Keplerian rotation of the disc. Assuming that the disc dissipates energy as a black body, $D(R)$ is the flux emitted at a radius R and can be written following the Stefan-Boltzmann law as $D(R) = 2\sigma_{\text{SB}}T^4$, where the factor 2 accounts for the emission of both sides of the disc. The profile of the surface temperature of the disc can be evaluated considering this last expression and Eq. (2.53)

$$T^4(R) = \frac{3GM\dot{M}}{8\pi\sigma_{\text{SB}}R^3} \left(1 - \sqrt{\frac{R_{\text{in}}}{R}} \right), \quad (2.55)$$

which does not directly depend on viscosity. However, viscosity indirectly affects the temperature profile as it influences the accretion rate.

An important contribution to the disc heating is provided by the radiation from the central star. It has been demonstrated that the temperature profile of a passive and flat disc scales as

$$T(R) = T_{\star} \left(\frac{R}{R_{\star}} \right)^{-3/4}. \quad (2.56)$$

However, this cannot be the temperature profile of a flat disc because the disc thickness H is not constant with R :

$$H = \frac{c_s}{\Omega} \propto T^{1/2} R^{3/2} \propto R^{9/8}, \quad (2.57)$$

which defines instead a flared disc. Self-consistent models show that if dust and gas are perfectly coupled in a disc, the temperature profile is described by the relation $T \propto R^{-1/2}$ (Adams & Shu 1986; Kenyon & Hartmann 1987).

2.4 Viscosity

From the previous theoretical description of gas evolution in protoplanetary discs, it is evident that viscosity plays a fundamental role in the accretion process. Performing a dimensional analysis of the diffusion equation for the disc surface density (Eq. 2.32), we find that the viscous evolution timescale is $\tau_{\nu} = R^2/\nu$. Comparing with the dynamical timescale, $\tau_{\text{dyn}} = \Omega^{-1}$, which is the disc orbital time at a radius R , we obtain

$$\frac{\tau_{\nu}}{\tau_{\text{dyn}}} = \frac{R^2 \Omega}{\nu}, \quad (2.58)$$

which defines the Reynolds number, Re , of the flow and measures whether a flow is dominated by turbulent, chaotic motions or a viscous, smooth behaviour: a low Re indicates that viscous forces are dominant and the fluid is characterised by a laminar flow, while a high Re suggests turbulent motions in the flow. In this section we investigate whether a collisional or turbulent viscosity could be responsible for the expected transport of angular momentum, and how to estimate the magnitude of viscosity.

2.4.1 Collisional viscosity

Considering the protoplanetary disc as a molecular disc, we can describe the macroscopic exchange of angular momentum as a result of the microscopic motion of particles. In this case, the origin of viscosity lies in the molecular collision between the particles. The collisional viscosity is proportional to the mean free path of the particles, and to their mean thermal velocity, $\nu_{\text{coll}} \propto \lambda_{\text{mfp}} v_{\text{th}}$. For $\lambda_{\text{mfp}} = \mu m_{\text{p}} / \rho \sigma_{\text{coll}}$ and $v_{\text{th}} = c_s$, we can write

$$\nu_{\text{coll}} = \frac{\mu m_{\text{p}}}{\rho \sigma_{\text{coll}}} c_s = \frac{\mu m_{\text{p}}}{\Sigma \sigma_{\text{coll}}} \Omega, \quad (2.59)$$

where μ is the average molecular weight, m_{p} is the proton mass, $\rho = \Sigma/H$ is the volume density of particles in the disc, σ_{coll} is the collision cross section, and we used $H = c_s/\Omega$. The viscous timescale for a typical disc where the viscosity is supposed to be collisional reads as

$$\tau_{\nu} = \frac{R^2}{\nu_{\text{coll}}} = \left(\frac{H}{R} \right)^{-2} \frac{\Sigma \sigma_{\text{coll}}}{\mu m_{\text{p}}} \frac{1}{\Omega}. \quad (2.60)$$

Evaluating the timescale for typical values of protoplanetary discs, $\Sigma \sim 10 \text{ g/cm}^2$, $H/R \sim 10^{-2}$, $\mu \sim 1$, the typical hydrogen cross section $\sigma_{\text{coll}} \sim 10^{-16} \text{ cm}^2$, $m_{\text{p}} \sim 10^{-24} \text{ g}$, and

$\Omega^{-1} \sim 100$ yr, we find $\tau_\nu \sim 10^{15}$ yr, which is larger than the Hubble time. So, since we have observational evidence of already formed planetary systems, the collisional viscosity cannot be responsible for the exchange of angular momentum in accretion discs, and we have to consider other processes.

2.4.2 Turbulent viscosity

Turbulence produces a violent mixing of fluid elements, leading to an exchange of angular momentum, and therefore this mechanism can be a potential source of the accretion process. We address this problem by assuming that there is a mechanism that extracts energy from the mean flow and injects it into the fluctuations, and we evaluate the torque exerted by these fluctuations to measure the flux of angular momentum. Specifically, we can write the Navier-Stokes equation considering the velocity fluctuations $\mathbf{v} + \delta\mathbf{v}$ and separating the mean flow motion $\mathbf{v}$ from the fluctuating component $\delta\mathbf{v}$

$$\frac{\partial}{\partial t}(\Sigma R v_\phi) + \frac{1}{R} \frac{\partial}{\partial R}[R v_R \Sigma R v_\phi] = - \sum_i \frac{1}{R} \frac{\partial}{\partial R}[\Sigma R^2 \langle \delta v_R^{(i)} \delta v_\phi^{(i)} \rangle], \quad (2.61)$$

where the angle brackets indicate a vertical and azimuthal average, and the sum in the right-hand term of the equation is over the fluctuating fields (e.g., fluctuations of the velocity field, the magnetic field, the gravitational field). Considering a hydrodynamic flow where only velocity fluctuations are relevant, the stress tensor is the Reynolds stress

$$T_{R\phi}^{\text{Re}} = -\Sigma \langle v_R^{\text{Re}} v_\phi^{\text{Re}} \rangle, \quad (2.62)$$

where v^{Re} is the velocity fluctuation. In the case of a magnetized disc, the stress tensor is the Maxwell stress

$$T_{R\phi}^{\text{M}} = -\Sigma \langle v_R^{\text{A}} v_\phi^{\text{A}} \rangle, \quad (2.63)$$

where $v^{\text{A}} = B/\sqrt{4\pi\rho}$ is the Alfvén velocity. Instead, if the disc is self-gravitating, the perturbations of the gravitational field provide a stress in the form

$$T_{R\phi}^{\text{g}} = -\Sigma \langle v_R^{\text{g}} v_\phi^{\text{g}} \rangle, \quad (2.64)$$

where $v^{\text{g}} = \Phi/\sqrt{4\pi G\rho}$ is the velocity associated with the gravitational potential. Turbulence is driven by disc instabilities such as hydrodynamic, gravitational, or magneto-hydrodynamic, and produces a stress that acts analogously to the viscous stress tensor $T_{R\phi}$. This means that the turbulent fluctuations effectively provide the mechanism for transporting angular momentum across the disc. Although the nature of turbulence and the processes dominating the angular momentum transport during disc lifetime are still debated, it is well established that the presence of turbulence in discs has strong implications on the planet formation process, influencing the radial diffusion of dust grains (Clarke & Pringle 1988), the collisional dust growth (Fromang & Papaloizou 2006; Birnstiel et al. 2010a), and the migration of planetary bodies (Nelson & Papaloizou 2004). The magnitude of the turbulent viscosity is estimated by Shakura & Sunyaev (1973) using an elegant parametrisation. From a dimensional analysis, the kinematical viscosity ν is proportional to the product of a length and a velocity. Therefore, we can write the turbulent viscosity as $\nu \sim lv$, where l is the length scale of turbulence, and v is the typical turbulent velocity. Since l has to be necessarily smaller than the disk thickness H ($l < H$), assuming isotropic turbulence, and v cannot be supersonic ($v < c_s$), otherwise

it would dissipate due to shocks, the turbulent viscosity can be expressed through the prescription

$$\nu = \alpha c_s H, \quad (2.65)$$

where α must verify $\alpha \leq 1$. Writing the viscous evolution timescale considering the α -prescription, we obtain

$$\tau_\nu = \frac{1}{\alpha} \left(\frac{H}{R} \right)^{-2} \tau_{\text{dyn}}, \quad (2.66)$$

where, under the assumption of a thin disc, $H/R \ll 1$, and considering $\alpha < 1$, we obtain that disc evolution happens on a much longer timescale than the orbital timescale, $t_\nu \gg t_{\text{dyn}}$.

Although the parameter α is usually assumed to be constant, in general it may depend on the physical properties of the disc (e.g., radius, density, temperature). Most observational constraints and population synthesis models agree that discs are not highly turbulent and that $\alpha < 10^{-2}$. Empirical and observational constraints lead to a typical values of α in the range $3 \times 10^{-3} - 3 \times 10^{-4}$ (see Rosotti 2023 for a review). Since the α -prescription of Shakura & Sunyaev (1973) is simply a parametrisation, it should not be considered as a theory of the nature of the viscosity as the unknown variable only shifts from ν to α .

2.5 Dust dynamics

Although the $\sim 99\%$ of the total protoplanetary disc mass consists of gas, dust particles play an extremely important role not only in the process of planet formation, but also in determining the structure, evolution, and chemical composition of the disc. Dust is the dominant source of opacity in discs, and therefore the easiest component to observe. The stellar radiation, which is reprocessed by dust grains and observed at different wavelengths, provides insight into the dimension and spatial distribution of dust in discs. Figure 2.4 summarises the main processes acting on dust grains in protoplanetary discs, and some of the observational facilities that are used (or will be used, in the case of the ELT) to trace dust of different size, and to probe the disc stratification. In order to form terrestrial planets, μm -size particles coming from the ISM need to grow of at least 12 orders of magnitude in size. Therefore, understanding the main processes governing dust dynamics in discs is essential.

In this Section we describe the main processes related to the dynamics of the dust in protoplanetary discs and the interaction between the gas and dust components.

2.5.1 Aerodynamic drag force

Dust particles moving through a fluid medium experience the aerodynamic drag from the gas (and vice-versa), which tends to decelerate dust particles until the relative velocity vanishes. The aerodynamic drag force for a spherical dust particle of radius a , density of the material ρ_m , and velocity relative to the gas $\mathbf{v}_{\text{rel}}$ can be written as

$$\mathbf{F}_{\text{drag}} = -\frac{C_D}{2} \pi a^2 \rho v_{\text{rel}} \mathbf{v}_{\text{rel}}, \quad (2.67)$$

where the drag coefficient C_D depends on the drag regime. In the Epstein regime, the size of the dust particles is smaller than the mean free path of the gas molecules,

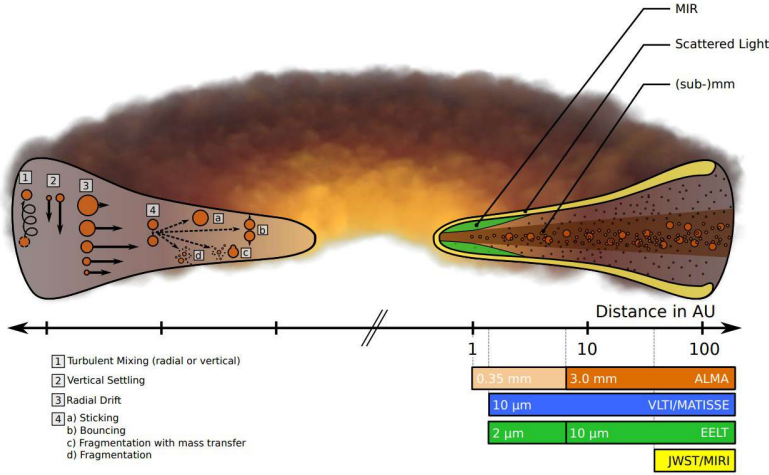


Figure 2.4: Illustration of the structure, grain evolution processes and observational constraints for protoplanetary disks. On the left side, we listed the main grain transport and collision mechanism properties. The different lengths of the arrows illustrate the different velocities of the different grains. On the right hand side, we show the areas of the disk that can be probed by the various techniques and facilities. The axis shows the logarithmic radial distance from the central star. The horizontal bars show the highest angular resolutions (left edge of the bars) that can be achieved with a set of upcoming facilities and for at the typical distance of the nearest star forming regions. From Testi et al. (2014).

$a \lesssim \lambda_{\text{mfp}}$, meaning that the gas and dust particles randomly collide and the velocity distribution of the fluid is Maxwellian. In this case, the drag coefficient is

$$C_D = \frac{8v_{\text{th}}}{3v_{\text{rel}}}, \quad (2.68)$$

where $v_{\text{th}} = \sqrt{8\kappa_B T / \pi \mu m_p}$ is the thermal speed of the gas molecules with temperature T . In the Stokes regime, where $s \gtrsim \lambda$, the gas flows around the dust particles and the drag coefficient depends on the Reynolds number associated with the particles, $\text{Re} = 2av_{\text{rel}}/\nu_{\text{m}}$, where ν_{coll} is the collisional viscosity. Specifically, the drag coefficient is (Weidenschilling 1977a)

$$C_D \simeq \begin{cases} 24\text{Re}^{-1} & \text{Re} < 1 \\ 24\text{Re}^{-0.6} & 1 < \text{Re} < 800 \\ 0.44 & \text{Re} > 800. \end{cases} \quad (2.69)$$

Epstein and Stokes drags have the same intensity for particles of size $a = 9\lambda/4$. An alternative empirical formulation for the drag coefficient is given by (Cheng 2009):

$$C_D = \frac{24}{\text{Re}} (1 + 0.27\text{Re})^{0.43} + 0.47[1 - \exp(-0.04\text{Re}^{0.38})]. \quad (2.70)$$

A comparison between the drag coefficient in various regimes is presented in Fig. 2.5. To quantify the coupling between gas and dust particles, we define a typical timescale, the

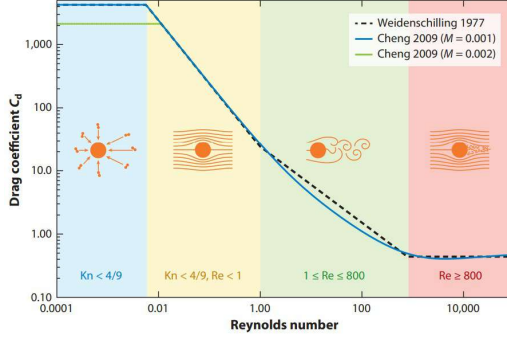


Figure 2.5: Drag coefficient in different drag regimes as a function of the Reynolds number. The drag regimes are separated on the basis of the ratio $Kn = \lambda/a$, which is known as the Knudsen number. Credits: Bagheri et al. (2016), Birnstiel (2024).

stopping time, which measures the time in which the drag modifies the relative velocity

$$\tau_{\text{stop}} = \frac{m|\mathbf{v}_{\text{rel}}|}{|F_{\text{drag}}|} = \frac{8a}{3C_D v_{\text{rel}}} \left(\frac{\rho_m}{\rho} \right), \quad (2.71)$$

where $m = (4/3)\pi a^3 \rho_m$ is the mass of the dust particle with density ρ_m , while ρ is the gas density. This timescale is short for small particles, meaning that they are strongly coupled to the gas. Indeed, for typical values of $\rho_m \sim 3 \text{ g cm}^{-3}$, $\rho \sim 10^{-9} \text{ g cm}^{-3}$, and in the Epstein regime with $v_{\text{th}} \sim 10^5 \text{ cm s}^{-1}$ and $a = 1 \text{ } \mu\text{m}$, the stopping time is $\tau_{\text{stop}} \sim 3 \text{ s}$, which is much lower than the dynamical timescale $\tau_{\text{dyn}} = \Omega^{-1}$. Thus, it is useful to define the Stokes number

$$\text{St} = \frac{\tau_{\text{stop}}}{\tau_{\text{dyn}}}. \quad (2.72)$$

Since in typical protoplanetary discs the gas mean free path is much larger than the dust particle size, we can write the Stokes numbers in the Epstein regime

$$\text{St} = \gamma^{1/2} \frac{\pi \rho_0 a}{2\Sigma(R)} \exp \left[\frac{z^2}{2H^2} \right] = \text{St}_{\text{mid}}(R) \exp \left[\frac{z^2}{2H^2} \right], \quad (2.73)$$

where γ is the adiabatic index, we used the relation $\rho = \Sigma/\sqrt{2\pi H^2} \exp[z^2/2H^2]$, and St_{mid} indicates the Stokes number at the disc mid-plane. For small dust particles and/or small gas density, the Stokes number is small and therefore dust dynamics follows the gas one. It is important to underline that dust is usually considered as pressureless, as the frequency of collisions between dust particles is much smaller compared to collisions between dust and gas particles. Moreover, we consider the drag force on the dust by the gas, but we neglect the back-reaction of the dust on the gas. Although being a good approximation for the small dust-to-gas ratios of $\sim 10^{-2}$ typically considered in discs, it is important to keep in mind that in specific cases the back-reaction can become non negligible for describing the evolution of discs (e.g., when the dust disc is settled into a thin midplane than the gas disc).

Considering a geometrically thin, axisymmetric, non-magnetic, non-self-gravitating, viscous, and vertically isothermal disc where dust and gas are well mixed, we can write

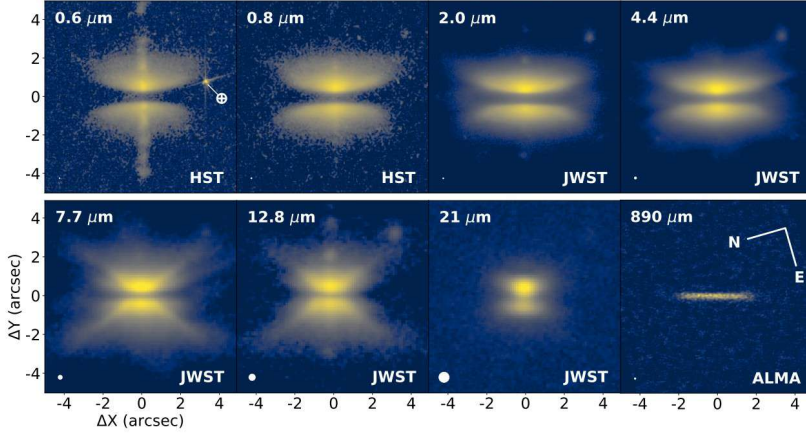


Figure 2.6: Images of the protoplanetary disc around the star 2MASS J04202144+2813491, seen in different wavelengths with HST, JWST, and ALMA, which are tracing different particle sizes. Dust settling for larger grains is evident comparing the images. Credits: Duchêne et al. (2024).

the fluid equations for dust and gas (Nakagawa et al. 1986)

$$\frac{\partial \Sigma_g}{\partial t} + \nabla \cdot (\Sigma_g \mathbf{v}_g) = 0 \quad (2.74)$$

$$\frac{\partial \Sigma_d}{\partial t} + \nabla \cdot (\Sigma_d \mathbf{v}_d) = 0 \quad (2.75)$$

$$\frac{\partial \mathbf{v}_g}{\partial t} + (\mathbf{v}_g \cdot \nabla) \mathbf{v}_g = -\frac{1}{\rho_g} (\nabla P - \nabla \cdot \sigma) - \nabla \Phi - \frac{1}{\rho_g} \mathbf{F}_{\text{drag}}^V, \quad (2.76)$$

$$\frac{\partial \mathbf{v}_d}{\partial t} + (\mathbf{v}_d \cdot \nabla) \mathbf{v}_d = -\nabla \Phi - \frac{1}{\rho_d} \mathbf{F}_{\text{drag}}^V, \quad (2.77)$$

where $\mathbf{F}_{\text{drag}}^V$ is the drag force per unit volume and we included in the Navier-Stokes equation for the gas component the back-reaction of the dust, which is negligible at smaller dust-to-gas ratios. The aerodynamic coupling in discs affects dust dynamics causing two main motions: vertical settling and radial drift.

2.5.2 Vertical settling

Dust affected by aerodynamic drag due to the presence of gas, tends to settle on the midplane. In order to explore this process, we consider the equation of motion for dust particles (Eq. 2.77) assuming that the back-reaction is negligible, in cylindrical coordinates

$$\frac{\partial v_{d,z}}{\partial t} = -\Omega_k^2 z + \frac{v_{g,z} - v_{d,z}}{\tau_{\text{stop}}}, \quad (2.78)$$

which is the Langevin equation and, considering a stationary flow (i.e., $v_{g,z} = 0$) and working close to the mid-plane ($z \ll H$), the above equation becomes

$$\frac{d^2 z}{dt^2} + \frac{1}{\tau_{\text{stop}}} \frac{dz}{dt} + \Omega_k^2 z = 0, \quad (2.79)$$

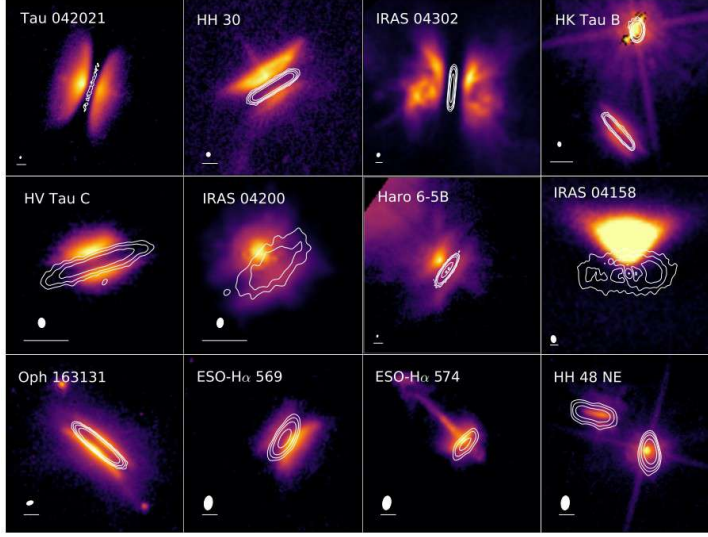


Figure 2.7: Gallery of edge-on discs as seen in scattered light (colours, in logarithmic scales) and ALMA band 7 continuum images (5, 10, 20, and 50 σ contours). Credits: Villenave et al. (2020).

which means that the motion in the vertical direction around the mid-plane is represented by a damped harmonic oscillator. We solve the equation by substituting $t = \tau_{\text{dyn}}/\Omega_k$ and using the ansatz $z \propto \exp(\lambda\tau)$, obtaining the two solutions

$$\lambda_{1,2} = -\frac{1}{2\text{St}} \left(1 \pm \sqrt{1 - (2\text{St})^2} \right), \quad (2.80)$$

where two classes of solutions can be distinguished. When $\text{St} < 1/2$, dust tends to be decoupled from gas, $\lambda_{1,2}$ are real and $z(t)$ is an over damped oscillation, while for $\text{St} > 1/2$, $\lambda_{1,2}$ are complex and $z(t)$ is an under damped oscillation. In the case $\text{St} = 1/2$, the motion is not oscillatory and reaches the stationary state on an orbital timescale. Considering the three cases, the settling timescale can be approximated as (e.g. Youdin & Lithwick 2007a)

$$\Omega_k t_{\text{settle}} \simeq \text{St} + \frac{1}{\text{St}}, \quad (2.81)$$

where in typical disc conditions, strongly coupled and decoupled particles will settle onto the mid-plane with over damped and under damped oscillations, respectively, in a few hundreds of years, while in the case of $\text{St} = 1/2$, the settling timescale is much shorter and particles will move faster towards the mid-plane. Considering Eq. (2.73), where z is now expressed as a function of time, since the gas density is lower at the surface layers and higher around the mid-plane, μm -size particles can present larger Stokes number at the disc surface and smaller Stokes number at the mid-plane. Therefore, from Eq. (2.73), while dust particles settle to the mid-plane, their Stokes number would decrease and they would decelerate their motion. We expect that, even in the presence of mixing due to turbulence in discs, the uppermost layers would be depleted quickly. Since the solution to Eq. (2.79) depends on the Stokes number, which depends on the grain size, which means that particles of different sizes will settle to different vertical heights. In particular, larger dust grains will settle closer to the midplane compared to smaller grains. Dust settling can be observationally verified by observing edge-on

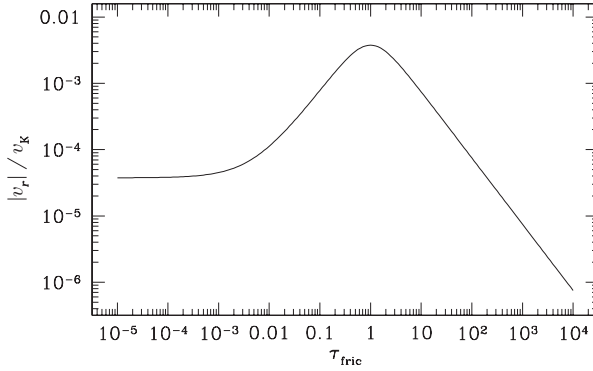


Figure 2.8: Radial velocity of dust particles at the mid-plane as a function of the stopping time ($\tau_{\text{fric}} = \Omega_k \tau_{\text{stop}}$), considering $\eta = 7.5 \times 10^{-3}$. The radial velocity reaches its maximum at $\text{St} = 1$.

protoplanetary discs at different wavelengths. Indeed, different wavelengths trace dust particles of different sizes and, considering that wavelength scales linearly with grain size, in Fig. 2.6 and 2.7 we show that larger dust grains are typically found closer to the mid-plane.

2.5.3 Radial drift

The aerodynamic drag force acting between gas and dust is responsible for the radial motion of the dust particles. This motion originates mainly as an implication of the sub-Keplerian rotation of the gas (Eq. 2.25), and the nature of the two components. Indeed, gas experiences the pressure, which makes the azimuthal velocity sub-Keplerian, and the viscosity, which produces a negative radial velocity. Dust is pressureless and inviscid, rotating in a Keplerian motion. Therefore, since solid particles are embedded in a fluid rotating slowly, they will experience a headwind of intensity that varies according to their size. The headwind will make the particles to lose angular momentum, and they will decelerate and migrate inward. Since this motion is a consequence of the drag force, which is proportional to the relative velocity between gas and dust, solid particles will migrate inward or outward depending on the sign of the relative velocity. The sub-Keplerian azimuthal velocity of the gas can be written as

$$v_{\phi,g} = v_k (1 - \eta)^{1/2}, \quad (2.82)$$

where $\eta = nc_s^2/v_k^2$. The problem can be described by two equations, in the radial and azimuthal direction

$$\frac{dv_R}{dt} = \frac{1}{R}(v_\phi^2 - v_k^2) - \frac{1}{\tau_{\text{stop}}}(v_R - v_{R,g}), \quad (2.83)$$

$$\frac{d(Rv_\phi)}{dt} = -\frac{R}{\tau_{\text{stop}}}(v_\phi - v_{\phi,g}), \quad (2.84)$$

where v_R and v_ϕ are the radial and azimuthal components of the dust velocity, and $v_{R,g}$ and $v_{\phi,g}$ are the radial and azimuthal components of the gas velocity. Assuming that the gas azimuthal velocity is nearly Keplerian, we can write

$$\frac{d(Rv_\phi)}{dt} \simeq v_R \frac{d(Rv_k)}{dR} = \frac{1}{2} v_R v_k, \quad (2.85)$$

and use this into the second equations. Moreover, considering the steady state solution in which the time derivative of the first equation goes to zero, and approximating $v_\phi + v_k \simeq 2v_k$, we obtain the equation for the radial dust velocity

$$v_R = -\frac{v_{R,g} + 2\eta v_k St}{1 + St^2}, \quad (2.86)$$

and the azimuthal dust velocity

$$v_\phi = \frac{St^2}{1 + St^2} \left[v_k - v_{\phi,g} \left(1 - \frac{1 + St^2}{St^2} \right) \right]. \quad (2.87)$$

For $St \ll 1$ (i.e., strongly coupled dust particles) the dust radial and azimuthal velocities tend to those of the gas, which means that dust grains will be transported with the gas and will migrate inward, accreting onto the central star. For large particles, $St \gg 1$, the dust radial velocity goes to zero, which means that the drag force becomes negligible as the dust is completely decoupled from the gas. If instead $St = 1$, the dust radial velocity is $v_R \simeq 2\eta v_k \simeq (H/R)2v_k$ and the drift timescale is

$$\tau_{\text{drift}} = \frac{R}{v_R} \simeq \left(\frac{H}{R} \right)^{-2} \frac{1}{\Omega_k} = 100 \left(\frac{H/R}{0.1} \right)^{-2} \left(\frac{R}{1 \text{ au}} \right)^{-3/2} \text{ yr}, \quad (2.88)$$

which is extremely small compared to the typical disc lifetime of a few Myr. Two fundamental conclusions follow from the results obtained. First of all, planetesimal formation must be rapid, otherwise most of the dust will migrate inward leaving not sufficient material to form planets. Additionally, radial redistribution of solids is very likely to occur. In particular, considering collisions between dust particles, it can be demonstrated that dust grains smaller than a certain critical size (which is on the order of ~ 2 m for typical disc parameters), would have time to grow via collisions before migrating inward (Armitage 2013). This problem is commonly known with the name *meter-size barrier* problem, and how to overcome radial drift and allow planets to form remains highly debated.

2.5.4 Smoluchowski equation for coagulation and fragmentation of dust grains

Dust particles in protoplanetary discs collide at a rate that depends on the kinetics and cross-sections. Dust collisions can lead to coagulation, bouncing, or fragmentation of the grains (for a review, see Birnstiel 2024). The collisional evolution of a distribution of grains with mass m and number density $n(m)$ can be described by the generalised Smoluchowski equation

$$\begin{aligned} \frac{\partial n(m)}{\partial t} = & \int_0^\infty \int_0^{m'} K(m, m', m'') R(m', m'') \times n(m') n(m'') dm' dm'' \\ & - n(m) \int_0^\infty R(m, m') n(m') dm', \end{aligned} \quad (2.89)$$

which has a positive and negative contribution. The positive contribution is the sum over all possible collisions of particles with masses m' and m'' at a collision rate $R(m', m'')$. The matrix $K(m, m', m'')$ holds information about how much of a single collision of particles with masses m' and m'' is added to the final $n(m)$. A perfectly sticking collision

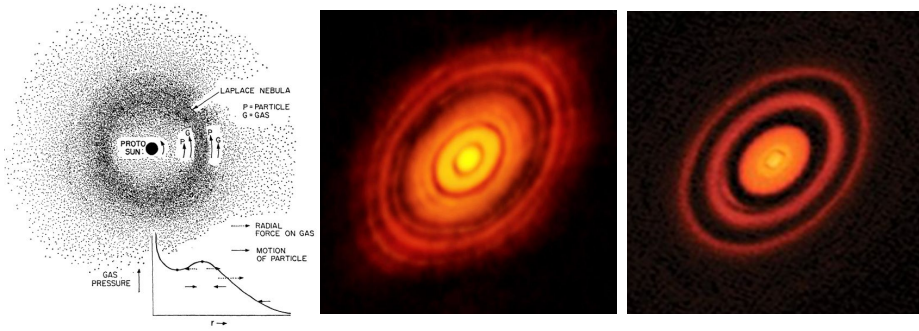


Figure 2.9: Left panel: representation of entrainment of dust grains in pressure/density maxima from Whipple (1972). Middle panel: ALMA continuum image of HL Tau. Right panel: ALMA continuum image of HD 163296. Credits: ALMA (ESO/NAOJ/NRAO).

would be described by a Dirac delta $K(m, m', m'') = \delta(m - (m' + m''))$. The negative term in Eq. (2.89) accounts for the particles that get removed from the distribution because they have collided with other particles. The collision rate $R(m', m'')$ is the product of the collision cross-section and the relative velocity of the colliding particles. The probability of having coagulation, bouncing, or fragmentation, is velocity dependent and is described by the form of K . Wada et al. (2013) provided the critical collision velocity above which collisions changes from growth positive (e.g., coagulation) to negative (e.g., fragmentation):

$$v_{\text{crit}} = C \left(\frac{a_0}{0.1 \mu\text{m}} \right), \quad (2.90)$$

where the constant C is $\sim 80 \text{ m s}^{-1}$ for water ice and $\sim 8 \text{ m s}^{-1}$ for silicates. In general, the number density of dust particles in discs depends not only on the mass, but also on the vertical height above the midplane. However, coagulation processes are most important near the midplane, which allows us to approximate the kernels as being vertically constant (using the values at the mid-plane). This motivates the shape of Eq. (2.89). For a detailed investigation on dust collision, and the multiple outcomes, we refer to Birnstiel et al. (2010a); Stammer & Birnstiel (2022); Birnstiel (2024).

2.5.5 Dust entrainment in pressure maxima

One possible solution to the meter-size barrier problem and the fast radial drift of dust is the presence of *traps* that allow dust particles to concentrate in the disc, in correspondence to pressure maxima. This idea was first supported by ALMA observations of protoplanetary discs presenting a ring-gap structure in their (sub-)mm dust grains (e.g., ALMA Partnership et al. 2015; Andrews et al. 2018a). Since then, various theories have been proposed to explain the mechanisms that may accumulate dust in pressure maxima, such as planet-disc interaction, spirals, vortices, and magnetic fields (e.g., Pinilla et al. 2012; Dipierro et al. 2015; Bae et al. 2017). The common feature is that dust particles migrate to high pressure regions. Specifically, assuming a barotropic equation of state, the pressure is proportional to the density, and therefore a density maximum corresponds to a pressure maximum. Then, considering a local pressure maximum at R_0 , the gas is sub-Keplerian at $R > R_0$, where $\nabla P < 0$, and super-Keplerian at $R < R_0$, where $\nabla P > 0$. Since dust particles follow a Keplerian motion, they will experience a

headwind at $R > R_0$ and a tailwind at $R < R_0$, and the net effect is that particles tend to concentrate in pressure maxima, corresponding to density inhomogeneities. The left panel of Figure 2.9 shows an illustration of the mechanism that causes dust grains to migrate toward pressure maxima, while the middle and right panels show a representative example of the ring-gap structure seen in the continuum emission from protoplanetary discs with ALMA.

Externally UV-driven disc winds



Like a proplyd - Comet C/2020 F3 (NEOWISE), July 2020.

The concept that protoplanetary discs exposed to external UV radiation experience photoevaporation of their outermost layers originated from observational evidence of ionized globules in the vicinity of the ONC, which hosts the brightest nearby HII region. These globules show cometary shapes with the cusp pointing toward the nearest massive star. In this Chapter, we first introduce the observations of proplyds in the ONC and in more distant star-forming regions (Sec. 3.1). In Sec. 3.2, 3.3, and 3.4 we describe the standard 1D model for the photoevaporative flow referring to the works of Johnstone et al. (1998); Störzer & Hollenbach (1999); Hollenbach et al. (2000). Subsequently,

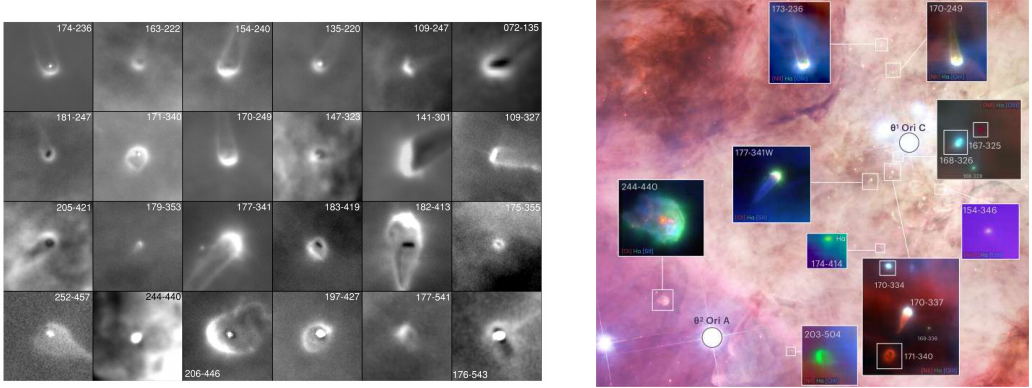


Figure 3.1: Left: from Bally et al. (2000), H_α images related to HST observations of 24 ONC proplyds. Right: from Aru et al. (2024), RGB images of continuum subtracted, single line integrated flux images of proplyds observed with MUSE in a variety of emission lines. The background image is a Hubble optical image of the ONC.

we discuss the behaviour of dust grains in presence of a photoevaporative wind (Sec. 3.5). Finally, we illustrate how to couple 1D disc viscous evolution model and external photoevaporation (Sec. 3.6).

3.1 The beginning of a new story: observations of proplyds

3.1.1 Proplyds in the ONC

The first images showing objects described as *ionized nebular condensations* refer to the work of Laques & Vidal (1979), where they observed the core of the ONC, illuminated by the stars of the Trapezium, in several optical emission lines (e.g., H_α , H_β , [OIII]) revealing elongated ionized globules. Subsequently, observations by Churchwell et al. (1987) performed by using the Very Large Array (VLA) confirmed that the previously observed nebular condensations have radio counterpart. Both the hypothesis that these objects are dense molecular globules embedded in diffuse HII regions with thin ionized envelopes, and the alternative that they are low-mass stars ($\sim 1 M_\odot$) surrounded by evaporating accretion discs, point toward a common conclusion: the peculiar structure is caused by the radiation from the O-type star $\theta^1 C$ and the B-type stars of the Trapezium. The fact that the average diameter of these globules was similar to the Solar System size, supported the hypothesis that the observed objects were evaporating protoplanetary discs. Strong evidence for their association with protoplanetary discs came from HST observations, which revealed emission lines and structures that were not previously detectable with ground-based telescopes. Based on these findings, O'Dell et al. (1993) introduced the term *proplyds* to describe the teardrop-shaped nebulae oriented toward the OB stars of the Trapezium. The hypothesis that the ionization front is caused by UV radiation from the surrounding OB stars is strongly supported by two observations. First of all, the radiation emitted by the central protostar is not sufficiently intense to ionize the outer gas layers and produce the observed optical line emission. Moreover, the tails of the cometary structures consistently point away from the nearest massive star, further reinforcing this interpretation. Six of the brightest and most distinguishable silhouettes seen in projection against the HII background were analysed by McCaughrean & O'Dell

(1996), who found that these objects have average sizes in the range $\sim 50 - 1000$ au. A more recent definition of proplyds reads (Winter & Haworth 2022):

Proplyd: a circumstellar disc with an externally driven photoevaporative wind composed of a photodissociation region and an exterior ionisation front with a cometary morphology.

This definition clarifies the difference between proplyds, which are evaporating protoplanetary discs, and teardrop globules that are seen on a larger scale (e.g., Smith et al. 2003; Guarcello et al. 2023; Reiter et al. 2019, 2020). Although young stars can be found in ionized globules, they can gain mass from the globule itself, contrary to proplyds that lose mass due to photoevaporation. Interestingly, Smith et al. (2005) found that $\sim 80\%$ of the observed proplyds show thermal infrared emission at $11.7 \mu\text{m}$, and an anti-correlation of the presence of proplyds showing infrared emission with distance from $\theta^1\text{C}$.

As will be detailed in the next Section, from the observed position of the ionization front in this objects, it is possible to estimate the associated mass loss rate (Johnstone et al. 1998; Haworth et al. 2021), which results to be on the order of $10^{-8} - 10^{-7} M_{\odot}/\text{yr}$. This high value for the mass loss rate, raised the puzzling problem of the survival of protoplanetary discs in the close proximity of bright OB stars: since the ONC has average age of a few Myr, why proplyds have not been already dispersed? This is known as the *proplyds lifetime problem* and remains not fully solved. One potential solution highlights that the estimate of the photoevaporation timescales has been made using measurements of the disc masses obtained by assuming optically thin emission at sub-millimetre wavelengths. However, this estimate of the mass may be a lower limit of the actual disc mass, due to optically thick emission, implying longer timescales than previously computed (Clarke 2007). Another solution suggests that dust extinction plays a key role in shielding discs from intense UV radiation during the early stages of their lifetime ($\sim 0.5\text{-}1$ Myr, Ali & Harries 2019). According to this scenario, proplyds would have only recently become exposed to extreme UV radiation fields, allowing detection of the characteristic elongated cometary morphology (Winter et al. 2019; Qiao et al. 2022).

An atlas of the observed ionized sources in the ONC was compiled by Bally et al. (2000) (see left panel of Fig. 3.1 for a gallery of H_{α} images of 24 proplyds) and enlarged by Ricci et al. (2008), who included multi-coloured observations performed using the Wide Field Camera and Advanced Camera for Surveys (ACS/WFC) on the HST. The impressive result emerging from this gallery of objects is that, if most stars in the nearby Galaxy form in massive clusters like the ONC, or in environments hosting even more OB stars, then the abundance of proplyds observed in the ONC suggests that this is not an exceptionally rare evolutionary pathway for discs, but rather the most likely one. Additional studies have been performed to probe the chemical and thermal processes in the photo-dissociation region (PDR) and the size distribution of the objects with HST observation (Vicente et al. 2013; Vicente & Alves 2005), while ALMA observations complemented the investigation observing the continuum emission of $\sim\text{mm}$ -dust grains (Mann et al. 2014; Eisner et al. 2016), showing hints of an anti-correlation between the dust disc mass (corresponding to the dust mm-flux emission) and the distance from $\theta^1\text{C}$ (Eisner et al. 2018a). Recent observations with the Multi-Unit Spectroscopic Explorer (MUSE) instrument on the Very Large Telescope (VLT) showed a spatially and spectrally resolved proplyds, providing deeper insight in the morphology of the ionization front (Haworth et al. 2023b; Aru et al. 2024). In the right panel of Fig. 3.1 we show the 12 proplyds investigated by Aru et al. (2024) in various emission lines detected by MUSE. Moreover, studies of the physical structure and chemical composition of proplyds are

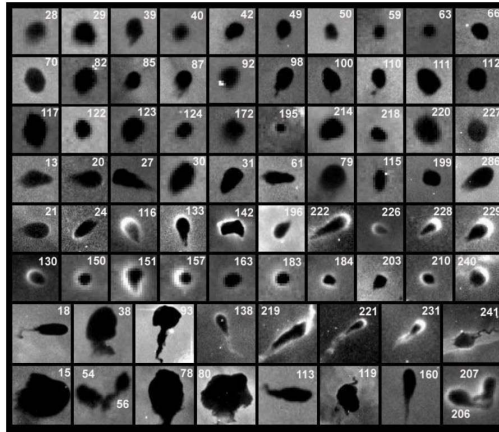


Figure 3.2: Dark globulettes in the Carina Nebula. From Grenman & Gahm (2014).

growing thanks to JWST (Habart et al. 2024). In particular, NIRCам and MIRI provide a large coverage at infrared wavelengths and a resolution similar to or better than HST, allowing the detection of fainter objects and the characterization of warm dust and PAH. Moreover, with JWST, it is now possible to study of the innermost disc regions through ionized gas tracers (e.g., [Ne II], [Ar III], H recombination lines), and therefore understand the effect of the external radiation on the innermost disc.

3.1.2 Proplyds in other star-forming regions

Detections of proplyds in the ONC opened the way to observations of these characteristic objects in other star-forming regions. In Orion, seven proplyds have been found in NGC 1977 (Kim et al. 2016), where the FUV radiation field impinging on the proplyds is supposed to be 10 to 30 times less than that incident on the classic proplyds in the ONC. Indeed, the dominant source of UV irradiation in NGC 1977 is 42 Ori, a B1-type star, and observations in this region set an estimate of the minimum FUV flux able to produce the proplyd-like shape, which is $\sim 3 \times 10^3 G_0$. Proplyds have also been detected in the NGC 2024 star-forming region by using HST (Haworth et al. 2021). This region is mainly illuminated by IRS 1 and IRS 2b, which are a B0.5V star and a O8V star, respectively. Young stars in this region are deeply embedded in the surrounding cloud, supporting the theory that the observed proplyds have been affected by photoevaporation only recently. Due to the high resolution and deep sensitivity requested, observations in more distant and irradiated regions ($\gtrsim 400 - 500$ pc) are challenging. However, proplyd-like objects have been observed in the Carina Nebula (NGC 3372, Smith et al. 2003) and the Pismis 24 region (NGC 6357, Fang et al. 2012). Examples of irradiated globules have been found in massive clusters such as the Carina Nebula (Grenman & Gahm 2014), which presents the famous globulettes shown in Fig. 3.2, in Westerhout 5 (Koenig et al. 2008), showing dusty elongated globules, and in the Cygnus OB2 association (Wright et al. 2012). However, it is currently unknown whether these objects contain a YSO or a protoplanetary disc that is losing mass.

Proplyds are particularly relevant to the study of planet formation because they are observed in very young star-forming regions, $\lesssim 3$ Myr, and planet formation is believed

to start early in the evolutionary history of discs in order to explain the observed population of planets (Tychoniec et al. 2020). This suggests that external photoevaporation may compete with, or even interfere with, the earliest stages of planet formation.

3.2 Basic wind geometry

The observational evidence of ionized globules close to UV sources, described in the previous Section, has inspired the formulation of physical models to explain these phenomena. These models also aim to predict the evolution of protoplanetary discs and newly born planets under the influence of the external environment. We present a simple geometrical model that describes the physics of a photoevaporative flow affecting the outer layers of a protoplanetary disc.

External photoevaporation of protoplanetary discs can be initialized by EUV photons (energies $h\nu > 13.6$ eV) or FUV photons (energies $6 < h\nu < 13.6$ eV), which heat the disc surface and drive thermal winds. As a result of the radial pressure gradient, the geometry of the flow, which is more cylindrical near the disc surface, becomes approximately spherically symmetric at a distance of the order of the disc size, and the flow is accelerated into a Parker wind (Parker 1958). In order to produce an evaporative flow, both mechanisms require that the thermal pressure must exceed the local gravity. Considering an isothermal gas in Keplerian motion around a star of mass M_* , the gravitational radius is defined by balancing the thermal velocity of the gas particles (i.e., the sound speed c_s) with the local escape velocity:

$$R_g = \frac{GM_*}{c_s^2} = \frac{GM_*\mu m_H}{k_B T} \sim 140 \left(\frac{M_*}{1M_\odot} \right) \left(\frac{T}{100 \text{ K}} \right)^{-1} \text{ au}, \quad (3.1)$$

where μ , the mean molecular weight of the gas is assumed to be 1.3. This radius corresponds to the location in the disc where the thermal energy balances the gravitational binding energy and beyond which the material is unbound. This radius will move closer to the central star in the presence of UV photons heating the disc material and hence increasing the sound speed. However, an evaporative flow can be launched from a radius smaller than R_g . Indeed, a Parker wind (i.e., in the case of a steady, isothermal, and spherical flow) satisfies the relation

$$(v^2 - c_s^2) \frac{dv}{dr} = \frac{2vc_s^2}{r} - \frac{GM_*v}{r^2}, \quad (3.2)$$

where v is the flow speed and r is the radius of a sphere centred on the star of mass M_* . The sonic point is the location where the pressure gradient balances local gravity and the flow passes from subsonic to supersonic, which is verified when $v = c_s$ and $r = R_s = GM_*/2c_s^2 = R_g/2$, which is smaller than R_g . The geometry of the flow in the EUV and FUV dominated regime is represented in Fig. 3.3. The boundary between ionized gas and neutral gas is known as ionisation front (IF). Since the temperature in the ionised gas is much higher than in the neutral gas, the change in pressure causes the formation of a shock front ahead of the IF, in the neutral medium. Across a shock point, the conditions of Rankine-Hugoniot for the conservation of mass and momentum must be satisfied, which in 1D are

$$\rho_1 v_1 = \rho_2 v_2, \quad (3.3)$$

$$\rho_1 v_1^2 + P_1 = \rho_2 v_2^2 + P_2, \quad (3.4)$$

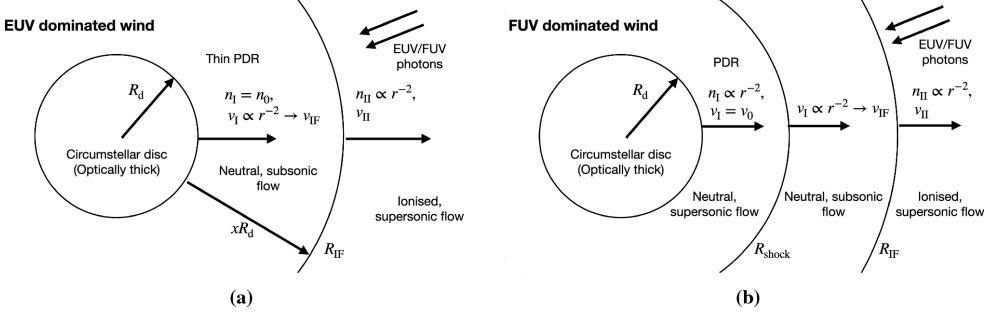


Figure 3.3: Schematic illustration of the geometry of a photoevaporative flow driven by EUV and FUV radiation fields. Panel (a): in the EUV-dominated regime the gas is ionized from the ionization front to the disc radius, with a thin PDR. Panel (b): in the FUV-dominated regime, the supersonic flow launched from the disc crosses a shock region becoming subsonic before reaching the ionization front. From Winter & Haworth (2022).

where ρ_1 , v_1 , and P_1 are the density, velocity, and pressure before crossing the shock, and ρ_2 , v_2 , and P_2 are evaluated beyond the shock point. The neutral material crossing the IF is subject to supersonic expansion. To ensure conservation of mass and angular momentum of the material that crosses the IF, the flow must be subsonic on the PDR side of the IF (i.e., D-type front), with velocity $v_{IF} \sim c_{IF}^2/2v_{II}$, where c_{IF} is the sound speed of the PDR and v_{II} is the flow speed on the external part of the IF, with typical values $v_{IF} \sim 1 \text{ km s}^{-1}$ in a neutral flow at $T \sim 10^3 \text{ K}$ and $v_{II} \sim 10 \text{ km s}^{-1}$ in an ionized medium.

3.3 EUV dominated flow

EUV photons penetrate the disc from the IR near the disc surface, ionizing the gas and raising its temperature to $T \sim 10^4 \text{ K}$. For strong EUV photons, the IF is located close to the disc surface and the PDR occupies only a thin layer. As shown in the left panel of Fig. 3.3, a neutral, subsonic flow is launched from the disc surface and crosses the PDR. Under assumed isothermal conditions, the density in the PDR is constant and, therefore, the velocity of the flow decreases as $v_I \propto r^{-2}$ to conserve the mass. At the IF, the mass and momentum conservation imply $v_{IF} = c_{s,I}^2/2c_{s,II} \sim 0.5 \text{ km s}^{-1}$, considering $c_{s,I} \sim 3 \text{ km s}^{-1}$, in the PDR, and $c_{s,II} \sim 10 \text{ km s}^{-1}$ beyond the IF. If the PDR thickness is set to xR_d , the density in the PDR is described by $n_0 = n_I = N_D/xR_d$, where N_D is the column density in the PDR, which depends only on the density at the base of the flow. The mass loss rate in the EUV-driven scenario can be written as

$$\dot{M} = \dot{M} = 4\pi\mathcal{F}(1+x)^2 R_d^2 n_I m_I \frac{c_{s,I}^2}{2c_{s,II}}, \quad (3.5)$$

where $\mathcal{F}$ is a geometrical correction (i.e., the fraction of the solid angle subtended by the evaporative flow), and m_I is the mean molecular mass in the PDR. This equation diverges in the limit of a very thin PDR, $x \rightarrow 0$, and must be associated with the additional condition that the EUV photons which are not absorbed by the dust can penetrate the disc

$$\frac{f_r \Phi}{4\pi d^2} = \int_{R_{IF}}^{\infty} \alpha_r n_{II}^2 dr = R_{IF}^4 \left(\frac{m_{IF} v_{IF}}{m_{II} v_{II}} \right)^2 \int_{R_{IF}}^{\infty} \alpha_r \frac{n_{IF}^2}{r^4} dr, \quad (3.6)$$

where f_r is the fraction of photons which are not attenuated by the ISM, Φ is the number of EUV photons emitted by an external source at a distance d , and $\alpha_r = 2.6 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ is the recombination coefficient for hydrogen heated at 10^4 K (e.g., Osterbrock 1989). In right-side term of Eq. (3.6), n_{II} is written considering mass conservation across the IF. Substituting $R_{\text{IF}} = (1 + x)R_d$, and solving in the limit of $x \ll 1$, the mass loss rate in the EUV-dominated regime for typical values is (Johnstone et al. 1998; Winter & Haworth 2022):

$$\dot{M} = 5.8 \times 10^{-8} \epsilon_{\text{EUV}} \mathcal{F} \left(\frac{d}{0.1 \text{ pc}} \right)^{-1} \left(\frac{f_r \Phi}{10^{49} \text{ s}^{-1}} \right)^{1/2} \left(\frac{R_d}{100 \text{ au}} \right)^{3/2} M_{\odot} \text{ yr}^{-1}, \quad (3.7)$$

which directly depends on the distance from the ionizing source, d .

3.4 FUV dominated flow

FUV photons dissociate molecules and heat the outer disc regions to a temperature of $T \sim 10^2 - 10^3 \text{ K}$, producing a predominantly neutral flow. A schematic view of the geometry of the flow in the FUV-dominated regime is presented in the panel (b) of Fig. 3.3. The neutral flow is launched from the disc radius $R_d > R_g$, with an escape velocity given by

$$v_0 = v_{\text{esc}} = \sqrt{\frac{2GM_{\star}}{R_d}} = 4.2 \left(\frac{M_{\star}}{1 M_{\odot}} \right)^{1/2} \left(\frac{R_d}{100 \text{ au}} \right)^{-1/2} \text{ km s}^{-1}, \quad (3.8)$$

which is supersonic and accelerates in the PDR. As a consequence of conservation of the mass in the PDR, since the velocity increases with radius, the gas density decreases with the distance from the star as $n_{\text{I}} \propto r^{-2}$. In order to satisfy the conservation of mass and momentum across the IF, the flow must be subsonic before crossing the IF, which implies that it must cross a shock front between R_d and the IF, at a radius

$$R_{\text{shock}} = R_{\text{IF}} \sqrt{\frac{v_0}{2c_{\text{s,II}}}} \sim 0.4 R_{\text{IF}} \sqrt{\frac{1}{2} \left(\frac{v_0}{3 \text{ km s}^{-1}} \right) \left(\frac{10 \text{ km s}^{-1}}{c_{\text{s,II}}} \right)}. \quad (3.9)$$

The position of disc radius, shock radius, and IF radius, and their separation, set the condition for the mass loss to be dominated by EUV or FUV driven winds. If $R_{\text{shock}} < R_d$ the flow is EUV-dominated. In the opposite case, the condition $R_{\text{IF}} \gtrsim 2.5 R_d$, which follows from Eq. (3.9), ensures an FUV-driven wind.

Across the shock front, under the assumption of isothermal shock, the gas density n_{I} increases by a factor of the order of $\sim \mathcal{M}^2 = (v_0/c_{\text{s,I}})^2$, where $\mathcal{M}$ is the Mach number. This shock jump results in strong compression of the gas. Between R_{shock} and R_{IR} , PDR models have shown that the temperature, T , and the thermal pressure, $n T$, are nearly constant for relatively small discs ($R_d \lesssim 50 \text{ au}$), implying a nearly constant density, and therefore a flow speed that decreases with distance from the central star as $v_{\text{I}} \propto r^{-2}$. In contrast, larger discs can present a lower temperature in the PDR than in correspondence to the IF.

In the FUV-dominated regime, the mass loss rate depends on the properties at the base of the wind and reads as

$$\dot{M} = 4\pi \mathcal{F} R_d^2 n_0 v_0 m_{\text{I}}, \quad (3.10)$$

where $\mathcal{F}$ is a geometrical correction ($\mathcal{F} = 1$ in the spherical case), and m_l is the mean molecular mass in the PDR. Although the mass loss in Eq. (3.10) does not directly depend on the FUV flux, a hidden dependence is in the PDR temperature, which can increase with increasing FUV flux. However, the contribution of this effect to the total mass loss is relatively small for FUV fluxes $> 10^4 G_0$ (Störzer & Hollenbach 1999). Moreover, the mass loss rate is a steeper function of the disc radius in the FUV-dominated regime compared to the EUV-dominated case. As a result, once the disc is sufficiently truncated by external radiation (which is a fast process), it is the FUV that determines the mass loss over the disc lifetime.

3.4.1 Subcritical regime

In the previous description of the FUV-dominated flow we considered that the disc is more extended than the gravitational radius, $R_d > R_g$, which is called the supercritical regime. Depending on the stellar mass and the FUV flux, this situation is mainly verified in the case of very extended and irradiated discs. In the case where $R_d < R_g$, part of the disc atmosphere can still expand beyond the disc radius R_d , reaching the gravitational radius and being subject to external photoevaporation. Discs in the Solar neighbourhood, which are at intermediate FUV flux ($\sim 10^2 - 10^4 G_0$, see Sec. 4.4), are expected to be mostly in this regime, known as subcritical.

In the subcritical regime, a subsonic flow (instead of supersonic as in Fig. 3.3) is launched from a disc radius R_d and increases its velocity until reaching the sound speed at the sonic point. The sonic point is smaller than or similar to the gravitational radius. Beyond the sonic point, the velocity of the flow is nearly uniform. As shown in the FUV-dominated supercritical regime, the mass loss rate depends on the properties at the base of the wind, which in this case is the disc radius R_d instead of the gravitational radius.

Disc evolution and external photoevaporation in the subcritical regime have been studied by Adams et al. (2004); Facchini et al. (2016); Haworth et al. (2018), which have demonstrated that the mass loss rate can still be very high and therefore the impact on disc evolution can still be significant. In particular, Facchini et al. (2016) has explored the behaviour of dust grains of different sizes in the subcritical regime, determining the grain size limit below which the particles are entrained in the wind. Moreover, Facchini et al. (2016) has shown the relation between efficiency of grain growth and mass loss rate. Haworth et al. (2018) ran a grid of disc evolution models varying disc and stellar parameters, and the FUV flux experienced at the outer disc radius, solving the PDR physics and including the dust entrainment in the wind. These models were used to compile the extremely useful grid of mass loss rates, the FRIED (FUV Radiation Induced Evaporation of Discs) grid, which was subsequently updated to the FRIEDv2 grid, which allows dust growth in the wind and includes various polycyclic aromatic hydrocarbon (PAH) abundances (Haworth et al. 2023a). Interpolating the FRIED grid, it is possible to verify that mass loss rates of $\sim 10^{-9} - 10^{-8} M_\odot/\text{yr}$ can be produced even at moderate FUV fluxes ($\sim 10 - 100 G_0$), and this clearly states the relevance of considering external photoevaporation in the subcritical regime.

3.5 Dust entrainments in external winds

The effect of external photoevaporation on discs strongly depends on the microphysics of the PDR, which affects the mass loss rate, as summarised by the term ϵ in Eq. (3.7) and (3.10). In particular, dust plays a key role by attenuating the external radiation,

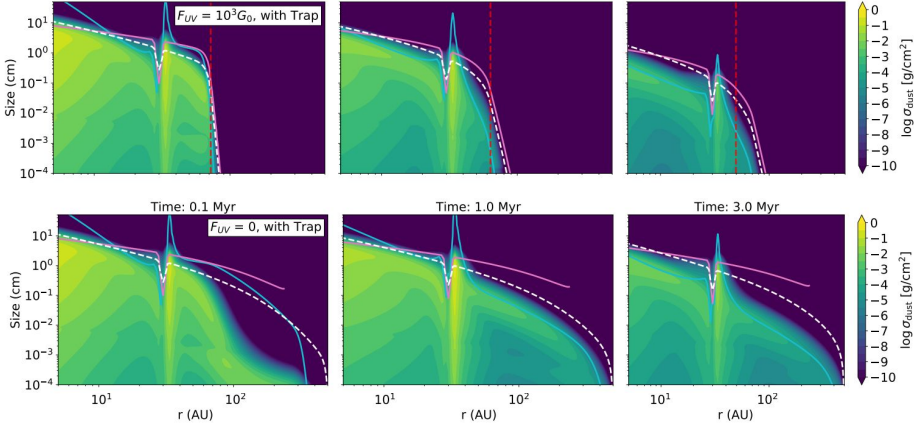


Figure 3.4: Dust size distribution at 0.1, 1, and 3 Myr (from left to right), for the simulations with/without photoevaporation, and with dust traps. The solid lines indicate the estimated fragmentation and drift growth limits (magenta and cyan, respectively). The grain sizes that correspond to $St = 0.1$ is indicated by a dashed white line, and the photoevaporation radius with a dashed red line. From Gárate et al. (2024).

thereby setting the penetration depth of UV photons and affecting the mass loss rate. Dust composition, metallicity, and polycyclic aromatic hydrocarbon (PAH) abundance significantly influence the PDR temperature, chemistry, and set the mass loss rate. Moreover, dust particles can be entrained in evaporative winds, decreasing the reservoir of solid material available for planet formation. Therefore, determining the amount of dust that can be entrained in evaporative winds and its size and composition is crucial. Following an analytic approach, Facchini et al. (2016) determined the maximum grain size which can be entrained in the wind, a_{ent} , which is defined by the balance between the drag force exerted on the dust grains by the wind and the gravity of the host star:

$$a_{\text{ent}} = \frac{1}{4\pi F} \frac{v_{\text{th}} \dot{M}_{\text{ext}}}{GM_{\star} \rho_{\text{d}}}, \quad (3.11)$$

where $4\pi F$ is the solid angle subtended by the wind, v_{th} is the velocity of the gas flowing outward, and ρ_{d} is the bulk density. Since this size limit is typically small (e.g., $\lesssim 10 \mu\text{m}$), the mass loss rate is extremely sensitive to the efficiency of grain growth and radial drift in the disc. Very efficient grain growth implies a smaller cross section for the FUV photons, allowing a larger penetration depth of the photoevaporative wind, and increasing the mass loss rate. Conversely, if a large amount of ISM-like dust is entrained in the wind, the extinction in the wind increases and the mass loss rate is reduced. In 1D disc evolution models, the fraction of grains entrained in the wind can be calculated as (Sellek et al. 2020b)

$$f_{\text{ent}} = \min \left[1, \left(\frac{a_{\text{ent}}}{a_{\text{max}}} \right)^{4-p} \right], \quad (3.12)$$

where a_{ent} and a_{max} are the dust sizes of the entrained grains and the maximum grain that can be entrained (Facchini et al. 2016), respectively. Typically, dust stops being entrained very early during disc evolution, in $\sim 10^5$ years, when it reaches a size $\sim$

10 – 100 μm . Afterward, dust moves inward driven by radial drift. These models show a higher dust depletion with a higher α -viscosity. If dust growth proceeds at a higher rate than depletion due to external photoevaporation, planetesimals may form in the outer disc regions (Throop & Bally 2005), but they are likely to produce lower-mass, predominantly rocky planets, rather than gas giants, since the gas is rapidly lost. Hutchison & Clarke (2021) studied the delivery and entrainment of dust grains in the photoevaporative wind using a semi-analytic model and focusing of a turbulent disc illuminated by an inclined EUV-driven flow. They showed, in particular, that diffusive mixing of dust grains close to the ionisation front can influence the rate of dust delivery to the ionisation front, and therefore the dust entrainment in the wind. 1D models including dust entrainment in the wind find that there is a radial gradient in maximum grain size close to the disc outer edge (Throop & Bally 2005; Facchini et al. 2016; Sellek et al. 2020b; Gárate et al. 2024). This is consistent with the behaviour of dust at the disc midplane found in 2D simulations of external photoevaporation by Paine et al. (2025). However, 2D simulations show lower dust-to-gas ratios and maximum entrained grain sizes above the midplane.

The formation of dust traps and substructures in discs is often linked to the process of planet formation. Therefore, understanding whether and under which conditions UV-irradiated discs can host substructures is key for characterising planet formation in typical protoplanetary discs. This topic is addressed by Gárate et al. (2024), who demonstrated that disc dust rings can survive external photoevaporation if they form beyond the effective truncation radius of photoevaporation (red line in Fig. 3.4), where the wind passes from being optically thin to optically thick. Figure 3.4 shows that, when an external FUV radiation field is included in the simulation, the disc is truncated and the smallest dust grains are lost. However, the dust trap survives because it is located within the effective radius of photoevaporation. ALMA high-resolution observations of mm-continuum dust emission in discs in the σ Orionis region have shown that discs subject to $10^2 - 10^3$ G_0 present gaps-rings structures (Huang et al. 2024). However, the abundance of the substructures decreases with decreasing distance from the most massive star in the region, suggesting the effect of external photoevaporation. Both observations and models indicate that substructures, and potentially planets, can form in irradiated discs. However, the truncation of their outer regions may affect where and when the formation of planetesimals occurs (e.g., Qiao et al. 2023).

3.6 Coupling disc viscous evolution and external photoevaporation

Modelling irradiated protoplanetary discs and their evolution requires solving the PDR microphysics, which is highly sensitive to temperature, while also building a reliable chemical network and coupling it with hydrodynamics. A 3D radiation hydrodynamic model is extremely challenging to build and computationally expensive. Therefore, it is illustrative to reduce the dimensions of the problem and start from a semi-analytic approach in 1D. Usually, the disc is modelled as a 1D radial grid, where the photoevaporative wind applies over a solid angle subtended by the outer radius of the disc. In this model, the mass loss rate at a radius R can be computed as

$$\dot{M} = 4\pi R^2 \mathcal{F} \rho v, \quad (3.13)$$

where ρ is the density of the disc, v is the velocity of the flow, and $\mathcal{F} = H_d / \sqrt{H_d^2 + R_d^2}$ is the fraction of 4π steradians subtended by the flow, where H_d and R_d are the disc height to the mid-plane and the disc radius. This simple geometry, where the flow solution is solved along the disc midplane using appropriate boundary conditions and the

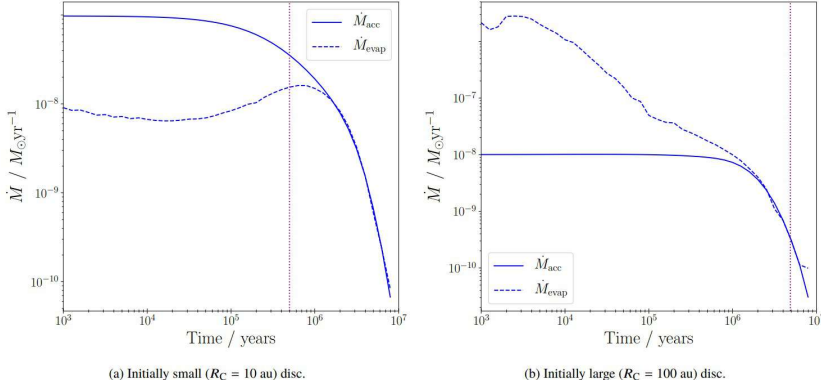


Figure 3.5: Results from 1D models of externally irradiated discs: mass loss rate due to external photoevaporation and mass accretion rate onto the star, as a function of time for an initially small ($R_c = 10$ au, left panel) and initially large ($R_c = 100$ au, right panel) disc. The vertical dotted line indicates the viscous timescale at R_c . From Sellek et al. (2020b), reproducing the results found by Clarke (2007).

outer disc radius delimiting the critical point of photoevaporation, is justified by the fact that in the disc material in the outer regions is less gravitationally bound. Moreover, for a geometrically thin disc, the vertical scale height is much smaller than the disc radial extent, $H_d \ll R_d$, which means that the density at the radial sonic point exceeds that at the vertical sonic point, and therefore the flow solution can be approximated to the radial component. Adams et al. (2004) compared analytically the mass loss due to photoevaporation from the disc surface, $\dot{M}_{\text{surf}}$, and from the radial edge, $\dot{M}_{\text{edge}}$, showing that

$$\frac{\dot{M}_{\text{surf}}}{\dot{M}_{\text{edge}}} \sim \left(\frac{R_d}{R_g} \right)^{1/2}, \quad (3.14)$$

where, in the limit $R_d \ll R_g$, the mass loss from the disc edge dominates over that from the surface. For large discs, where R_d is comparable to R_g , the mass loss rate approaches its supercritical value and the contribution of the disc surface cannot be neglected.

Considering a viscous disc where the surface density evolves following the Lynden-Bell & Pringle (1974) equation (Eq. 10.4), we can add an extra term accounting for a generic loss of material caused by external winds, $\dot{\Sigma}_{\text{ext}}$, such that

$$\frac{\partial}{\partial t} \Sigma_g = \frac{3}{R} \frac{\partial}{\partial R} \left(R^{1/2} \frac{\partial}{\partial R} (\nu \Sigma_g R^{1/2}) \right) - \dot{\Sigma}_{\text{ext}}. \quad (3.15)$$

The main challenge in coupling viscous disc evolution with the physics of a photoevaporative flow is the enormous difference between the time-scales involved, where wind dynamics operate on timescales which are orders of magnitude smaller than viscous timescale. This makes it computationally impossible to model both processes self-consistently in one calculation. For this reason, the process of external photoevaporation is typically included into viscous disc evolution models using a numerical or parametric approach. Clarke (2007) coupled disc viscous evolution and external photoevaporation by describing the mass loss rate as a function of the outer disc edge. Specifically, the disc is depleted outside-in to mimic the effect of external photoevaporation: starting from

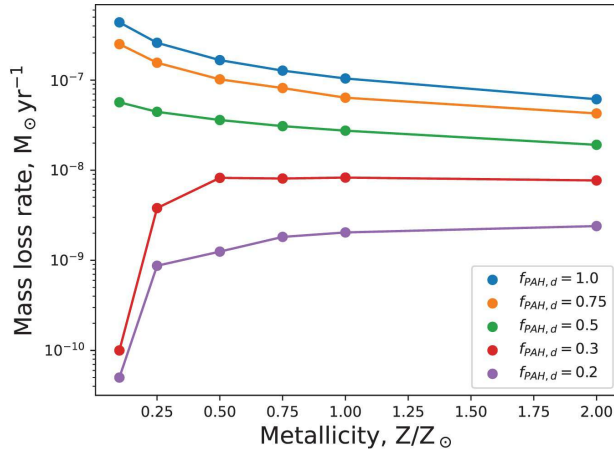


Figure 3.6: External photoevaporation mass loss rate as a function of the metallicity and the PAH-to-dust mass ratio (f_{PAH}), for a 100 au disc around a $1 M_{\odot}$ star irradiated by $1000 G_0$. Credits: Haworth et al. (2023a).

the outer disc edge, a fraction $1-\epsilon$ of mass is removed from each grid cell while proceeding inward, until the mass removed at that timestep exceeds $\dot{M}_{\text{wind}} \times dt$. The results of this work showed that the disc evolves toward a quasi-steady configuration where the photoevaporation mass loss rate balances the stellar mass accretion rate, on a timescale of ~ 1 Myr, as shown in Fig. 3.5. As the photoevaporation rate exceeds the accretion rate, the viscous spreading is suppressed, and the disc outer edge moves inward. The comparison between the predictions of these models and the disc sizes of the sample of proplyds in the ONC, which are expected to be smaller than $\sim 10 - 20$ au, suggests that discs started experiencing external photoevaporation only recently, and the mass loss rate increases after this exposure. A diagnostic for verifying whether discs are dominated by viscous spreading or outside-in depletion is the dimensionless term (Rosotti et al. 2017)

$$\eta = \frac{\tau_{\text{age}} \dot{M}_{\text{acc}}}{M_{\text{disc}}}, \quad (3.16)$$

where τ_{age} is the age of the disc, $\dot{M}_{\text{acc}}$ is its accretion rate, and M_{disc} is its gas mass. This term is ~ 1 in discs that are driven by viscosity, whereas it drops below unity when outside-in depletion dominates. We emphasize that this type of analysis relies on accurate estimates of disc age, accretion rate, and gas mass, which are often subject to high uncertainty (particularly disc age and gas mass). Using a 2D radiation hydrodynamic model, Haworth & Clarke (2019) validated the 1D model, showing that the mass loss rate is slightly higher when passing from 1D to 2D.

The PAH abundance is a crucial parameter in determining the mass-loss rate due to external photoevaporation, as the dominant gas-heating mechanism is photoelectric heating by PAHs. These are carbon-ring molecules where electrons are easily emitted upon photon absorption (Tielens 2008). Haworth et al. (2023a) used a 1D locally isothermal model based on disc hydrodynamics where the temperature is set by the PDR physics, to explore the role of metallicity and PAH-to-dust mass ratio in determining the mass loss rate. Figure 3.6 shows that the mass loss rate is mainly influenced by a variation in PAH abundance rather than in metallicity. However, observational constraints

on PAH abundance in discs are challenging and very limited to a few objects showing levels below the ISM. Noteworthy in this context is the protoplanetary disk HST 10 (Vicente & Alves 2005), for which the PAH-to-gas ratio was measured to be a factor 50 lower than the surrounding Orion Bar, and a factor 90 lower than in NGC 7023 (Berné & Tielens 2012). The advent of JWST enables a more precise investigation of PAH molecular structures and abundances (e.g., Khan et al. 2025). Until new observational constraints become available, it is clear that model predictions remain highly sensitive to this parameter.

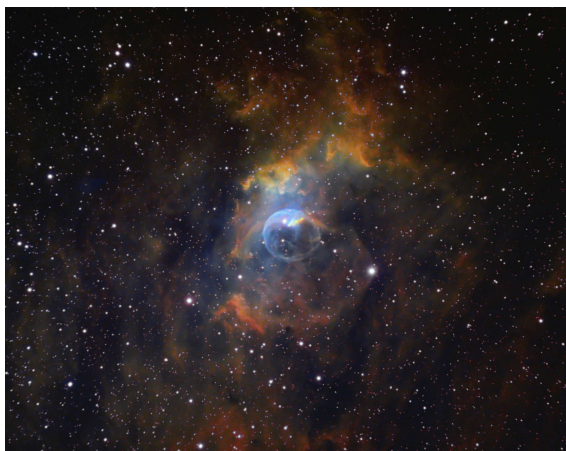
A numerical implementation of 1D disc viscous evolution and photoevaporation, using the mass loss rate from the FRIED grid, was performed by Sellek et al. (2020b) and subsequently used in several works (Gárate et al. 2024, Anania et al. 2025a, included in this thesis). The evolution of the disc surface density can be described by the diffusion equation 2.32 and following the Lynden-Bell & Pringle (1974) solution with an extra term accounting for the mass removal due to photoevaporation, $\dot{\Sigma}_{\text{ext}}$. The mass loss rate can be computed by performing a bilinear interpolation in log-space of the FRIED grid, using the stellar mass and FUV flux, and then interpolating the grid using each disc radius as the outer disc radius to set the corresponding mass loss rate. The radius corresponding to the transition between optically thin and thick wind is called *truncation radius*, R_t , and corresponds approximately to the radius where the mass loss rate is maximum (Sellek et al. 2020b). In the numerical model, disc material is removed outside the truncation radius, where the wind is supposed to be optically thin. In order to ensure smooth mass removal, a fraction of the mass in each cell is removed based on the total mass loss rate, so that the effective mass loss in each grid cell (i) at $R > R_t$ is

$$\dot{M}_{\text{ext},i} = \dot{M}_{\text{tot}} \frac{M_i}{M(R > R_t)}, \quad (3.17)$$

where $\dot{M}_{\text{tot}}$ is the total mass loss rate outside the truncation radius. The result obtained by Clarke (2007) that the accretion rate and the mass loss rate evolve toward a condition of equilibrium was confirmed by these models (Fig. 3.5). An alternative numerical approach consists of modelling a smooth transition from a rotationally supported accretion disc to a pressure driven outflow using a slim disc formalism, where fluid parcels conserve their specific angular momenta (Owen & Altaf 2021). This approach avoids the brutal change in the angular velocity profile when crossing the radial location from which the wind is launched. This approach is particularly useful for modelling FUV-irradiated discs in the sub-critical regime, where the absence of an ionization front precludes the formation of a sharp cut in the angular velocity profile. However, this model currently lacks a self-consistent treatment of FUV heating and the relevant cooling processes.

In this Chapter, we conducted a theoretical analysis of externally irradiated discs. A natural follow-up question is: What levels of UV flux are actually present in nearby star-forming regions? To address this question, it is necessary to identify the locations and properties of massive OBA-type stars in the solar neighbourhood.

Rising under the spotlight: nearby OB associations



A bubble of light - NGC 7635 (Bubble Nebula, ~ 2.2 kpc from the Sun); the radiation from the O-type star SAO 20575 ionised the surrounding gas.

Mapping the spatial distribution of massive O-, B-, and early A-type stars in relation to young stars is the first step to understand the role of UV radiation in affecting protoplanetary disc evolution and planet formation in our Galaxy. Indeed, approximating stellar emission to a blackbody, the FUV flux is a steep function of the stellar mass for $M_{\star} > 2M_{\odot}$, and therefore OBA-type stars are those emitting the most intense FUV radiation. For this purpose, a comprehensive study of the distribution, geometry, and kinematics of OB associations is crucial. It is well established from observational evidence and numerical N-body simulations that most young stars in star-forming regions are clustered (Lada & Lada 2003; Bressert et al. 2010), and comparison of embedded star-forming regions to open cluster statistics suggests that most stars form in unbound grouping hosting OB stars. Therefore, these regions are the most suitable places to study typical protoplanetary disc evolution and planet formation in the solar neighbourhood, in connection with the environment (i.e., particularly the UV radiation from massive stars).

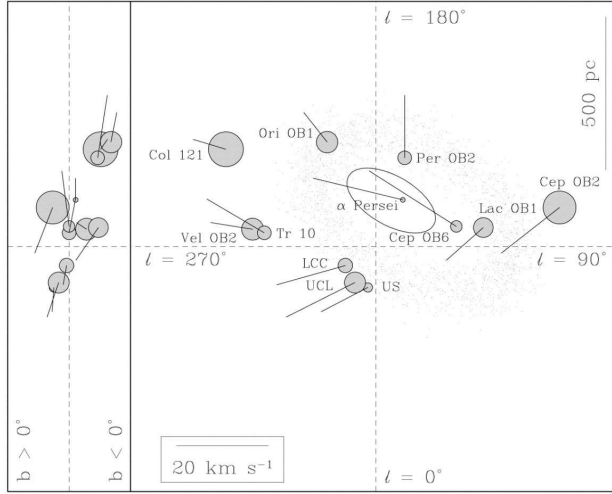


Figure 4.1: Locations of the kinematically detected OB associations projected onto the Galactic plane (right) and a corresponding cross section (left). The gray circles indicate the physical dimensions as obtained from the angular dimensions and mean distances, on the same scale. The lines represent the streaming motions, derived from the average proper motions, mean distances and median radial velocities of the secure members, corrected for "standard" solar motion and Galactic rotation. The ellipse around the α Persei cluster indicates the Cas–Tau association. From de Zeeuw et al. (1999).

In this Section we summarize the main properties of the OB associations (Sec. 4.1), with a particular focus on the Scorpius–Centaurus OB association (Sec. 4.2) and the Orion OB1 association (Sec. 4.3). Subsequently, we illustrate the distribution of the UV radiation field in the solar neighbourhood (Sec. 4.4).

4.1 Main properties

OB associations are gravitationally unbound groups of young stars, typically containing many OB stars and numerous low-mass stars. The presence of short-lived OB stars, and dense and kinematically linked sub-groups (despite being unbound), is an indicator of the young age and that star-formation is still proceeding. Due to the high luminosity of OB stars with respect to lower-mass stars, OB associations are optimal tracers of populations of young stars. In particular, they allow for the study of the distribution and kinematic configuration of the birthplaces of stars and their protoplanetary discs, as well as the characterization of the stellar initial mass function. Furthermore, they provide an opportunity to study star formation over time and in connection with stellar feedback and processes that trigger and regulate it. The term *association* was originally associated with groups of stars that are very young compared to the age of the Galaxy and a low stellar density, with typical values of $0.001 - 0.1 \text{ M}_{\odot} \text{ pc}^{-3}$ if compared to the Galactic field (Ambartsumian 1947). Historically, they were divided into OB associations and T associations, depending on the most prominent members (Ambartsumian 1968). OB associations are located in the Galaxy with a scale-height similar to the young open clusters, and have total mass similar to the open clusters.

The fundamental difference between open clusters and OB associations is that the first systems are gravitationally bound, whereas associations are unbound. However,

OB associations are highly substructured and host dense regions of star-formation of a few pc in size, which are bound systems. Therefore, it seems that OB associations are the inevitable consequence of star-formation going on in molecular clouds, with global density which is too low to remain bound. It is still debated whether OB associations are the result of multiple star-formation events happened over large spatial scales (Miller & Scalo 1978), or are instead remnants of dense molecular clouds (Kroupa 2001; Lada & Lada 2003). Following the first hypothesis, Miller & Scalo (1978) claimed that the number of low-mass stars hosted in OB associations is too low to explain the field star population, and therefore an additional birthplace is required. On the other hand, Lada & Lada (2003) argued that embedded star clusters can evolve into gravitationally bound open clusters or unbound and low-density associations.

The advent of *Hipparcos* tremendously improved the identification of stellar members and the knowledge of the systematic motions of OB associations. In Fig. 4.1, we show a map of the locations and kinematics of OB associations identified by de Zeeuw et al. (1999). However, due to apparent brightness, *Hipparcos* was unable to reliably provide kinematics and membership for stars with spectral type later than A. Subsequently, comprehensive censuses of OBA stars and low-mass stars hosted in associations have become possible thanks to Gaia, with the added advantage of resolving the internal motions of clouds and subdividing associations into numerous bound clusters (e.g. Ratzenböck et al. 2022).

Since OB associations are low-density and unbound systems, they are unstable against tidal gravitational forces, and therefore it is reasonable to assume that they are expanding over time. Ambartsumian (1947) estimated an average expansion velocity of $\sim 5 \text{ km s}^{-1}$. The origin of the expansion is still debated, with theories asserting that unbound stars are born into a state of expansion (Oort 1954), and other theories explaining how stellar feedbacks in bound systems can make the systems unbound and trigger their expansion, which are supported by N-body simulations (e.g. Tutukov 1978; Lada 1987).

Among the OB associations catalogued during the last years (e.g. de Zeeuw et al. 1999; Blaauw 1964; Wright 2020), the most extensively studied are Scorpius-Centaurus, Orion OB1, Vela OB2, and Cygnus OB2. In the next Sections we briefly describe Sco-Cen and Orion OB1, which are the closest to the Sun ($\sim 150 \text{ pc}$ and $\sim 400 \text{ pc}$, respectively) and play an important role in the context of the study of disc evolution. Indeed, Scorpius-Centaurus hosts the largest population of old protoplanetary discs in the solar neighbourhood, which is fundamental to link the protoplanetary disc phase with the debris disc phase and the formation of planetary systems. Conversely, Orion OB1 is the closest region presenting active star-formation and a young population of OBA-type stars, particularly enabling the study of the early phases of disc evolution in connection to the stellar formation environment. We refer to Wright (2020) for a more detailed review on the nearby OB associations.

4.2 The Sco-Cen association

The Scorpius-Centaurus (Sco-Cen or Scorpius OB2) association is the nearest OB association to the Sun at a distance of 100-150 pc (de Zeeuw et al. 1999). It covers a large portion of the sky, extending across the constellations Scorpius, Lupus, Norma, Centaurus, Circinus, Musca, and Crux. There are ~ 150 B-type stars hosted by this association, and only one O-type star, the runaway O9V ζ Ophiuchi (Blaauw 1964). Historically, the association has been divided into three groups with different average age: Upper Scorpius (Upper Sco), which is the youngest ($\sim 10 \text{ Myr}$), Upper Centaurus-Lupus (UCL), containing the oldest stars, average age of $\sim 16 \text{ Myr}$, and Lower Centaurus-Crux (LCC),

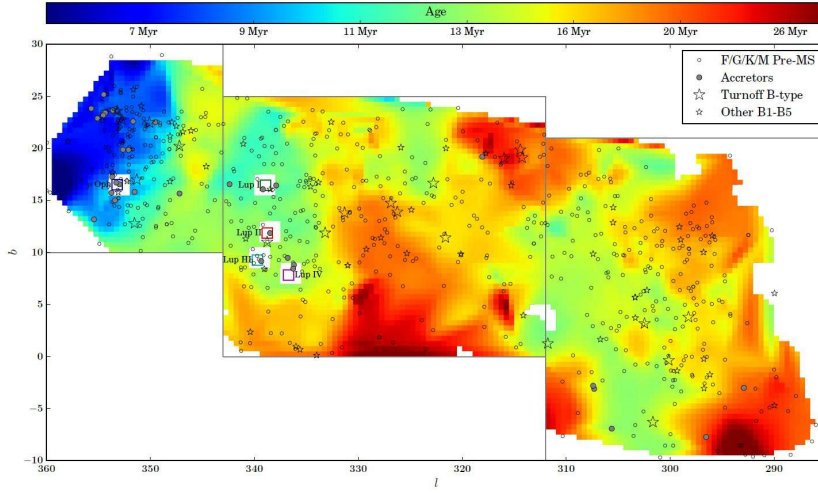


Figure 4.2: A spatially averaged age map of Sco-Cen constructed from the ages of 657 F/G/K/M-type pre-MS members of the association, with the luminous early-type stars also shown. The three boxes delineate the classical boundaries of the three subgroups Upper Sco (left), UCL (centre) and LCC (right) that have median ages of 10, 16 and 15 Myr, respectively. The very young regions of Ophiuchus and the Lupus clouds, which are projected against Sco-Cen, are masked over in this map. From Pecaut & Mamajek (2016).

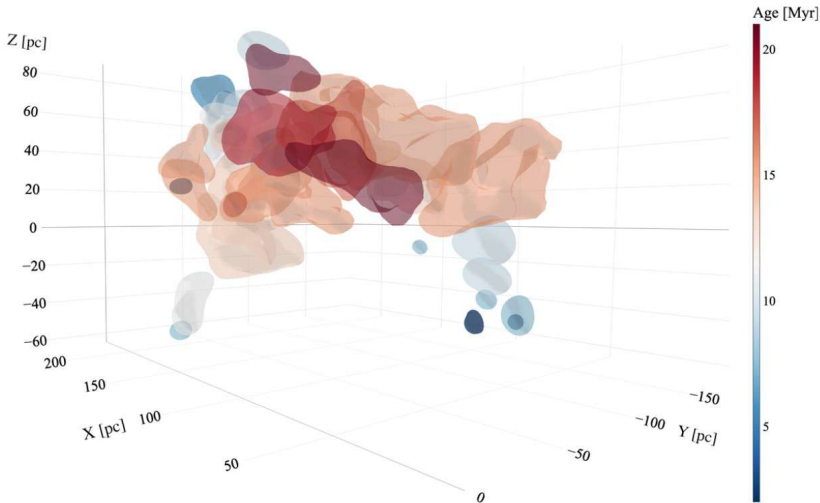


Figure 4.3: 3D distribution of 34 clusters in the Sco-Cen association found by SigMA (Ratzenböck et al. 2023). The Sun is at (0,0,0) and the $Z=0$ plane is parallel to the Galactic plane. The surfaces of the cluster volumes are shown, colour-coded by age, from dark blue (2 Myr) to dark red (21 Myr). From Ratzenböck et al. (2023).

with average age of ~ 15 Myr (Pecaut & Mamajek 2016; Ratzenböck et al. 2023), as shown in Fig. 4.2. Although across the sub-groups there is an age spread of $\pm 6 - 7$ Myr, Fig. 4.2 shows a clear relationship between average age and position of Sco-Cen sub-groups: the older clusters, > 15 Myr, are located closer to the centre, while the younger occupy the outskirts of the association. This spatial configuration was confirmed, and further investigated using Gaia data and 3D-mapping algorithms by Ratzenböck et al. (2023), which allowed the division of the associations in even more sub-groups (Fig. 4.3). This study, and the previous work by Ratzenböck et al. (2022), divided Upper Sco into up to nine sub-clusters with ages between 3 and 19 Myr. This division has important implications in the investigation of protoplanetary discs and planet formation, particularly because discs in Upper Sco should be regarded more cautiously as representative of *old* systems due to the presence of a large age spread. Moreover, when interpreting observations and comparing with disc evolution models, it should be considered that the properties of discs in different sub-groups may vary as a result of their distinct evolutionary histories.

The peculiar age gradient across the Sco-Cen association has been interpreted as the sequential dissolution of unbound systems, and a direct consequence of the star-formation process triggered by stellar feedbacks. Ratzenböck et al. (2023) identified four stages during the evolution history of Sco-Cen, which are associated with bursts of elevated star formation activity at ~ 20 Myr, 15 Myr, 10 Myr, and 5 Myr. Multiple events of supernovae explosion were proposed as a trigger mechanism for the star-formation in the Sco-Cen associations, and de Geus (1992) and Preibisch & Mamajek (2008) proposed a number of ~ 7 past supernovae. The strong candidate supernovae which presumably started star-formation in Upper Sco is the progenitor of the pulsar PSR J1932+1059 and the runaway star ζ Ophiuchi, which has been supposed to be a O5-O6 star that exploded 1.5 Myr ago (Hoogerwerf et al. 2001).

The Sco-Cen association serves as an important laboratory for studying the latest stages of the disc lifetime. In particular, most of our knowledge of disc behaviour at ages > 3 Myr comes from observations in the Upper Sco region.

4.3 The Orion OB1 association

The Orion OB1 association is the closest to the Sun (300 – 450 pc, Brown et al. 1994; Zari et al. 2018) that is still closely associated with a giant molecular cloud, the Orion Molecular Cloud (OMC), and that presents a vast population of young early-type OB stars and active star-formation. It contains at least 60 OB-type stars (Quintana et al. 2025) and is historically divided into four main groups based on the projected position on the sky (Blaauw 1964): Ori OB1a, Ori OB1b, Ori OB1c, and Ori OB1d. In the left panel of Fig. 4.4 we show the location of the four subgroups and the λ Orionis (λ Ori) region. Ori OB1a includes the 25 Orionis (25 Ori) region and is the oldest part of Orion, with an average age $\sim 8 - 12$ Myr (Brown et al. 1994; Briceño et al. 2005), and presents little or no remnants of the original giant molecular cloud. Ori OB1b is visually associated with the three bright stars of the Orion Belt (ϵ , δ , and ζ Ori, respectively of spectral type B0I, O9.5II, O9.7I+B0III), which are located at a smaller radial distance from the Sun ($\sim 200 - 300$ pc) with respect to the average Orion population. This region contains some among the youngest stellar clusters in Orion, which are NGC 2024 and NGC 2068. Ori OB1d contains the core of the Orion Nebula Cluster (ONC), which is the densest cluster in Orion and hosts a group of massive stars in the centre, the stars of the Trapezium, including the bright θ^1 C. Ori OB1c is separated from the ONC core due to his older age of $\sim 2 - 5$ Myr, similar to the age of the σ Ori cluster. A more detailed division of the stellar population

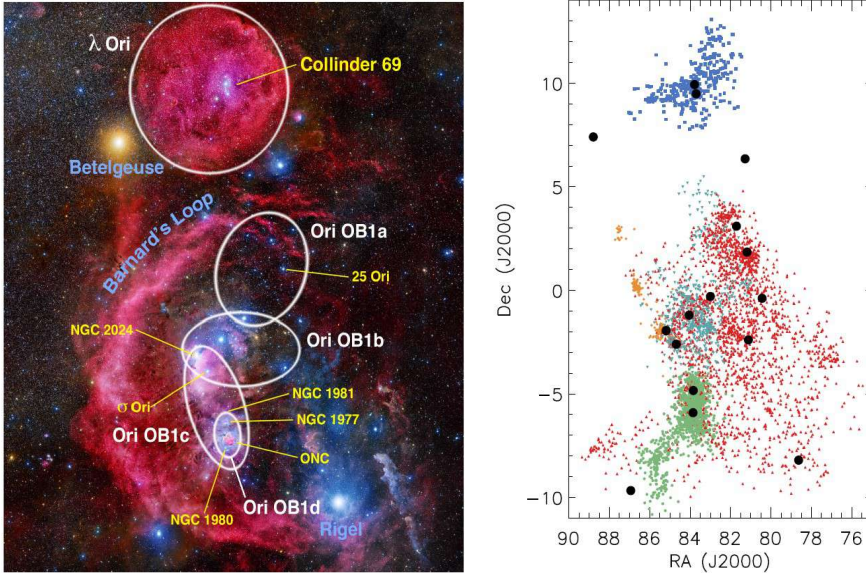


Figure 4.4: Left: Wide-field map of the Orion constellation showing the locations of the four sub-groups of Orion OB1 and the λ Ori region (white ellipses and labels). Prominent star clusters across the region are shown in yellow, Barnard’s Loop and the bright stars Betelgeuse and Rigel in light blue. From Wright (2020). Right: Division of young stars in Orion OB1 into five large structures: Orion A (green), Orion B (orange), Orion C (cyan), Orion D (red) and λ Ori (blue). From Kounkel et al. (2018a).

of Orion in sub-clusters was performed by Kounkel et al. (2018a) using clustering algorithms and the stellar sample selected from APOGEE and Gaia DR2. They kinematically divided the stellar population in 190 small sub-groups and five larger structures shown in the right panel of Fig. 4.4. The five larger structures are known as Orion A, containing the ONC and extending toward L1641 and L1647, Orion B, which includes the young NGC 2024 and NGC 2068, Orion C, in the central part of Orion, Orion D, consisting of the extended and old stellar population, and λ Ori in the north part. Orion D includes the foreground Orion group, Orion X, investigated by Bouy et al. (2014).

Contrarily to Sco-Cen, Orion does not show evidence of a clear age gradient indicating the star-formation history. A gradient in age is seen only in the ONC, where the stellar population in the core is younger than the more expanded population (e.g., Becchari et al. 2017). As shown in Becchari et al. (2017), the color-magnitude diagram (CMD) of the stars surrounding the ONC presents multiple pre-main sequence stars that are suggested to be the results of different stellar formation events.

The bright arc in H_α emission shown in the left panel of Fig. 4.4, which surrounds the west part of Orion and is known as Barnard’s Loop, is part of a larger bubble incorporating the Eridanus Loop (Sivan 1974; Ochsendorf et al. 2015). This huge bubble is surrounded by a shell of ionised gas that is expanding at velocity $\sim 20 \text{ km s}^{-1}$ and is suggested to have hosted $\sim 10 - 20$ supernovae explosions in the last 12 Myr (Bally 2008).

Orion OB1 is the closest region to the Sun that allows us to investigate protoplanetary disc properties in a typical environment, where discs can be significantly influenced by the UV radiation from the neighbouring OBA stars.

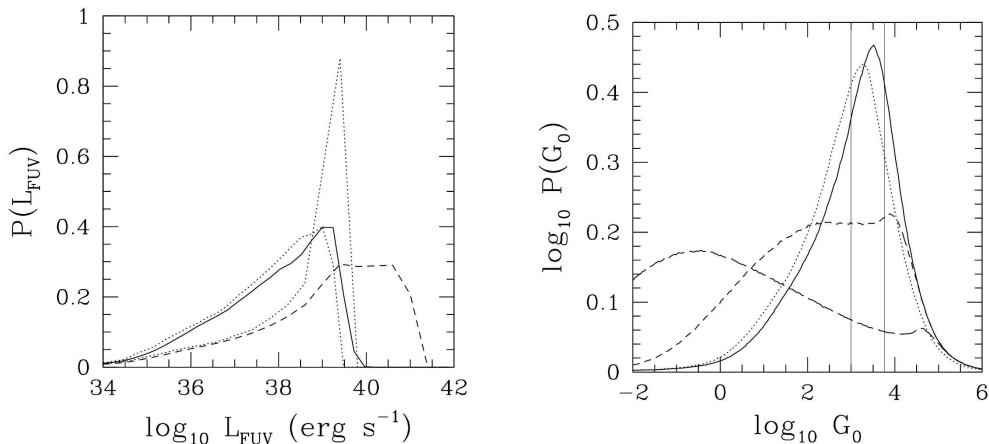


Figure 4.5: Left: Probability distribution of the FUV luminosity for the sample of clusters observed in the solar neighbourhood (Lada & Lada 2003) (solid curve), the corresponding distribution for the extended cluster distribution to $N = 10^5$ (dashed curve), the analogous distributions for the case in which only the most massive star is allowed to contribute to the FUV luminosity of the cluster (dotted curves). Right: Distribution of the FUV flux determined by sampling over the Lada & Lada (2003) distribution of cluster sizes. The solid curve shows the full distribution, the dotted curve shows the reduced distribution using only the radiation from the most massive star, the dashed curves result from the inclusion of a strong and weak extinction (long-dashed and short-dashes, respectively). The vertical lines mark the benchmark flux values for which the background cluster radiation dominates that of the central star (left) and for which the disk is evaporated in 10 Myr (right). From Fatuzzo & Adams (2008).

4.4 UV radiation field of young clusters in the solar neighbourhood

Investigating the distribution of UV radiation fields in young clusters in the solar neighbourhood is essential to access the importance of the environment in shaping the evolution of nearby protoplanetary discs and their planetary systems. Once the locations of the OB associations are identified, it is useful to perform a statistical analysis to investigate the average FUV flux experienced by protoplanetary discs in the nearby star-forming regions. This requires defining the distribution of members in star-forming regions, dn_{SFR}/dN , where N is the number of stellar members. N is fundamental, as it determines the number of massive stars in the region, but it is also highly uncertain, since establishing reliable memberships of star-forming regions is challenging (particularly at larger distance from the Sun, due to uncertainty in parallax and proper motion measurements). It has been shown that, given a certain cluster size, clusters with more than 1000 members (e.g., the Trapezium) can be disruptive to the star formation process (e.g., Störzer & Hollenbach 1999), whereas in small clusters with fewer than 100 members the cluster dynamics and environment have little impact (e.g., Adams & Myers 2001). Young clusters in the solar neighbourhood contain $N \sim 100 - 1000$ stellar members (Lada & Lada 2003) and therefore the impact of dynamical encounters and background radiation need to be considered. Using N-body simulations, Adams et al. (2006) showed that in clusters with $N \sim 100 - 1000$ members the rate of stellar interactions is low, where a solar system is suggested to experience a single encounter at a

distance of ~ 1000 au across 10 Myr of evolution. Fatuzzo & Adams (2008) investigated the distribution of cluster size and UV luminosity of the observed clusters in the solar neighbourhood. They adopted as dn_{SFR}/dN the one observed by Lada & Lada (2003), which is biased due to a smaller statistics in massive star-forming regions, and then used an extrapolation from a log-normal distribution up to a membership sizes $N = 10^5$. For a given initial mass function (IMF), the mean total luminosity of a star-forming region with number of members N is

$$\langle L_{\text{FUV}} \rangle = N \int \frac{dn}{d \log M_{\star}} L(M_{\star}) d \log M_{\star}, \quad (4.1)$$

where $L(M_{\star})$ is the luminosity of a single star of mass $M_{\star}$. In order to derive the distribution of FUV fluxes, we can convolve the distribution of FUV luminosities with a distribution of radial positions in the clusters assuming a density of stars $\rho_{\star} \propto r^{-1}$ and, equivalently, a distribution of masses $dm \propto \rho r^2 dr \propto r dr$. This stellar density distribution is equivalent to what was found in N-body simulations by Adams et al. (2006) of clusters considering a similar number of members. The resulting distributions of the FUV luminosity and FUV flux in the solar neighbourhood are shown in Fig. 4.5, where the dotted curves isolate the contribution of the most massive stars to the global distribution. The contribution of an extinction factor to shielding UV radiation is shown in the right panel of Fig. 4.5, considering a stronger and weaker extinction (long-dashed and short-dashed curves, respectively). The FUV flux distribution of young clusters in the solar neighbourhood peaks at $\sim 3 \times 10^3$ G₀. However, Fatuzzo & Adams (2008) emphasized that 90% of the stars in the observed sample in the solar neighbourhood are born in clusters with $N < 1000$, which is the regime of incomplete sampling of the IMF, and that the UV luminosity is affected by incomplete sampling. Therefore, it is necessary to account for the full distribution of possible FUV luminosities, rather than relying solely on mean or median values, when assessing the role of radiation fields in these environments. A critical point in this analysis is that, in order to characterise the FUV flux experienced by protoplanetary discs, we need to estimate the typical distance from OBA-type stars, which requires knowledge of the 3D geometry of the star-forming region, which is often affected by large uncertainties. Fatuzzo & Adams (2008) used the Larson (1981) relation, that the surface density of the cluster is independent of the cluster size, $R \propto \sqrt{N}$. However, for very young and massive, or very old clusters, this relation can be steeper or shallower. Winter et al. (2020b) reproduced a stellar density distribution in a given galactic environment by using a log-normal density distribution of turbulent flows and a theoretical star formation efficiency. Therefore, the distribution of the FUV flux can be estimated given the mass function for the SFRs in the solar neighbourhood.

The results from Fatuzzo & Adams (2008) and Winter et al. (2020b) consistently find that the majority of young clusters are found in the range $3 \times 10^2 - 3 \times 10^4$ G₀ (right panel of Fig. 4.5). Therefore, characterising disc evolution and planet formation in this range of FUV fluxes means accessing the role of the environment in typical birthplaces in the solar neighbourhood.

Part II

External photoevaporation of moderately irradiated discs: Upper Scorpius and Lupus

Extrnal photoevaporation shaping disc masses and radii in Upper Scorpius



A Scorpion was sent to punish Orion for his arrogance, that is why they never share the sky, forever avoiding each other. When Orion was slain by the Scorpion, Ophiuchus, which is placed close to it in the sky, restored him to life - The picture shows a portion of the sky close to ρ Ophiuchi and Antares. [Sampeyre, July 2025]

This Chapter summarises the main results of the AGE-PRO ALMA Large Program and focuses, in particular, on Anania et al. (2025a), which explored the role of external photoevaporation in shaping protoplanetary disc masses and sizes in the Upper Scorpius star-forming region.

The ALMA survey of Gas Evolution in PROtoplanetary discs (AGE-PRO; ID 2021.1.00128.L) has been designed to investigate the evolution of gas discs across ages, targeting 30 objects in three nearby star-forming regions: Ophiuchus, Lupus, and Upper Scorpius. In this Chapter, we present the main results of AGE-PRO, and we emphasise the importance of accounting for the environmental irradiation when comparing star-forming regions of different ages, even when moderate FUV radiation fields are experienced, as in the case of Upper Scorpius. In Anania et al. (2025a), we showed that the process of external photoevaporation, which causes an outside-in depletion of disc material due to environmental UV radiation, can lead to a significant decrease in disc size, mass, and

lifetime even at moderate irradiation levels (1-10 G_0). We present the comparison between the disc masses and sizes resulting from 1D numerical viscous evolution models in which the effect of external photoevaporation is included, and the values retrieved from the AGE-PRO observations.

5.1 The ALMA survey of Gas Evolution in PROtoplanetary discs (AGE-PRO)

Over the past decade, ALMA has revolutionized our understanding of *dust* in protoplanetary discs by revealing rings, gaps, and spirals, linked to planet formation (e.g., Andrews et al. 2016; Long et al. 2018; Cieza et al. 2019; Andrews 2020) and by showing that dust disc mass decreases over time, based on statistical studies of ~ 1500 nearby discs (Manara et al. 2023a). However, the lack of understanding of the global evolution of *gas* discs has become a primary bottleneck in the field of planet formation (Morbidei & Raymond 2016; Mordasini & Burn 2024). Gas disc mass and size are fundamental properties to determine in order to understand the basic architecture and evolution of discs. The ALMA survey of Gas Evolution in PROtoplanetary discs (AGE-PRO) is an ALMA Cycle 8 Large Program (2021.1.00128.L, PI Zhang) that aimed to systematically trace the evolution of gas disc mass and size throughout the lifetime of protoplanetary discs. The two primary goals of AGE-PRO are to: (i) provide a legacy library of high-quality line observations that enable robust measurements of gas masses and sizes of 30 protoplanetary discs across the typical disc lifetime (0.5-6 Myr); (ii) constrain global disc evolution mechanisms by comparing model predictions with mass and size measurements of discs at different ages. The results provide a baseline test for the viscous and MHD disc wind models of protoplanetary discs, serving as an essential context for future in-depth studies of planet formation processes in discs. The disc sample was selected from three nearby star-forming regions of different ages: Ophiuchus (embedded discs, $\lesssim 1$ Myr), Lupus (middle age, $\sim 1 - 3$ Myr), and Upper Sco (the end of gas disc lifetime, $\sim 4 - 6$ Myr). All three regions have been characterized by ALMA shallow surveys (Ansdell et al. 2016; Barenfeld et al. 2016; Cieza et al. 2019; Carpenter et al. 2025), and are nearby (140-160 pc), providing high sensitivity and spatial resolution. The AGE-PRO sample consists of Class I objects in Ophiuchus and Class II objects in Lupus and Upper Sco. The sample was selected on the basis of three main criteria: (i) stars with known stellar spectral type ranging between M3 and K6 (i.e., stellar masses in the range $\sim 0.3-0.8 M_\odot$), to minimise the correlation between disc properties with stellar mass; (ii) stars that have no stellar companions or are wide-separation binaries (>600 au); (iii) stars from previous detections of mm continuum and CO line emission.

5.1.1 Main results

The first results of AGE-PRO have been collected in 12 papers published in a dedicated focus issue by the Astrophysical Journal (ApJ). A description of the observations, the methodology for the self-calibration, the imaging processes, and the calculation of the mm continuum and CO fluxes can be found in Ruiz-Rodríguez et al. (2025) for the Ophiuchus sample, Deng et al. (2025) for the Lupus targets, and Agurto-Gangas et al. (2025) for the Upper Sco sources. In Fig. 5.1 we show the 1.3 mm continuum and ^{12}CO (2-1) moment zero images of the AGE-PRO sample. The median disc properties among the three investigated regions are shown in Fig. 5.2, and the main findings of AGE-PRO are summarised as follows.

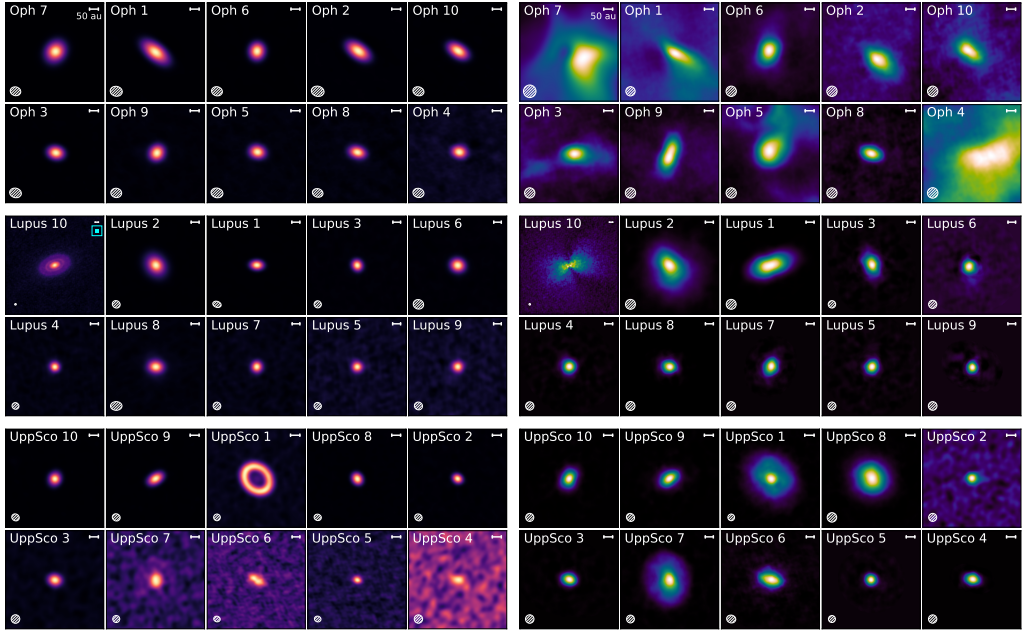


Figure 5.1: Gallery of the AGE-PRO observations: 1.3 mm (Left panels) and ^{12}CO (2-1) moment zero images (Right panels). The orders are from the brightest 1.3 mm continuum source to the faintest one within each star-forming region. Images are $2'' \times 2''$ and colours are peak normalized and shown with a linear colour stretch. The exception is Lupus 10 (V1094 Sco) which is shown in a $5'' \times 5''$ image and its continuum image is shown with an asinh stretch to accentuate fainter details. The difference in image size between Lupus 10 and the other sources is visualized in cyan in its continuum image. The beam sizes are shown in a circle on the lower-left of each panel and a 50 au size bar is shown in the upper right corner of each panel. From Zhang et al. (2025).

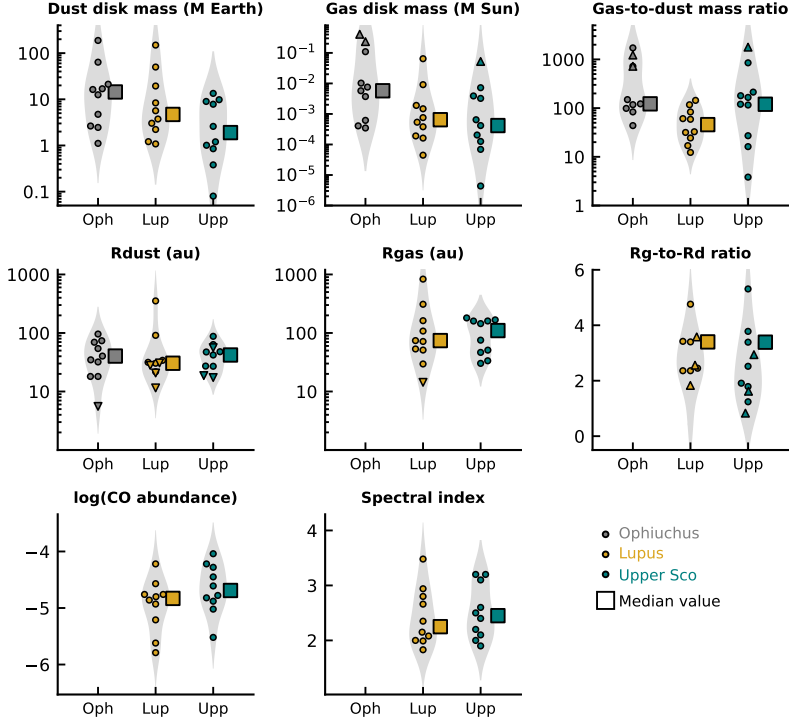


Figure 5.2: Comparison of disc properties in the Ophiuchus, Lupus, and Upper Sco samples in AGE-PRO. The circles are individual values and the squares are median values. The gas disc sizes, $R_{\text{CO},90\%}$ are measured as radius that encloses 90% of the ^{12}CO (2-1). Due to the severe cloud/envelope contamination in the ^{12}CO emission of the Ophiuchus discs, we only measure $R_{\text{CO},90\%}$ for the Lupus and Upper Sco discs. The CO abundances of Ophiuchus discs are assumed to be at the ISM level, and therefore are not plotted. The spectral index of Ophiuchus discs is not constrained due to the lack of Band 7 observations. From Zhang et al. (2025).

Gas disc mass

The gas disc masses were estimated by comparing the line fluxes with a large grid of thermochemical models (Trapman et al. 2025b). Specifically, the approach was to simultaneously fit line observations of CO isotopologues (^{12}CO , ^{13}CO , C^{18}O , C^{17}O 2-1) and N_2H^+ 3-2, a complementary molecule produced when CO freezes out, to a large grid of protoplanetary disc models, using an MCMC routine to derive gas masses and their associated uncertainties for the twenty AGE-PRO discs in the Lupus and Upper Sco star-forming regions. For the ten AGE-PRO discs in the Ophiuchus star-forming region, the observed C^{17}O 2-1 line fluxes were compared to a wide range of models to estimate their gas masses. In each star-forming region, the gas disc masses show a wide spread of 2-3 orders of magnitude. The median gas mass of the Ophiuchus disc sample is one order of magnitude higher than that of the Lupus and Upper Sco regions. There is no significant difference between the masses of the Lupus and Upper Sco surviving discs. This suggests that most of the gas disc mass evolution occurs at early times (between <1 Myr to 1-3 Myr).

Gas disc sizes

The gas disc sizes were computed by fitting 2D Nuker and Sersic profiles convolved with the clean beam to ^{12}CO moment zero maps (Trapman et al. 2025a). Using this approach, we measured the gas disc radius as the location in the disc enclosing the 90% of the ^{12}CO flux emission. The median disc size increases from Lupus (~ 74 au) to Upper Sco (~ 110 au). Although such an increase is in line with expectations for viscously driven disc evolution, it might also be explained by sample selection effects and survivorship biases. Indeed, in Anania et al. (2025a) we showed that a pure viscous evolution model is not able to explain the gas disc sizes that have been observed in Upper Sco, and that discs in this region may have experienced some degree of external photoevaporation during their lifetime due to the presence of massive OB-type stars in the region. With the inclusion of the effect of external photoevaporation to a viscous model we were able to reproduce most of the gas sizes for the AGE-PRO discs in Upper Sco (see next Section). The increase in the median gas disc size from Lupus to Upper Sco does not contradict the presence of external photoevaporation. As we show in Sec. 6.2, although external photoevaporation truncates discs in their outer regions, the median gas disc size in a population of moderately irradiated discs ($1\text{--}100\text{ G}_0$, the range of FUV flux in Upper Sco) can increase over time as a result of survivorship bias.

Comparison with population synthesis models

In Tabone et al. (2025), the gas disc masses and gas disc sizes obtained from AGE-PRO are compared to a set of population synthesis models: viscous evolution models including also the effect of internal photoevaporation and MHD wind-driven accretion models. The aim of this work was to constrain the model and initial conditions that best reproduce the disc properties of the three star-forming regions and the evolution of the median disc properties with ages. For the Upper Sco region, gas disc sizes were not used to draw conclusions on the best-fit model and initial conditions, as in Anania et al. (2025a) we demonstrated that gas disc radii are significantly affected by external photoevaporation. In this region, the observed gas disc masses and accretion rates were compared with the results of the models. MHD wind-driven models can reproduce disc fraction and the overall properties of the Ophiuchus, Lupus, and Upper Sco populations. It requires initially compact ($\sim 5\text{--}10$ au) discs with a moderate mass-loss rates $\dot{M}_W/\dot{M}_* \lesssim 1$, and an accretion time scale of about $t_{\text{acc},0} \simeq 0.4\text{--}0.8$ Myr ($\alpha_{\text{DW},0} = 5 \times 10^{-4}\text{--}10^{-3}$). This corresponds to discs with an initial magnetization of about $\beta \simeq 10^4\text{--}10^5$. Turbulence-

driven models with internal photoevaporation can reproduce disc dispersal and gas size but fail to reproduce simultaneously the accretion rate and the disc mass. The short apparent disc lifetimes of $M_D/\dot{M} \sim 0.1 - 1$ Myr for half of the middle-aged discs is the major tension with turbulence-driven models. Still, detailed studies of compact discs which are the lowest disc mass should be conducted to confirm their low masses.

Dust disc mass

The dust disc mass was inferred from the mm-fluxes (assuming optically thin emission and the same temperature and dust opacity), and decreases from the youngest to the oldest region, in agreement with previous works (e.g., Ansdell et al. 2016; Manara et al. 2023a).

Gas-to-dust mass ratio

Although both the median gas and dust masses decrease with age, the decrease rate of gas and dust appears to vary at different times. Consequently, the gas-to-dust mass ratio evolves from ISM values (~ 122) in Ophiuchus, to lower values in Lupus (~ 46), to ISM values in the surviving Upper Sco discs (~ 120). The drop in Lupus arises from a faster decline in gas mass than dust mass, whereas the increase in Upper Sco is driven mainly by dust mass loss.

Dust disc size

The dust disc sizes were computed by performing a visibility fitting to the dust continuum Band 6 1.3 mm data (Vioque et al. 2025). The median disc dust sizes are similar across the three regions (30-42 au), with the medians in Lupus and Upper Sco consistent with their values within 3σ , contrarily to what expected from a classic disc evolution theory. Sensitive and larger sample observations are needed to improve the statistics on the evolution of dust disc size.

Dust substructures

The continuum visibility fitting performed by Vioque et al. (2025) indicates that dust substructures are present in half of the AGE-PRO sample, with a tentative increase in the detection of large cavity-type structures in older discs. Substructures are detected in the young sources as well, suggesting that these can form in older embedded discs. This is consistent with the ALMA Large Program eDISK, which found more substructures in their Class I discs compared to Class 0 ones (Ohashi et al. 2023). Comparison with gas and dust disc evolution models showed that most of the discs in the AGE-PRO sample are consistent with simulations that have either weak or strong dust traps. An increase in spectral index as a function of age in the AGE-PRO sample is also suggestive of an unresolved population of dust traps (Kurtovic et al. 2025).

The AGE-PRO ALMA Large Program highlighted the importance of measuring gas discs and investigating disc properties on a population basis, across ages, to characterise the typical evolutionary path of protoplanetary discs. This is fundamental to understand the process of planet formation and the resulting planetary population. Expanding AGE-PRO to include a statistically larger sample of YSOs will enable more robust conclusions about how disc properties evolve with age. One of the key messages of AGE-PRO is that the stellar formation environment is a crucial factor that must be considered when comparing average disc properties across regions of different ages. As we will show in the next Sections, external photoevaporation has probably played a significant role in shaping the masses and radii of discs in Upper Sco.

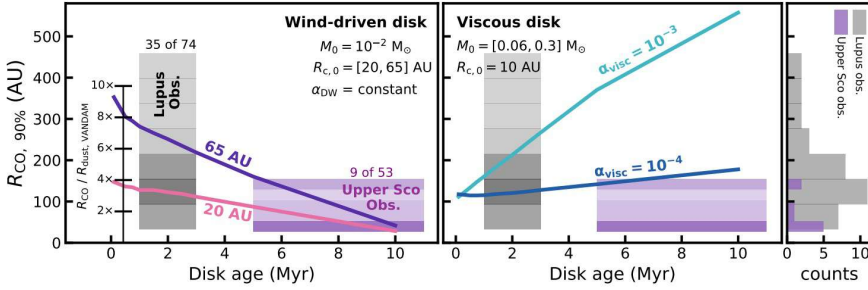


Figure 5.3: Comparison between gas disc sizes of an MHD wind-driven disc model (left panel), viscously evolving disc models (middle panel) and observations in Lupus (gray, Sanchis et al. 2021, Ansdell et al. 2018) and Upper Sco (purple, Barenfeld et al. 2016). Viscously evolving disc models were obtained from Trapman et al. (2020). Top-down views of the histograms on the left are included in the left and middle panel, where a darker shade in a given bin corresponds to a higher count in said bin. The x-axis and width of histograms corresponds to the age range of both star-forming regions. The ticked vertical line shows the mean R_{dust} , from the VANDAM survey (Tobin et al. 2020). The width of the horizontal lines is the average age range of Class I sources. From Trapman et al. (2022a).

5.2 Why Upper Scorpius

Upper Sco, located at ~ 142 pc from the Sun (Fang et al. 2023), is the oldest nearby Star-Forming Region (SFR), aged ~ 3 -14 Myr (e.g. Ratzenböck et al. 2023), which includes a large population of disc-hosting stars¹. It exemplifies the late stages of (Class II) disc evolution with discs that have survived both accretion and environmental processes. Upper Sco includes ~ 300 stars presenting full, evolved, transition or debris discs (Carpenter et al. 2025). This region is a sub-group of the Scorpius-Centaurus (Sco-Cen) OB association, which counts ~ 150 OB stars, ~ 50 of which are in Upper Sco (Luhman & Esplin 2020). Therefore, Upper Sco discs may have been strongly influenced by the prolonged exposure to external UV across their evolution. As a consequence, a sample of discs in the Upper Sco region may not be representative of the evolved phase of discs in low-irradiated SFRs, such as Taurus or Lupus, and should be modeled separately (Zagaria et al. 2023). Moreover, the work of Ratzenböck et al. (2023) has shown that the Upper Sco stellar population can be divided in nine sub-groups of different age, covering from ~ 3 to ~ 20 Myr, based on astrometric parameters and kinematics. Therefore, since part of the Upper Sco stellar population may be as old as the Lupus population, the large age spread must be taken into account when comparing disc properties across regions on the basis of the age.

The study of Trapman et al. (2020) presented preliminary evidence that the gas discs in the Upper Sco region appear to be smaller than predicted by viscous models (even when varying the level of viscosity). Moreover, the work of Trapman et al. (2022a) showed that even MHD wind-driven models fail to fully explain gas disc sizes in Upper Sco. As shown in Fig. 5.3, for the MHD wind-driven scenario to reproduce the observed Upper Sco disc sizes, discs at younger ages would need to have larger gas-to-dust size ratios than the values typically observed in young regions (e.g., a factor 8 larger than what observed in Class 0/I). A potential solution to this problem might involve includ-

¹In this work we refer to Upper Sco as a star-forming region for the presence of young stars even if, due to its age, the process of star formation has presumably stopped

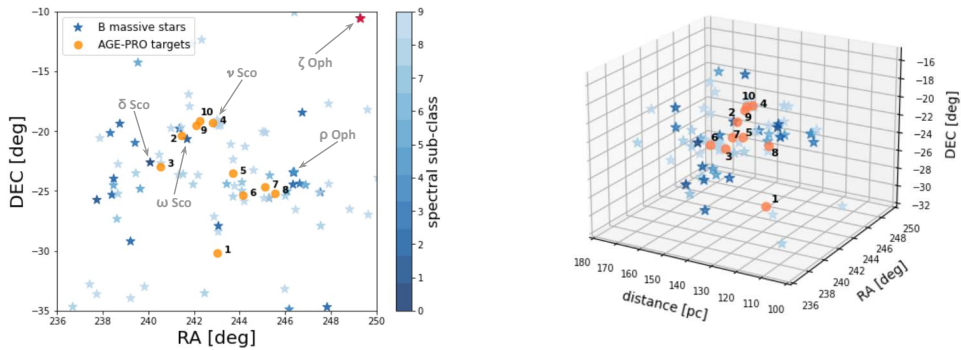


Figure 5.4: Distribution of the ten AGE-PRO sources in Upper Sco and the O and B stars closest to them. B-type stars are color-coded based on their spectral sub-class (from B0 to B9). The only O-type star survived in the considered region is ζ Oph, flagged in red. *Left:* projected-sky map. *Right:* 3D map.

ing external photoevaporation in the used disc evolution models. Indeed, in a pure viscous evolution framework, gas disc size is predicted to increase over time as a consequence of the re-distribution of angular momentum across the disc that drives accretion. In contrast, if the disc is truncated by external photoevaporation, the gas disc size will decrease with time, potentially at a rate even greater than that predicted by the MHD wind-driven framework. In order to investigate the role of external photoevaporation in shaping the disc masses and radii, we compared the observed disc properties of the ten AGE-PRO targets in Upper Sco, with those predicted by models that incorporate both viscous evolution and external photoevaporation. The methodology used in this work, the results obtained, a discussion on the results, and our conclusions are presented in the following Sections of this Chapter.

5.3 FUV flux calculation

The source sample investigated in this study consists of the ten Upper Sco protoplanetary discs targeted by the AGE-PRO ALMA Large Program. Table 5.1 summarises the stellar parameters and the dust and gas disc properties as retrieved from AGE-PRO. Since the majority of the massive stars in Upper Sco are late-type B, with no early-type O star in the proximity, we expect the FUV flux to dominate mass loss over the disc lifetime against EUV flux (Winter & Haworth 2022). Therefore, we focus only on this irradiation regime, where the range of wavelengths corresponding to far-ultraviolet is [912 - 2400] Å. Firstly, we select the O and B stars in the neighbourhood of each source (Sec. 5.3.1). Then, we calculate the FUV flux as the contribution of all the selected field stars (Sec. 5.3.2).

5.3.1 Field stars selection

From Luhman & Esplin (2020) and Luhman (2022a) we can retrieve a list of OB stars members of the Upper Sco region (49 stars). To avoid excluding any relevant stars from the calculation, we enlarge the sample to cover all OB stars in a ~ 55 degree-wide box that is centred on our source sample. This box size is large enough to capture meaningful FUV flux contributions (i.e. $> 0.1 G_0$) from nearby and distant OB stars and contains

Table 5.1: Upper Sco stellar and disc parameters

2MASS id	AGE-PRO id	dist [pc]	$M_{\star}$ [$M_{\odot}$]	age Myr	$\log M_{\text{gas}}$ [$M_{\odot}$]	F_{mm} [mJy]	$R_{12\text{CO},90\%}$ [AU]	R_{dust} [AU]
J16120668-3010270	UppSco 1	131.9	0.51	$4.37^{+4.7}_{-2.2}$	$-2.49^{+0.15}_{-0.11}$	34.27 ± 0.05	$166.9^{+5}_{-1.8}$	87.6 ± 0.6
J16054540-2023088	UppSco 2	137.6	0.13	$2.09^{+3.8}_{-1.3}$	$-3.9^{+0.38}_{-0.24}$	7.03 ± 0.03	$50.6^{+10.7}_{-23.4}$	< 17
J16020757-2257467	UppSco 3	139.6	0.37	$3.72^{+5.1}_{-2.1}$	$-3.19^{+0.24}_{-0.43}$	2.99 ± 0.04	$34.1^{+3.1}_{-1.9}$	27 ± 6.6
J16111742-1918285	UppSco 4	136.9	0.50	$2.09^{+1.8}_{-1.0}$	$-3.69^{+0.92}_{-1.33}$	0.208 ± 0.02	$46.3^{+1.1}_{-1.4}$	< 56
J16145026-2323297	UppSco 5	143.9	0.29	$3.8^{+6.9}_{-2.4}$	$-5.36^{+0.27}_{-0.2}$	0.68 ± 0.03	$30.8^{+1.7}_{-3.4}$	< 19
J16163345-2521505	UppSco 6	158.4	0.52	$5.89^{+7.3}_{-3.2}$	$-4.16^{+0.33}_{-0.33}$	1.91 ± 0.03	$159.7^{+15.2}_{-10.3}$	63 ± 19
J16202863-2442087	UppSco 7	152.7	0.34	$1.78^{+1.7}_{-0.9}$	$-3.38^{+0.29}_{-0.23}$	3.14 ± 0.03	$160.5^{+1.2}_{-1.7}$	47 ± 22
J16221532-2511349	UppSco 8	138.9	0.30	$2.4^{+3.1}_{-1.4}$	$-2.41^{+0.27}_{-0.23}$	22.21 ± 0.05	$144.1^{+1.5}_{-2.8}$	27 ± 0.2
J16082324-1930009	UppSco 9	136.9	0.56	$4.68^{+4.9}_{-2.3}$	$-1.28^{+0.2}_{-0.38}$	34.86 ± 0.12	$180.8^{+4.6}_{-8.7}$	47.8 ± 0.3
J16090075-1908526	UppSco 10	136.9	0.53	$2.34^{+1.9}_{-1.1}$	$-2.14^{+0.43}_{-0.45}$	38.04 ± 0.13	$74.6^{+1.8}_{-3.7}$	42 ± 1.5

Notes. For each of the AGE-PRO targets in Upper Sco, the table indicates (from left to right) 2MASS identifier, AGE-PRO identification name, distance (Bailer-Jones et al. 2021a), stellar mass, isocronal age (Zhang et al. 2025), gas disc mass (?), mm flux (?), gas disc radius (?), and dust disc radius (?) for which UppSco 2, UppSco 3, and UppSco 5 are upper limits.

in total 174 objects (see Anania et al. 2025a for the complete list of OB stars). Early A-type stars do not contribute significantly to the total FUV flux (Cleeves et al. 2016) and therefore are not included in our calculation. The stellar parameters are taken from Gaia DR3 (Gaia Collaboration et al. 2023b, Gaia Collaboration et al. 2023a)² and Hipparcos catalog (Perryman et al. 1997)³ when Gaia data are not available for the brightest sources. Afterwards, we searched in the ALS II catalog (Pantaleoni González et al. 2021) for any star included the previously defined box, but missing in the previous selection (due to unclear or undefined spectral type). We retrieve the spectral type and luminosity class of the stars from the SIMBAD database (Wenger et al. 2000). Moreover, we added to our selection the hot sources according to the Gaia Blue-Red pass band spectra (those with reported effective temperature in Gaia but missing spectral type in SIMBAD). The extended list of selected massive stars, along with their parameter, is available as supplementary material in Anania et al. (2025a).

Figure 5.4 illustrates the distribution of the ten AGE-PRO sources in Upper Sco and the OB stars closest to them, using a projected sky map (left panel) and a 3D sky map (right panel). The field is dominated by B9/B8 stars. ζ Oph, the only O-type star (in red in Figure 5.4), contributes with less than 0.5 G_0 to the total FUV flux on the AGE-PRO sources. The distance of the considered objects is determined through parallax inversion (we motivate this approach in Appendix 5.9.1).

5.3.2 FUV flux experienced by the sources

The FUV flux experienced by each disc is evaluated by adding the contribution of all the selected field stars. Spectral types and luminosity classes of the OB stars are derived from the SIMBAD database and subsequently linked to effective temperatures, thereby distinguishing between Main-Sequence stars, Giant and Super-Giants using the tables from Pecaut & Mamajek (2013a) and Gray & Corbally (2009a). For simplicity we are assuming a blackbody emission for the stars. While this tends to underestimate the total stellar flux, we expect the uncertainty to be dominated by the parallax measurements (see Appendix 5.9.1). Given the effective temperature, we fit the stellar masses using the stellar evolution models of Schaller et al. (1992). Subsequently, we derive the FUV luminosity from the stellar mass using the empirical relationship derived by Winter & Haworth (2022). Finally, the FUV flux experienced by each disc is determined by the contribution of the overall field stars:

$$F_{\text{FUV}} = \sum_B \frac{L_{\text{FUV},B}}{4\pi|x_{\text{disc}} - x_B|^2}, \quad (5.1)$$

where $L_{\text{FUV},B}$ is the FUV luminosity of the massive star and $|x_{\text{disc}} - x_B|$ is the relative distance of the disc from the massive star. It is worth noticing that in our approach we are neglecting the influence of the extinction from residual gas and dust on shielding protoplanetary discs from the FUV irradiation. On the other hand, recent studies have shown that in massive star forming regions where O-type stars are present, this effect can be relevant only within the first half million year of stellar evolution (Ali & Harries 2019, Wilhelm et al. 2023). Since the AGE-PRO sample in Upper Sco has age spread ranging from ~ 2 to ~ 6 Myr (Table 5.1), we do not expect this effect to have a significant impact on our flux calculation. In Section 5.7 we analyse the case in which the external irradiation acts on discs after a certain shielding time. In order to account for and quantify the

²<https://www.cosmos.esa.int/web/gaia/dr3>

³<https://www.cosmos.esa.int/web/hipparcos/home>

error on the flux value arising from uncertainties in parallax measurements of the stellar objects, we use a Monte Carlo method. Specifically, the parallaxes of the single objects are randomly sampled assuming a normal distribution around the Gaia measurements, with standard deviation given by Gaia parallax uncertainties, and recompute the flux using the new values. We repeat this operation enough times to extract the best flux value as the median of the distribution and the associated uncertainty (i.e. we stopped the iteration once the median flux stabilized, even with an increased number of iterations). A further discussion about the results obtained using this approach, as well as an estimate of the upper limit of the FUV irradiation field experienced by each source, is given in Appendix 5.9.1. Figure 5.5 shows for each source the median of the flux value and the relative uncertainties. The error bars reach down to the 16th percentile and up to the 84th percentile of the flux distribution. The extended distribution of FUV fluxes is shown in Appendix 5.9.1. In this calculation we are employing the current 3D separation between discs and massive stars (obtained by inverting the parallax from Gaia measurements), which is not a static parameter in time and might have influenced the experienced FUV field. We discuss this last topic in Section 5.7.3.

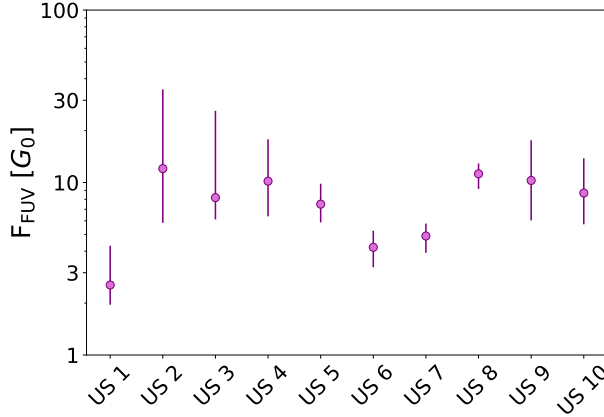


Figure 5.5: FUV fluxes experienced by the ten AGE-PRO targets in Upper Sco. Error bars extend from the 16th percentile to the 84th percentile of the distribution, while purple dots refer to the median values.

5.4 Disc evolution model

5.4.1 Gas evolution

We consider a geometrically thin and axisymmetric disc undergoing viscous accretion and photoevaporative mass loss in its outer region. The gas surface density $\Sigma_g(R, t)$ is obtained by solving the 1D standard diffusion equation (Lynden-Bell & Pringle, 1974; Pringle, 1981):

$$\frac{\partial}{\partial t} \Sigma_g = \frac{3}{R} \frac{\partial}{\partial R} \left(R^{1/2} \frac{\partial}{\partial R} (\nu \Sigma_g R^{1/2}) \right) - \dot{\Sigma}_{\text{ext}}, \quad (5.2)$$

where $\nu = \alpha c_s H_g$ is the kinematic viscosity, depending on the gas scale height H_g and the sound speed c_s , while $\dot{\Sigma}_{\text{ext}}$ accounts for the rate of mass lost in external photoevaporative winds. We assume the constant α viscosity prescription of Shakura & Sunyaev (1973).

The radial temperature profile is set considering the approximation for a flared disc, passively heated by the central star under an angle ψ :

$$T(R) = \left(\frac{\psi L_\star}{8\pi R^2 \sigma_{\text{SB}}} \right)^{1/4}, \quad (5.3)$$

where σ_{SB} is the Stefan-Boltzmann constant, and $L_\star$ is the stellar luminosity. We use a constant flaring irradiation angle, $\psi = 0.05$, leading to the temperature scaling with radius as $T(R) \propto R^{-1/2}$.

In our model, we neglect the effects of internal photoevaporation, since the goal of this work is to focus on the impact of a moderate external irradiation. Furthermore, external photoevaporation impacts disc evolution across the entire disc life, while internal photoevaporation is particularly relevant for disc dispersal at the end of disc life, on a timescale that is a small fraction of the disc lifetime (e.g. Clarke et al. 2001a).

In order to estimate the mass loss rate arising from external photoevaporation, we perform a bi-linear interpolation of a sub-set of the FRIEDv2 grid (Haworth et al. 2023a) (i.e. the updated version of the FRIED grid for mass loss rate Haworth et al. 2018), considering a fiducial ratio between the Polycyclic Aromatic Hydrocarbon (PAH) abundance and the dust abundance of 1, corresponding to ISM-like PAHs for ISM-like dust, and the *grain growth* model as grain growth is expected to happened in Upper Sco and wind is dust-depleted (Facchini et al., 2016; Haworth et al., 2018). The bilinear interpolation is performed using, for a given stellar mass and FUV flux, the disc surface density at each radial position to interpolate through the grid. We identify as *truncation radius*, R_t , the radius associated with the maximum mass loss rate, which corresponds to the position of the optically thin/thick transition of the wind at the FUV wavelengths. Results from numerical simulations support the fact that the photoevaporation rate increases in the outer, weakly bound, disc regions if the wind is optically thick in the FUV, and decrease with radius in the optically thin region (Sellek et al. 2020b, Haworth et al. 2023a). Following the numerical approach of Sellek et al. (2020b), we implement the outside-in depletion of gas and dust at $R > R_t$ weighting on the total mass external to the truncation radius, $M(R > R_t)$, such that the mass loss at a grid cell i is:

$$\dot{M}_{\text{ext},i} = \dot{M}_{\text{tot}} \frac{M_i}{M(R > R_t)}, \quad (5.4)$$

where $\dot{M}_{\text{tot}}$ is the total mass loss rate outside the truncation radius.

5.4.2 Dust evolution

The dust evolution is modeled considering a distribution of particle species, with different mass and size, undergoing the processes of dust transport, dust growth and dust entrainment with photoevaporative wind.

Referring to a particular dust species, the dust transport is described by the advection-diffusion equation (Birnstiel et al. 2010b):

$$\frac{\partial}{\partial t}(R\Sigma_d) + \frac{\partial}{\partial R}(R\Sigma_d\nu_d) - \frac{\partial}{\partial R}\left(RD_d\Sigma_g\frac{\partial}{\partial R}\left(\frac{\Sigma_d}{\Sigma_g}\right)\right) = -\dot{\Sigma}_{d,\text{ext}}, \quad (5.5)$$

where Σ_d is the dust surface density, $D_d = \nu/(1 + \text{St}^2)$ is the diffusivity (Youdin & Lithwick 2007b), which depends on the kinematic viscosity ν and the Stokes number

St , and $\dot{\Sigma}_{d,ext}$ is the rate of dust entrained and lost in photoevaporative winds. The Stokes number St characterizes how much dust particles couple with the gas, and hence it determines the radial drift velocity. It is defined as the ratio of the dust stopping time and the dynamical timescale, which in the Epstein regime (grain size $\ll$ particles mean free path) is given by (Birnstiel et al. 2012):

$$St = \frac{\pi}{2} \frac{a \rho_m}{\Sigma_g}, \quad (5.6)$$

where we are assuming spherical particles with radius a and material density ρ_m . At fixed dust and gas density, small grains ($St \ll 1$) follow the gas motion, while large particles ($St \gg 1$) move slowly in the radial direction and settle rapidly onto the mid-plane. At $St \sim 1$ dust particles drift quickly towards the star following the pressure gradient (Weidenschilling 1977b; Whipple 1972).

Dust growth through collisions is included in our model by solving the full coagulation equation known as Smoluchowski equation (Smoluchowski 1916) for a distribution of particle masses. Both radial drift and fragmentation are accounted for, considering that shorter drift timescales compared to growth timescales result in the removal of particles before they can achieve substantial growth. On the other hand, fragmentation occurs when dust grains collide at velocities exceeding their fragmentation threshold, v_{frag} . This results in the outer regions being dominated by drift, while fragmentation becomes important at smaller radii (Birnstiel et al. 2010a).

As for the gas evolution model, we use the FRIEDv2 grid to estimate the rate of dust lost in photoevaporative winds, considering that dust particles smaller than a certain critical size, a_{ent} , are entrained in the wind. The work of Facchini et al. (2016) demonstrates that this critical size is defined by the balance between the drag force exerted on dust grains by the wind and the gravity from the host star:

$$a_{ent} = \frac{1}{4\pi F} \frac{v_{th} \dot{M}_{ext}}{GM_\star \rho_d}, \quad (5.7)$$

where $4\pi F$ is the solid angle subtended by the wind, v_{th} is the velocity of the gas flowing outward, and ρ_d is the bulk density.

5.4.3 Simulation setup

Our numerical simulations are performed using `DustPy` (Stammler & Birnstiel 2022) along with an integrated module that incorporates the effects of external photoevaporation (Gárate et al. 2024) adapted to support the interpolation of the FRIEDv2 grid. The gas surface density is initialized by the Lynden-Bell & Pringle self-similar solution, considering a linear dependence of the viscosity from the radius $\nu \propto R^\gamma$ with $\gamma = 1$:

$$\Sigma_g(R) = \Sigma_0 \left(\frac{R}{R_c} \right)^{-1} \exp \left(-\frac{R}{R_c} \right) \quad (5.8)$$

where R_c is the characteristic radius, which corresponds to the exponential cut off and set the initial extent of the disc, and Σ_0 is set by the total disc mass.

The parameter space explored in our simulations is summarized in Table 5.2. We studied discs with initial different masses and radial extent. The values assumed by the α viscosity parameter are motivated by the empirical constraints required to explain disc accretion (Rosotti 2023).

Table 5.2. Parameter space

Parameter	Value
α	$10^{-2}, 10^{-3}, 10^{-4}$
$R_{c,0}$	[10, 40, 100] AU
$M_{\text{disc},0}$	[0.01, 0.1] $M_{\star}$

Note. — Values for the α viscosity parameter, initial characteristic radius $R_{c,0}$, and disc mass $M_{\text{disc},0}$ explored in our simulations.

We assume that each target is subject to the FUV irradiation field evaluated as in Section 5.3. The temperature profile is based on the stellar luminosity (Eq. 5.3) which, for the sake of simplicity and in consideration of the stellar masses of our targets, we set to a representative value of $0.3 L_{\odot}$.

For all our simulations, we use a radial grid comprised of 250 cells equispaced in $R^{1/2}$. The radial extent of the grid is established based on the boundaries of the FRIEDv2 grid, with the inner and outer edge set to 5 AU and 500 AU respectively. We note that the grid outer edge is much larger than the observed extent of discs in Upper Sco ($\lesssim 200$ au, Trapman et al. 2025a). An important constraint is imposed by the minimum mass loss rate allowed by the FRIEDv2 grid. To ensure the reliability of our results, the mass loss rate must not drop below the lower limit set by the grid, which is $\sim 10^{-14} M_{\odot} \text{ yr}^{-1}$. Therefore, we stop our simulations if the computed mass loss rate falls below this threshold. This approach has no influence on the final result since by the time the disc reaches the minimum mass loss rate, the disc is nearly completely dispersed.

For the dust we consider a logarithmic grid in particle mass going from 10^{-12} g to 10^5 g, corresponding to a grain size distribution covering from $0.5 \mu\text{m}$ to 24 cm. The initial dust-to-gas ratio for the distribution of particles is 0.01 and the fragmentation velocity is set to $v_{\text{frag}} = 10 \text{ m s}^{-1}$ (Wada et al. 2011, Gundlach et al. 2011, Gundlach & Blum 2015).

In order to constrain the age up to which we should let the discs evolve in our simulations, we take into account the isochronal ages of the Upper Sco AGE-PRO sources derived from the stellar parameters (Zhang et al. 2025), which spans between ~ 2 and ~ 6 Myr, on average (see Table 5.1). Therefore, we compare the resulting disc masses and sizes at 2 and 6 Myr with the AGE-PRO observed values. Whenever the photoevaporative mass loss rate falls below the minimum allowed before 6 Myr, we stop the simulations at that time, as discs are nearly completely dispersed after that threshold and numerical results not reliable.

5.4.4 Evaluation of disc radius and mass

We consider as gas disc radius from the AGE-PRO observations the radius in the disc enclosing the 90% of ^{12}CO line flux (see Table 5.1 and Trapman et al. 2025a for the derivation). This radius is a good approximation of the position in the disc where the ^{12}CO

column density drops below a critical threshold, inhibiting self-shielding and causing photo-dissociation. In our models, this radius is evaluated considering that the gas column density at the radius enclosing the 90% of ^{12}CO line flux, $N_{\text{gas}}(R_{12\text{CO}})$, which refers to column density of H_2 , can be analytically prescribed as function of the disc mass and stellar luminosity (Trapman et al. 2023):

$$N_{\text{gas}}(R_{12\text{CO},90\%}) \approx 10^{21.27-0.53 \log(L_*)} \left(\frac{M_{\text{gas}}}{M_{\odot}} \right)^{0.3-0.08 \log(L_*)} \text{ cm}^{-2}, \quad (5.9)$$

where M_{gas} is the gas mass of the disc at a fixed time, and the stellar luminosity is set to $0.3 L_{\odot}$ in our model. The above expression is retrieved from fitting the results of DALI thermochemical models and depends on the initial carbon abundance and the disc temperature. Given the critical column density, $R_{12\text{CO}}$ is computed in our model as the radius corresponding to the critical value of the disc surface density, at each time-step:

$$\Sigma(R_{12\text{CO}}, t) = \Sigma_{\text{crit}}(t) = N_{\text{gas}}(R_{12\text{CO},90\%}, t) \mu_{\text{gas}}, \quad (5.10)$$

where μ_{gas} is the mean molecular weight of the gas in the disc, which we set to the standard value $\mu_{\text{gas}} \sim 2.3 \mu_{\text{H}} = 1.15 \mu_{\text{H}_2}$.

An external UV radiation heats the disc in its outer region, leading to an increase in temperature and subsequently altering the shape of the critical column density. In particular, as shown in Trapman et al. (2023), $N_{\text{gas}}(R_{12\text{CO},90\%})$ increases in smaller discs ($\leq 10^{-4} M_{\odot}$) subject to a $30 G_0$ FUV flux, gradually approaching the level observed in discs irradiated with $1 G_0$ for larger masses. The expression for the critical column density shown in equation (5.9) is valid when the external FUV flux is $1 G_0$. We interpolated Figure 2 of Trapman et al. (2023) to take into account the (constant) FUV flux applied. A higher N_{gas} due to higher FUV flux leads to higher Σ_{crit} , which depending on the shape of the surface density can result in smaller or similar $R_{12\text{CO},90\%}$.

For what concerns the dust component, from an observational point of view a measure of the disc mass is inferred from the sub-mm flux density (e.g. Ansdell et al. 2016, Barenfeld et al. 2016). To compare models and observations, we compute the integrated flux density from the disc accounting for the contribution of all the dust grains included in the simulations at a given wavelength. If we neglect scattering and assume a face-on⁴ and vertical isothermal disc, the sub-mm flux density from the disc is

$$F_{\nu} = \frac{1}{d^2} \int_{R_{\text{in}}}^{R_{\text{out}}} 2\pi R B_{\nu}(T) (1 - e^{-\tau_{\nu}}) dR, \quad (5.11)$$

where d is the distance to the source, which we consider ~ 142 pc for Upper Sco, $B_{\nu}(T)$ is the Planck function at temperature T and $\tau_{\nu} = \kappa_{\nu}(a) \Sigma_{\text{dust}}$ is the vertical optical depth. The dust opacity $\kappa_{\nu}(a)$ is computed using the DSHARP Mie-Opacity Library (Birnstiel et al. 2018), employing the volume fractions and the properties of the dust grains of the model by Ricci et al. (2010), for each particle size a . The Planck function $B_{\nu}(T)$ is evaluated assuming a constant dust temperature $T = T_{\text{dust}} = 20$ K, and all the dust properties are estimated at wavelength 1.3 mm, which corresponds to the continuum Band 6 observations of AGE-PRO (Agurto-Gangas et al. 2025). The flux estimate is obtained by taking into account that the sensitivity of the AGE-PRO survey is $\sim 3 \times 10^7$ Jy sr^{-1} (Agurto-Gangas et al. 2025). The dust radius is determined as the position in the disc that encompasses 90% of the continuum flux evaluated as in equation (5.11).

⁴The inclination do not influence the total flux as long as the disc is optically thin, but can lead to an overestimate of the flux (and the disc mass) if the disc is optically thick. Our models are always in the optically thin regime.

Table 5.3. Initial parameters of the test simulations

Parameter	Value
$M_{\star}$	$0.6 M_{\odot}$
$M_{\text{disc},0}$	$0.06 M_{\odot}$
$R_{\text{c},0}$	10 AU
α	10^{-3}
FUV	[0, 1, 10, 100, 1000] G_0

Note. — Stellar and disc parameters chosen for the test simulation. The FUV flux experienced varies from 0 to 1000 G_0 .

5.5 Testing the effect of external photoevaporation on disc properties

In order to explore the effect of external photoevaporation on disc mass and radius, we propose here an illustrative example in which we numerically evolve a protoplanetary disc, keeping the initial parameters constant except for varying the FUV flux. We use `DustPy` simulations and the initial parameters listed in Table 5.3. We let the simulations evolve until reaching 10 Myr, with the condition of stopping them earlier whether the photoevaporative mass loss drops below the minimum allowed by the `FRIEDv2`.

In Figure 5.6 we show in colored lines the time evolution of the gas surface density when the disc is subject to 1, 10, 100 and 1000 G_0 , compared to the pure viscous framework (FUV = 0 G_0) in gray dashed lines. We observe that under low irradiation conditions (1 G_0 and 10 G_0), the early evolution of the disc, spanning the initial 2-3 million years, is primarily influenced by viscosity, resulting in spreading. However, as the mass loss rate becomes significant, due to the increase in disc size via viscous spreading, the subsequent evolution is driven by external photoevaporation, leading to disc shrinking. Higher FUVs cause a more important loss of material in winds, and the disc is immediately truncated. In the 1000 G_0 scenario, the simulation stops before 5 Myr, when the mass loss rate reaches the `FRIEDv2` limit, coinciding with the near-complete dispersal of the disc.

It is crucial to point out that external photoevaporation is influencing disc evolution, leading to a substantial decrease in disc size, and in disc mass, even at low G_0 , as shown in Figure 5.7. Under a low/moderate irradiation (1-10 G_0), disc mass at 10 Myr may decrease by one order of magnitude compared to the prediction of the pure viscous model. However, considering that through direct observations disc sizes are usually known with higher precision than disc masses, a slight decrease in size (of few AU) may be more evident than a greater decrease in mass (hundreds of $M_{\odot}$). This is evident even in the results of the high-resolution observations of *AGE-PRO*, where disc masses show large uncertainties compared to disc sizes (see Table 5.1 and the following Figure 5.8 and 5.10). Furthermore, in the range [1-10] G_0 , disc sizes are significantly reduced by the first 2-4 Myr, whereas the difference in disc mass becomes relevant after 6-8 Myr. Therefore, particular attention should be directed towards size-related studies, especially if

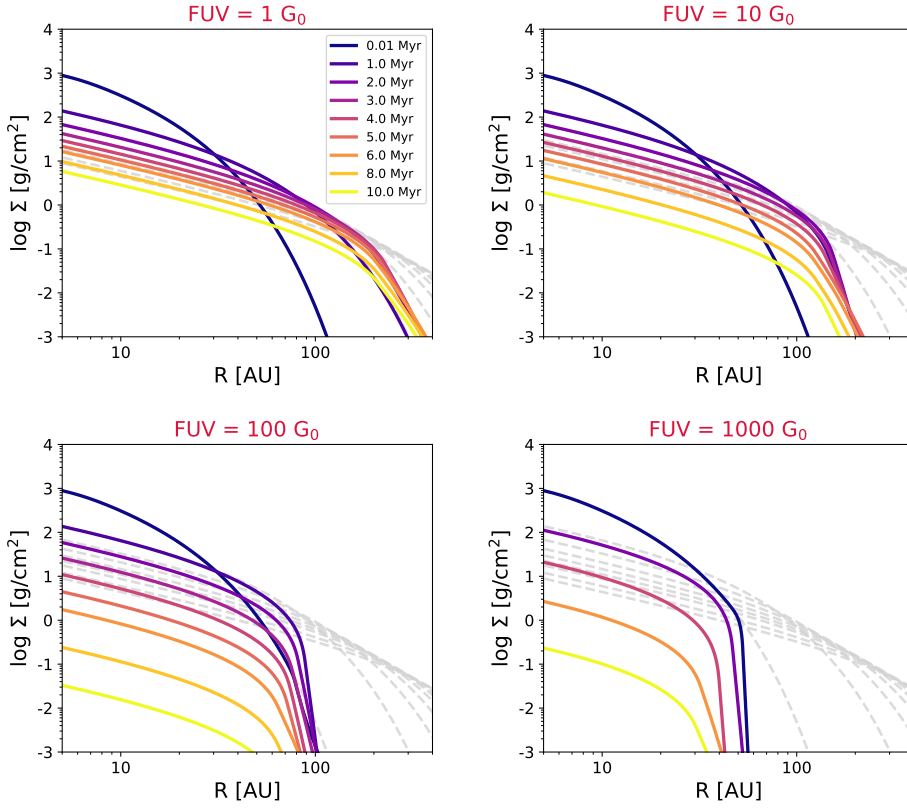


Figure 5.6: Evolution of the disc surface density when an external FUV field of 1, 10, 100 and 1000 G_0 is applied. Lines are color-coded based on the age, while the dashed gray lines corresponds to the $0 G_0$ pure viscous evolution from 0 to 10 Myr.

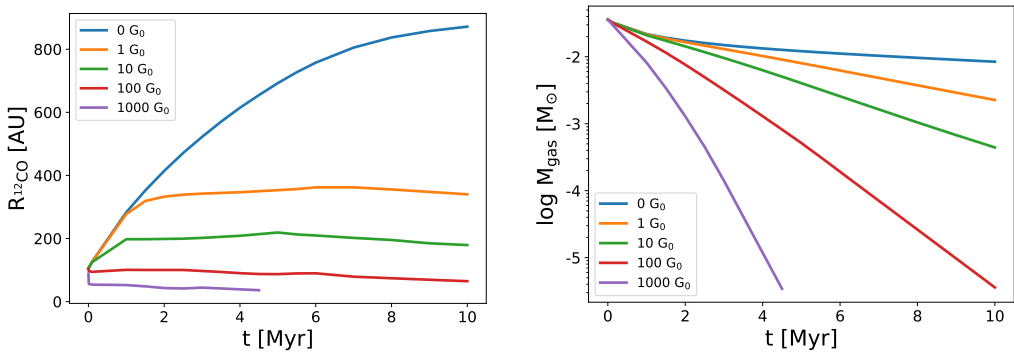


Figure 5.7: Left panel: time evolution of the disc radius when the FUV flux experienced is $[0, 1, 10, 100, 1000] G_0$. In the pure viscous model ($0 G_0$) the radius increases with time, but even low/moderate FUV fluxes (1-10 G_0) can halt viscous spreading. In the case of 1000 G_0 , we stopped the simulation when the minimum mass loss rate allowed by the FRIEDv2 grid is reached, and the disc has almost completely dispersed in ~ 5 Myr. Right panel: Time evolution of the total gas mass when the disc is subject to $[0, 1, 100, 1000] G_0$.

we want to investigate the role of external photoevaporation in moderately irradiated environments such as Upper Sco. We highlight that extreme caution should be applied if no viscous spreading is observed in mildly irradiated regions, as our results show that even moderate levels of irradiation can suppress disc spreading.

Since the FUV fluxes for the AGE-PRO targets in Upper Sco span (on average) from 2 to 12 G_0 (Figure 5.5), the results presented above motivate our investigation into disc masses and sizes accounting for external photoevaporation. We emphasize that the consequences on the overall disc evolution will be more or less significant depending on the stellar mass and initial disc parameters. These dependencies are explored in the following sections.

5.6 Results: gas and dust disc masses and radii

5.6.1 AGE-PRO gas results

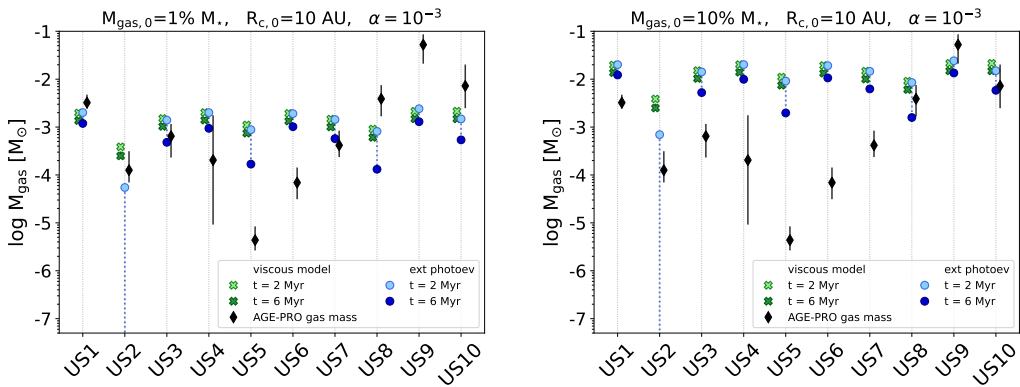


Figure 5.8: Gas disc masses in the purely viscous scenario (green crosses) and adding the effect of external photoevaporation (blue dots), compared with the gas mass from AGE-PRO (black diamonds). *Top:* Initial disc mass set to 1% $M_\star$. *Bottom:* Initial disc mass set to 10% $M_\star$. The dotted line associated with UppSco 2 is not connected to any blue dot, as it is predicted to be almost fully dispersed due to photoevaporation by 6 Myr.

Focusing for the moment on the gas disc component, we study the evolution of the ten discs in both the purely viscous scenario and when subject to an external FUV radiation field. In this Section we provide the results of the models in which we set the initial characteristic radius, $R_{c,0}$, to 10 AU, $\alpha = 10^{-3}$, and the initial disc mass to either 1% and 10% of the stellar mass, for each source.

In Figure 5.8 the disc mass values predicted by our models, at 2 Myr and 6 Myr, are compared with the disc mass retrieved from the AGE-PRO observations (Trapman et al. 2025b). The inclusion of an external radiation field causes a reduction in disc mass compared to the viscous framework, which becomes more evident at older ages (see green and blue markers in Figure 5.8). The observed gas masses of eight out of ten targets are consistent within the error bars with the results of the model incorporating external photoevaporation, if the initial disc mass is between 1% and 10% of the stellar mass, within a factor of 2 (see Figure 5.9). UppSco 5 and UppSco 6 are closer to the predictions made by the model with initial disc mass $0.01 M_\star$, but too low-mass to be well represented by the model used, as shown in Figure 5.9. Among the other eight targets, four are represented

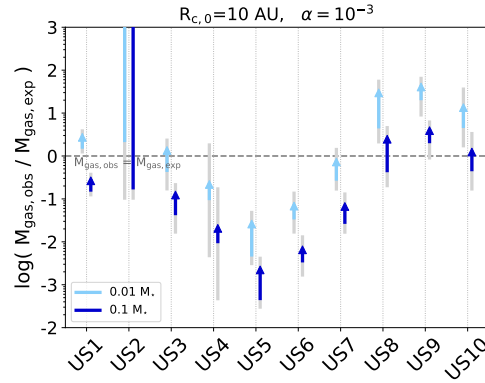


Figure 5.9: Ratio between the observed gas masses and the those predicted when external photoevaporation is included in the model. The colored lines refer to the values in the range of ages [2 - 6] Myr (with arrows pointing from younger to older ages) considering an initial disc mass of $0.01 M_{\star}$ and $0.1 M_{\star}$, respectively shown in light blue and dark blue. The shaded gray regions refer to the uncertainties in the observed gas masses.

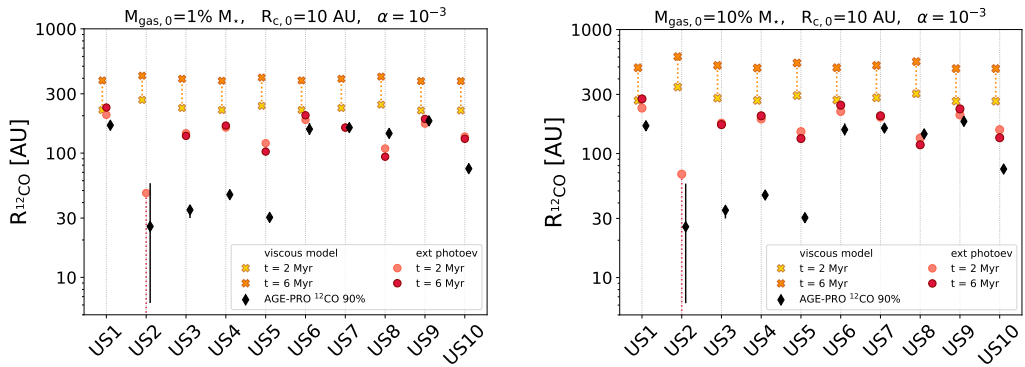


Figure 5.10: $R_{12\text{CO}}$ from our simulations in the purely viscous scenario (orange crosses) and adding external photoevaporation (red dots). Comparison with the radius enclosing the 90% of the ^{12}CO flux, observed by AGE-PRO. *Top* and *Bottom* panels refer to models with initial disc mass set to $1\% M_{\star}$ and $10\% M_{\star}$, respectively. The line related to UppSco 2 is not connected to any red dot, as it is predicted to be almost fully dispersed due to photoevaporation in < 6 Myr.

better by the case where the initial disc mass is $0.01 M_\star$ (UppSco 1, UppSco 3, UppSco 4, and UppSco 7, as shown in the top panel of Figure 5.8 and in Figure 5.9), and four sources (UppSco 2, UppSco 8, UppSco 9, and UppSco 10) prefer an higher initial disc mass of $0.1 M_\star$ (see the bottom panel in Figure 5.8 and Figure 5.9). However, relying solely on the analysis of the gas mass is insufficient for quantifying the impact of the external photoevaporation, as well as constraining the initial conditions. Therefore, in this study we intend to correlate this result with the information provided by gas radii and dust properties to derive our conclusions.

UppSco 2 is predicted to be almost entirely dispersed in less than 6 Myr due to external photoevaporation, considering it to be exposed to $\sim 12 G_0$. This result remains unchanged even when considering an initial higher mass for the disc ($0.1 M_\star$ in our case) and is caused by the combination of a small stellar mass ($\sim 0.13 M_\odot$, Table 5.1) against an experienced FUV flux of $\sim 12 G_0$ (Fig. 5.5). UppSco 2 is the only disc for which we stopped the stimulation including external photoevaporation before reaching 6 Myr.

The actual detectability of UppSco 2 may be attributed to a lower FUV exposure than what we assumed. Notably, we evaluated a significant uncertainty in the external flux experienced by this target, as shown in Figure 5.5. Moreover, as often discussed in the context of proplyds, the dynamics of the cluster can alter the relative distances between discs and massive stars, leading to variations in the FUV flux with time. Additional discussion on this topic will be carried out in Section 5.7.

In Figure 5.10, the disc radii resulting from our simulations are compared with the AGE-PRO $R_{12\text{CO},90\%}$ retrieved from the data analysis (Trapman et al. 2025a). As highlighted in previous works (Trapman et al. 2020), the observed radii (black diamonds in Figure 5.10) are highly inconsistent with viscous models (orange and yellow dots in Figure 5.10). The disparity between model and observations significantly decreases with the inclusion of the external photoevaporation. This result definitively shows the importance of accounting for the process of external photoevaporation even in regions where the FUV radiation field is moderate, as observed in Upper Sco.

Although the inclusion of photoevaporation drastically reduces gas disc sizes, not all our targets shrink over time (within the time range considered): between 2 and 6 Myr, UppSco 1, UppSco 6, and UppSco 9 are still predicted to experience viscous spreading, although the observed spreading is less significant compared to the pure viscous scenario.

The observed disc sizes of UppSco 3, UppSco 4, UppSco 5, and to a lesser extent, UppSco 10 are even smaller than predicted when accounting for external photoevaporation. This discrepancy might be caused by additional processes contributing to further disc depletion (e.g. MHD-driven winds, which could be relevant especially for UppSco 3 and UppSco 10, Fang et al. 2023), as well as a higher FUV flux than initially estimated, as we discuss in Section 5.7.3. Moreover, tidal truncation produced by the presence of a stellar or planetary companion results in a smaller disc radius.

5.6.2 Varying the disc viscosity

In this Section we discuss the impact of varying the α viscosity parameter, studying disc evolution when $\alpha = 10^{-4}$ and $\alpha = 10^{-2}$. The comparison between models with $\alpha = 10^{-4}$ and $\alpha = 10^{-2}$, both with and without external photoevaporation, and the observed masses and radii from AGE-PRO is presented in Figure 5.11 and 5.12. We show only the case in which $M_{\text{gas},0} = 1\% M_\star$ when $\alpha = 10^{-4}$, and $M_{\text{gas},0} = 10\% M_\star$

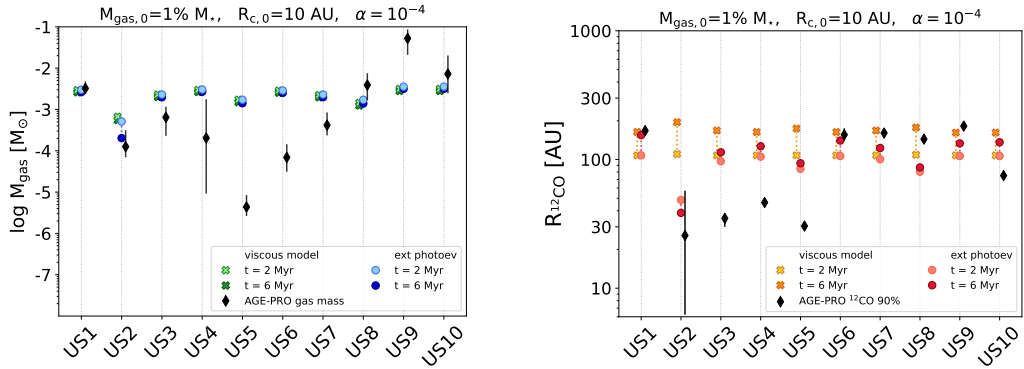


Figure 5.11: Gas disc masses (*top panel*) and radii (*bottom panel*) resulting from models where $\alpha = 10^{-4}$. The results from using the viscous model and adding external photoevaporation are compared to the AGE-PRO observations.

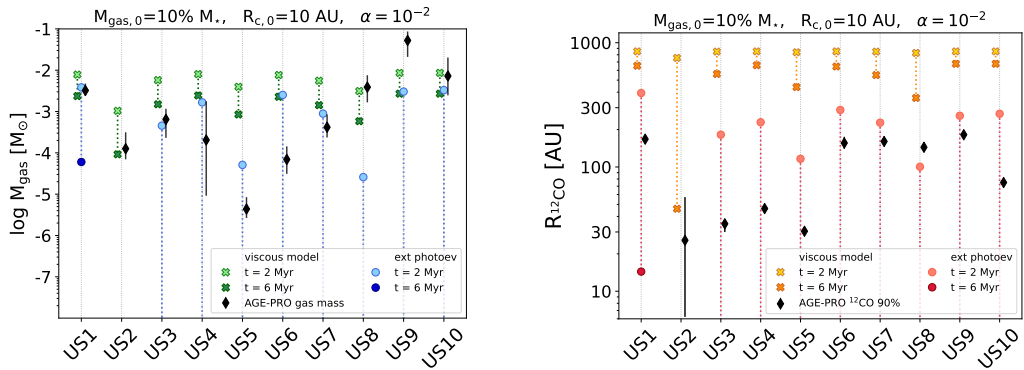


Figure 5.12: Gas disc masses (*top panel*) and radii (*bottom panel*) considering $\alpha = 10^{-2}$ in our simulations. We compare the results from the viscous model with those including external photoevaporation and AGE-PRO observations. Dotted lines without end point means that discs are dispersed in less than 6 Myr.

when $\alpha = 10^{-4}$, as the other models disagree more from observations.

According to the viscous theory, a higher viscosity results in a greater mass accretion rate, implying a shorter disc evolution timescale. As a consequence, after the same amount of time the disc experiences a more significant mass reduction and spreads more efficiently. Consequently, the effect of external photoevaporation is expected to be more relevant if the disc is more viscous. However, relying solely on the analysis of the resulting gas radii, and comparing them with the observations (Figure 5.11) is insufficient to establish a preference between $\alpha = 10^{-4}$ and $\alpha = 10^{-3}$. While UppSco 3, UppSco 4, and UppSco 5 still present too small gas disc sizes to be explained by the inclusion of photoevaporation, the other targets show gas sizes that could be consistent with the models. However, the large discrepancies in disc mass (see Figure 5.11) disfavour the model with $\alpha = 10^{-4}$ for the majority of the sources. An exception is UppSco 2, if we assume an the initial disc mass of $0.01 M_{\star}$. We remark that due to the substantial uncertainties in the FUV flux experienced by this target, it is challenging to make reliable claims.

In the case of $\alpha = 10^{-2}$ (Figure 5.12), the viscous model completely fails in reproducing gas disc sizes and the majority of gas disc masses. When external photoevaporation is included, UppSco 2 is dispersed in 1 Myr, UppSco 5 and UppSco 8 in 3.5 Myr, while the other sources are dispersed between 5 and 6 Myr. The sole survivor is UppSco 1 if $M_{\text{gas},0} = 0.1 M_{\odot}$. The observed gas mass and size of UppSco1, UppSco 4, UppSco 5, UppSco 6, and UppSco 7 may be well reproduced by this model if the age of the those sources is bigger than 2 Myr but smaller than 6 Myr (the other sources present inconsistency between masses and sizes). Given the large uncertainties in the isocronal age estimate of the targets (see Table 5.1), only UppSco 4 and UppSco 7 present a range of ages covering values smaller than 6 Myr that make them potentially consistent with $\alpha = 10^{-2}$. Assuming UppSco 5 and UppSco 6 to be younger than 6 Myr, the model with $\alpha = 10^{-2}$ may explain the observed properties better than the $\alpha = 10^{-3}$ model. However, the large error bars in disc age prevent us from favouring $\alpha = 10^{-2}$ against $\alpha = 10^{-3}$ for these two sources.

We highlight that the claim that external photoevaporation is affecting disc masses and sizes in Upper Sco discs is supported by AGE-PRO observations even when a higher α viscosity is assumed.

As final remark, it is worth noticing that the α viscosity parameter is expected to vary over time and across the disc (e.g., Delage et al. 2022), while we assume it as a constant. Given our incomplete understanding of disc accretion mechanisms, we can consider the selected value as a representative average across the disc extent and lifetime.

5.6.3 AGE-PRO dust results

In Figure 5.13 we compare the integrated emission at 1.3 mm observed by AGE-PRO with the integrated flux resulting from our simulations, considering both scenarios with and without the contribution of the external FUV field. In the case of an initial disc mass set to $0.01 M_{\star}$, the targets exhibit millimeter fluxes above (or slightly consistent with) the model's predictions, even when considering the purely viscous framework. A similar pattern is observed for the dust radii, as shown in Figure 5.14. Increasing the initial disc mass leads to an overestimate of the dust disc sizes, with few sources being slightly consistent with the external photoevaporation model. In our models we adopted as dust radius the position in the disc enclosing the 90% of the millimeter continuum flux, evaluated using equation (5.11). This analytic value is compared with the corresponding

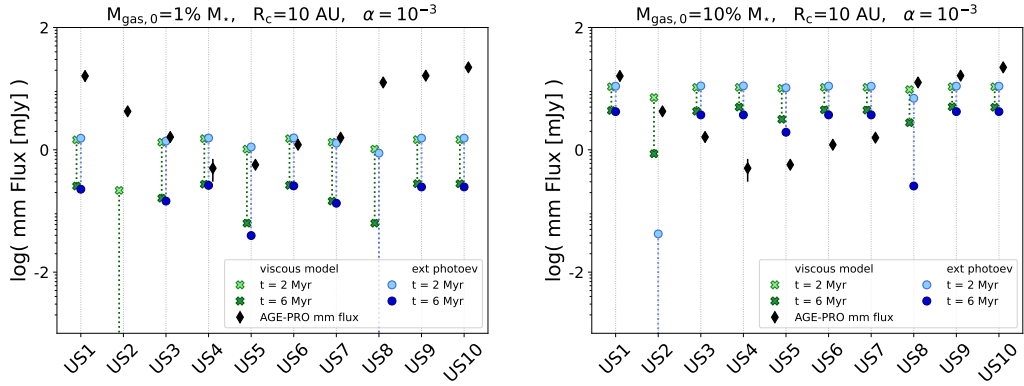


Figure 5.13: Comparison between the observed 1.3 mm continuum flux density of the ten AGE-PRO targets and the simulation results at 2 and 6 Myr, considering the purely viscous model (green crosses) and adding external photoevaporation (blue dots). The panels show the outcomes considering an initial disc mass of 1% and 10% of the stellar mass, respectively shown on *top* and *bottom* panel.

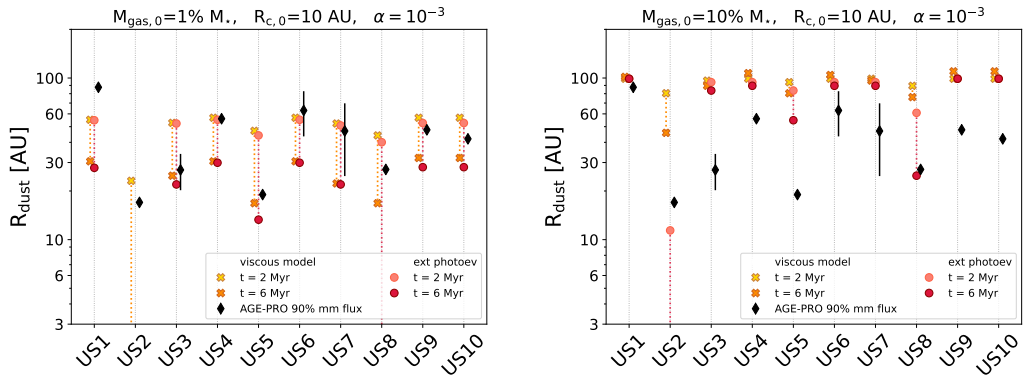


Figure 5.14: Dust radius: enclosing the 90% of the 1.3 mm continuum flux, determined through AGE-PRO visibilities (black diamonds, where the values for UppSco 2, UppSco 4 and UppSco 5 are upper limits), and from our simulation with and without the effect of external photoevaporation (in orange crosses and red dots). *Top* and *bottom* panels show the results for an initial disc mass of 1% and 10% of the stellar mass, respectively.

$R_{\text{dust},90\%}$ retrieved from the AGE-PRO visibility fitting of continuum Band 6 observations (Vioque et al. 2025).

In general, the models are not able to reproduce consistently both dust radii and mm fluxes. A potential solution to this is the role of dust substructures (Kurtovic et al. 2025), which are not included in our models.

5.6.4 Gas and dust relative sizes

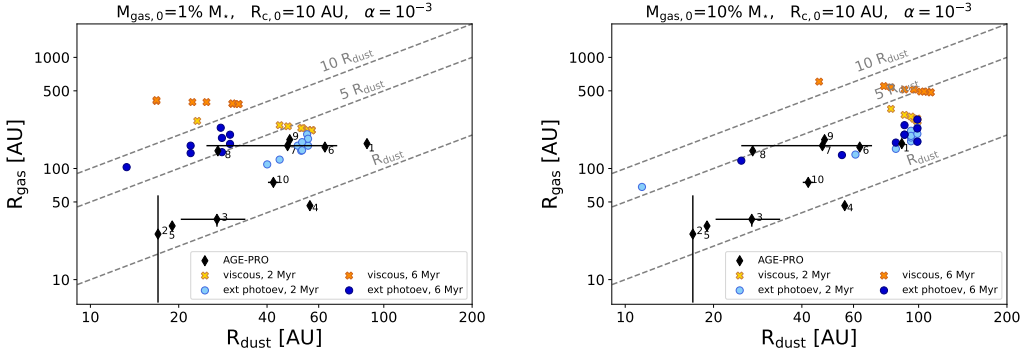


Figure 5.15: The gas and dust sizes detected by AGE-PRO in Upper Sco (black diamonds) are compared with the simulations results after 2 Myr (in light colors) and 6 Myr (in dark colors). The simulations refer to an initial disc mass of 1% and 10% of the stellar mass on the top and bottom panels, respectively. Orange and yellow dots are related to the sole viscous evolution model, while blue and light blue dots refer to the models where external photoevaporation is also included.

Gas and dust relative sizes resulting from our models are compared with the AGE-PRO observations. For the bulk population of discs in non-irradiated star-forming regions, it is known that smooth model fails in reproducing the observed gas-to-dust relative sizes (Toci et al. 2021), specifically producing a dust disc that is too small for a given gas disc. Moreover, the size ratio observed in Upper Sco is not too different from Lupus (Sanchis et al. 2021, Zhang et al. 2025). Because we showed that gas disc sizes are reduced by external photoevaporation, it is worth investigating whether our models yield similar conclusions. We verify that this discrepancy holds in Upper Sco as well, as can be seen in Figure 5.15.

The viscous (smooth) model predicts higher gas and dust relative sizes (identified as orange and yellow dots in Figure 5.15) with respect to the AGE-PRO observations (black diamonds in Figure 5.15), as already found by other authors (e.g. Toci et al. 2021). If an external FUV field is experienced, discs are significantly truncated in the gas (see Figure 5.10). However, dust disc sizes are also expected to decrease with time, due to both photoevaporation and radial drift. While the gas-to-dust size ratio does decrease and bring our models closer to the observations, the decrease is not significant enough to solve the discrepancy, which needs a different explanation.

Considering that the AGE-PRO targets in Upper Sco exhibit gas sizes ranging from one to five times larger than the corresponding dust sizes, the results shown in Figure 5.15 suggest that the sources are better represented by the scenario that includes external photoevaporation and $M_{\text{disc},0} = 0.1 M_*$. However, this apparent correlation with an high initial disc mass is primarily due to the models inability to reproduce dust radii (see Figure 5.14). Indeed, the presence of substructures is known to slow down the dust

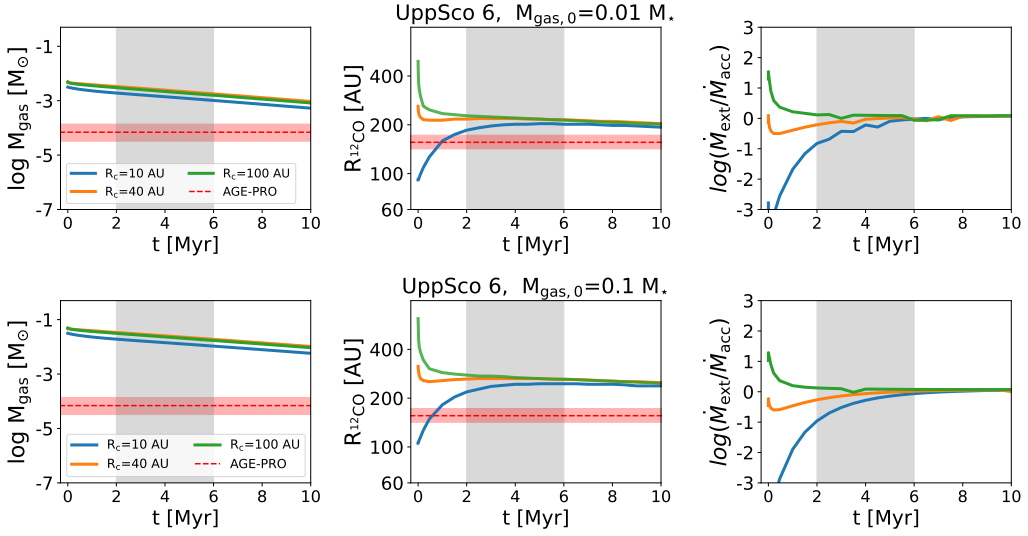


Figure 5.16: UppSco 6. Time evolution of the disc mass, radius, and $\dot{M}_{\text{ext}}/\dot{M}_{\text{acc}}$, when the initial R_c is set to 10 AU, 40 AU, and 100 AU. The top and bottom rows refer to simulations in which the initial disc mass is 1% and 10% of the stellar mass, respectively. The AGE-PRO observations and uncertainties are shown in red.

inward drift, leading to changes in the disc size (Stadler et al. 2022, Toci et al. 2021). This process persists also when external photoevaporation is included (Gárate et al. 2024), highlighting the necessity of substructures to account for the observed gas-to-dust size ratios.

5.6.5 Varying the initial disc extent: UppSco 6 and UppSco 8

We want to investigate how the initial disc extent influences the final value of disc mass and size, when an external FUV radiation field is introduced, and whether this parameter can be constrained by AGE-PRO observations. In this section we limit our analysis to just two sources from our sample, UppSco 6 and UppSco 8, which are among the disc with the lowest uncertainty on the evaluated FUV flux (see Figure 5.5), and allow the investigation of two different regimes of stellar mass and flux ($\sim 0.50 M_\odot$ and FUV flux $\sim 4 G_0$ for UppSco 6, while $\sim 0.30 M_\odot$ and $\sim 10 G_0$ for UppSco 8).

In Figure 5.16 and 5.17 we show the time evolution of disc mass, radius, and ratio between the mass loss rate due to external photoevaporation and the accretion rate onto the star ($\dot{M}_{\text{ext}}/\dot{M}_{\text{acc}}$), given different initial values for the characteristic radius R_c . We found that discs with different initial extent evolve to a similar size in the estimated age of the region. On the other hand, our analysis reveals that the AGE-PRO gas disc masses are not precise enough to be used as potential constraint for the initial disc extent. The differences in final disc mass and final disc size for various initial disc extents, at older ages are too small to provide a robust constraint of the initial R_c . In order to obtain reliable constraints on this parameter, the difference in radius and/or mass caused by varying the initial disc extent should be at least larger than the uncertainty on the observed values. This justifies our approach in which we set a fixed $R_{c,0}$ in our simulations.

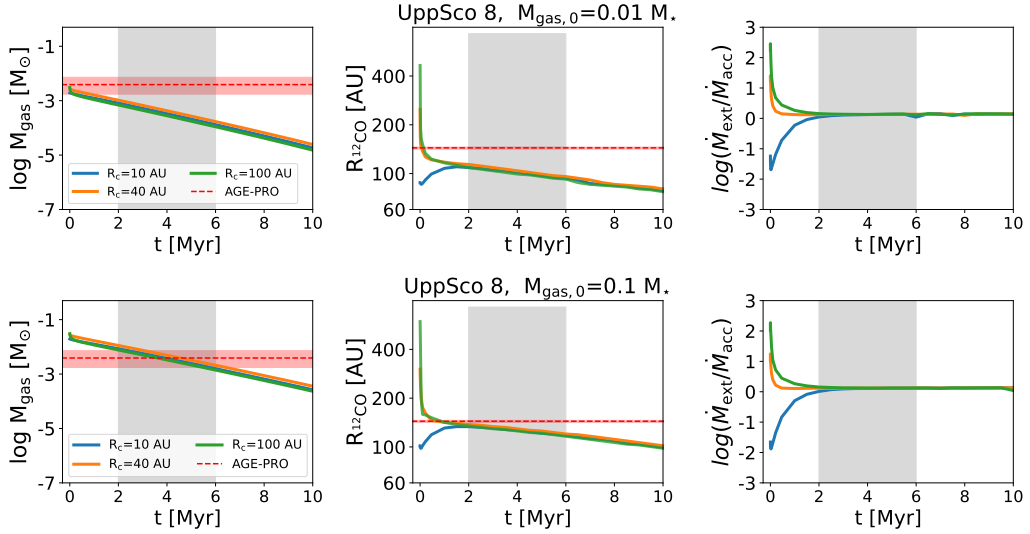


Figure 5.17: UppSco 8. Time evolution of the disc mass, radius, and $\dot{M}_{\text{ext}}/\dot{M}_{\text{acc}}$, when the initial R_c is set to 10 AU, 40 AU, and 100 AU. The top and bottom rows refer to simulations in which the initial disc mass is 1% and 10% of the stellar mass, respectively. The AGE-PRO observations and uncertainties are shown in red.

We verified that, after a few million years the disc achieves a state of equilibrium in which disc material is dispersing in winds at the same rate at which it is accreting onto the star. Although this behavior is well known from previous works (e.g. Clarke, 2007, Sellek et al. 2020b), we demonstrate that discs with identical properties but distinct initial extents converge to a similar size, and then shrink, within the same timescale required for the disc to achieve the state of equilibrium. In particular, initially large discs are easily truncated by external photoevaporation (the weakly gravitational bound material at larger radii is easily removed, and the disc shrinks from the beginning of its lifetime). On the other hand, initially compact discs viscously spread until reaching the equilibrium between the photoevaporation rate and the mass accretion rate, then slowly decrease. We provided a representative result based on only two sources; however, the same conclusions can be drawn for the entire sample.

5.6.6 Correlations: FUV flux and disc properties

In this Section we search for potential correlations between the FUV flux experienced by the ten targets in the Upper Sco region, and the disc masses and radii observed by AGE-PRO. In order to do this, we use the Spearman's coefficients, where the ρ parameter refers to steepness of the correlation, which can be positive or negative, and the p -value indicates the strength of the correlation, being considered strong or weak depending on whether it is smaller or greater than 0.05, respectively. The Spearman's parameters, resulting from applying a Monte Carlo method, and the corresponding 16th and 84th percentile of the distributions, are listed in Table 5.4. We find a weak negative correlation between the disc sizes, both gas and dust, and FUV fluxes, whereas disc masses show no apparent correlation with the external radiation (where p -values are not smaller than 0.05), as can be seen also from the values plotted in Figure 5.18. However, our disc

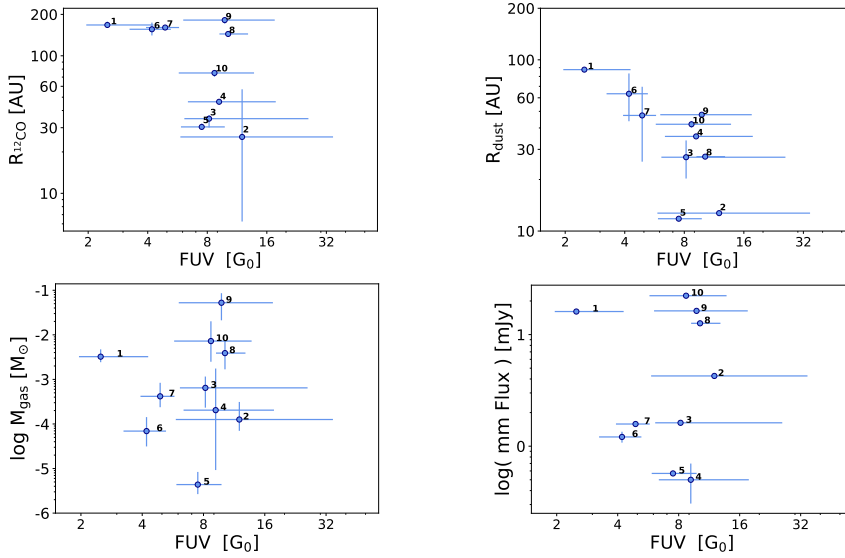


Figure 5.18: The estimated FUV flux experienced by the ten AGE-PRO sources in Upper Sco is plotted against the gas radius (*top left*), dust radius (*top right*), gas mass (*bottom left*), 1.3 mm flux (*bottom right*). The disc properties are those retrieved from the AGE-PRO observations. The correlation between size and FUV flux is marginal, while no correlation is found with the mass.

sample is too small to offer robust statistical evidence for the presence of correlations. Therefore, this analysis will be extended in future with a larger sample of discs.

5.7 Discussion

5.7.1 Gas results: the importance of considering external photoevaporation

The main outcome arising from the comparison between models and observations is that the sole reliance on a purely viscous model is insufficient to account for the observed disc masses and radii in Upper Sco. This result agrees with the findings from the population synthesis investigation conducted by Tabone et al. (2025). The process of external photoevaporation is relevant in this region, influencing mostly disc radii over disc masses, even if only a moderate level of FUV radiation is present. Discs in lower and higher irradiated star forming regions have different evolutionary history, and therefore must be modelled separately. Considering that the average FUV flux experienced by discs in the Upper Sco region is higher than in the Lupus region, the average properties of Upper Sco discs cannot be regarded as representative of old evolutionary stages of Lupus discs, as pointed out also in the work of Zagaria et al. (2023).

From the comparison between models and observations we note a general preference for initially low-mass discs. A more detailed analysis can be conducted as follows. From the observed gas masses and the outcomes of the model (Figure 5.8 and 5.9), we deduce that six targets out of ten show a preference for initially low-mass discs (closer to $0.01 M_{\star}$ than $0.1 M_{\star}$). These are UppSco 1, UppSco 3, UppSco 4, UppSco 5, UppSco 6, and UppSco 7. UppSco 2, UppSco 8 UppSco 9 and UppSco 10 are better described instead

Table 5.4. Spearman’s correlation parameters

Quantity	$\langle \rho \rangle$	$\rho, 16\%$	$\rho, 84\%$	$\langle p \rangle$	$p, 16\%$	$p, 84\%$
$R_{12\text{CO}}$	-0.41	-0.63	-0.18	0.12	0.02	0.31
R_{dust}	-0.52	-0.69	-0.28	0.064	0.01	0.21
M_{gas}	-0.07	-0.28	0.12	0.53	0.25	0.61
M_{dust}	-0.12	-0.40	0.05	0.41	0.22	0.55

Note. — Median, 16th and 84th percentiles related to the ρ and p Spearman’s values for testing a potential correlation between FUV flux and disc radius (gas and dust) and mass (gas and dust).

by initially more massive discs (closer to $0.1 M_{\star}$). Models overestimate the observed disc mass and size of UppSco 5, when external photoevaporation is included. Gas size is overestimated for UppSco 3 and UppSco 4 as well, with the latest showing gas mass consistent with the model within the error bars (see top panel in Figure 5.8). The comparison of gas sizes, shown in Figure 5.10, is crucial to investigate the impact of external photoevaporation. However, sizes do not serve as a reliable discriminant for the initial disc mass. The difference between the two considered evolutionary scenarios is indeed minimal.

Considering the current debate regarding the main driving mechanism of disc accretion (viscous vs MHD winds), the failure of the viscous model to explain the small disc sizes observed in Upper Sco should not be interpreted as a demonstration that accretion in this region is driven by MHD winds, although winds may be common (see Fang et al. 2023, Tabone et al. 2025). Rather, it underlines the importance of considering external effects when describing disc evolution, and adds another parameter (the irradiation level) to the comparison of disc properties across different SFRs, which are typically compared on the basis of the age alone. On the other hand, the relative success of the viscous external photoevaporation model in reproducing disc sizes in Upper Sco should not be interpreted as a proof that disc accretion is driven by viscosity, since we did not test alternative models in which accretion is driven by winds. Furthermore, we are aware of the results previously shown by Trapman et al. (2022a), and more recently by Tabone et al. (2025), which highlight that a pure MHD wind-driven model faces challenges in reproducing the observed disc sizes in this particular region, if we refer to population synthesis models set to match disc properties at younger ages as seen in the Lupus region.

5.7.2 Dust Results: the Necessity of Including Unresolved Substructures

The results from both mm fluxes and dust radii suggest a higher preference for initially more massive discs (see Figure 5.13) compared to what deduced from the gas-only simulations. This result is likely caused by the absence of substructures in our prescription. Substructures are known to impede the radial drift of dust, thereby preventing the expected decrease in dust mass and size. In this work we consider smooth discs and, from the work of Sellek et al. (2020b) we know that dust drifts even more efficiently in

discs subject to external photoevaporation and Kurtovic et al. (2025) found that weak or strong substructures are needed to explain AGE-PRO observations. From the comparison between gas and dust sizes (see Figure 5.15), we conclude that, even if discs are truncated in gas due to external photoevaporation, substructures are still needed to explain the observed gas-to-dust size ratios. Planetesimals formation, gravitational instabilities (Birnstiel et al., 2010a, Pinilla et al., 2012), and planet formation (which is reasonable to be occurred for, at least, some of our targets, Sierra et al. 2024), can cause disc pressure bumps trapping dust grains and delaying radial drift. In presence of an external FUV radiation field, the study of Gárate et al. (2024) suggests that dust traps only survive photoevaporation if they are located inside the truncation radius. Otherwise these disperse along with the gas even in cases where the experienced radiation increases from milder to stronger values.

Among the ten discs in our sample, UppSco 1 and UppSco 7 (see Miley et al. 2025) exhibit clear substructures in the continuum at the current resolution, while a detailed study in the visibilities, conducted in the Fourier space by Vioque et al. (2025), reveals the presence of substructures in UppSco 6, UppSco 8, UppSco 9 and UppSco 10 as well. Given that the spatial resolution of AGE-PRO is ~ 30 AU (Agurto-Gangas et al. 2025), we expect all (or most) of our targets to show substructures at smaller spatial scales than those that could be accessed by our observations. Future high-resolution observations will be able to confirm this hypothesis. Moreover, in Vioque et al. (2025) is shown that no evolution of the dust disc radius with age is detected, while we observe a potential correlation with the FUV flux (Section 5.6.6), which increases the strength of our argument that external photoevaporation cannot be neglected in Upper Sco.

We want to highlight that, while an underestimate of the observed dust radius can be explained with the fact that we did not include substructures able to stop the dust drift, an overestimate of dust disc size cannot be explained without changing global parameters (e.g. the alpha viscosity parameter, though this presents the additional problem of how to explain the observed disc masses, as described in 5.6.2). Dust results are particularly useful in constraining the initial disc mass in the case of UppSco 1. This is the most extended disc in the continuum among the AGE-PRO targets in Upper Sco, showing a bright ring-like substructure (Sierra et al. 2024). Since our models do not include substructures, we expect the actual dust size to be underestimated. Consequently, from the plots in Figure 5.14, this target is unlikely to have a high initial disc mass.

5.7.3 Additional factors impacting disc masses and sizes

Several processes, including variations in the FUV flux experienced by the disc, and the loss of material in photoevaporative winds, can impact the resulting disc masses and sizes. In this Section we briefly discuss some of the main caveats and their consequences on disc evolution.

Time-varying FUV radiation field. The FUV flux experienced by a disc during its lifetime is not constant in time, especially due to the evolution in FUV luminosity of the massive stars and the dynamics of the cluster.

The work of Kunitomo et al. (2021) shows that intermediate-mass stars, such as those found around Upper Sco, exhibit a sharp increase in FUV luminosity after a period of time that depends on the stellar mass, and then remains approximately constant. Given that the average mass of the B stars in Upper Sco is $\sim 3 M_{\odot}$, from the stellar evolution models of Kunitomo et al. (2021) based on the effective temperature of the stars (see Figure 3 in Kunitomo et al. 2021), we expect the stellar FUV luminosity to increase from

$\sim 10^{30} \text{ erg s}^{-1}$ to $\sim 10^{34} \text{ erg s}^{-1}$ after the first million year. This means that, at a distance of 0.2 pc, the FUV flux experienced by the disc will shift from $\sim 10^{-4} G_0$ to $\sim 7 G_0$. In addition, during (approximately) the first million year of its lifetime the disc is well shielded against external UV radiation by interstellar gas (Qiao et al. 2022, Ali & Harries 2019). In order to evaluate the effects of this delayed exposure to an external radiation, we perform a test simulation in which the external FUV flux is turned on after 1 Myr. The resulting evolution of the disc radius is then compared to that of a simulation in which the FUV radiation field is present from the beginning. We focus on the comparison of disc radii. As a representative example, we perform this test on UppSco 6 and UppSco 8. The results are shown in Figure 5.19. We find that when external photoevaporation is activated the radius rapidly evolves to the same value predicted if the disc experiences the same external radiation from the beginning. Therefore, while this process may be influential in younger star-forming regions, in the context of discs in Upper Sco, they are not affected by the timing of the actual exposure to the estimated FUV flux.

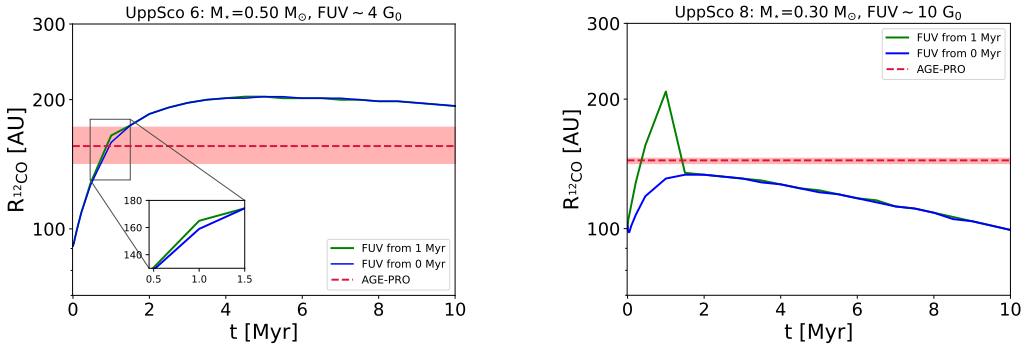


Figure 5.19: Time evolution of the gas radius when a constant FUV flux is experienced from the beginning of the disc lifetime, and after 1 Myr (shown in blue and green lines, respectively). Application to the case of UppSco 6 (*top panel*) and UppSco 8 (*bottom panel*), with initial parameters: $R_{c,0}=10 \text{ AU}$, $\alpha = 10^{-3}$, and $M_{disc,0} = 0.01 M_{\star}$ for UppSco 6, and $M_{disc,0} = 0.1 M_{\star}$ for UppSco 8.

The external radiation experienced by discs can fluctuate over time due to the stellar cluster dynamics (e.g. Wilhelm et al. 2023), as the distance between objects in the same cluster does not remain fixed in time. As the cluster expands, stars are exposed to a weaker external radiation with time. Furthermore, it is reasonable to infer that the average FUV radiation field was stronger in the past, given the presence of O-type stars (now turned into neutron stars). Nevertheless, discs detected in a high irradiated environment, which are expected to be smaller and less massive or even dispersed, could have migrated from a low irradiated region (Winter et al. 2019), and vice-versa. This last scenario may be involved in the explanation for the detectability of UppSco 2, which is predicted to be fully dispersed in less than 6 Myr. On the other hand, weakly irradiated discs might have been formed near O-type stars, experiencing high irradiation, and then drifted away over time. Future investigations of the 3D geometry and kinematics of the region could address this topic in more detail.

Winds. High-energy radiation from the central star (mostly EUV and X-ray) and magnetized winds launched from the disc (MHD) may contribute significantly to the disc dispersal.

EUV photons from the host star drive late-stage disc evolution, causing gap formation when accretion matches photoevaporative mass loss, leading to rapid disc dispersal

in $\sim 10^5$ yr (Clarke et al. 2001b). Picogna et al. (2021) showed that the X-ray irradiation from the central star produces a mass-loss rate which scales linearly with the stellar mass. Our AGE-PRO sample covers stellar masses between $\sim 0.2 M_{\odot}$ and $\sim 0.6 M_{\odot}$, corresponding to a luminosity $L_X \sim 10^{29}$ erg s $^{-1}$ and mass loss rates between $\sim 5 \times 10^{-9}$ and $\sim 10^{-8} M_{\odot} \text{yr}^{-1}$. Our models do not include the inside-out depletion of material due to internal photoevaporation, which may impact disc mass, accretion rate, and cavity formation.

In three of our AGE-PRO discs (UppSco 1, UppSco 3 and UppSco 10), Fang et al. (2023) detected emission of the [OI] λ 6300 line, tracing the presence of MHD-driven winds which can lead to an additional decrease of disc masses and sizes. The impact of this process on the dynamics of the analysed sources is still to be quantified.

5.8 Conclusions

In this work we showed that external photoevaporation can significantly impact the evolution of moderately irradiated protoplanetary discs, experiencing FUV fluxes greater than $1 G_0$. We compared the observed disc masses and sizes of the ten AGE-PRO targets in Upper Sco, with those predicted by models that incorporate both viscous evolution and external photoevaporation. The actual FUV flux experienced by each disc was computed accounting for the contribution of all the OB massive stars in the region. We performed numerical simulations varying the initial disc mass, the initial disc extent, and the viscosity. Then, we compared disc masses and radii at 2 and 6 Myr (i.e. average range of ages for the selected Upper Sco targets), with the AGE-PRO observations. In particular, we focused on ^{12}CO emission and millimeter flux to investigate respectively the gas and dust component. The main outcomes of this work are summarized as follows:

1. The AGE-PRO sources in Upper Sco experience median FUV fluxes between 2 and $12 G_0$.
2. Protoplanetary discs are truncated by external photoevaporation, and consequently shrink, even at moderate levels of irradiation ($1 - 10 G_0$). The process of external photoevaporation is not negligible in Upper Sco, where we demonstrated a substantial decrease in gas disc sizes. As a consequence, the pure viscous framework is unable to reproduce the observed disc properties in this region.
3. The relative success in reproducing gas sizes through the inclusion of external photoevaporation to our model underlines that different SFRs cannot be compared on the sole basis of the age. Indeed, we showed that the irradiation level is a crucial parameter even in a moderate irradiated environment. In other words, the average disc properties of the Upper Sco region cannot be treated as an old version of the Ophiuchus and Lupus regions.
4. The initial disc extent cannot be constrained by comparing models that include external photoevaporation and observations. Discs subject to the same FUV flux and with different initial extent converge to a similar size by the age of Upper Sco. On a similar timescale discs reach the condition in which the mass loss rate due to external photoevaporation equals the accretion rate onto the star.
5. Dust masses and radii are in general not well reproduced by our models, indicating the presence (and the relevance) of unconsidered processes that are influencing the

radial drift, such as dust traps and substructures. Even if the inclusion of photoevaporation results in smaller discs in gas, substructures are still needed to explain the observed gas-to-dust size ratios.

6. Models with the inclusion of external photoevaporation provide a good representation of the actual gas size observations (within a factor <2) in 7 out of 10 discs (our model predicts larger radii for UppSco 3, UppSco 4 and UppSco 5), considering initial disc masses between 1% and 10% of the stellar mass. In general, initially low-mass discs ($0.01 M_{\star}$) are preferred.

5.9 Appendix

5.9.1 Uncertainty on the FUV flux calculation

The FUV flux at the position of the ten AGE-PRO targets in Upper Sco is computed using a Monte Carlo approach. Specifically, we sampled from parallax distributions of the considered targets and the massive stars. The full FUV flux distribution for each of the AGE-PRO targets in Upper Sco is shown in Fig. 5.20. The error bars in the provided FUV flux estimates result from the uncertainties on the parallax measurements, which we used to compute the distance between the T Tauri and massive stars.

Despite the precision of the Gaia and Hipparcos data, large uncertainties on the parallax can arise when dealing with very bright stars (O and B stars) and/or faint stars (faint T Tauri stars). When the uncertainties on the line-of-sight distances of the objects are comparable to the physical separations between the OB stars and discs, which can occur when a massive star is close to a disc, the inferred median FUV flux may tend to be underestimated. A good example is provided by UppSco 2, which is ~ 0.37 deg away from the B1 star ω Sco. Using a Monte Carlo sampling of parallax distributions, the median relative distance (in three-dimensional space) is estimated to be ~ 6 pc, with 16% and 84% of the distribution at ~ 2 pc and ~ 12 pc, respectively. This yields the FUV flux to fluctuate between $\sim 5.8 G_0$ and $\sim 31 G_0$, with median of the distribution peaked at $\sim 12 G_0$. Remarkably, ω Sco contributes to the $\sim 99\%$ of the total FUV flux experienced by UppSco 2.

Previous works state that reliable line-of-sight distances cannot be obtained by a rough inversion of the parallax (e.g. Schönrich et al. 2012; Bailer-Jones et al. 2021a). However, Upper Sco is close enough to us to ensure good Gaia and Hipparcos parallax measurements, and we verified that our method and Bailer-Jones (Bailer-Jones et al. 2021a) method for distance evaluation yield the same result, confirming the validity of our approach. We tested the astrometric goodness-of-fit associated with the parallax measurements considering the Renormalised Unit Weight Error (RUWE). The vast majority of stars in our sample exhibit $\text{RUWE} < 1.4$, which indicates a satisfactory fit.

It is important to note that in our utilization of the Monte Carlo method, we randomly selected values from a normal distribution, despite the fact that the distribution may not be perfectly Gaussian. We can provide an upper limit of the FUV flux using the following method. We consider the OB-type stars members of Upper Sco (Luhman & Esplin 2020) and, for the closest pairs disc-massive star showing uncertainties on line-of-sight distances comparable or larger than their separation, we use the 2D projected separation (and the average distance to Upper Sco, 142 pc) to evaluate the flux as in equation (8.1). For the other pairs, we use the same Monte Carlo sampling method described in Section 5.3. In Figure 5.21 we compare the resulting FUV flux values obtained from applying the two calculation methods described above to the AGE-PRO targets in

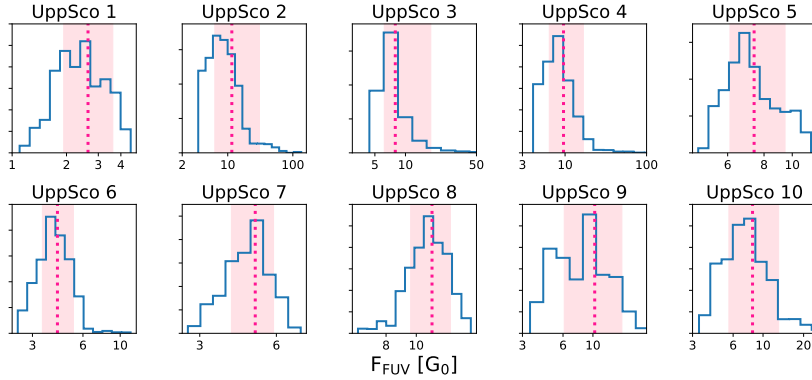


Figure 5.20: For each AGE-PRO target in Upper Sco, we show the distribution of FUV fluxes as resulting from applying the method detailed in Sec. 5.3.

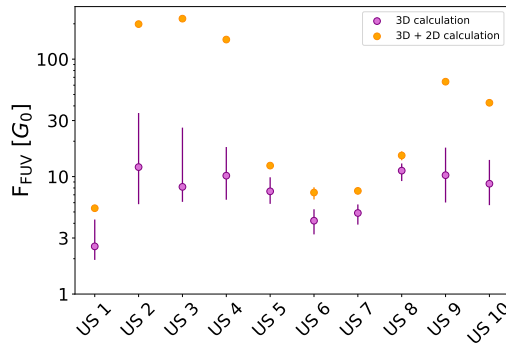


Figure 5.21: FUV fluxes and uncertainties for the ten AGE-PRO targets in Upper Sco, evaluated using Monte Carlo sampling of parallaxes (*purple*, same as in Figure 5.5), and 2D projected separation between Upper Sco OB stars and the AGE-PRO discs (*orange*).

Upper Sco. The best estimate of the FUV flux is presumably between the two provided estimates. Adopting the flux upper limits in our simulations, instead of the median values, would result in more dispersed discs (less massive and extended). Consequently, higher initial disc masses would be required to explain the observations. However, as discs shrink slowly when the equilibrium between the mass lost in wind and the mass accreting onto the star is reached, we expect the final sizes not to be significantly smaller than what predicted by our models.

The extended Upper Scorpius disc sample



(A different kind of) Discs interacting with the environment - M51 (Whirlpool Galaxy) and its smaller companion, NGC 5195, are being distorted and truncated due to their interaction.

This Chapter includes results from publications to which I have actively contributed during my PhD and for which I am a co-author. Section 6.1 summarised the results presented in Zagaria et al. (2025, in prep.). In this work, I evaluated the FUV flux at the position of the Upper Sco targets on which the paper is focused, and compared gas disc sizes and FUV fluxes drawing the main conclusions. Section 6.2 contains the results of the Master Thesis of Jingyi Ping, a Master student that I co-supervised with Prof. Paola Pinilla. This project involves a population synthesis study of the Upper Sco protoplanetary discs that will be released in Ping, Anania, Pinilla, Vioque (2025, subm.). Section 6.3 summarises the main findings by Pinilla et al. (2025). This work uses the FUV fluxes that I evaluated for the Upper Sco stars in Carpenter et al. (2025), and the implementation of external photoevaporation in the code `DustPy`, to which I have contributed.

Upper Sco is the closest region to the Sun that hosts a large stellar population with ages ranging from ~ 3 to ~ 19 Myr (e.g. Luhman 2022c; Ratzenböck et al. 2023), making it a standard reference for studying the latest stages of protoplanetary disc evolution. An initial census of the properties of protoplanetary discs in Upper Sco was conducted by Carpenter et al. (2014); Barenfeld et al. (2016, 2017), which analysed a sample of 106

stars with spectral type between G0 and M5. These observations have shown that the disc millimetre fluxes and dust disc sizes in Upper Sco are lower compared to younger star-forming regions such as Taurus, Lupus, Chamaeleon, and Ophiuchus (Manara et al. 2023a; Hendler et al. 2020). The ALMA Large Program AGE-PRO opened the way to a deeper investigation of gas disc properties. This program investigated 30 discs in Ophiuchus, Lupus, and Upper Sco, showing that the median gas disc mass is similar between Lupus and Upper Sco (Trapman et al. 2025b), and that the average gas disc size increases slightly from Lupus to Upper Sco (Trapman et al. 2025a). Although this result is extremely interesting, we emphasise that, in order to draw robust conclusions about the properties of the disc population in Upper Sco, and to enable a reliable comparison with younger disc populations, it is imperative to investigate a larger and statistically more representative sample of discs. A key observable to test our understanding of angular momentum transport in protoplanetary discs, as well as the aerodynamic evolution of disc solids, is the extent of gas and dust in a population of protoplanetary discs. A comparison between the distribution of gas and dust disc sizes in Upper Scorpius and the predictions of disc evolution models is therefore crucial for identifying the dominant process shaping disc evolution in this region (e.g. viscous spreading, MHD wind-driven evolution, internal or external photoevaporation) and for assessing the influence of the stellar formation environment.

Over the last years, the Upper Sco membership has been expanded and refined thanks to photometric and spectroscopic surveys, and with high-precision astrometry from the Gaia mission. Carpenter et al. (2025) presented an ALMA Band 7 continuum (340 GHz) and CO J=3–2 survey of a sample of 284 stars in Upper Sco observed at moderate resolution ($\sim 0.1'' - 0.4''$) and sensitivity (~ 0.15 mJy/beam). The resolution of this survey is three times higher than the previous (Hendler et al. 2020). Among the 284 targets, 208 have kinematics that are consistent with Upper Sco alone or Upper Sco and other Sco-Cen groups, while 76 are not members of Upper Sco or lack the kinematic data from Gaia DR3 for membership classifications. For six of the 208 sources, no continuum emission was detected around the target star, but a continuum source was detected nearby that could be responsible for the observed infrared excess. Excluding these six stars, the final Upper Sco sample includes 121 sources where the disc has been classified (Esplin et al. 2018) as *full* (where the disc is optically thick at infrared wavelengths and lack significant clearing of dust in the inner disc), 15 as *transitional* (where the discs have inner holes of radius $\gtrsim 1$ au), 28 as *evolved* (the disc is becoming optically thin but no significant clearing of the inner disc has occurred), and 38 as *debris, evolved transitional* or Class III, where the disc is optically thin and have inner holes of radius $\gtrsim 1$ au. For full discs, Carpenter et al. (2025) found that the sub-millimetre luminosities show a trend with stellar spectral type: discs around later-type stars have fainter sub-millimetre continuum emission than discs around earlier-type stars, for spectral types between K0 and M5. This difference is not explained by the differences in stellar luminosities, rather it indicates a reduction in dust mass for the later type stars (Andrews et al. 2016; Ansdell et al. 2016). In this Chapter, we focus on the gas and dust disc sizes of this sample, and we present a population synthesis model for gaseous discs.

6.1 Gas disc sizes

As introduced in the previous Chapter, the gas disc size is a fundamental parameter to shed light on the main process governing disc evolution. Indeed, comparing the observed gas disc sizes with those predicted by disc evolution models including viscous evolution or MHD winds, we would be able to characterise which process dominates the

evolution of angular momentum in the disc. Moreover, this comparison would allow us to access the role of internal and external processes (e.g., internal and external photo-evaporation) in shaping disc properties. We considered the ALMA 0.9 mm continuum and CO $J = 3 - 2$ rotational line emission survey of Carpenter et al. (2025) and estimated the gas disc sizes for the 90 sources where CO emission was detected with $> 5\sigma$ significance, which is the minimum signal-to-noise ratio (SNR) to provide reliable gas disc size estimates (e.g., Trapman et al. 2019).

6.1.1 Fitting pipeline

The approach adopted to estimate the gas disc sizes is similar to that proposed by Trapman et al. (2025a, see also Sanchis et al. 2021). This approach consists of generating 2D versions of the Nuker or Gaussian intensity models (see below), applying a convolution with the synthesised beam of the observations and fitting to the integrated intensity maps under the assumption that emission is azimuthally symmetric, and using the noise maps generated by `bettermoments` (Teague 2019) to estimate the fitting weights (cf., Trapman et al. 2025a).

The Nuker profile can be described by five free parameters: γ , the slope of the power law inside the transition radius, R_t , β , the slope of the power law outside it, α , describing the smoothness of the transition across R_t , and a normalisation constant, A :

$$I_{\text{Nuker}}(R) = A \left(\frac{R}{R_t} \right)^{-\gamma} \left[1 + \left(\frac{R}{R_t} \right)^{\alpha} \right]^{\frac{\gamma-\beta}{\alpha}}, \quad (6.1)$$

where R is the cylindrical disc radius in astrometric coordinates. The Nuker profile provides an ideal functional form to fit the CO emission morphology: its inner power law well approximates the optically thick inner CO emission region (whose slope follows the radial decay of the temperature profile in the CO emitting layer) and the outer one describing the emission drop beyond the radius where the CO column density drops below the photodissociation threshold. In fact, the Nuker profile generally provides good-quality fits to high peak SNR data with radially-extended moment zero maps, but, in the presence of fainter and more compact CO emission, it struggles to reproduce the integrated intensity profiles and, due to its flexibility, the estimated uncertainties on its best-fit parameters are too large to provide reliable gas size estimates. In these cases, we preferred a Gaussian profile that, instead, only depends on two parameters: a scale radius (i.e., the standard deviation), σ , and a normalisation constant, A :

$$I_{\text{Gauss}}(R) = A \exp \left(-\frac{R^2}{2\sigma^2} \right). \quad (6.2)$$

Other than these model-dependent free parameters we also fitted for the disc geometrical parameters, i.e., its centre position, (x_0, y_0) , position angle, PA, and inclination, i . Due to the even more compact nature of continuum emission (compared to CO), robust constraints on the disc geometry were obtained only for the largest sources. Therefore, we decided to leave PA and i as free parameters in our fits and used the best-fit continuum estimates, when available, as initial guesses. Two additional sources of uncertainty are worth mentioning. First, CO emission is expected to originate from elevated layers rather than the disc mid-plane (e.g., Law et al. 2021; Paneque-Carreño et al. 2023; Galloway-Sprietsma et al. 2025; Pezzotta et al. 2025). Even though this effect is not included in our (geometrically thin) models, we do not expect it to significantly bias our results (see Long et al. 2022). Secondly, our choice of fitting integrated intensity maps

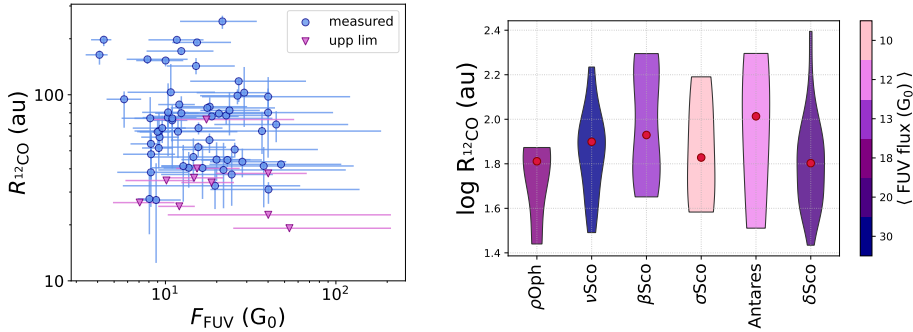


Figure 6.1: Left: CO disc radius and FUV flux for the protoplanetary discs members of Upper Sco in Carpenter et al. (2025) that have gas radius estimate in Zagaria et al. (2025, in prep.). Right: Distribution of the CO disc radii in the various Upper Sco sub-groups, ordered by increasing age from left to right. We also ordered the clusters by median FUV flux experienced by the disc sample, with colours from darker to lighter that indicate increasing FUV flux.

could also introduce systematics in our sizes. In fact, by construction, moment zero maps weight the inner disc regions preferentially more compared to the outermost ones because they span a larger velocity range. Peak intensity maps, instead, are less affected by this issue. Although this could potentially lead to underestimate of the gas disc size, Trapman et al. (2019) showed that disc sizes measured using curve of growth techniques on the moment zero and peak intensity maps were consistent among each other.

Gas disc sizes, defined as the radius enclosing 90% ($R_{\text{CO},90\%}$) of the total CO flux, were measured on the unconvolved best-fit models with the curve-of-growth method. Uncertainties were estimated by using a Monte Carlo approach and evaluated as the range between the 16th and 84th percentiles (1σ) of the size distributions obtained by measuring the sizes of 3000 model profiles generated drawing sets of model parameters from a multivariate Gaussian distribution with means equal to the best-fit parameters and uncertainties (assumed uncorrelated) from the covariance matrix of the fit (cf. Trapman et al. 2025a, see also Sanchis et al. 2021).

6.1.2 Dependence on the environmental FUV radiation field

The FUV flux at the position of the Upper Sco targets is evaluated using the method detailed in Chapter 8. This calculation uses the 2D geometry of the Upper Sco stellar cluster to account for the uncertainty in the separation from massive stars in 3D space, and provides the median FUV flux (dots in the left panel of Fig. 6.1) and its uncertainty (represented, in the left panel of Fig. 6.1, as a line for each dot, which connects the 16th and the 84th percentiles). There is no evident correlation between the CO disc size, $R_{12\text{CO}}$, and the FUV flux evaluated at the position of the target stars, as shown in the left panel of Fig. 6.1. We considered that the Upper Sco stellar population can be divided in several sub-groups based on the astrometry, kinematics, and age (Ratzenböck et al. 2023). Specifically, the discs with measured gas disc size in this work are consistent with being part of six clusters in total, which are ρOph (~ 3 Myr), νSco (~ 4 Myr), βSco (~ 5 Myr), σSco (~ 6 Myr), Antares (~ 6.5 Myr), and δSco (~ 9 Myr). In the right panel of Fig. 6.1, we show the distribution of the gas disc sizes in the different Upper Sco sub-groups, which are ordered on the x-axis by increasing average age. We do not detect a correlation between median gas disc size and cluster age. However, we emphasise that the age value

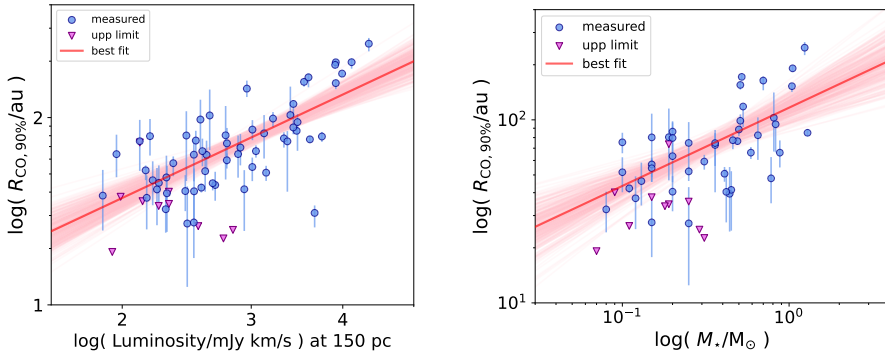


Figure 6.2: Left: Correlation between the gas disc radius and the CO luminosity found in Upper Sco. The red line indicates the best fit relation, which is $\log_{10}(R_{\text{CO},90\%}/\text{au}) = (1.83 \pm 0.25) \log_{10}(L_{\text{CO}}/\text{mJy km s}^{-1}) + (1.02 \pm 0.12)$. Right: Correlation between the gas disc radius and the stellar mass, where the best fit line (in red) indicates the relation $\log_{10}(R_{\text{CO},90\%}/\text{au}) = (0.43 \pm 0.08) \log_{10}(M_*/M_{\odot}) + (2.01 \pm 0.04)$.

considered here is the average in each sub-groups Ratzenböck et al. (2023), and presents a spread of at least $\sim 1\text{--}3$ Myr, which covers the typical lifetime of protoplanetary discs (e.g. Fedele et al. 2010). Therefore, the lack of correlation between gas disc size and sub-group age does not directly imply conclusions on disc evolution theory, because the considered age may not be representative of the individual ages of the considered targets. We also considered the FUV flux experienced by protoplanetary discs in each of the Upper Sco sub-groups and, in the right panel of Fig. 6.1, we colour-coded the disc size distributions based on the FUV flux strength. No correlation between average disc size and FUV flux in the various clusters is detected. On one hand, the large uncertainty in gas disc radius and FUV flux makes it extremely challenging to detect a correlation. On the other hand, the FUV flux range covered by the disc sample is extremely small ($\sim 2 - 40 G_0$ on average), and has presumably been subject to variations over time due to cluster dynamics.

The absence of a detectable correlation between gas disc radii and FUV fluxes should not be interpreted as evidence against the external photoevaporation influencing disc evolution in Upper Sco. Instead, it reflects the difficulty in assessing the role of external photoevaporation in moderately irradiated discs in the absence of direct tracers.

6.1.3 Gas size - Luminosity space and correlation with stellar mass

To our knowledge, a correlation between CO disc sizes and line luminosities has been so far reported only by Long et al. (2022). The presence of correlations between CO line luminosities and CO-based gas disc sizes with host star properties (M_* , L_*) has so far been investigated only in Lupus (Ansdell et al. 2018) and the heterogeneous sample of Long et al. (2022). The first could not find any correlations between $R_{90,\text{CO}}$ and stellar masses most likely because of their small sample sizes. The latter, instead, reported tentative correlations between $R_{90,\text{CO}}$ and both stellar masses and luminosities with a large dispersion of ≈ 0.3 dex. For the Upper Sco targets investigated in this work, we show in the left panel of Fig. 6.2 that the relation linking the gas radius and the CO luminosity is $\log_{10}(R_{\text{CO},90\%}/\text{au}) = (1.83 \pm 0.25) \log_{10}(L_{\text{CO}}/\text{mJy km s}^{-1}) + (1.02 \pm 0.12)$ with a dispersion of ~ 0.2 dex. In the right panel of Fig. 6.2 we show that the gas disc

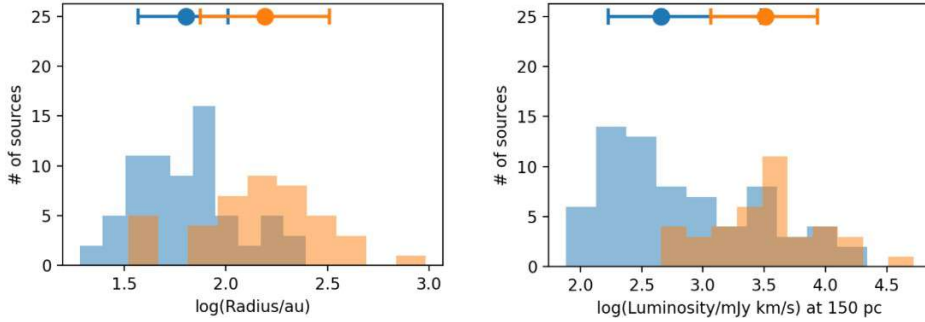


Figure 6.3: Left: Histogram of the best-fit $R_{90,\text{CO}}$ in Upper Sco (blue) and Lupus (orange). Right: Histogram of the Upper Sco (blue) and Lupus (orange) CO luminosity at 345.795 GHz.

radius correlates with the stellar mass and that the relation between the two quantities is $\log_{10}(R_{\text{CO},90\%}/\text{au}) = (0.43 \pm 0.08) \log_{10}(M_{\star}/M_{\odot}) + (2.01 \pm 0.04)$ with a dispersion of ~ 0.2 dex.

6.1.4 Comparison with Lupus and AGE-PRO

In Fig. 6.3 we compare the distributions of gas disc sizes and CO luminosities between our Upper Sco sample and the sample presented in Sanchis et al. (2021). Contrarily to what was found in the AGE-PRO sample (Trapman et al. 2025a), the median gas disc size in Lupus is higher than in Upper Sco. The fact that in Upper Sco we do not detect the large discs that are present in Lupus might suggest the presence of additional processes that shrink discs over time, and external photoevaporation is a potential candidate. As shown in Trapman et al. (2022b); Zagaria et al. (2023), assuming that Upper Sco is an older instance of Lupus, an MHD wind-driven evolution scenario is unable to predict disc sizes much smaller than those detected in Lupus, and therefore fails to explain Upper Sco disc sizes. Indeed, the MHD wind-driven model predicts that the discs remain the same size or shrink slightly, whereas external photoevaporation truncates the discs more efficiently.

However, when comparing disc samples derived from different surveys, it is essential to take into account the observational constraints specific to each dataset. First of all, the Upper Sco and Lupus surveys do not cover the full population of disc-bearing stars in their respective regions, and their levels of completeness differ significantly. Therefore, comparing the average disc properties between Lupus and Upper Sco using the above-mentioned surveys can lead to wrong conclusions if parts of the population are excluded, particularly when the selection criteria differ between the surveys. Moreover, the Lupus and Upper Sco surveys include observations performed with different sensitivity and resolution, and gas disc sizes are derived from different rotational line emission (CO $J = 3 - 2$ for Upper Sco, CO $J = 2 - 1$ for Lupus). This can significantly influence data analysis, leading to larger (or smaller) gas disc sizes. The effect of sensitivity in the calculation of the gas disc sizes is evident in Fig. 6.4, where we compare the Upper Sco gas disc radii retrieved from AGE-PRO (see Chap. 5) and those evaluated in this work. Although the method used to evaluate gas disc sizes is similar between this work and AGE-PRO (Trapman et al. 2025a), there are discrepancies among the results: three objects in Upper Sco exhibit AGE-PRO gas disc sizes significantly larger than the radii inferred from CO observations of the shallower survey of Carpenter et al. (2025).

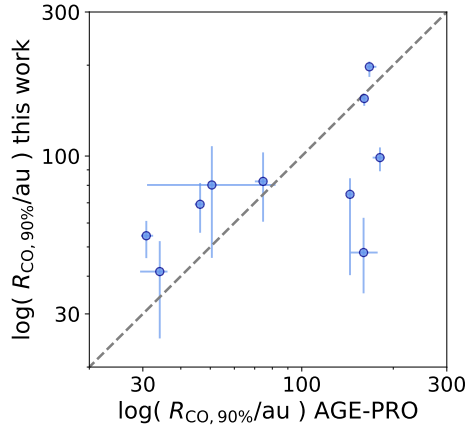


Figure 6.4: Comparison between the gas disc sizes evaluated in this work and those resulting from AGE-PRO observation in Upper Sco. The dotted gray lines refer to the 1:1 relation.

Therefore, observing discs with deeper sensitivity can reveal larger gas disc sizes.

6.1.5 Conclusions

We provided measurements of gas disc radii and FUV fluxes for a large sample of protoplanetary discs in Upper Sco. The gas radii, defined as the radii enclosing the 90% of ^{12}CO line flux emission, were evaluated for the 90 Upper Sco discs in Carpenter et al. (2025) that present CO detection with $\text{SNR} > 5\sigma$. The main conclusions can be summarised as follows.

- There is no detectable correlation between the gas disc sizes and the FUV fluxes at the position of the Upper Sco sources, not even an evident relation between the distribution of gas disc sizes, FUV flux, and average age of the Upper Sco sub-groups. However, the range of FUV fluxes covered by this sample of discs is small (i.e., spans only one order of magnitude) and the presence of a correlation can easily be obscured by observational uncertainties. This result highlights the current difficulties in providing evidence of the effect of external photoevaporation on moderately irradiated discs in the absence of direct tracers.
- Comparing the distributions of gas disc sizes and CO luminosities between Upper Sco and Lupus, we found that discs in Lupus are in median larger and more luminous in CO than those in Upper Sco. However, this result should be taken carefully, as the two compared disc samples differ in completeness, sensitivity, and resolution.
- The comparison between the gas disc radii resulting from AGE-PRO observations and those computed from shallower CO observations (i.e., this work for Upper Sco, but applicable also to Sanchis et al. (2021) for Lupus), shows that the sensitivity of the observations can deeply influence gas disc size measurements. Therefore, deeper observations can reveal larger discs.

The Upper Sco gas disc radii presented in this Section and the FUV fluxes at the position of the Upper Sco stars in Carpenter et al. (2025), were fundamental in the works presented in Sec. 6.2 and 6.3.

Table 6.1: Initial parameters of the disc population

Parameter	Description	Range	PDF
α	Viscosity	$10^{-5} - 10^{-2}$	Log uniform
$M_{\star} (M_{\odot})$	Stellar mass ($M_{\odot}$)	0.1, 0.3, 0.6, 1.0, 1.5, 3.0	IMF
$\epsilon = M_{\text{disk}}/M_{\star}$	Disk-to-stellar mass ratio	$10^{-3} - 0.5$	Log uniform
R_c (AU)	Characteristic radius	10 – 200	Log uniform
$F_{\text{UV}} (G_0)$	External FUV flux	1, 10, 100	Log normal

6.2 Population synthesis model of gaseous discs

In Anania et al. (2025a), we compared the sizes and masses of simulated protoplanetary discs experiencing viscous evolution and external photoevaporation, with those measured for the ten Upper Sco discs in AGE-PRO. We showed that the pure viscous model is not able to reproduce the observed properties, and that external photoevaporation can efficiently truncate discs even in moderately irradiated regions. In Ping et al. (2025, subm.), we expand this work to a population level, modelling a large sample of viscous protoplanetary discs experiencing external photoevaporation. We assume that discs are irradiated by the moderate FUV flux level characteristic of the Upper Sco star-forming region. The aim of this work is to investigate the time evolution of the median gas disc mass and radius within a disc population, and to verify whether, and for which initial conditions, the distribution of gas disc sizes reproduces that observed in the recent Upper Sco survey (Zagaria et al., 2025 in prep.). Additionally, we compare the time evolution of the disc fraction, resulting from the population synthesis model, with existing empirical relations.

A total of 5000 simulations are performed using the 1D disc evolution code `DustPy`. Since the focus of this work is gas evolution, we modelled gas-only discs neglecting the dust component. Table 6.1 shows the parameter space considered in the population synthesis model. We explored a grid of stellar masses $M_{\star}$ that are sampled with probabilities proportional to the IMF of Maschberger (2013). Similarly, the representative FUV flux values are sampled with a probability given by a log-normal distribution of the values computed for the observed disc sample. The FUV flux values of the observed disc sample in Carpenter et al. (2014) are computed using the method presented in Anania et al. (2025b), and detailed in Chap. 8. The α -viscosity, the disc-to-stellar mass ratio ϵ , and the disc characteristic radius R_c are randomly drawn from a log-uniform distribution in the range of values indicated in Table 6.1. We define the gas disc radius as the location in the disc enclosing the 90% of ^{12}CO flux emission, which can be described approximately as the location where the gas surface density of the disc falls below a critical value for self-shielding of ^{12}CO (see Sec. 5.4.4). This definition enables a direct comparison between models and observations. In our models, we adopt three thresholds to define disc dispersal, corresponding to when the gas disc mass decreases below 90%, 95%, or 99% of its initial value.

6.2.1 Gas disc mass distribution

Figure 6.5 shows the time evolution of the median gas disc mass of the population of surviving discs. Alternatively assuming that the discs are dispersed when they lose 90%, 95%, or 99% of the initial mass, we obtained a similar trend: the median gas disc mass decreases in the first 0.2 Myr and then remains approximately constant, increasing slightly. This is similar to the trend found in population synthesis models of viscous discs

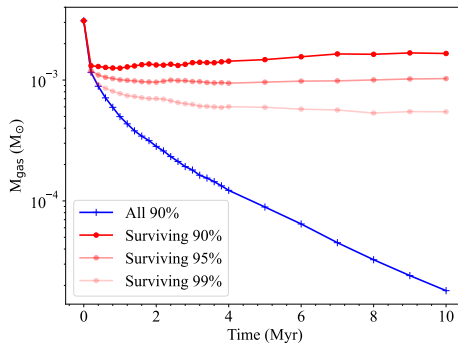


Figure 6.5: Time evolution of the median gas disc mass M_{gas} for the simulated population of discs. The red lines consider the surviving discs, while the blue line refers to the whole disc population. The dark, medium and light red represent disc dispersal when losing 99, 95, 90% of the initial mass. From Ping et al. (2025, *subm.*).

experiencing internal photoevaporation (Malanga et al. 2025). In contrast, the median gas mass value of the entire population continues to decline over time (blue line in Fig. 6.5). This result highlights the importance of survivorship bias among the observed disc population. The evolution of the gas disc mass is mainly influenced by the mass that is accreted onto the star and the mass that is removed by external photoevaporative winds. Our results suggest that even a moderate FUV radiation field can cause a rapid decline of the median disc mass in a relatively short timescale. Indeed, in the simulated population, initially $\sim 80\%$ of discs show a mass loss rate at least one order of magnitude higher than the mass accretion rate. As the discs evolve, the accretion rate onto the star gradually reaches an equilibrium with the rate of materials lost in winds (Clarke 2007; Sellek et al. 2020b). After 0.2 Myr, only $\sim 40\%$ of the surviving discs exhibit a mass loss rate at least one order of magnitude higher than the mass accretion rate. Moreover, the surviving discs do not show a significant preference for higher initial disc-to-star mass ratios ϵ .

6.2.2 Gas disc size distribution

The left panel of Fig. 6.6 illustrates the time evolution of the median gas disc radius for the simulated populations. Similarly to the trend shown for the median gas disc mass evolution, the median gas disc radius decreases significantly in the first 0.2 Myr, then increases slightly. The gas radius at 10 Myr is $\sim 18\%$ larger than that at 5 Myr. As verified for the gas disc mass, there is a survivorship bias for the gas disc radius as well. Discs that are initially larger are easily truncated by external photoevaporation, while initially compact discs spread viscously until the mass loss rate becomes significant, and then shrink. The increase in the average gas disc radius is likely to be attributed to the viscous spreading of the surviving discs. We verified that the gas disc radius, $R_{12\text{CO},90\%}$, is comparable to or larger than the radius that encloses the 90% of the gas disc mass, and the ratio between these two size-scales can reach up to five.

We compared the distribution of the gas disc sizes resulting from our population synthesis model, with the distribution of $R_{12\text{CO},90\%}$ measured by Zagaria et al. (2025, *in prep.*) for the Upper Sco disc sample in Carpenter et al. (2014). The resulting CDFs

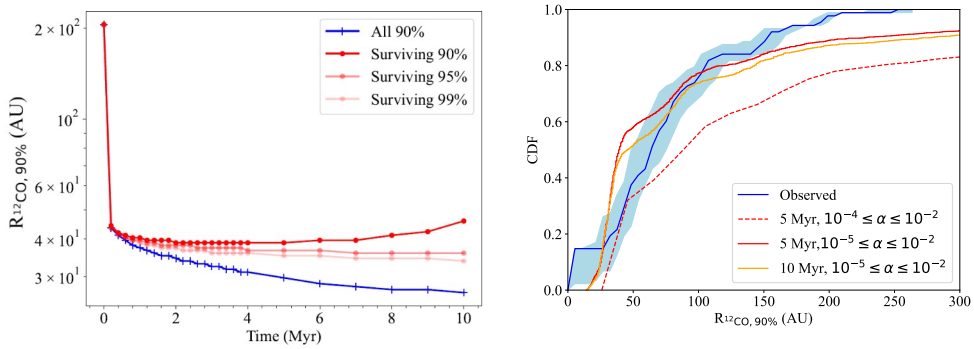


Figure 6.6: Left: Time Evolution of the median gas disc radius $R_{12\text{CO},90\%}$. The red lines indicates the population of survived discs (i.e., considering disc dispersal when losing 90%, 95%, and 99% of the initial disc mass), while the blue line shows the whole population. Right: CDF of the gas disc radius of surviving discs at 5 (red line) and 10 Myr (yellow line) compared to the observation (blue line). The dotted line refers to the restricted population where we excluded the less viscous discs, considering only $10^{-4} \leq \alpha \leq 10^{-2}$. From Ping et al. (2025, subm.).

are shown in the right panel of Fig. 6.6. In general, the CDF of the entire simulated disc population, with viscosity $10^{-5} \leq \alpha \leq 10^{-2}$, aligns well with the CDF of the observed population. This result shows that, contrarily to pure viscous or MHD wind-driven models, which were not able to explain the distribution of observed gas disc sizes in Upper Sco (Tabone et al. 2025), the inclusion of external photoevaporation in a pure viscous framework allows us to predict a distribution of gas disc sizes which is statistically more similar to that observed. However, our model fails to reproduce the population of the largest discs observed in Upper Sco ($\gtrsim 200$ au). This result may be due to the efficiency of external photoevaporation, which does not allow discs to expand to larger sizes. A potential solution to this problem involves the time variation of the FUV flux experienced by protoplanetary discs, which can be the result of dust extinction or cluster dynamics. Dust extinction can efficiently shield discs from external UV radiation for the first $\sim 0.5 - 1$ Myr (Ali & Harries 2019; Winter et al. 2019). However, a protoplanetary disc where external photoevaporation is turned on at 1 Myr, is expected to rapidly (i.e., in 0.5–1 Myr) converge toward the size it would have reached if photoevaporation acted since the beginning (Anania et al. 2025a). Consequently, this delayed effect is likely to have only a minor impact on the current disc population in Upper Sco. The FUV flux variation can instead be caused by cluster dynamics, which changes the relative distance from massive stars.

6.2.3 Disc fraction

We compared the disc fraction (i.e., the fraction of discs that are not dispersed at each time-step), with the empirical relations assuming an exponential fit. Specifically, we considered an empirical relation that follows the form $f_{\text{disk}} = \exp(-t/\tau)$, where $\tau = 3$ Myr is derived from measuring the fraction of accreting stars in young stellar clusters and associations with a wide range of ages (Fedele et al. 2010), while $\tau = 7$ Myr is based on the results of (Pfalzner 2022), which considered only nearby clusters (< 200 pc) that do not host O-type stars, doubling the typical lifetime of discs. An investigation of the fraction of accreting stars in the Sco-Cen region performed by Delfini et al. (2025) has found $\tau \sim (2.7 \pm 0.4)$ Myr, in agreement with Fedele et al. (2010). Figure 6.7 shows that the disc

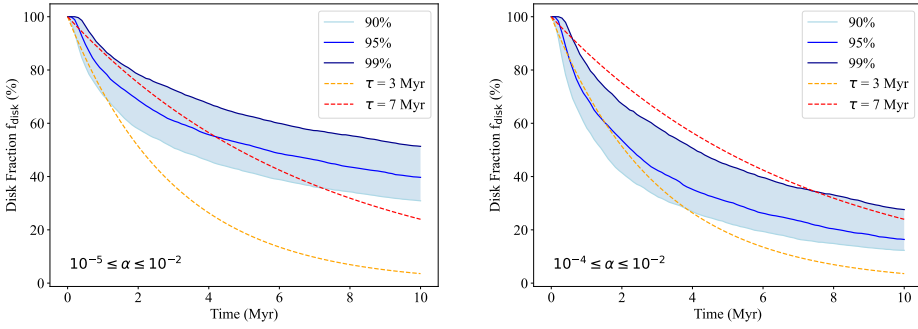


Figure 6.7: Disk fraction f_{disk} as a function of age. The blue lines show the disk fraction obtained from our simulations, considering disc dispersal when losing 90% (light blue), 95% (blue) and 99% (dark blue) of the initial mass. The dashed lines are the exponential relations with $\tau = 3$ Myr (yellow) and $\tau = 7$ Myr (red). Top and bottom panel show the results when including and removing the less viscous discs, respectively considering the ranges $10^{-5} \leq \alpha \leq 10^{-2}$ and $10^{-4} \leq \alpha \leq 10^{-2}$. From Ping et al. (2025, *subm.*).

fraction resulting from the population synthesis model decreases slower than predicted by the empirical relations when considering the full range of α -viscosity included in this work (top panel). At 10 Myr, the disc fraction obtained when assuming disc dispersal corresponding to the loss of 90% of the initial mass is $\sim 30\%$ larger than the empirical relation with $\tau = 7$ Myr. However, if we remove from the sample the less viscous discs, and we consider the range $10^{-4} \leq \alpha \leq 10^{-2}$, the time evolution of the disc fraction is more consistent with the empirical relations (bottom panel of Fig. 6.7) because a higher viscosity increases the accretion rate onto the central star, resulting in shorter disc lifetimes. Moreover, a higher viscosity drives faster viscous spreading, making discs more vulnerable to external photoevaporation. We highlight that this investigation was meant to explore the role of external photoevaporation in varying the properties of a population of discs, and therefore the effect of internal photoevaporation is not included in the population synthesis. Internal photoevaporation may cause additional disc depletion, accelerating disc dispersal, particularly in the last stages of disc lifetime, and therefore leading to a different disc fraction over time (e.g., Clarke et al. 2001a; Malanga et al. 2025).

6.2.4 Conclusions

We performed a population synthesis model of protoplanetary discs evolving under the effect of viscosity and experiencing external photoevaporation due to moderate FUV fluxes ($\sim 1 - 100 G_0$), with the aim of investigating the evolution of gas disc mass, gas disc size, and disc fraction in Upper Sco. We found that the median gas disc mass and the median gas disc size of the population of survived discs (i.e., those that have lost a fraction of mass equal to or greater than 90% of the initial disc mass, at each timestep) decreases sharply in the first fraction of Myr, and then increases slightly. This is a result of survivorship bias, where low-mass and extended discs around lower-mass stars are more affected by external photoevaporation, while more massive and compact discs are able to spread viscously.

Our population synthesis model is able to reproduce at a statistical level the distribution of gas disc sizes observed in Upper Sco, where the gas disc sizes have been

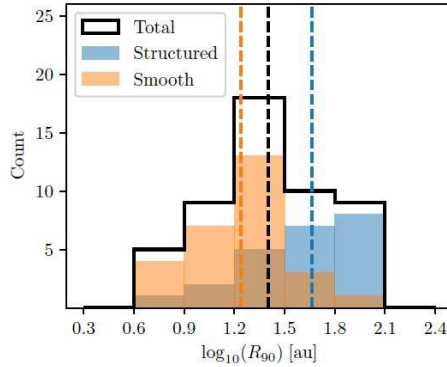


Figure 6.8: Distribution of the dust radius, R_{90} , inferred from dust continuum visibility modelling for the resolved discs with structures (blue) smooth discs (orange), and all resolved discs (in black). The vertical line indicates the median value of each distribution.

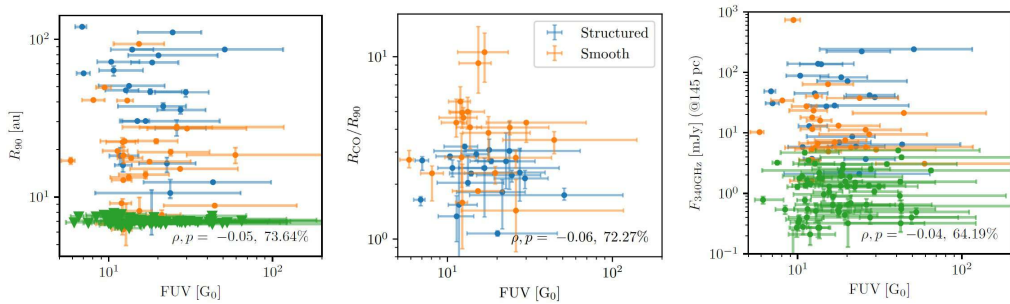


Figure 6.9: Left: Dust disc radius and FUV flux. Middle: Gas-to-dust size ratio and FUV flux. Right: continuum flux emission at 340 GHz and FUV flux. The blue, orange, and green points indicate the substructured, smooth, and unresolved discs. From Pinilla et al. (2025, under review).

evaluated as detailed in Sec. 6.1. Despite the fact that the population of large discs observed in Upper Sco ($\gtrsim 200$ au) cannot be reproduced by our model, our results show that a viscous external photoevaporation model is preferred to a pure viscous model or an MHD wind-driven model to explain the observed disc size distribution in Upper Sco. Indeed, in Tabone et al. (2025) it has been shown that, although an MHD wind-driven model can reproduce the distribution of disc properties in Lupus, it fails to explain gas disc sizes in Upper Sco. Adding external photoevaporation to the model is a potential solution to this discrepancy.

We compared the evolution of the disc fraction derived from our models with the empirical relations found in the literature. However, we do not have enough observational constraints on the Upper Sco disc fraction to determine the best-fit initial conditions and make arguments about the Upper Sco stellar population.

6.3 Dust disc sizes

The work by Pinilla et al. (2025, under revision) presented a model of the dust continuum visibilities of the 120 discs from the sample of Carpenter et al. (2025) which are confirmed members of Upper Sco and present $\text{SNR} > 3$. The focus of this work is to constrain

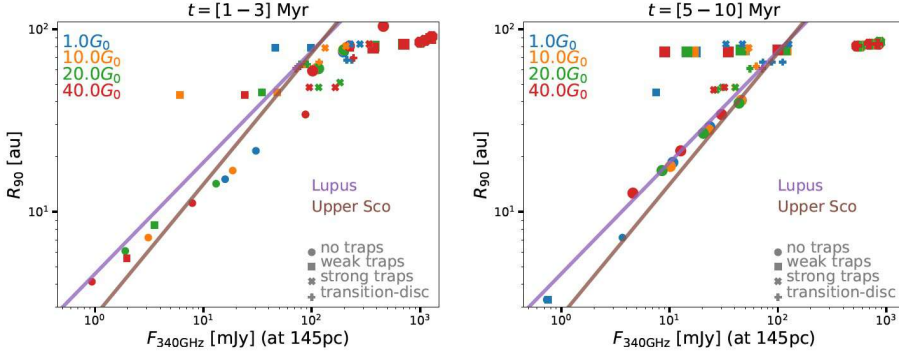


Figure 6.10: Disc millimetre luminosity vs. dust disc radius R_{90} from the models, for different values of FUV flux and traps, at 1-3 Myr (left panel) and 5-10 Myr (right panel). The colours represent different FUV fluxes, while the shape of the points the type of pressure traps. The size of the points symbolises different stellar masses (small size for $0.3 M_{\odot}$ and larger sizes for $1 M_{\odot}$). As a reference, we included the best observational fit found for Lupus (Guerra-Alvarado et al. 2025) and Upper Sco (this work). From Pinilla et al. (2025, under review).

the flux, size, and geometry of the discs in the selected sample. The observations are compared to gas and dust disc evolution models that include external photoevaporation.

The dust continuum visibilities are modelled using *Galarío* (Tazzari et al. 2018) and assuming an axisymmetric disc, and an intensity radial profile described by a Gaussian function. For the faintest discs, which are unresolved in the model, the brightness is modelled as a point source and the free parameters are the flux and offsets. For the resolved discs, the amplitude, width, and peak position of the Gaussian are taken as free parameters of the fit. Discs were classified in structured and smooth, depending on whether they present a resolved ($R_{\text{peak}} > 0.05''$) or unresolved ($R_{\text{peak}} < 0.05''$) peak of the best-fitting Gaussian function, where $0.05''$ is an average value of the half of the beam size of the observations. The dust disc radius is defined as the location that encloses the 90% of the total continuum flux emission. The results presented in Fig. 6.8 show that the substructured discs are more extended in dust than the smooth discs, which is the same trend found in the Lupus region (Guerra-Alvarado et al. 2025). Figure 6.9 shows that there is no evident dependence of the dust disc size (left panel), gas-to-dust size ratio (middle panel), and continuum flux (right panel) on the FUV flux. The gas-to-dust size ratio is evaluated using the gas disc sizes as detailed in Sec. 6.1 and is on average higher in smooth discs than in substructured discs.

The 1D disc evolution code *DustPy* is used to investigate the evolution of the dust disc size considering a viscous disc experiencing external photoevaporation, and including dust traps. The *DustPy* viscous external photoevaporation model is detailed in Chapter 5, while dust traps are modelled as a perturbation in the α -viscosity profile:

$$\alpha_{\text{visc}}(r) = \alpha_0 \left(1 + A \exp \left(-\frac{(r - r_{\text{gap}})^2}{2w_{\text{gap}}^2} \right) \right), \quad (6.3)$$

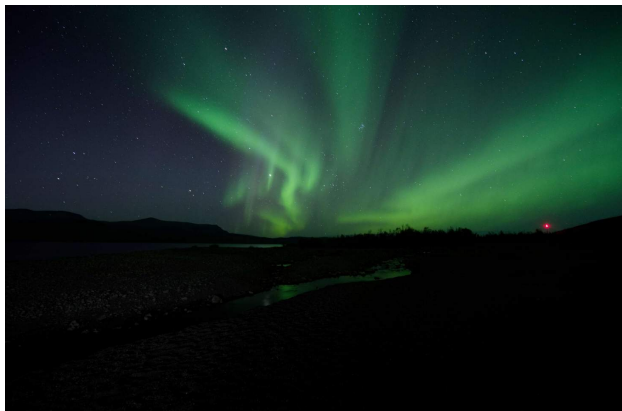
where α_0 is the global disc viscosity parameter (Shakura & Sunyaev 1973) and is assumed to be 10^{-3} , A sets the strength of the pressure bump, w_{gap} is the width of the gap, which is taken equal to the local pressure scale height, and r_{gap} is the radial position of the gap, which is taken equal to 10, 40, and 70 au, and represents the case of three pressure bumps included simultaneously in the same disc. We defined the dust traps

with $A = 1$ as weak, and those with $A = 4$ as strong. A representative set of test discs is simulated, considering two values for the stellar mass $M_\star = [0.3, 1] M_\odot$, the initial disc mass which scales with the stellar mass $M_{\text{disc},0} = 0.05 M_\star$, a fixed initial characteristic radius $R_c = 80$ au, the viscosity parameter $\alpha = 10^{-3}$, a fragmentation velocity $v_{\text{frag}} = 10 \text{ m s}^{-1}$, three positions of dust traps $r_p = [10, 40, 70]$ au, and four values of the FUV flux $F_{\text{FUV}} = [1, 10, 20, 40] G_0$. The main findings of this work can be summarised as follows.

- Discs around low-mass stars, $0.3 M_\odot$, should have already dispersed by the age of Upper Sco. The gas-to-dust size ratio (i.e., the ratio between the gas radius defined as the location enclosing the 90% of ^{12}CO flux emission, and the dust radius that enclosed the 90% of dust continuum flux) is predicted to increase with age in the absence of pressure traps. For a $1 M_\odot$ star, the presence of weak traps is enough to keep the size ratio nearly constant over ten million years of the disc evolution. However, for discs around a $0.3 M_\odot$ star, only strong traps keep the size ratio low and constant over the last million years of evolution. Comparing with observations, this scenario suggests that, for the surviving discs in Upper Sco, most of the traps should be stronger than in earlier ages.
- The correlation between disc luminosity and dust disc size is $R_{90} \propto L_{\text{mm}}^{0.72}$, which is steeper than what was previously found (Barenfeld et al. 2016; Hendler et al. 2020). Figure 6.10 shows that this correlation is not significantly affected by variations in FUV flux, but rather by drift and the presence of traps. However, the FUV flux does have an effect on what discs populate this relationship at older ages, as many are dispersed due to photoevaporation. A more detailed investigation of the survival rate of discs in Upper Sco can be obtained in the future by performing population synthesis models with a more dense parameter space of disc and stellar conditions.

From the investigation presented in this Chapter we can derive two important conclusions. First, the effect of external photoevaporation needs to be included in the current disc evolution models to explain the disc properties observed in Upper Sco. However, less model-dependent evidence that external photoevaporation is playing an important role in shaping disc evolution in Upper Sco is still missing. The disc radius is the most accessible disc property that may provide an indirect indication of the effect of external photoevaporation on discs. However, the investigation performed in this Chapter demonstrates that even with the best set of data currently available in Upper Sco, it is challenging to find strong evidence (i.e., correlations with the FUV flux) of the effect of photoevaporation in this moderately irradiated region. Moreover, the history of the Upper Sco sub-clusters and their dynamics may have played a key role in shaping disc evolution and planet formation Upper Sco over the last $\sim 2 - 3$ Myr. Indeed, stellar relative motions and, particularly, supernovae explosions may have caused a rapid and strong change in the FUV flux (at least for part of the Upper Sco stellar population), contributing to a rapid increase in mass loss rate. We showed, in Chap. 5, that the closest runaway star, which is the sole O-type star in the Upper Sco region, is not significantly impacting the current FUV flux at YSOs. However, a further investigation of proper motions and history on the FUV flux in Upper Sco would better address the impact of cluster dynamics on disc evolution in this region.

Substructures and weak FUV flux: application to Lupus



(A different kind of) Irradiation and substructures - The aurora borealis illuminating the night and evolving in rapidly changing substructures of light. [Abisco - September 2024]

This Chapter summarises the results presented in Delussu, Anania et al. (2025, under revision). My involvement in the project consisted of implementing external photoevaporation in the latest version of the disc evolution code `DustPy`, running a subset of the models, providing expertise on the evolution of gas-to-dust size ratios thanks to my work in AGE-PRO, and contributing to the preparation of the manuscript text.

7.1 A tension between gas and dust disc radii

The presence of local maxima in the disc pressure profile can mitigate and even stop the migration of solid particles, that get trapped in correspondence to the pressure maxima (e.g., Pinilla et al. 2012, 2013). An efficient dust trapping leads to an enhancement of the dust-to-gas ratio, thus creating a favourable region for planetesimal formation (Drażkowska et al. 2016). Different disc evolution frameworks predict distinct evolution of disc properties over time. Dust and gas radii, in particular, are extremely sensitive to the evolution framework, and therefore comparing observed values with model

predictions offers a key test for identifying the most likely evolutionary pathway. Observationally, the disc gas-to-dust size ratio is often evaluated as the ratio between the radius enclosing 90% of the CO line flux emission, and the radius enclosing 90% of the dust continuum emission. Studies performed on a disc population in the Lupus and Taurus star-forming regions (Sanchis et al. 2021; Long et al. 2022) have shown that the vast majority of discs present gas-to-dust size ratio, $R_{\text{CO}}/R_{\text{dust}}$, between two and four, with the $\sim 15\%$ of the sources showing $R_{\text{CO}}/R_{\text{dust}} > 4$. This result hints at a similar evolution mechanism for the two star-forming regions. However, the theoretical analysis performed by Toci et al. (2021) by assuming disc viscous evolution, grain growth and pure radial drift has shown that, for a broad range of disc initial conditions, $R_{\text{CO}}/R_{\text{dust}}$ becomes rapidly larger than five. Dust substructures have been proposed as a potential solution to this discrepancy between models and observations, as they are able to stop the inward dust radial drift, trapping grains at pressure maxima.

The work by Delussu et al. (2024) has shown that substructures are needed to reproduce the distributions of the spectral index and the relation between dust disc size and luminosity observed in the disc sample in the Lupus region presented in Sanchis et al. (2021). In this work, we extended the population synthesis study of Delussu et al. (2024) by investigating whether substructures can provide a solution to the gas-to-dust size ratio problem, as proposed by Toci et al. (2021). Moreover, we investigated whether it is possible to match at the same time the distributions of gas and dust size, size-luminosity, and spectral index, and what initial disc parameters are required for that. Furthermore, we considered the impact of the stellar formation environment, exploring whether external photoevaporation caused by the UV radiation emitted by OBA-type stars may explain the $R_{\text{CO}}/R_{\text{dust}}$ observed in the population of discs in the Lupus region. This last scenario is motivated by the fact that gas discs can be efficiently truncated by external photoevaporation even in low UV-irradiated environments (van Terwisga & Hacar 2023; Anania et al. 2025a), and therefore the combination of forming substructures and this process may provide the conditions for reproducing the observed gas-to-dust size ratio.

7.2 Method

With the aim of investigating whether the presence of multiple disc substructures could reproduce the gas-to-dust size ratio observed in the Lupus disc population, we performed a population synthesis study of substructured discs using a 1D viscous evolution model for dust and gas in protoplanetary discs. Subsequently, we explored the behaviour of the gas-to-dust size ratio for a population of discs exposed to the average FUV flux for the Lupus region.

7.2.1 Two-pop-py

Following Delussu et al. (2024), we used the Two-pop-py tool to study a population of discs considering the dust surface density evolution, the viscous evolution of the gas and accounting for different particle sizes (Birnstiel et al. 2012). A summary of the key elements of two-pop-py is outlined as follows.

- The gas surface density of the disc evolves following the viscous model, accordingly to the (Lynden-Bell & Pringle 1974) equation. The initial gas surface density

follows the Lynden-Bell & Pringle (1974) self-similar solution

$$\Sigma_g(R) = \Sigma_0 \left(\frac{R}{R_c} \right)^{-p} \exp \left[- \left(\frac{R}{R_c} \right)^{2-p} \right], \quad (7.1)$$

where the normalization parameter $\Sigma_0 = (2 - p)M_{\text{disc}}/2\pi R_c^2$ is set by the initial disc mass M_{disc} , R_c denotes the so-called characteristic radius of the disc and p the initial surface density power law exponent. p has been set to 1 for the initial profiles of all discs.

- Adopting the two-population model described in Birnstiel et al. (2012), the dust surface density evolves assuming that the small dust is tightly coupled to gas while the large dust particles can decouple from it and drift inward.
- We considered a passive irradiated disc, with a disc floor temperature set to 10 K.
- We modelled smooth and substructured discs. A substructure is modelled as a gap caused by the presence of a planet in the disc. To mimic the presence of a planetary gap, we subdivided the α -viscosity parameter into its two contributions: α_{gas} and α_{dust} . The presence of the planet is modelled as a local variation of the α_{gas} parameter. Given the inverse proportionality between α_{gas} and Σ_g in a steady-state regime, a bump in the α_{gas} profile results in a gap in the Σ_g profile, effectively simulating the presence of a planetary gap. This approach preserves the viscous evolution of Σ_g . The parameter α_{dust} enters linearly in the diffusion coefficient of the evolution of the dust surface density (see Sec. 2.5) and, other than in proximity of the gap boundary, it is set $\alpha_{\text{gas}} = \alpha_{\text{dust}}$. We adopted the Kanagawa et al. (2016) prescription to model the planetary gaps.

We modelled a 1D disc using a logarithmic radial grid that ranges from 0.05 au to 10000 au. The main parameters of the grid model are presented in Table 7.1. We performed 10^5 simulations for each population synthesis. To map the entire parameter space, the set of initial conditions adopted for each disc is selected by randomly drawing each parameter from a probability distribution function (PDF). The main parameters that describe both smooth and substructured discs are: the disc mass (M_{disc}), the stellar mass (M_{star}), the disc characteristic radius (R_c), the viscosity parameter (α) and the fragmentation velocity (v_{frag}). The stellar mass values are drawn from the functional form of the IMF proposed by Maschberger (2013). Substructured discs are characterised by three additional parameters: the mass of the planet that creates the gap (m_p), the time at which the planet is inserted (t_p), and the position of the planet within the disc (r_p). Table 7.2 shows the range that we considered for each disc parameter and the corresponding PDFs adopted.

7.2.2 DustPy and external photoevaporation

To perform numerical simulations of viscous discs undergoing external photoevaporation, we used the code `DustPy` (Stammler & Birnstiel 2022) integrated with an external module that includes the effect of an external FUV field (Gárate et al. 2024; Anania et al. 2025a). This choice is dictated by the fact that the effect of external photoevaporation has been previously implemented and already tested in `DustPy`. Moreover, `DustPy` allows us to simulate the radial evolution of gas and dust in a viscous disc, considering a distribution of particle species subject to advection, diffusion, grain growth by coagulation

Table 7.1: Fixed parameters adopted for two-pop-py model.

Parameter	Description	Value or Range
Σ_d/Σ_g	initial dust-to-gas ratio	0.01
ρ_s [g/cm ³]	particle bulk density (no porosity)	1.7 (Ricci opacity)
p	initial surface density power law exponent	1
R [au]	grid extent	0.05-10000
n_r [cells]	grid resolution	2000
t [Myr]	duration of each simulation	3

Table 7.2: Disc initial parameters adopted for two-pop-py population synthesis.

Parameter	Description	Range	PDF
α	viscosity parameter	$10^{-4} - 10^{-2}$	log uniform
$M_{\text{disk}} [M_{\text{star}}]$	initial disc mass	$10^{-3} - 0.5$	log uniform
$M_{\text{star}} [M_{\odot}]$	stellar mass	0.2 – 2.0	IMF
R_c [au]	characteristic radius	10 – 230	log uniform
v_{frag} [cm/s]	fragmentation velocity	200 – 2000	uniform
$m_p [M_{\oplus}]$	planet mass	1 – 1050	uniform
$R_p [R_c]$	planet position	0.05 – 1.5	uniform
t_p [Myr]	planet formation time	0.1 – 0.4	uniform

Notes. Disc initial parameters and corresponding probability distribution function (PDF) from which their value. The drawing of the m_p value was performed after the M_{disk} to impose the further physically reasonable restriction of $m_p < M_{\text{disk}}$. In the case of multiple substructures, we applied the following constraint in a planet i : $\sum_i m_{p,i} < M_{\text{disk}}$.

Table 7.3: Disc initial parameters adopted for DustPy simulations.

Parameter	Description	Value
$M_{\text{star}} [M_{\odot}]$	stellar mass	0.3, 1
$M_{\text{disk}} [M_{\text{star}}]$	initial disc mass	0.01, 0.1
R_c [au]	characteristic radius	20, 50, 100
α	viscosity parameter	$10^{-3.5}, 10^{-2.5}$
v_{frag} [cm/s]	fragmentation velocity	1000
$m_p [M_J]$	planet mass	1
$R_p [R_c]$	planet position	0.5, 0.7, 1, 1.2
$F_{\text{FUV}} [G_0]$	FUV flux	4

Notes. Disc initial parameters used in DustPy simulations of discs subject to viscous evolution and external photoevaporation. The FUV flux corresponds to the average value in the Lupus region, and is a constant during the simulation.

and fragmentation based on Birnstiel et al. (2010a), while two-pop-py evolves only two species of particles at the same time. The gas surface density, Σ_g , evolves accordingly with the advection-diffusion equation where we included an extra term to account for the loss of material in external winds:

$$\frac{\partial \Sigma_g}{\partial t} = \frac{3}{r} \frac{\partial}{\partial r} \left[r^{1/2} \frac{\partial}{\partial r} (\nu \Sigma_g r^{1/2}) - \frac{2\Lambda \Sigma_g}{3\Omega_K} \right] - \dot{\Sigma}_{\text{ext}}, \quad (7.2)$$

where $\nu = \alpha c_s^2 / \Omega_K$ is the viscosity, Λ is the injection rate of specific angular momentum, and $\dot{\Sigma}_{\text{ext}}$ accounts for the rate of mass loss in external photoevaporative winds. The photoevaporative mass loss rate is evaluated performing a bi-linear interpolation of the FRIEDv2 grid (Haworth et al. 2023a), where we assumed ISM-like Polycyclic Aromatic Hydrocarbons (PAHs) for ISM-like dust. Specifically, following the numerical implementation of Sellek et al. (2020b), the grid is interpolated fixing the stellar mass and the FUV flux at the outer disc edge, and using the disc surface density at each radial position. At each time step, the disc radial position corresponding to the maximum mass loss rate defines a truncation radius (which corresponds to the optically thin/thick wind transition in the disc). To simulate the outside-in disc depletion caused by external photoevaporation, the disc material outside the truncation radius is removed accordingly to Sellek et al. (2020b). The model takes into account that external photoevaporation influences the dust component, where dust grains smaller than a certain size limit, which depends on the stellar and disc parameters, are entrained in the wind (Facchini et al. 2016). The study of Gárate et al. (2024) has shown that dust substructures that may form in discs survive the outside-in depletion only if they are placed at a smaller distance from the central star compared to the truncation radius of external photoevaporation.

Due to the higher computational costs of `DustPy` compared to two-pop-py, instead of a population synthesis study, we opted for a grid of test discs based on the constraints on the parameter space placed by two-pop-py simulations of not-photoevaporating discs (Delussu et al. 2024), considering the initial conditions needed to reproduce the observed distributions of spectral index and size-luminosity in Lupus. The extent of the radial grid, the dust particle distribution, and the fragmentation velocity are set as in two-pop-py simulations. In total, we explored the parameter space by performing 27 simulations, considering the combinations of the parameter values listed in Table 7.3. In the simulations including external photoevaporation, we assumed that the discs are subject to a constant external FUV flux of $4 G_0$, which is the average FUV flux experienced by disc-hosting stars in the Lupus star-forming region (Anania et al. 2025b). We let the simulated discs evolve up to the age of 3 Myr in order to compare the outcomes with the observables in the Lupus disc population.

In `DustPy`, the substructures are modelled as a gap arising from the presence of a planet inserted in the disc. We imposed the planetary gap through the radial gas flux induced by a torque profile, rather than modifying the viscosity α -parameter. This approach avoids the issue of unrealistically high gas velocities with a deep planetary gap, which could arise from low gas surface density caused by disc dissipation due to external photoevaporation. The angular momentum injection rate Λ in Eq. (7.2) is derived by assuming a steady-state disc. Neglecting the photoevaporation term, Eq. (7.2) can be rewritten as

$$\frac{\partial \Sigma_g}{\partial t} = \frac{3}{r} \frac{\partial}{\partial r} [r \Sigma_g (v_\nu + v_\Lambda)], \quad (7.3)$$

with the accretion velocity

$$v_\nu = -\frac{3\nu}{r} \frac{\partial \log}{\partial \log r} (\Sigma_g r^{1/2} \nu), \quad (7.4)$$

and the additional velocity due to the torque injection

$$v_{\Lambda} = \frac{2\Lambda}{\Omega_K r}. \quad (7.5)$$

The condition for the planetary gap to not impede disc accretion can be expressed as

$$\Sigma_{g,\text{eq}}(v_{\nu,\text{eq}} + v_{\Lambda}) = \Sigma_{g,0}v_{\nu,0}, \quad (7.6)$$

where $\Sigma_{g,0}$ is the unperturbed gas surface density, $v_{\nu,0}$ is the unperturbed accretion velocity, $\Sigma_{g,\text{eq}}$ and $v_{\nu,\text{eq}}$ denote the gas surface density and the accretion velocity in a steady state, with the imposed gap. Starting from Eq. (7.4), we define the target surface density profile as

$$f(r) \equiv \frac{\Sigma_{g,\text{eq}}}{\Sigma_{g,0}}. \quad (7.7)$$

Given that $\partial(\Sigma_{g,0}\nu)/\partial r = 0$, under the assumption of a constant disc accretion rate, $v_{\nu,\text{eq}}$ is given by

$$v_{\nu,\text{eq}} = -\frac{3\nu}{r} \frac{\partial \log}{\partial \log r} \left(f \Sigma_{g,0} r^{1/2} \nu \right) = -\frac{3\nu}{r} \left(\frac{\partial \log f}{\partial \log r} + \frac{1}{2} \right), \quad (7.8)$$

and $v_{\nu,0}$ is

$$v_{\nu,0} = -\frac{3\nu}{2r}. \quad (7.9)$$

Considering Eq. (7.6), it follows that

$$v_{\Lambda} = \frac{v_{\nu,0}}{f(r)} - v_{\nu,\text{eq}} = -\frac{3\nu}{r} \left(\frac{1}{2f} - \frac{\partial \log f}{\partial \log r} - \frac{1}{2} \right). \quad (7.10)$$

To implement this process in `DustPy`, an external source is applied and calculated with v_{Λ} , following the procedure described in Lau (2024). For the profile f , we adopted the planetary gap profile by Kanagawa et al. (2016), as for the two-pop-py model.

7.2.3 Observables

We derived the distributions of dust disc radii, gas disc radii, millimetre fluxes, and spectral indices from the simulations. Each of these observables is evaluated as follows.

Millimetre flux

Dust fluxes are evaluated following the procedure described in Delussu et al. (2024), that is, adopting the Miyake & Nakagawa (1993) scattering solution of the radiative transfer equation and the modified Eddington-Barbier approximation (Birnstiel et al. 2018). We adopted the Ricci compact opacity (Rosotti et al. 2019a), which is the best opacity model to reproduce the observed spectral index distribution in Lupus (Delussu et al. 2024).

Dust radius

We defined an empirical radius, commonly referred to as the effective disc radius R_{eff} , which encloses a specified fraction of the total continuum flux emitted by the dust disc. The two most commonly adopted choices for the dust disc radius are $R_{68\%}$ and $R_{90\%}$, which are the radii enclosing 68% and 90% of the total disc continuum flux, respectively. As shown in Sanchis et al. (2021), $R_{68\%}$ is a safer choice because it is usually characterized by a smaller dispersion. Therefore, we evaluated $R_{68\%}$ of the simulated discs

to allow a comparison with the observed size-luminosity distribution (Andrews et al. 2018b). Instead, we used $R_{90\%}$ for the comparison with the observed gas and dust radii distributions (Sanchis et al. 2021).

Gas radius

Given their brightness and their optically thick emission, CO rotational emission lines are the most commonly used tracers for gas disc radii. Specifically, we considered ^{12}CO as gas disc size tracer and $R_{\text{CO},90\%}$ as the gas radius, which is the location in the disc that encloses the 90% of the ^{12}CO flux emission. As shown in Toci et al. (2023), $R_{\text{CO},90\%}$ is well-approximated by the radial location where the disc surface density equals a certain critical value, inhibiting ^{12}CO self-shielding.

To compute the gas disc radius, $R_{\text{CO},90\%}$, we evaluated the critical value of the gas column density, which corresponds to the column density of H_2 evaluated at a radius $R_{\text{CO},90\%}$, and has been obtained by Trapman et al. (2023)

$$N_{\text{gas,crit}} = 10^{21.27-0.53\log_{10}L_{\star}} \left(\frac{M_{\text{gas}}}{M_{\odot}} \right)^{0.3-0.08\log_{10}L_{\star}} \text{ cm}^{-2}. \quad (7.11)$$

The critical surface density of ^{12}CO , corresponding to the critical column density defined above, is given by

$$\Sigma_{\text{crit}} = 10\mu_{\text{gas}}N_{\text{gas,crit}} \quad (7.12)$$

where μ_{gas} is the mean molecular weight of the gas in the disc, and we adopted the standard value for protoplanetary discs (i.e., $\mu_{\text{gas}} \sim 2.3\mu_{\text{H}} = 1.15\mu_{\text{H}_2}$). In Eq. (7.11), the initial carbon abundance is assumed to be 10^{-4} . We accounted for additional carbon depletion due to CO freeze-out and photodissociation by multiplying the critical surface density by a factor of 10. The gas disc radius $R_{\text{CO},90\%}$ is evaluated as the location where the gas surface density profile of each simulated disc is equal to its critical surface density value.

Spectral index

The spectral index is defined as the slope of the (sub-)mm SED of the disc dust emission

$$\alpha_{mm} = \frac{d \log F_{\nu}}{d \log \nu}, \quad (7.13)$$

where F_{ν} is the disc-integrated flux at a given frequency ν^1 , thus α_{mm} is the disc-integrated spectral index. Considering very close frequency, Eq. (7.13) can be written as

$$\alpha_{mm} = \frac{\log(F_{\nu 2}/F_{\nu 1})}{\log(\nu_2/\nu_1)}. \quad (7.14)$$

We applied Eq. (7.14) to determine the spectral index of each simulated disc, particularly considering ν_1 and ν_2 as the frequencies corresponding to the wavelengths $\lambda_1 = 3.10$ mm and $\lambda_2 = 0.89$ mm. These wavelengths refer to the observations of the disc sample in Lupus by (Ansdell et al. 2016) and Tazzari et al. (2021b), which we used for comparison with our simulations.

¹All fluxes will be given in frequency space (F_{ν} , typically in units of Jy), even though the subscript is stating the corresponding wavelength instead of the frequency.

7.3 Results

Our simulated gas and dust size distribution is compared with the observed sample reported in Long et al. (2022) for the Lupus region, while the simulated spectral index distribution is compared to the observed sample from Tazzari et al. (2021b). The simulated size-luminosity distribution is compared to that presented in Andrews et al. (2018b). In order to consider the age-spread of Lupus in the simulated population, we selected randomly the observables from the snapshots at 1 Myr, 2 Myr, and 3 Myr.

The main results obtained from the population synthesis of smooth and substructured discs, in the absence of an external UV radiation field, are presented in Fig. 7.1 and are summarised as follows.

- Smooth discs produce a large $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ compared to observations. This result confirms and extends to the broader level of a disc population synthesis the result previously obtained by Toci et al. (2021).
- Substructured discs, either with one or two substructures, produce a size distribution shifted toward lower $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$. The bulk of the population shows $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)} \sim 4$.
- The presence of substructure(s) in discs helps to narrow the gap between theoretical models and observations, however it does not fully resolve the discrepancy.

For completeness, in Fig. 7.1 we also present the spectral index and size-luminosity distributions associated with smooth and substructured discs. As previously shown in Delussu et al. (2024), smooth discs fail to reproduce the spectral index and size-luminosity distributions observed in Lupus, whereas using substructured discs we are able to populate the parameter space region delimited by the observations. However, the results in Fig. 7.1 highlight a tension for substructured discs between the favourable results for the spectral index and size-luminosity distributions, which can be simultaneously reproduced by restricting the parameter space of the initial conditions of the simulated discs, and the discrepancy between the simulated and observed gas and dust size distributions.

We then included the effect of an external FUV flux of $4 G_0$ in a population of viscous discs. In Fig. 7.2, we show the behaviour of the ratio $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ as a function of $R_{\text{dust}(90\%)}$ in the regime of low viscosity (i.e., $\alpha = 10^{-3.5}$) for substructured discs. Specifically, the right panels of Fig. 7.2 show that discs orbiting solar mass stars evolve toward high $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$. The disc dust radius decreases, settling rapidly to a fixed value corresponding to the location of the substructure. Nevertheless, $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ increases over time because these discs experience viscous spreading or reduce their $R_{\text{gas}(90\%)}$ only slightly due to a small effect of external photoevaporation. The left panels of Fig. 7.2 show that discs around stars of mass $0.3 M_\odot$ and with a small characteristic radius $R_c = 20$ au evolve toward high $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ as they experience viscous spreading because their external photoevaporation truncation radius is placed too far away (~ 100 au) for impacting on disc evolution. Moreover, medium ($R_c = 50$ au) and large ($R_c = 100$ au) discs surrounding a star of mass $0.3 M_\odot$ evolve toward small $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$, falling close to or inside the region delimited by the observations.

Having established that small-size discs (i.e., $R_c = 20$ au) and discs around solar-mass stars do not reproduce the observed distribution of $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$, we fo-

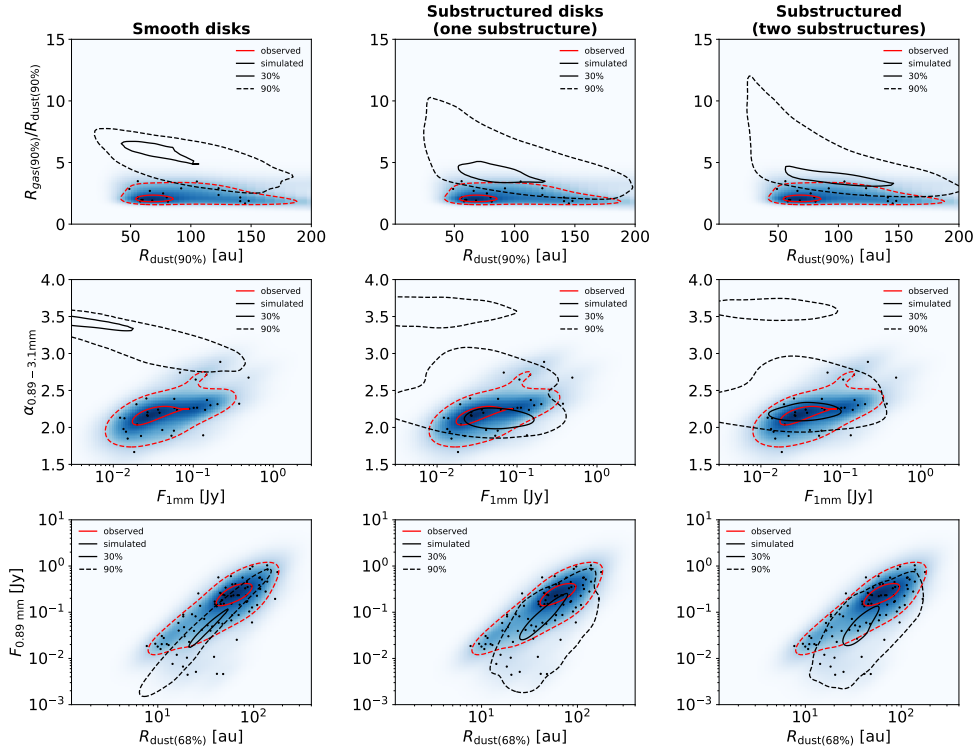


Figure 7.1: Gas-dust size distribution (first row), spectral index distribution (second row) and size-luminosity distribution (third row) for the parameter space of the initial conditions selecting discs with a spectral index $0 \leq \alpha_{0.89-3.1\text{mm}} \leq 4$, $10^{-3} \text{ Jy} \leq F_{1\text{mm}} \leq 10 \text{ Jy}$, $10^{-3} \text{ Jy} \leq F_{0.89\text{mm}} \leq 10 \text{ Jy}$, $10^{0.1} \text{ au} \leq R_{\text{dust}(68\%)} \leq 10^{2.6} \text{ au}$, $1 \text{ au} \leq R_{\text{dust}(90\%)} \leq 200 \text{ au}$ and $0.1 \leq R_{\text{gas}(90\%)/R_{\text{dust}(90\%)}} \leq 20$. Left plots: smooth disks. Middle plots: substructured disks with one planet randomly inserted in a range between 0.1-0.4 Myr from the start of the disc evolution. Right plots: substructured disks with two planets randomly inserted in a range between 0.1-0.4 Myr (innermost planet) and between 0.5-0.8 Myr (outermost planet) from the start of the disc evolution. Heat-map of the observed discs with the black dots representing each single observed disc. The black and red lines refer to the simulated results and the observational results respectively. In particular, the continuous lines encompass 30% of the cumulative sum of the discs produced from the simulations or observed. The dashed lines encompass instead the 90%.

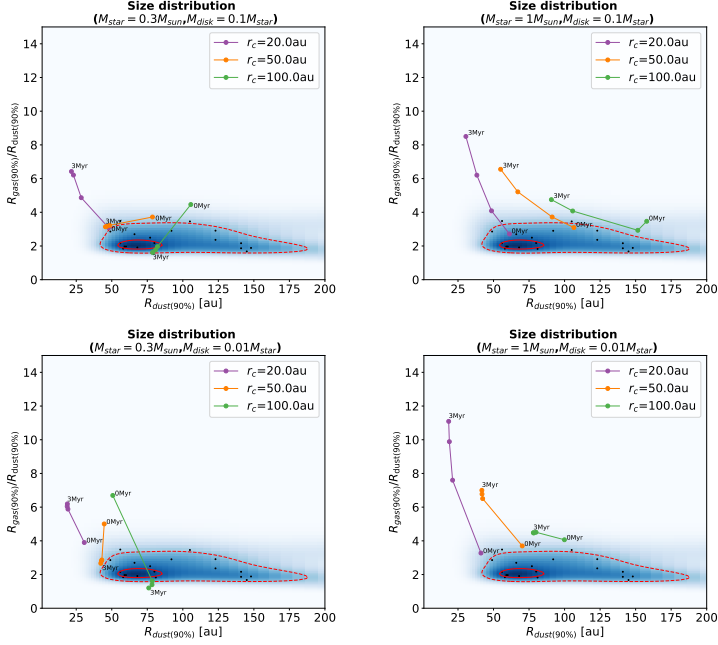


Figure 7.2: Models including external photoevaporation ($F_{\text{FUV}} = 4 \text{ G}_0$): Evolution of $R_{\text{gas}}(90\%)/R_{\text{dust}}(90\%)$ vs $R_{\text{dust}}(90\%)$ for some test substructured discs evolved with `DustPy` in the low viscosity regime ($\alpha = 10^{-3.5}$) and for different values of the characteristic radius r_c . The points associated with each trajectory represent the snapshots taken at 0 Myr, 1 Myr, 2 Myr, and 3 Myr, respectively.

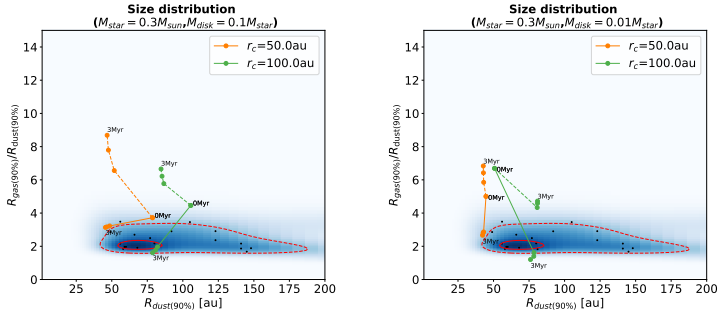


Figure 7.3: Models that do not include external photoevaporation: Evolution of $R_{\text{gas}}(90\%)/R_{\text{dust}}(90\%)$ vs $R_{\text{dust}}(90\%)$ for some test substructured discs evolved with `DustPy` in the low viscosity regime ($\alpha = 10^{-3.5}$) and for different values of the characteristic radius r_c . The points associated with each trajectory represent the snapshots taken at 0 Myr, 1 Myr, 2 Myr, and 3 Myr, respectively. Solid line: external photoevaporation ($F_{\text{FUV}} = 4 \text{ G}_0$). Dashed line: no external photoevaporation ($F_{\text{FUV}} = 0 \text{ G}_0$).

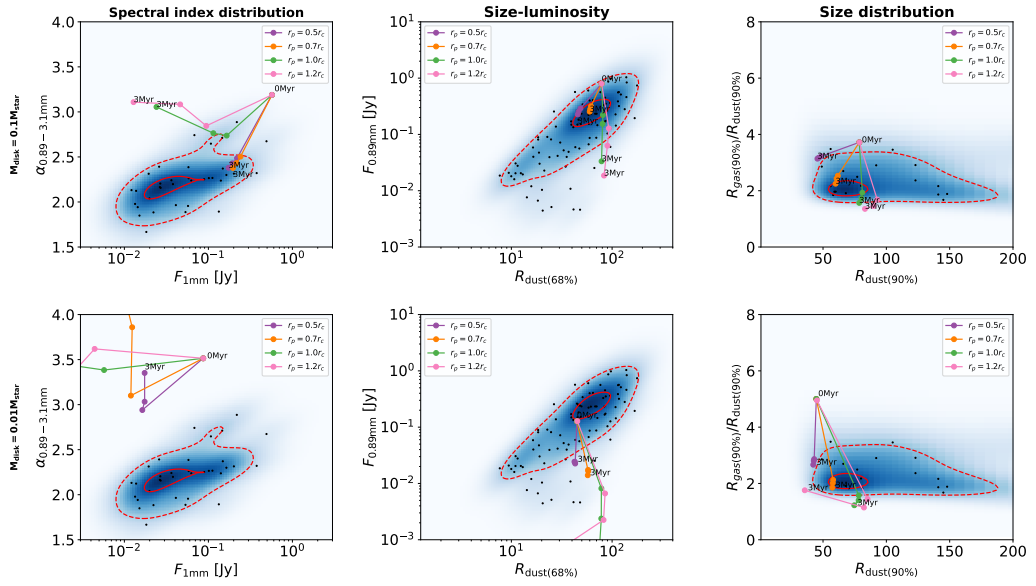


Figure 7.4: Evolution in the spectral index, size-luminosity, and $R_{\text{gas}}(90\%)/R_{\text{dust}}(90\%)$ vs $R_{\text{dust}}(90\%)$ spaces of substructured discs evolved with DustPy code with external photoevaporation ($F_{\text{FUV}} = 4G_0$). Low viscosity regime ($\alpha = 10^{-3.5}$) and disc characteristic radius fixed to $R_c = 50$ au. Different values of the position of the inserted substructure r_p have been explored. The points associated with each trajectory represent the snapshots taken at 0 Myr, 1 Myr, 2 Myr, and 3 Myr, respectively. Top row: $M_{\text{disk}} = 0.1 M_{\text{star}}$. Bottom row: $M_{\text{disk}} = 0.01 M_{\text{star}}$.

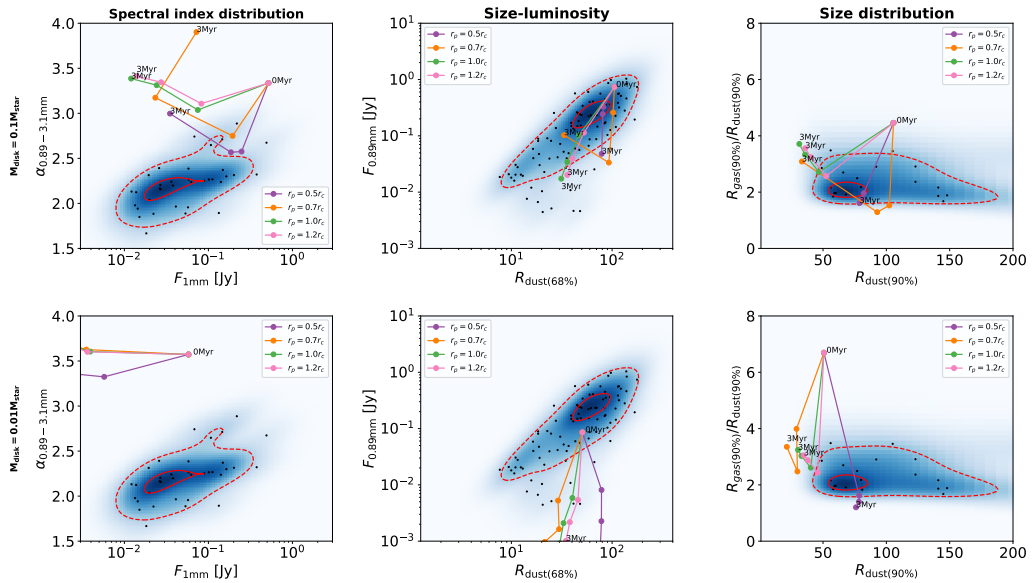


Figure 7.5: Same as Figure 7.4, but discs with characteristic radius fixed to $R_c = 100$ au.

cused on discs with initial characteristic radius of 50 au and $R_c = 100$ au orbiting stars of mass $0.3 M_\odot$.

Figure 7.3 shows that, in the absence of external photoevaporation, the discs move toward high ratios $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ as they experience viscous spreading in $R_{\text{gas}(90\%)}$. The decrease of $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ verified in discs with initial mass $M_{\text{disk}} = 0.01 M_{\text{star}}$ and $R_c = 100$ au is due to the fact that $R_{\text{dust}(90\%)}$ settles toward the location of the substructure, which in this case happens to produce an increase in $R_{\text{dust}(90\%)}$. However, also this disc is affected by spreading in $R_{\text{gas}(90\%)}$, which combined with the stable $R_{\text{dust}(90\%)}$ produced by the presence of the substructure, leads to an increase of $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ over time. In Fig. 7.4 we show the effects of varying the location of the disc substructure on the spectral index distribution, the size–luminosity relation, and the gas-to-dust size ratio. We found that discs with initial extent $R_c = 50$ au experience a continuous reduction in $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$, even when changing the location of the disc trap. However, considering the spectral index and size-luminosity distributions, we noticed that lighter discs (i.e., initial disc mass of $0.01 M_{\text{star}}$) produce large values of spectral indexes and extremely low fluxes. A similar situation is verified for more massive discs (i.e., initial disc mass of $0.1 M_{\text{star}}$) if the substructure is placed at or beyond the characteristic radius. However, placing the substructure too far inside, as in the case where $R_p = 0.5 R_c$, produces smaller $R_{\text{dust}(90\%)}$, which leads to $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ which are larger than what observed. If the substructure is placed in an intermediate position (i.e., $R_p = 0.7 R_c$) and the initial disc mass is set to $0.1 M_{\text{star}}$, the disc falls in the region delimited by the observations in all parameter spaces considered. However, these results also introduce a warning that we may face a fine-tuning problem.

Figure 7.5 shows that initially large discs experience a reduction of $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ only in the case where the substructure is placed at $R_p = 0.5 R_c$. Indeed, placing the substructure far away causes it to be gradually removed by external photoevaporation as it gets closer to the truncation radius (~ 110 au). Furthermore, the erosion of the substructure due to external photoevaporation translates into severe discrepancies with respect to the observed spectral indices and fluxes as these discs start to behave as smooth discs. Despite its good behaviour in the size and size-luminosity space, the $R_p = 0.5 R_c$ fails to produce a spectral index in the observed range.

7.4 Discussion and future steps

Including external photoevaporation does not fully resolve the discrepancy between the gas-to-dust size ratios of the discs predicted by the models and those observed in the Lupus population. Therefore, it may be necessary to reconsider the assumptions underlying the adopted disc evolution model.

A fundamental parameter is the level of viscosity that we assumed in the model. We verified that, while a lower α -viscosity value compared to those considered in this work (e.g., $\alpha = 10^{-5}$) helps in mitigating the disc viscous spreading, this scenario leads to discs with very low fluxes and overly small rings. Moreover, we highlight that we considered a constant α value across the disc radial extent, while the dependence of α on the distance from the central star might alter the overall disc evolution, including how the disc radius varies over time. This radius-dependent α profile may arise, for example, due to the presence of dead zones within the disc (e.g., Tong et al. 2024).

Another important constraint that we imposed in our model is the initial surface density power law exponent, p in Eq. (7.1). As shown by Toci et al. (2021), p (in their notation γ) strongly impact the dust and gas disc radii. The gas disc radius is significantly affected by the shape of the outer part of the disc surface density, which is ultimately

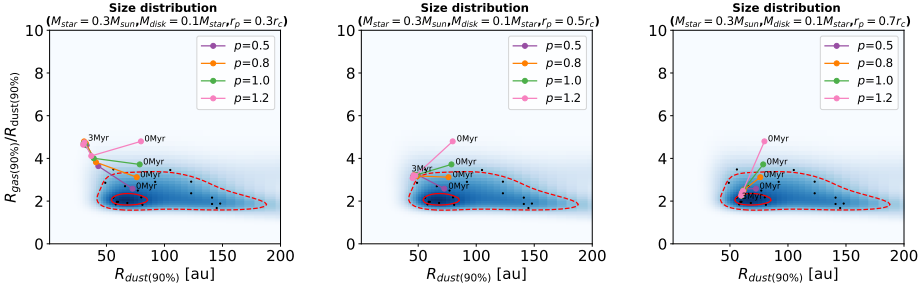


Figure 7.6: Evolution in the $R_{\text{gas}}(90\%)/R_{\text{dust}}(90\%)$ vs $R_{\text{dust}}(90\%)$ space of substructured discs evolved with `DustPy` code with external photoevaporation ($F_{\text{FUV}} = 4 G_0$). Low viscosity regime ($\alpha = 10^{-3.5}$) and disc characteristic radius fixed to $R_c = 50 \text{ au}$. Different values of the position of the inserted substructure R_p and p initial surface density power law exponent values have been explored. The points associated with each trajectory represent the snapshots taken at 0 Myr, 1 Myr, 2 Myr, and 3 Myr, respectively.

determined by p . Lower values of p imply a sharper outer edge, leading to a smaller gas radius. Nevertheless, the effect of this parameter on the dust radius is mitigated after 2-3 Myr due to dust radial drift. Fig. 7.6 shows that, in the case of combined substructures and external photoevaporation, varying p results in evolutionary tracks that rapidly converge towards similar behaviours. This is caused by the fact that changing the initial gas density profile, while adopting a radially constant α leads to an out-of-equilibrium condition. As a result, discs tend to the $p = 1$ configuration after a few viscous timescales. However, Fig. 7.6 shows a promising result as reducing p reduces the initial gas-to-dust size ratio due to the expected reduction in gas radii for $p < 1$.

Alternatively to the pure viscous accretion framework, where angular momentum is re-distributed across the disc, accretion can be driven by MHD-winds that subtract angular momentum (and disc material) from the disc (e.g., Lesur et al. 2014; Tabone et al. 2022). The last scenario, which was not investigated in this work, can prevent disc spreading, and therefore it may help keeping gas disc sizes small enough to reproduce the observed distributions. The discrepancy between the observed and predicted gas and dust disc radii may offer a way to constrain the relative contributions of viscous- and wind-like frameworks. An important future step for the presented work consists in using an MHD-wind driven model instead of a viscous model, with and without the inclusion of external photoevaporation, to investigate disc evolution in the Lupus region and to compare with the observed properties.

Our viscous external photoevaporation models of substructured discs prefer in general initially larger discs (50 – 100 au) than obtained from AGE-PRO ($\lesssim 20 \text{ au}$, Tabone et al. 2025, see Chap. 5). However, it is fundamental to highlight that the two works differ in main goals, comparing observed sample and disc observables, and disc evolution model used. The main goal of the disc population synthesis conducted in AGE-PRO was to constrain the disc initial conditions to explain the median disc mass, radius, accretion rate, and the disc fraction across Ophiuchus, Lupus, and Upper Sco by using a total sample of only 10 discs in each region (excluding the match with Upper Sco when comparing the median gas disc sizes). They explored and compared the viscous and internal photoevaporation model, and the MHD wind-driven model, applied to gaseous-only discs. Instead, in this work we aimed to check whether external photoevaporation in

addition to a pure viscous model is able to explain the distribution of gas-to-dust size ratio observed in Lupus, by using a larger disc sample than AGE-PRO, different observables, and a viscous external photoevaporation model that includes dust traps. Since the observational biases of the two comparing samples and the evolution model used are different, the constraints derived by these two works should not be compared. Moreover, the constraints on the initial conditions derived from both this model and AGE-PRO should be carefully taken as representative of the entire Lupus population. This work demonstrates that including external photoevaporation to a pure viscous model cannot solve the discrepancy in gas-to-dust size ratio we observe in Lupus. Both the results in Tabone et al. (2025) and this work open the way to further and refined disc evolution models that would place stronger constraints on the initial conditions of the Lupus disc population.

7.5 Conclusions

In this work, we conducted a study to determine whether substructured discs can reproduce the observed gas-to-dust size ratio of the protoplanetary disc population observed in the Lupus star-forming region. Firstly, we performed a population synthesis study of substructured discs, following the work of Delussu et al. (2024). Subsequently, we explored the dust-gas size behaviour of discs exposed to a moderate environmental FUV radiation field of $4 G_0$, the average value registered in the Lupus region. We compared the outcomes of the simulations with the observed distributions of dust and gas sizes for discs of the Lupus forming region, complementing this investigation with the comparison of the spectral index and size-luminosity distributions. The main results are summarised as follows.

- We confirmed that smooth discs lead to larger $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ compared to observations in the Lupus region (Fig. 7.1). This result extended the work of Toci et al. (2021) to the broader level of disc population synthesis.
- We showed, on a population synthesis level, that substructured discs, whether with one or two substructures, produce lower $R_{\text{gas}(90\%)} / R_{\text{dust}(90\%)}$ compared to smooth discs (Fig. 7.1). However, although the presence of substructures helps in mitigating the discrepancy between simulation and observation, it does not completely resolve it.
- We identified a tension for substructured discs between the favourable results in Delussu et al. (2024), which show that the spectral index and size-luminosity distributions can be reproduced by simple constraints on their initial conditions, and the difficulty in matching the observed gas-to-dust ratios (Fig. 7.1). This problem is not fully solved even when including the effect of external photoevaporation (Fig. 7.2, 7.4, 7.5).
- Specific combinations of initial conditions can produce results consistent with observations for discs undergoing viscous evolution and external photoevaporation (Fig. 7.4). However, the restricted range of initial conditions needed to obtain these outcomes introduces a fine-tuning problem.
- Both population studies exposed the difficulty of reproducing the observed gas-to-dust size ratios. The presence of substructures can reproduce the spectral index and dust size but leads to an overestimate of the gas radii. However, when gas

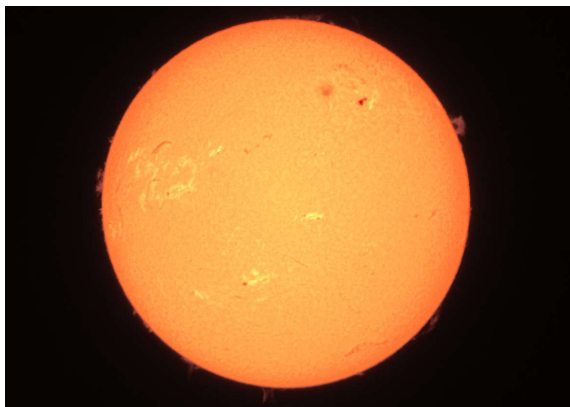
radii are reproduced, dust radii are underestimated. This ultimately emphasizes that the main issue lies in simultaneously reproducing both gas and dust sizes. One possible explanation is that the outermost substructure is linked to the disc truncation radius, which ultimately determines the gas radius. Alternatively, substructures could be so frequent within each disc that one is always found near the gas outer radius. This may point towards an inability of viscous evolution to reproduce observations if paired with constraints of dust evolution.

Future improvements of the model (e.g., including the effect of internal photoevaporation, and exploring an MHD-wind driven disc model instead of a viscous evolution scenario), and the possibility of enlarging the sample of discs with observed properties, are fundamental steps to provide better constraints on the initial conditions of protoplanetary disc populations.

Part III

FUV flux in the nearby star-forming regions

A novel method for evaluating the FUV flux



Refining methods drives progress - Aristarchus first estimated the distance to the Sun by measuring the angle between the Sun and Moon from Earth, when the Moon was half-illuminated and spanned a 90° angle with the Sun from Earth, and applied trigonometry.

This Chapter presents the method to evaluate the FUV flux and its application to nearby star-forming regions included in Anania et al. (2025b). Part of the content of Anania et al. (2025b), regarding the analysis of dust disc masses and FUV fluxes, is presented in Chapter 10.

Understanding the link between the environmental UV radiation and the observed disc properties requires accurately evaluating the FUV flux at disc-hosting stars, which can be challenging because of the uncertainty in stellar parallax. We addressed this issue in Anania et al. (2025b) by proposing a novel approach: using the local density distribution of a star-forming region (i.e. 2D pairwise star separation distribution) and assuming isotropy, we inferred the 3D separation between disc-hosting stars and massive stars. We used this method to compile a publicly accessible catalogue of the FUV flux and its uncertainty for a large sample of disc-hosting stars in seven nearby regions, covering those within ~ 200 pc that have been well-investigated by protoplanetary disc surveys (i.e. Taurus and Lupus), some clusters in the Orion star-forming region (i.e. ONC core, λ Orionis), and Serpens sub-regions at an average distance of ~ 500 pc. We found that discs in regions hosting late-type B and early-type A stars can reach non-negligible FUV

radiation levels for the disc evolution (10-1000 G₀). The method used and the results obtained are detailed and discussed in this Chapter.

8.1 Target regions and YSOs

We evaluated the FUV flux at the position of disc-hosting stars located in nearby star-forming regions with distance ranges from Taurus (~140 pc) to Serpens (~500 pc). In order to explore the level of FUV irradiation experienced by discs in various environmental conditions, we included regions hosting a different number of massive stars covering a wide range of spectral types and featuring well-known populations of disc-bearing stars with estimated ages $\lesssim 10$ Myr. Our investigations of the regions are thus motivated by previous studies of disc populations. The rationale for this selection was to present and apply our method for calculating the FUV flux and its associated uncertainty. By doing so, we access irradiation on individual discs and statistically investigate different regions. In particular, we selected discs from seven of the most nearby star-forming regions that have been intensively investigated by disc surveys (Manara et al. 2023b and references therein). These are Upper Scorpius (Upper Sco), Lupus, Taurus, ρ Ophiuchi (ρ Oph), Chamaeleon I (ChamI), Chamaeleon II (ChamII), and Corona Australis (CrA). In addition, we included discs from six star-forming regions in Orion, which are the Orion nebula cluster (ONC), σ Orionis (σ Ori), λ Orionis (λ Ori), 25 Orionis (25 Ori), NGC 2024, and NGC 1977, where more luminous O-type and early B-type stars are located. Furthermore, the Serpens region was divided into four disc clusters based on Anderson et al. (2022).

We refer to Manara et al. (2023b) for the disc sample in the most nearby regions ($\lesssim 200$ pc) and to Anderson et al. (2022) for the ClassII Serpens discs. The disc sample in the 25 Ori region consists of the sources in Briceño et al. (2019), who presented counterparts in the AllWISE catalogue (Cutri et al. 2021) and infrared excess according to Luhman (2022b), which is consistent with the classification of full, transitional, or evolved discs (Esplin et al. 2018). For the ONC, we selected the discs located in the innermost region around the Trapezium cluster, as included by Eisner et al. (2018b), and we completed this initial catalogue with the proplyds listed in the catalogue by Ricci et al. (2008). We refer to the next chapter for an expansion of the target sample in Orion.

8.2 Massive hot stars and their properties

The total FUV flux in a certain position is usually computed by adding the contribution of individual neighbouring massive stars,

$$F_{\text{FUV,disc}} = \sum_m \frac{L_{\text{FUV,m}}}{4\pi |x_{\text{disc}} - x_m|^2}, \quad (8.1)$$

where $L_{\text{FUV,m}}$ is the FUV luminosity of the massive star, and $|x_{\text{disc}} - x_m|$ is the separation between the location at which we wish to evaluate the flux (in our case, the disc position) and the massive star. In the calculation of the FUV flux, the selected massive stars are often limited to the nearest (or a few nearest) massive stars in the field, assuming that the closest stars dominate the flux, while the influence of more distant stars is typically neglected. However, including high- and intermediate-mass stars that are not members of a specific star-forming region increases the completeness of the analysis. In our study, we used a large sample of massive stars, which enabled us to explore the impact of

irradiation on discs in regions with fewer massive stars. The criteria we employed to select the massive stars that contribute to the overall FUV flux on individual discs are discussed in Sect. 8.2.1, and Sect. 8.2.2 describes how we retrieved the FUV luminosity.

8.2.1 Selection of massive stars

In order to perform a nearly comprehensive selection of massive stars that influence the total FUV flux on discs in specific regions, we cross-matched multiple stellar catalogues as described below.

- **ALS III catalogue:** We selected from the ALS III catalogue (Pantaleoni-Gonzalez et al., in prep.) all the massive stars within 2 kpc of the Sun with good *Gaia* DR3 (Gaia Collaboration et al. 2023b)¹ photometric and spectrometric measurements². This first selection provided a pure list of the known confirmed most massive stars with accurate *Gaia* measurements. It includes 5791 stars covering spectral type O, B2, and earlier for main-sequence stars, B5 and earlier for giants, B9 and earlier for super-giants, Wolf-Rayet (WR) stars, and stars lacking spectral type that were classified as massive since they are situated above the 20 kK extinction track in the Hertzsprung-Russell (HR) diagrams.
- **Zari et al. (2021) catalogue:** We cross-matched the initial ALSIII sample with the massive stars included in the catalogue of hot stars of Zari et al. (2021) that are located closer than 2 kpc to the Sun, and we added stars with effective temperatures greater than ~ 8000 K (corresponding approximately to an A5V star) that were not included in our first sample. In total, we included 218648 hot stars from this catalogue.
- ***Gaia* DR3 Extended Stellar Parametrizer for Hot Stars (ESP-HS) catalogue:** We added the stars (if they were not already included in the previous catalogues) that are contained in the *Gaia* DR3 ESP-HS pipeline (Creevey et al. 2023, Foesneau et al. 2023), which includes stars classified as A, B, and O based on the Extended Stellar Parametrizer for Emission-Line Stars classifier (ESP-ELS). This sample also includes stars that we excluded from the ALSIII initial selection due to their uncertainty or unknown parallax measurement, but that are classified as OBA-type. From this last sample, we excluded the outliers, which are stars whose effective temperature is associated with a higher parameter `teff_esphs` than 23000 K, which present large error bars and are not reliable measurements.
- **Hipparcos catalogue:** Because some of the most luminous stars are not included in the *Gaia* DR3 catalogue because the astrometric and photometric measurements are challenging, we included stars of type O, B, A0, and A1 that are contained in the Hipparcos catalogue (Perryman et al. 1997, van Leeuwen 2007)³ and were not included in the previous selection. The sample of massive stars selected from Hipparcos consists of ~ 6000 stars.

¹<https://www.cosmos.esa.int/web/gaia/dr3>

²Stars that show nebular contamination in the *Gaia* colour-colour diagrams, lack data in one of the three *Gaia* bands, with RUWE > 1.4, or lack parallax values were excluded from this sample

³<https://www.cosmos.esa.int/web/hipparcos/home>

- SIMBAD database: For each star-forming region, we checked and included the stars without a temperature estimate in *Gaia*, but classified as O, B, A0, or A1 in SIMBAD with a well-defined luminosity class and reference.

It is particularly important to include not only the most luminous and massive stars in the FUV flux calculation, but also late B-type and early A-type stars in regions where no O-type and early B-type stars are found, but where the environmental irradiation field is not negligible (e.g. Upper Sco). While the average FUV flux is dominated by the most massive and luminous stars, late B-type and early A-type stars in the close proximity of protoplanetary discs might also be relevant in determining the FUV flux they experience, and consequently, in shaping the disc properties.

8.2.2 Calculation of the FUV luminosity

The FUV luminosity of the massive stars, which is a crucial parameter for evaluating the FUV flux, was evaluated employing the following procedure. We derived the effective temperature from the spectral type and luminosity class of the massive stars using the spectral classification given by the Galactic O-Star Catalogue (GOSC; Maíz Apellániz et al. 2017), when available. This classification is based on high-resolution spectra and not only on photometric measurements. For the stars without a GOSC spectral classification that are contained in the ESP-HS *Gaia* DR3 pipeline, we used the effective temperature addressed by the parameter `teff_esphs` and the associated uncertainty. This is a more reliable estimate of the effective temperature for OBA stars, which is performed by the ESP-HS package, and is characterised by the omission of the regions of the BP/RP and RVS spectra dominated by emission lines that influence the calculation of the temperature. Alternatively, we used the parameter `teff_gspphot`, which corresponds to a temperature based on the assumption that the entire spectrum has a temperature-based origin in the stellar photosphere. The temperature of stars without these two parameters and the Hipparcos stars was retrieved from their spectral classification through tables referring to main-sequence stars, giant, super-giant, and WR stars (Pecaut & Mamajek 2013b, Gray & Corbally 2009b). After evaluating the effective temperature, we used the isochrones provided by the MIST models (Dotter 2016, Choi et al. 2016), assuming a fixed age (we used 1 Myr), to obtain the stellar parameters (stellar mass M_* , radius R_* , luminosity L_* , and the surface gravity g) needed to extract the stellar flux from the Castelli & Kurucz (2004) ATLAS9 stellar atmosphere models. Finally, the stellar flux was integrated in the FUV range of wavelengths [912 - 2400] Å to retrieve the FUV luminosity.

The method described above requires assuming an age for the massive stars because the stellar FUV luminosity increases during the stellar lifetime. However, Kunitomo et al. (2021) demonstrated that intermediate- and high-mass stars present a constant FUV luminosity after ~ 1 Myr of evolution. Therefore, setting the stellar age to 1 Myr does not significantly influence the final result.

8.3 Accounting for the uncertainty on the 3D separation

The FUV flux calculation is significantly influenced by the uncertainty on the true position of the stellar objects in 3D space. Even a small error on the relative separation between discs and massive stars can result in large uncertainties on the final flux values because the FUV flux strongly depends on the relative distance. This is particularly relevant the closer the discs are located to luminous stars, where errors on the relative distance between the two objects (caused by uncertainties in the parallax measurements)

can be comparable to or exceed their separation. We addressed this issue by presenting and comparing three approaches to evaluate the FUV flux that take the uncertainty in the line-of-sight distance of the stars from the observer (measured from parallax) into account, which is known with a far lower precision than the RA and Dec components.

Method 1: Distance uncertainty sampling. The radial distance of each stellar object was evaluated by performing a Monte Carlo sampling of the distance distribution based on the *Gaia* DR3 and Hipparcos parallax measurements and uncertainties, as described in Sect. 8.3.1. In general, this method tends to underestimate FUV fluxes.

Method 2: 2D projected separation and distance uncertainty sampling. For close pairs of disc-massive star members of the same star-forming region, we evaluated the FUV flux using their (2D) projected separation, which resulted in a general overestimate of the actual flux. More details of this approach are provided in in Sect. 8.3.2.

Method 3: Local density function and distance uncertainty sampling. We assumed spatial isotropy throughout a star-forming region, and we used the 2D geometry of the region to make arguments about the most probable 3D separation between stars. Specifically, we defined the probability of finding each pair of stars at a certain (3D) separation, given their (2D) projected distance on the sky plane, and we sampled from the corresponding cumulative distribution function to estimate the corresponding FUV flux, as detailed in Sect. 8.3.3. This approach provided our best estimate of the FUV flux and the associated uncertainty.

8.3.1 Method 1: Distance uncertainty sampling

To account for the uncertainty in the relative distance affecting the final FUV flux value, a standard approach involves using a Monte Carlo sampling of the distance distributions from Earth (or parallax values) of the single stars. This method is commonly used in nearby regions and where parallaxes are known with relative small uncertainties. The ALS III catalogue, as well as the catalogue of massive stars by Zari et al. (2021), provides the 50th, 16th, and 84th percentiles of the posterior distribution of distances. For the other stars contained in the *Gaia* catalogue, the median of the geometric distance posterior (Bailer-Jones et al. 2021b), with 16th and 84th percentiles can be accessed and evaluated for the the remnant stars. Larger uncertainties are registered in particular for faint objects, which are often embedded in gas clouds. This makes the detection more challenging (e.g. discs in the ρ Oph region). Larger uncertainties are also obtained for very luminous stars (e.g. σ Ori or θ^1 C). For objects without parallax measurements, we assigned parallax and parallax error as the median values of the nearest neighbour members of the same star-forming region with the same spectral type. The FUV flux at each disc position was evaluated by randomly assigning the radial distance of individual objects based on their provided distributions, and then repeating this process multiple times to determine the median, 16th, and 84th percentiles of the FUV flux. We show in Sects. 8.3.4 and 8.4 that this method in general tends to underestimate the FUV flux because the range of values that the 3D separation between objects can assume when randomly sampling from individual extended distributions of line-of-sight distances is large, and the probability of retrieving the correct separation is low. This method may yield unreliable FUV flux estimates when the distance uncertainties are comparable to or larger than the separation between an OBA star and the star that hosts the disc. In this scenario, even a small change in the relative separation between the disc and the mas-

sive star (due to large uncertainties in the line-of-sight position compared to the other two components on the sky plane) can lead to substantial variations in the final flux value.

8.3.2 Method 2: 2D projected separation and distance uncertainty sampling

The highest contribution to the FUV flux at the position of a certain disc-hosting star is given by massive star members of the same star-forming regions because they are closest to the object that is examined. However, the flux induced by these stars is subject to a large uncertainty due to the problem described above, where a small uncertainty in the line-of-sight position of two close objects can propagate into a large uncertainty in the FUV flux estimate. Therefore, for each star-forming region, we focused on these massive stars defined as members of the region. When we assume spatial isotropy within a given region, all directions have the same properties, meaning that stars with closer projected separations are more likely to be physically close (i.e. in 3D space). Hence, an upper limit of the FUV flux induced by massive star members of a certain region can be obtained considering the separation between them and the disc-hosting stars (members of the same region) projected on the sky plane at a line-of-sight distance corresponding to the object presenting the smallest distance uncertainty. For the massive stars outside the a certain region, we applied Method 1, described in the previous section, where an estimate of the uncertainty on the FUV flux is provided by randomly sampling the distribution of line-of-sight distances of single objects. While this approach offers an initial estimate of the flux when parallax measurements are not available, it has limitations. Specifically, as the primary source of the flux uncertainty for discs arises from the nearest massive stars, for which we only considered a 2D separation and ignored the errors (even when available) in the third dimension, we lose information about the accuracy of the flux calculation. The only source of the flux uncertainty in this method is produced by the separation between discs and the massive stars located outside their star-forming region, which is small because more distant stars influence the flux calculation less.

8.3.3 Method 3: Local density function and distance uncertainty sampling

For the massive stars and disc-hosting stars that are members of the same star-forming region, which can present errors on their 3D relative distance comparable to or larger than their 3D separation, we can improve Method 2 (Sect. 8.3.2). We can do this by appealing to the geometry of the region to find the best estimate of the 3D separation between stars given their 2D relative distance. We assumed that (i) the stellar members of a given star-forming region have well-defined separations on the sky plane, meaning that we neglected errors in the RA and Dec positions, as these are minimal compared to the separations between stars and are smaller than the uncertainties in parallax, and (ii) spatial isotropy applies throughout the region. Considering a stellar member i , we employed the Bayes theorem to calculate the differential probability to find a second member j at a certain 3D separation r_{ij} , assuming to know the 2D separation R_{ij} ,

$$dP(r_{ij}|R_{ij}) = \frac{dP(R_{ij}|r_{ij}) dP(r_{ij})}{dP(R_{ij})}. \quad (8.2)$$

In order to derive an analytic expression for Eq. (8.2) (i.e. likelihood and prior), we started by defining the differential probability to find a star j at a projected separation R_{ij} from a star i in a small area $dA = 2\pi R_{ij} dR$,

$$dP(R_{ij}) = \hat{\Sigma}_{\text{pairs}}(R_{ij}) dA, \quad (8.3)$$

where $\hat{\Sigma}_{\text{pairs}}(R)$ is the normalised probability density function of all the projected separations between the members of the star-forming region. Under the assumption of isotropy, the Abel theorem (Abel 1826) states that considering the (2D) surface density of neighbouring stars at distance R , $\hat{\Sigma}_{\text{pairs}}(R)$, in the form

$$\hat{\Sigma}_{\text{pairs}}(R) = 2 \int_R^\infty \hat{\rho}_{\text{pairs}}(r) \frac{R}{\sqrt{r^2 - R^2}} dr, \quad (8.4)$$

the corresponding (3D) volume density, $\hat{\rho}_{\text{pairs}}(r)$, is uniquely defined as

$$\hat{\rho}_{\text{pairs}}(r) = -\frac{1}{\pi} \int_r^\infty \frac{d\hat{\Sigma}_{\text{pairs}}(R)}{dR} \frac{1}{\sqrt{r^2 - R^2}} dR, \quad (8.5)$$

which is valid if $\hat{\rho}_{\text{pairs}}(r) \rightarrow 0$ as $r \rightarrow \infty$. As Abel's inversion requires spherical symmetry to project a surface density in a volume density, we applied the theorem to the distribution of separations between pairs of stars in the isotropic case and to the separations from the centre of the region for a centrally concentrated region. Using Eq. 9.3, we rewrote Eq. 8.2 for a single pair of stars ij as

$$dP(r_{ij}|R_{ij}) = dP(R_{ij}|r_{ij}) dP(r_{ij}) = \begin{cases} \frac{4\pi R_{ij} r_{ij}}{\sqrt{r_{ij}^2 - R_{ij}^2}} \hat{\rho}_{\text{pairs}}(r_{ij}) & R_{ij} < r_{ij} \\ 0 & \text{otherwise} \end{cases}, \quad (8.6)$$

which defines our probability distribution function (PDF), and where the factor multiplying the volume density was included to account for the geometrical effect when passing from 2D to 3D geometry. Using the precise RA and Dec measurements provided by *Gaia* DR3 and Hipparcos, we defined the separation R as the angular separation between two stars on the celestial sphere (evaluated using the Vincenty formula Vincenty 1975) projected to the median distance of the considered region. For each star-forming region, we used the membership census available in the literature to evaluate $\hat{\Sigma}_{\text{pairs}}(R)$ (hereafter $\hat{\Sigma}$) as the distribution of the projected separations between pairs of stars in the same region. Subsequently, we estimated the 3D density profile $\hat{\rho}_{\text{pairs}}(r)$ (hereafter $\hat{\rho}$) using Eq. (9.3), where r is the 3D (spherical) separation between stars. Finally, we sampled the probability distribution defined in Eq. (9.2) to derive the 3D separation between close stellar pairs, and we used this parameter to evaluate the FUV flux and the corresponding uncertainty.

When the geometry of the region was nearly centrally concentrated, we assumed spherical isotropy throughout the entire region and still used the expression in Eq. (9.2), but considering r_{ij} and R_{ij} as the 3D and 2D separations from the centre of the distribution (i.e. r_{ij} is the radius of the sphere centred in the centre of the distribution, and R_{ij} is radius in polar coordinates when neglecting the line-of-sight component). This method can be applied in regions such as the ONC and σ Ori, where stellar members are distributed almost symmetrically around the most luminous star, which in these cases are θ^1 C and σ Ori, respectively.

8.3.4 Testing the three methods for evaluating the FUV flux on synthetic stellar clusters

The three approaches for evaluating the FUV flux described in the previous section, in particular, the effectiveness of using the local density function to infer the 3D separation between stars in the same star-forming region, were examined through synthetic stellar

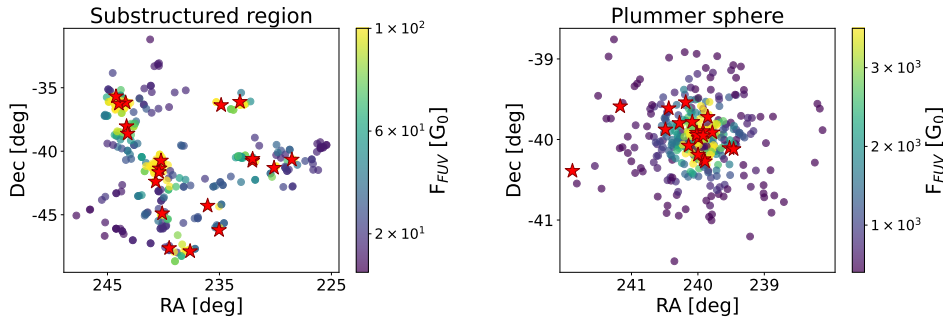


Figure 8.1: Position of the low- and high-mass stars in the synthetic stellar clusters: Substructured region in the top panel, and Plummer sphere in the bottom panel.

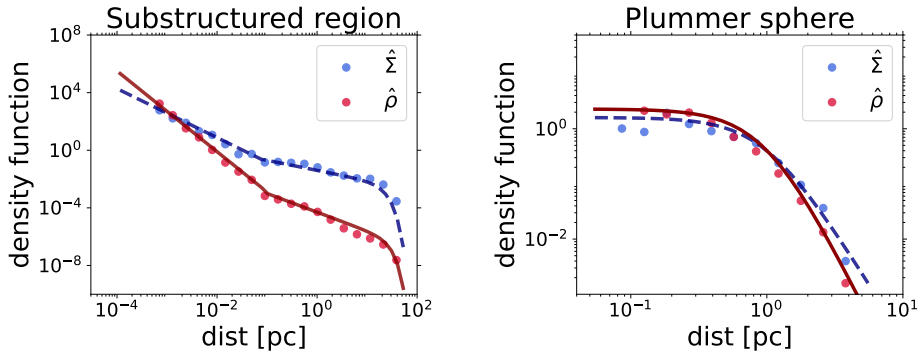


Figure 8.2: Surface density ($\hat{\Sigma}$) and volume density ($\hat{\rho}$) profiles related to the simulated clusters. The x-axis represents the separation between pairs of stars in the substructured region (top panel) and the distance from the cluster centre in the Plummer sphere case (bottom panel).

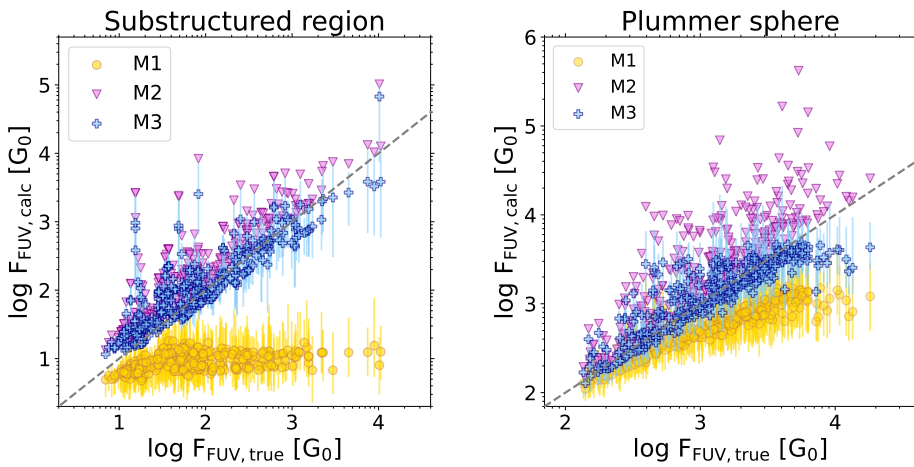


Figure 8.3: Comparison between the true FUV flux experienced by the low-mass stars as evaluated from the simulated clusters, and the value resulting from applying the three approaches described in Sects. 8.3.1 (M1), 8.3.2 (M2), and 8.3.3 (M3).

clusters. In order to highlight the dependence of the density profile on the geometry of the star-forming region, we simulated two regions with identical numbers of high-mass (20) and low-mass (340) stars, but with differing geometries: The first region presented internal substructures, and the second region was modelled as a Plummer sphere (to resemble a spherically symmetric cluster). They are shown in the top and bottom panels of Fig. 8.1. The substructured synthetic cluster was based on a random subset from the spatial distribution of stars in Winter et al. (2024b), considering the simulation snapshot at 1 Myr. Assuming isotropy, the density profile associated with the substructured region depends on the separation between pairs of stars. Conversely, under the condition of spherical symmetry around the centre of the region, as in the case of the Plummer sphere, the density profile is a function of the distance from the centre. The surface and volume density profile of the two simulated regions are shown in Fig. 8.2. For the substructured region, $\hat{\rho}$, which consists of the Abel transform of $\hat{\Sigma}$, can be analytically prescribed by a double power law that is truncated exponentially,

$$\hat{\rho}(r) = \begin{cases} \alpha_1 r^{-\beta_1}, & r \leq r_{\text{lim}}, \\ \alpha_2 r^{-\beta_2} \exp[-(r/r_b)^{-\gamma}], & r > r_{\text{lim}}, \end{cases} \quad (8.7)$$

where r is the separation between pairs of stars in 3D space (spherical coordinate), normalized to 1 pc in our expressions, and the parameters adopted in the presented case are $\alpha_1 = 2 \times 10^{-6}$, $\beta_1 = 2.8$, $\alpha_2 = 5 \times 10^{-5}$, $\beta_2 = 1.3$, the length-scale of the cluster $r_b = 25$ pc (which sets the boundary of the cluster), $\gamma = 2.6$, and the limit separation $r_{\text{lim}} = 0.1$ pc, which sets the change of the power-law steepness. These parameters vary based on the specific geometry of the region. The Plummer volume density profile was

$$\hat{\rho}(r) = \alpha (1 + r^2)^{-5/2}, \quad (8.8)$$

where r is the distance from the centre of the cluster in 3D space, and we assumed a Plummer radius of 1 pc and $\alpha \simeq 2.4$. The density profiles adopted for the various star-forming regions we investigated are summarised in Appendix 8.7.1.

We evaluated the FUV flux experienced by the low-mass stars due to the irradiation of the high-mass stars using the three approaches described in Sects. 8.3.1, 8.3.2, and 8.3.3. In particular, in order to apply the distance sampling method, we introduced an error on the distance of the stars comparable to the length scale of the region (~ 25 pc for the substructured region, and ~ 1 pc for the Plummer sphere), which is relatively large compared to the median parallax error in the most nearby regions (e.g. Taurus and Lupus), but is reasonable in more distant regions (e.g. Orion). In Fig. 8.3 the FUV fluxes resulting from using the three approaches are compared with the true values. Method 1 (Sect. 8.3.1), shown in yellow in Fig. 8.3, tends to significantly underestimate the actual flux by some orders of magnitude for some stars. This result highlights the error in the FUV flux that can result from sampling *Gaia* parallaxes and parallax uncertainties, and the power of our proposed approach. The use of Method 2 (Sect. 8.3.2) leads to a general overestimate of the flux. Specifically, the fraction of total FUV fluxes reproduced (within 1σ) by Method 1 is 1.3% for the substructured cluster and 21% for the Plummer sphere, while Method 2 reproduces the $\sim 7.5\%$ of the true values in both cases. The local density function (Method 3) yields FUV fluxes that are consistent with the true values within 1σ error bars for 48% of the low-mass stars in the substructured region and 68% of the low-mass stars in the Plummer sphere case. For the substructured region, we obtained a smaller fraction of successfully reproduced fluxes with respect to the Plummer sphere. This is caused by the fact that we neglected the correlation between stellar pairs

when employing the Bayes theorem. Specifically, in our approach, the 3D separation of each pair of low-mass - high-mass star is independently sampled from a cumulative distribution function (CDF). However, the separation between each pair of stars is not completely independent from the others, and a covariance term should be included in the analytic formulation. When this term is not accounted for, the flux uncertainty is underestimated in general. However, the uncertainty in the flux is more strongly underestimated as the number of massive stars that contribute to the flux rises. Therefore, if a star-forming region hosts a small number of massive stars, the error caused by neglecting the covariance term is relatively small. A more detailed analytic work is required to account for this term, together with further testing on simulated clusters. As the resulting fluxes reproduce within the error bars a substantially larger fraction of true values, this study is left to future investigation. Conversely, in the Plummer sphere, as all the separations are defined with respect to the fixed centre of the cluster, we did not investigate each stellar pair independently, but the entire cluster: We sampled from a CDF the 3D separation of all the stars from the centre, generating entire random clusters, and then we evaluated the FUV flux within each cluster.

The fact that Method 3 (Sect. 8.3.3) yields more than twice the percentage of FUV fluxes consistent with true values compared to other methods highlights the strength and advantage of the approach we propose over the investigated alternative methods.

8.4 A catalogue of the FUV flux for disc-hosting stars in nearby star-forming regions

In this section, we present the FUV flux estimate resulting from employing the three approaches introduced in the previous section, at the position of the disc-hosting stars in the selected star-forming regions. We refer to Sects. 8.4.1 and 8.4.2 for the investigation of the most nearby regions and regions in Orion and Serpens, respectively. The extended list of FUV fluxes and uncertainties for the considered discs in the various star-forming regions is available at the CDS.

8.4.1 Regions within 200 pc of the Sun

The nearest star-forming regions included in this work contain only a few late B-type and early A-type stars. Therefore, we expect the FUV irradiation to be lower than in more distant and massive regions. Moreover, due to their proximity to Earth, the line-of-sight distance of these stellar objects is known with greater precision, which should result in negligible differences when applying the three different methods for the flux calculation. However, if one or more of the few massive stars are in the close proximity of discs, the resulting FUV irradiation may not be negligible and could significantly influence the disc evolution.

In Fig. 8.4 we show the FUV flux resulting from applying the three approaches described in Sect. 8.3 to the discs in the most nearby regions. The three methods proposed for the calculation of the FUV flux provide a conservative estimate (Method 1), an overestimate (Method 2), and our best estimate (Method 3). In order to apply Method 3 (Sect. 8.3.3), we derived the local density distribution function associated with each region by using the membership catalogues from Luhman (2022c) for Upper Sco and Lupus, Luhman (2020) for Lupus, Luhman (2023) for Taurus, Galli et al. (2021) for Cham I and Cham II, Grasser et al. (2021) for ρ Oph, and Esplin & Luhman (2022) for CrA. The analytic expression that better describes the local density profile of these regions is a double power law truncated close to the length scale of the star-forming region (approximately the

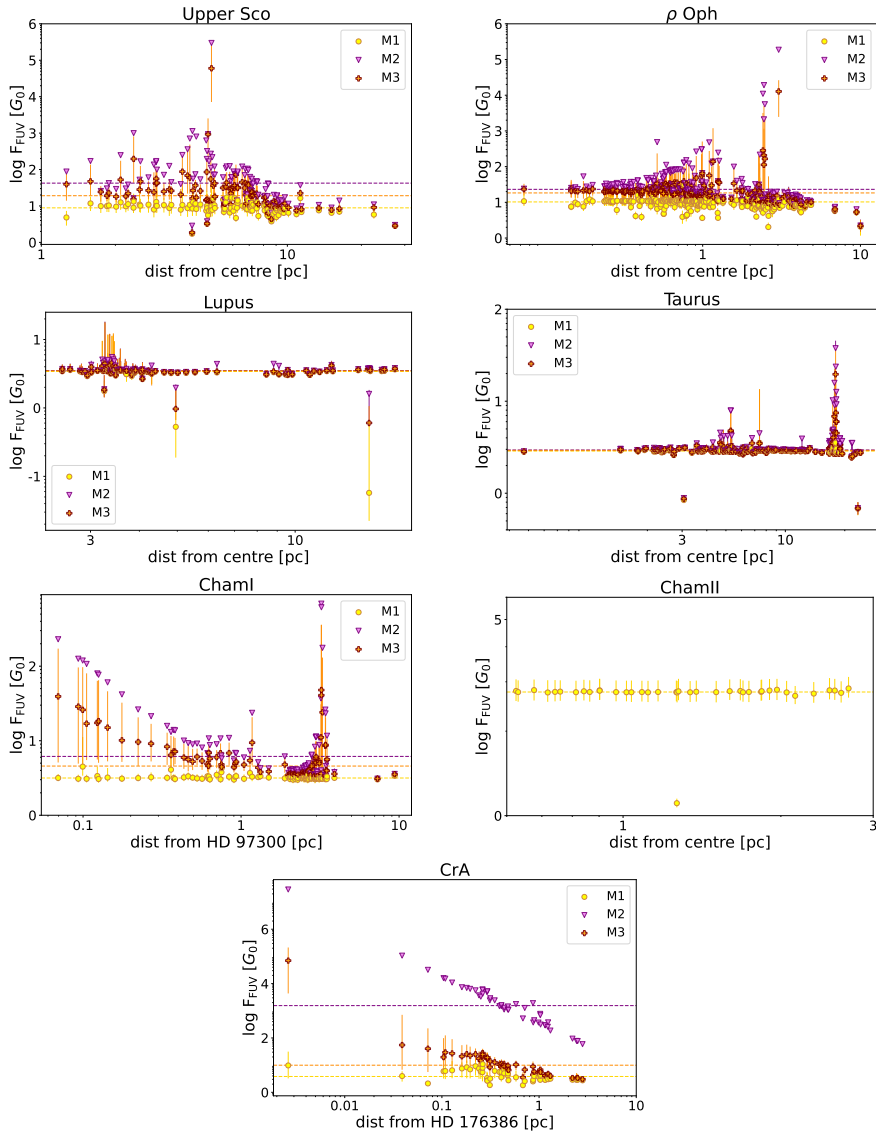


Figure 8.4: FUV flux experienced by discs in seven regions located $\lesssim 200$ pc from the Sun. The three approaches we used to evaluate the FUV flux are presented in yellow (Method 1), purple (Method 2), and orange (Method 3). The dashed lines show the median FUV flux value for each calculation method and region. x-axis: Distance from the most massive star, or the centre of the region (when the distribution of OBA stars, and therefore FUV fluxes, is approximately uniform in the region).

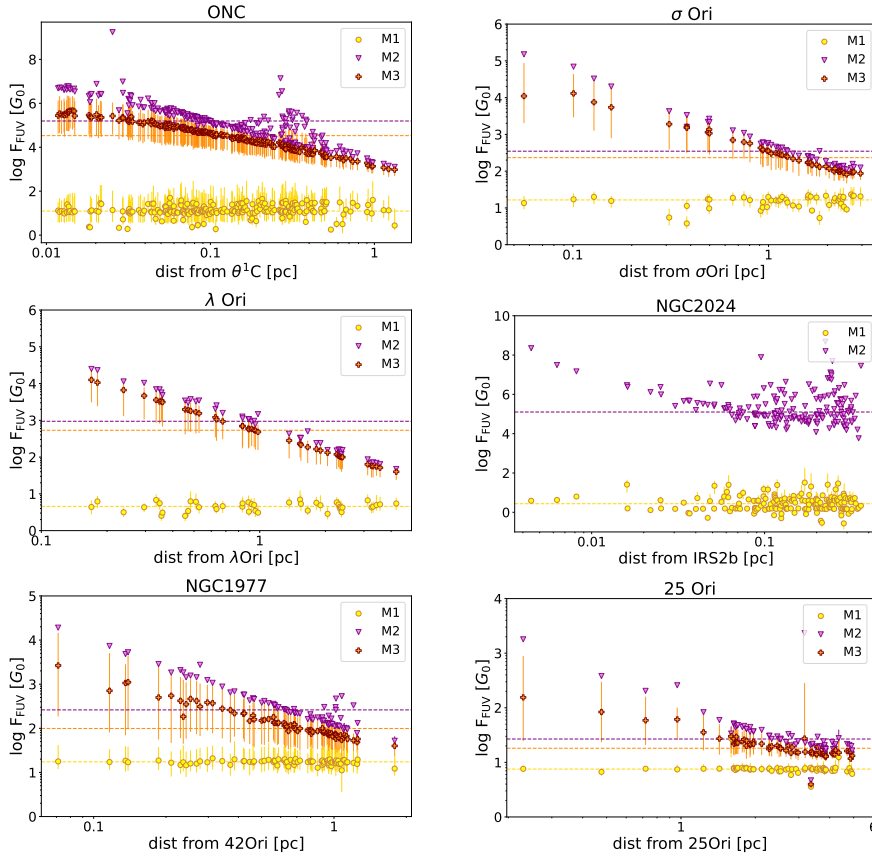


Figure 8.5: FUV fluxes experienced by discs in six star-forming regions in Orion. The different colours correspond to the different methods for the calculation of the FUV flux, as introduced in Sect. 8.3. The dashed lines show median FUV flux values for each region and for the calculation method. The x-axis contains the distance from the most massive star in the region. In NGC 2024, the local density distribution method cannot be applied (and therefore, no orange points lie in that panel), as an accurate density profile of the region cannot be currently retrieved.

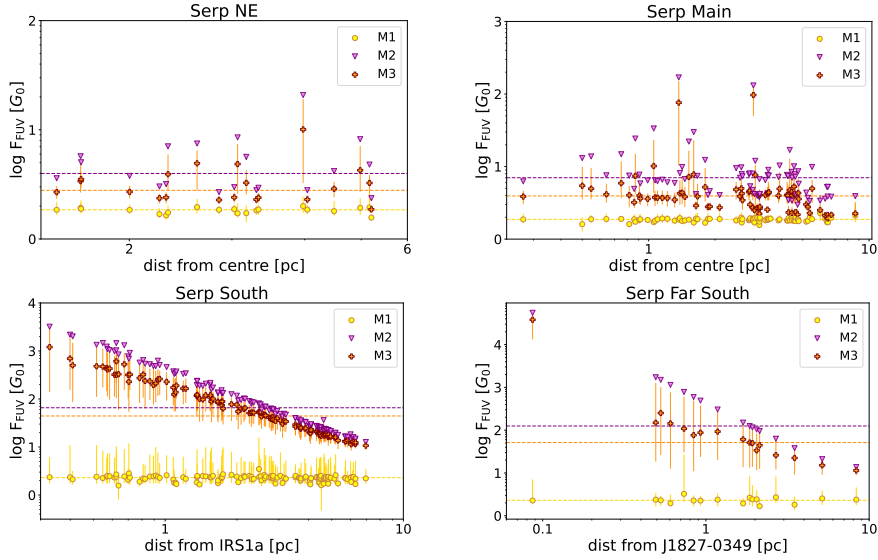


Figure 8.6: FUV flux experienced by discs in the Serpens region. Based on the sub-region in Serpens where they are located, the discs were divided into NE, Main, South, and the most southern part, Far South. Our flux estimates using Method 1, Method 2, and Method 3 (Sect. 8.3) are shown in yellow, purple, and orange, respectively. The lines show the median fluxes resulting from the three approaches. x-axis: Distance from the most massive star, or the centre of the region (if no strong dependence on the most massive star was found).

width of the region on the sky plane). The parameters of the expression for each region are listed and discussed in Appendix 8.7.1.

The difference between the three methods is evident in Upper Sco, ρ Oph, Cham I, and CrA. Upper Sco presents the highest number of B-type stars of the considered regions, ~ 50 (Luhman 2022c), and where external irradiation was demonstrated to substantially contribute to shaping disc properties (Anania et al. 2025a). The massive and hot stars in this region are mostly B8V and B9V, and the only O9 is ζ Oph, which is located at a large separation from the considered discs and therefore does not influence the total flux significantly. The median FUV flux in Upper Sco provided by our best estimate is $\sim 20 G_0$. A similar median is found in ρ Oph, which is located close to the Upper Sco region and contains younger protoplanetary discs with large parallax uncertainties (or no parallax measurements) due to the presence of dense material part of the natal molecular cloud. In this region, $\sim 90\%$ of the total irradiation at the position of the disc-hosting stars is produced by the multiple system of massive stars ρ Oph itself (B2IV+B2V, Abt 2011). At lower irradiation levels, the presence of massive stars in the close proximity of discs is relevant in Cham I and CrA, where the disc irradiation presents a clear dependence on the separation from the closest massive stars, which are HD 97300 (B9V, Houk & Cowley 1975) and HD 96675 (B5V, Siebenmorgen et al. 2020) in ChamI, and HD 176386 (B9V, Murphy et al. 2020) in Cra. These stars cause the high flux registered for a large sample of discs, with respect to the median, as shown in Fig. 8.4. In these cases, it is evident that the distance uncertainty sampling approach tends to underestimate the flux, especially when discs are close to massive stars. This occurs even in clusters with only a few massive stars and where the line-of-sight distances are known with higher precision than in more distant regions. The disc-hosting star with the highest FUV flux in CrA is

2MASS J19013912-3653292 (HD 176386B), which forms a binary system with the B9 star HD 176386. The close proximity to this massive star results in the high observed flux.

Conversely, in Cham II, no significant B-type stars are found near the considered discs, and therefore, the overall flux is determined by field stars, non-members of the region, and the best estimate coincides with the result provided by the sole distance uncertainty sampling approach. Taurus and Lupus show similar results, with few discs being very close to B9 stars.

8.4.2 Orion and Serpens

In Fig. 8.5 and Fig. 8.6 we present the FUV flux evaluated at the position of the disc-hosting stars in the selected star-forming regions in Orion and Serpens. The flux estimate provided by using our novel approach (Method 3, Sect. 8.3.3), shown in orange in Figs. 8.5 and 8.6, was evaluated considering the local density distribution function derived by using known stellar membership of each star-forming region from various catalogues retrieved from the literature. Specifically, we used Kounkel et al. (2018b) for the ONC, σ Ori, λ Ori, and 25 Ori, Da Rio et al. (2012) and Fang et al. (2021) for the ONC, Žerjal et al. (2024) for σ Ori and NGC 2024, Suárez et al. (2019) for 25 Ori, and the updated sample of members of NGC 1977 by Kim et al. (in prep.). In regions presenting a 2D nearly centrally concentrated geometry, which in our sample correspond to the ONC, σ Ori, λ Ori, and 25 Ori regions, the local density profile was defined as the function of the separation from the centre of the region, as discussed in Sect. 8.3.4. The analytic expression that better describes this profile is an exponentially truncated single power law, which was already known in previous works for the ONC (e.g. Da Rio et al. 2012). In the case of NGC 2024, the confirmed stellar census contains too few stars that complement the disc-hosting stars included in our study, spread across various separations, which makes it difficult to recover enough stars per unit separation to derive a reliable density profile. Therefore, our statistical approach cannot be fully trusted, and we only present the FUV fluxes obtained through Method 1 (Sect. 8.3.1), and Method 2 (Sect. 8.3.2). Additional details on the derivation and reliability of the density profile, along with the specific parameters used in the calculation, are discussed in Appendix 8.7.1.

For the Serpens region, we used the *Gaia* DR2 membership census by Herczeg et al. (2020) and the young stellar objects (YSOs) catalogue of Anderson et al. (2022) to derive the local density profile of the three main extended sub-regions, which are NE, Main, and South. In Fig. 8.6, the investigated ClassII discs in the southernmost region of Serpens South, referred to as Far South, are shown separately from the others. This distinction highlights the dependence of the flux on the separation from the most massive stars, which are IRS1a (O9.5V) and 2MASS J18273952-0349520 (B1.5V, Drew et al. 1997), located in Serpens South (near the HII region Westerhout 40, Shenoy et al. 2013), and Far South, respectively. The highest level of irradiation is found near the HII region Westerhout 40 (W40), where a small group of massive stars is concentrated (Anderson et al. 2022). We refer to Appendix 8.7.1 for the 2D sky maps of the selected disc sample and neighbouring massive stars included in this work, corresponding to each star-forming region.

The average FUV flux is significantly higher in the considered Orion and Serpens regions (with the exception of Serp NE and Serp Main) compared to the closest regions (Fig. 8.6). The flux estimate provided by involving Method 3 reaches $\sim 10^5 - 10^6 G_0$ in the ONC, where the total flux is dominated by the most luminous O-type and early B-type stars, while field stars only contribute $\sim 1\%$ of the total flux on average. Method 1 (Sect. 8.3.1) results in a nearly flat distribution of the FUV fluxes in all the regions, with

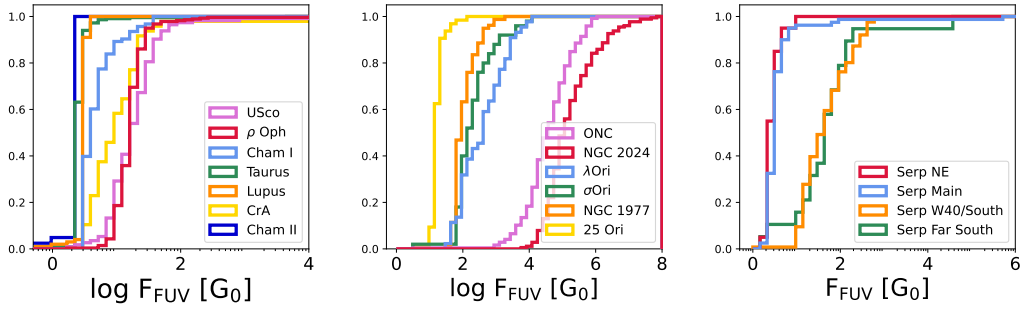


Figure 8.7: Cumulative distributions of the median FUV fluxes experienced by discs in the regions selected in $\lesssim 200$ pc (left panel), Orion (central panel), and Serpens region (right panel). The flux values are the best estimate evaluated using Method 3 (Sect. 8.3.3). The exception is NGC 2024, where we evaluated the FUV fluxes using Method 2 (Sect. 8.3.2). We consider them as overestimates of the actual fluxes.

a median value $\lesssim 10 G_0$ (as shown in yellow in Figs. 8.4, 8.5, and 8.6).

Method 2 and Method 3 both demonstrate a clear dependence of the flux on the distance from the most massive and luminous star in the region, as shown on the x-axis of the panels in Fig. 8.5.

As for the synthetic cluster, we show that Method 1 using real parallax errors from *Gaia* DR3 and Hipparcos severely underestimates the FUV flux, while placing constraints based on the 2D geometry of a star-forming region can result in a more accurate FUV estimate.

8.5 Discussion

We used the available information on the 2D geometry of a star-forming region to compute the best estimate of the FUV flux at the position of a sample of disc-hosting stars (Method 3, Sect. 8.3.3). In this section, we discuss the FUV fluxes obtained for each region (Sect. 8.5.1), the uncertainties, limitations, and future improvements of our proposed method (Sect. 8.5.2).

8.5.1 FUV flux in the star-forming regions

Figure 8.7 presents the cumulative distributions of the resulting FUV fluxes in the various regions, divided into the most nearby regions (left panel), regions in Orion (central panel), and regions in Serpens (right panel), for better visualisation. In Fig. 8.8 we provide for each region the distribution of the median FUV fluxes at the position of disc-hosting stars. Discs in Orion experience the highest levels of irradiation of the studied regions, particularly those in the ONC and NGC 2024⁴. However, we verified that even in regions in which the median irradiation is one hundred times lower than in the ONC, the flux distributions still reach moderate and high levels ($10^2 - 10^3 G_0$). This is due to late B-type and early A-type stars, which are closely located to disc-hosting stars, and it underlines the importance of considering the external irradiation even in moderately irradiated regions. The median incidental external FUV flux on disc-hosting stars in the

⁴We recall that for NGC 2024, we used the flux values derived with Method 2 (Sect. 8.3.2), because the local density profile cannot be reliably derived in this region. These values are overestimates of the actual FUV flux.

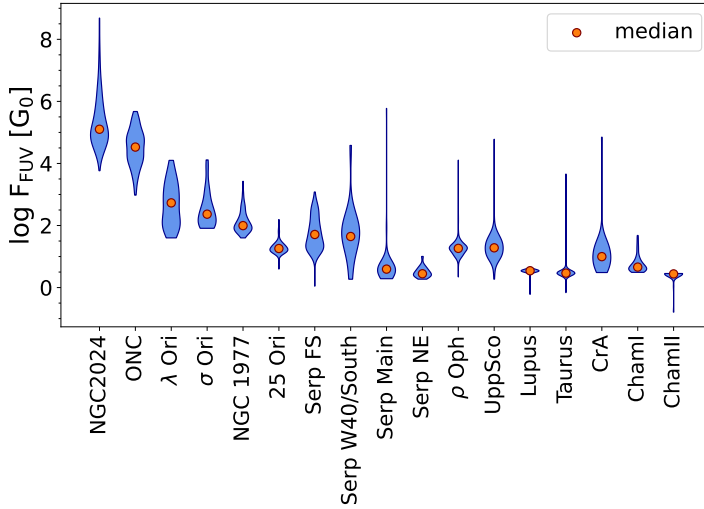


Figure 8.8: Distributions of the FUV fluxes for the star-forming regions. The median of the distributions is shown in orange. NGC 2024 fluxes are overestimates of the actual values because we used Method 2 (Sect. 8.3.2) to compute the FUV flux as we lacked information to accurately derive the local density profile.

lowest irradiated regions, which are ChamII, Taurus, Serp NE, and Serp Main, is ~ 2 – $3 G_0$, which is induced by field massive stars that are not members of the star-forming regions themselves. Upper Sco and ρ Oph present in median the highest FUV irradiation field in the most nearby regions because of the numerous massive stars in the Sco-Cen OB association (Wright 2020). The disc truncation due to external photoevaporation can efficiently reduce the disc mass and size (with consequences for the planet formation and composition), even below $100 G_0$, as demonstrated by numerical models and by a comparison with observations (Anania et al. 2025a). Therefore, it is essential to be aware of the irradiation level in the most nearby regions for interpreting the observations. The median FUV fluxes on disc-hosting stars in star-forming regions in Orion range from $\sim 20 G_0$, which is registered in 25 Ori, to $\sim 5 \times 10^5 G_0$ in the ONC. The FUV flux in Serpens is dominated by the irradiation produced by the O-type star and the early B-type stars close to Serp South and the HII region W40.

8.5.2 Uncertainties, limitations of the model, and future improvement

The primary sources of uncertainty in the flux calculation are the relative (3D) distances between massive stars and disc-hosting stars and the FUV luminosity of massive stars. We neglected the latter in most cases, and we now justify this and show that this causes a small effect.

In the flux calculation, the uncertainty in FUV luminosity is caused by the uncertainty in the effective temperature of massive stars. Because the effective temperature of most of the OBA stars is defined in the Astrophysical Parameters (ESPHS) pipeline provided by *Gaia* DR3, we have access to the temperature uncertainty, and we used it to investigate the implications on the final FUV flux value. In the ESPHS pipeline, the error on the effective temperature increases with spectral type. It reaches unreliable temperature values and uncertainties above 25000 K (Fouesneau et al. 2023). From stellar evolution

Table 8.1: Median FUV flux for the regions

Region	$F_{\text{FUV,median}}$ [G_0]	$(\sigma_{\text{FUV}}/F_{\text{FUV}})_{\text{median}}$
UppSco	$20.11^{+21.03}_{-10.7}$	0.35
Lupus	$3.48^{+21.03}_{-0.31}$	0.06
ρ Oph	$18.33^{+7.63}_{-7.12}$	0.30
Taurus	$2.92^{+0.16}_{-0.14}$	0.05
Cham I	$4.57^{+4.3}_{-1}$	0.19
Cham II	$2.76^{+0.03}_{-0.02}$	0.08
CrA	$9.94^{+14.83}_{-5.69}$	0.39
ONC	$33713.93^{+97361.13}_{-25803.39}$	0.96
σ Ori	$232.54^{+946.43}_{-103.7}$	0.48
λ Ori	$536.86^{+2602.54}_{-436.50}$	0.57
NGC 2024	126462.61	-
NGC 1977	$100.11^{+195.16}_{-32.07}$	0.73
25 Ori	$18.14^{+8}_{-3.86}$	0.26
Serp Main	$3.93^{+1.02}_{-1.42}$	0.22
Serp NE	$2.79^{+1.46}_{-0.49}$	0.13
Serp South	$44.27^{+190.37}_{-26.61}$	0.37

Notes. For each region listed in Column 1, we indicate the median FUV flux and uncertainties (16th and 84th percentiles of the distributions of the median individual FUV fluxes) and the median relative uncertainty. The FUV flux was evaluated using Method 3 (Sect. 8.3.3), except for NGC 2024, for which we indicate the upper-limit flux evaluated using Method 2 (Sect. 8.3.2), and Cham II, which does not contain any relevant OBA star, and therefore, the FUV flux estimate refers to Method 1 (Sect. 8.3.1).

models, we retrieved that effective temperatures between 8000 and 25000 K correspond to stellar masses between ~ 2 and $\sim 10 M_{\odot}$ (assuming an age of 1 Myr) and to FUV luminosities of $\sim 10^{34} - 10^{37} \text{ erg s}^{-1}$. The typical errors on the effective temperatures of massive stars below 23000 K are a few dozen Kelvin, which translates into errors of a few percent in the FUV luminosity and FUV flux (at a fixed separation). These are negligible errors compared to the uncertainty associated with relative separations. Massive stars in the temperature range [23000 - 25000] K reach temperature uncertainties of a few thousand Kelvin, which result in a FUV flux variation of $\sim 80\%$. In the last case, where errors cannot be neglected, a more precise temperature estimate is provided by spectral classification tables (Gray & Corbally 2009b). Therefore, in order to reduce the impact of the uncertainty in FUV luminosity on the total flux, we retrieved the effective temperature of massive stars above 23000 K in our calculation (which might present the highly uncertain *Gaia* ESPHS effective temperature) from a spectral classification, as introduced in Sect. 8.2.

We proposed the use of the local density profile of a star-forming region for the uncer-

tainties in parallax measurements and for determining the 3D separation between pairs of stars in a star-forming region. However, an intrinsic limitation of Method 3 (Sect. 8.3.3) is the shape of the local density function itself, which is individually derived (and analytically prescribed) for each region. Specifically, the density profile was used in the probability function in Eq. 9.2, from which we sampled the cumulative distribution function (CDF) to determine the best estimate of the 3D separation between stars. Therefore, an extremely flat density profile results in a nearly uniform probability of 3D separations across a wide range of values (an approximately flat CDF). As a consequence, the uncertainty on the final flux will be large. Conversely, when the initial density profile is extremely steep, the corresponding inverse CDF will be steep as well, presenting its maximum close to the 2D separation value. This leads to a high probability of obtaining fluxes similar to those predicted using the pure 2D separation. Table 8.1 lists median FUV flux values, with 16th and 84th percentiles of the distribution of the median FUV fluxes and the median relative uncertainty for each region. The relative uncertainties in the various regions generally do not depend on G_0 . However, in the case of centrally symmetric regions, high FUV flux values can present large error bars because a higher G_0 corresponds to a smaller separation between disc and massive stars, which corresponds to the approximately flat part of the spherically symmetric local density function (see the Plummer sphere in Sect. 8.3.4).

While errors produced by the shape of the local density distribution cannot be removed, future improvements can be made for a better identification of the density profile of star-forming regions. The derivation of accurate local density profiles requires a membership census of star-forming regions that are numerous enough to present a substantial number of stars per unit separation, which permits us to resolve stellar overdensities that influence the shape of the local density function. Therefore, future stellar membership studies in the regions included in this work will improve the precision of the FUV flux calculation method. Moreover, new stellar membership catalogues in additional regions will allow us to apply the method described in this study to a larger stellar sample.

8.5.3 The role of the extinction

Interstellar gas and, more importantly, dust can efficiently shield protoplanetary discs from an incoming UV radiation. Quantify the relevance of this process is challenging, as it requires knowledge of the internal geometry of a star-forming region and the column density between discs and massive stars. Moreover, gas outflows produced by massive stars can reduce the attenuating material, contributing to the variation in the irradiation experienced by protoplanetary discs with time (e.g. Bania & Lyon 1980, Jeffreson et al. 2021). Simulations of molecular clouds show that discs in high-mass clouds containing several O-type stars, can be efficiently shielded by FUV radiation during the first ~ 0.5 Myr of their evolution (Ali & Harries 2019, Qiao et al. 2022). After this shielding time, discs are rapidly subject to higher FUV fluxes. In order to investigate the role of the interstellar dust in attenuating the FUV irradiation, we used the 3D dust extinction map provided by Lallement et al. (2022), which uses *Gaia* and 2MASS photometric data to covers a volume of ~ 6 kpc $\times$ 6 kpc $\times$ 800 pc around the Sun, reaching resolution of few pc. As the resolution limit of this map is too large to resolve cloud substructures and accurately quantify the column density of material between disc-hosting stars and massive stars in a same star-forming region, where the separation between them is smaller than ~ 5 pc, we retrieve from the 3D dust map an average extinction magnitude value for each region. This value is included in the calculation of the FUV flux produced by

the massive stars internal to a certain region to the disc-hosting stars of the same region. We converted visual magnitudes, provided by the dust map, in FUV magnitudes using a factor $A_{\text{FUV}}/A_V \simeq 2.7$ (Cardelli et al. 1989, Salim & Narayanan 2020). For the field massive stars (those outside the considered region) we assumed an average galactic extinction value of $\sim 3A_V \simeq 2.1 \text{ mag kpc}^{-1}$, as in the work of Cleeves et al. (2016). The individual FUV flux corrected for the UV extinction magnitude is:

$$F_{\text{FUV, corr}} = F_{\text{FUV, comp}} 10^{-\frac{A_{\text{FUV}}}{2.5}}, \quad (8.9)$$

where $F_{\text{FUV, comp}}$ is our flux best estimate and A_{FUV} is the average extinction magnitude in the FUV range of wavelengths. In Table 8.2 we show the average extinction magnitude for each region, derived from the 3D dust extinction map and converted to FUV wavelengths, the median FUV flux computed using the local density distribution (Method 3, Sect. 8.3.3), which provides our best estimate of the flux, and the flux value corrected by the UV extinction. We distinguish between the percentage of flux reduction caused by massive hot stars internal the star-forming region (members) and external (field stars). In general, the median FUV flux per region is not drastically reduced by extinction. Since massive stars internal to a region are located closer to discs than field massive stars, their irradiation is less attenuated. In regions like the ONC, where the FUV flux is dominated by few massive stars very close to discs, the extinction correction to the total FUV flux is minimal. In these cases, even though the FUV flux reduction from field stars can be significant ($\sim 25\%$ for the ONC, $\sim 40\%$ for λ Ori), the total FUV flux decrease is negligible ($\lesssim 1\%$).

We provided a tentative estimate of the impact of the dust extinction in attenuating FUV photons incident on disc-hosting stars assuming, for the disc-hosting stars of a certain region, that the FUV radiation emitted by massive stars members of that region is attenuated by the average UV extinction magnitude provided by dust extinction maps at that location, and applying an average galactic extinction to the FUV radiation emitted by massive stars outside the region. The results presented in Table 8.2 show that the median FUV flux per region is not significantly reduced by extinction. However, we highlight that the extinction magnitudes used are average values based on a dust map with low resolution for our scopes, meaning that intra-star extinction due to dust material within a single region might be higher than what assumed. Future improvement will be possible with the availability of 3D high-resolution dust extinction maps of star-forming regions.

8.6 Conclusions

We proposed a novel approach to evaluating the FUV flux, along with its uncertainty, at the position of disc-hosting stars in nearby star-forming regions. We analysed three methods for the uncertainty in the separation between massive stars and low-mass stars in 3D space, which is the main source of the uncertainty in flux calculations. These methods consist of a distance uncertainty sampling (Method 1, Sect. 8.3.1), a 2D projected separation and distance uncertainty sampling (Method 2, 8.3.2), and a local density distribution and distance uncertainty sampling (Method 3, Sect. 8.3.3). Method 3, our novel approach, involves the use of the local density distribution of a star-forming region under the assumption of isotropy to derive the separation between stars in 3D space, given their 2D separation on the sky plane. The 3D separation was used to compute the FUV flux and its uncertainty. The procedure and main results obtained in this study are summarised below.

Table 8.2: Average interstellar extinction used for each region, and percentage of decrease in FUV flux due to attenuation of radiation from stars members of the region and field stars.

Region	A_{FUV} [mag kpc ⁻¹]	median $F_{\text{FUV, comp}}$ [G ₀]	median $F_{\text{FUV, corr}}$ [G ₀]	Reduction internal FUV %	Reduction field FUV %
UppSco	5.31	20.11	17.69	3.0	24.6
Lupus	6.15	3.48	2.15	13.3	43.4
ρ Oph	11.59	18.33	16.81	1.86	19.7
Taurus	0.56	2.92	1.54	0.5	48.5
ChamI	1.21	4.57	3.19	0.19	45.3
ChamII	1.12	2.76	1.37	0.0	50.5
CrA	1.47	9.94	8.36	0.12	43.3
ONC	1.23	33713.9	33380.1	0.022	25.02
σ Ori	1.28	232.5	227.9	0.22	13.6
λ Ori	1.03	536.9	531.5	0.12	41.6
NGC 1977	1.24	100.1	98.2	0.12	10.1
25 Ori	0.83	18.14	14.75	0.42	44.7
Serp Main	0.56	3.93	2.66	0.13	70.6
Serp NE	0.89	2.79	1.51	0.37	75.2
Serp South	1.12	44.27	42.98	0.40	72.2

Notes. For each region in the first column, the average interstellar dust extinction magnitude converted to the FUV band is presented in the second column. Column 3 and 4 contain the median FUV flux as computed in this work and corrected by the extinction. Column 5 and 6 distinguish between the decrease in FUV flux due to dust extinction between the internal and external OBA stars of single regions. An average value of ~ 2.1 mag kpc⁻¹ is applied to the incident FUV flux from field stars.

- We selected a large sample of OBA-type stars and cross-referenced various stellar catalogues to include the contribution of late-type B and early-type A stars to the total FUV radiation, as described in Sect. 8.2.
- Using synthetic clusters, we verified that Method 1 can significantly underestimate FUV flux, while Method 2 leads to upper-limit FUV fluxes (Sect. 8.3.4). Method 3, which consists of analytically prescribing the local density profile of star-forming regions and sampling the derived PDF to obtain 3D separations given 2D separations (Sect. 8.3.1), yields a significantly larger fraction of FUV fluxes that are consistent within 1σ error bars with true values compared to the other methods.
- We provided a catalogue of the FUV flux (and uncertainty) at disc-hosting stars in Taurus, ρ Oph, UppSco, CrA, Lupus, ChamI, ChamII, ONC, σ Ori, λ Ori, NGC 1977, 25 Ori, NGC2024, and the sub-clusters of Serpens. This catalogue is available at the CDS.
- In regions without O-type and early-type B stars, the irradiation induced by less massive stars (late-type B and early-type A) can even reach $\sim 10^2 G_0$ (Sect. 8.4). These FUV radiation levels are not negligible for the disc evolution and planet formation, and therefore, the impact of the environment must be taken into account when discs in different regions are compared.
- We included a study of the FUV attenuation caused by interstellar dust extinction (Appendix 8.5.3). We assumed that the FUV radiation emitted by the massive star members of a certain region is attenuated by an average FUV extinction magnitude retrieved from dust extinction maps, while using an average galactic extinction value for the radiation from field massive stars. The resulting decrease in the median FUV flux per region is not significant (a few G_0 , as shown in Table 8.2). However, interstellar dust inside individual regions might be not resolved by dust maps and might therefore produce a higher extinction than is assumed.

This study shows that information provided by the 2D geometry of a star-forming region can be used to constrain 3D separations between star pairs within that region. We applied these estimates to compute the FUV flux at the position of nearby disc-hosting stars. This approach can be extended to further regions and stellar samples in future studies.

8.7 Appendix

8.7.1 Gallery of investigated star-forming regions and adopted density profiles

The positions on the disc-hosting stars considered in each star-forming region included in this work, and the location of their closest massive stars, are shown in Fig. 8.9, 8.10, and 8.10, divided in regions in $\lesssim 200$ pc, Orion regions, and Serpens regions. The OBA stars are colour-coded based on their FUV luminosity, where more massive stars are associated with higher FUV luminosity.

In this work, we employed the Abel's theorem to derive the volume local density profile of star-forming regions, given an analytical prescription for the surface local density profile (method 3 in Sect. 8.3.3). As demonstrated in Sect. 8.3.4, the local density profile can be analytically described by a double power law with exponential cut off for substructured regions, or by a truncated single power law (or a Plummer profile) for centrally concentrated regions. The parameters for the power law profiles, as applied to

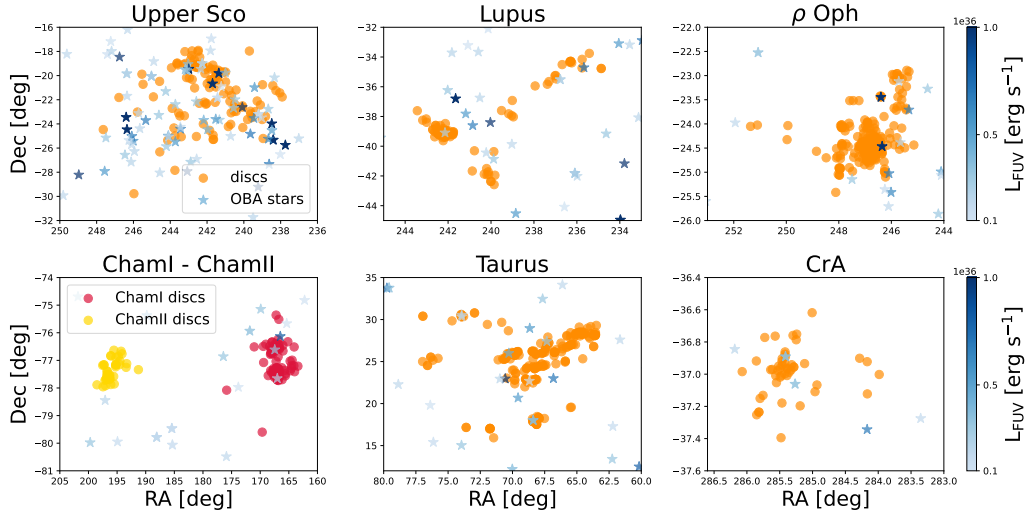


Figure 8.9: 2D sky maps showing the positions (RA, Dec) of discs and neighbouring massive stars in the nearby star-forming regions included in this work. ChamI and ChamII are shown together due to their small relative distance.

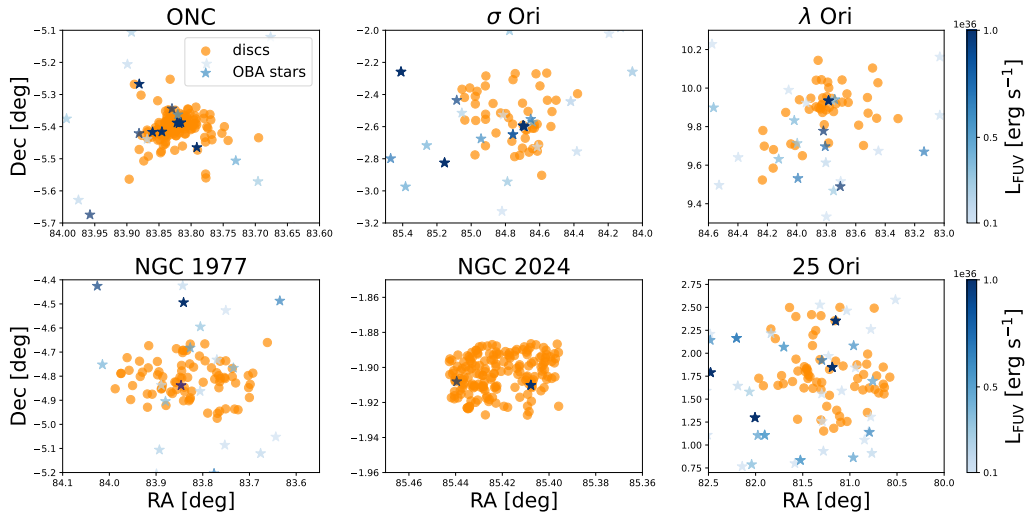


Figure 8.10: 2D sky maps showing the RA and Dec position of the disc sample and neighbouring massive stars in the Orion regions.

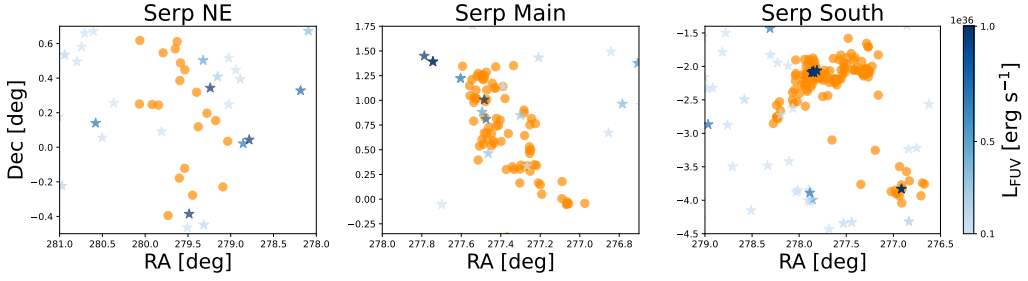


Figure 8.11: 2D sky maps presenting the location (RA, Dec) of discs and neighbouring massive stars in the Serpens sub-regions.

the star-forming regions in this study, are listed in Table 8.3. The terms reported in the Table refers to the expressions:

$$\hat{\rho}(r) = \begin{cases} \alpha_1 r^{-\beta_1}, & \text{if } r \leq r_{\text{lim}}, \\ \alpha_2 r^{-\beta_2} \exp[-(r/r_b)^{-\gamma}], & \text{if } r > r_{\text{lim}}, \end{cases} \quad (8.10)$$

where the 3D separation r is normalized to 1 pc, r_{lim} is the limit distance around which the steepness of the power law profile changes, and r_b is the boundary radius setting the length scale of the star-forming region. This is valid for a region with internal substructures. For the ONC, σ Ori, λ Ori, and 25 Ori regions, we employed a centrally concentrated geometry, with density profile that is function of the distance from the centre of the region, written as:

$$\hat{\rho}(r) = \alpha_2 r^{-\beta_2} \exp[-(r/r_b)^{-\gamma}], \quad (8.11)$$

where r is normalized to 1 pc and in Table 8.3 the terms α_1 , β_1 , and r_{lim} assume null values as they do not enter in this expression.

The accuracy in estimating the volume density profiles requires having a substantial number of stars per unit separation (i.e. a nearly complete stellar census of the star-forming regions), to properly account for the 2D internal geometry. In this study, we retrieved membership catalogues for each region from those available in literature. Assigning the stellar membership requires accurate measurements of positions and proper motions, which becomes increasingly difficult in more distant regions, such as Orion and Serpens in our case. Therefore, the defined density profiles may need refinement as future, updated stellar censuses become available. The confirmed stellar membership of NGC 2024 (Kounkel et al. 2018b, Žerjal et al. 2024) consists of stars too spatially separated with respect to the compact sample of discs considered in this work, to reliably make claims about the density profile per unit separation. Thus, our new proposed approach to evaluate the FUV flux cannot be applied in this region, which is not included in Table 8.3. However, we provided FUV flux estimates using method 2 (Sect. 8.3.2), being aware that this is an overestimate of the true value. The density profiles of NGC 1977 and Serpens sub-regions are described by single power law relations (where $\hat{\rho}$ is function of the separation between pairs of stars) for distances larger than a defined limit separation, r_{lim} , due to the lack of a complete stellar membership at smaller separations, where the shape of $\hat{\rho}$ is primarily influenced by binary systems. However, we verified that no pairs of disc-hosting star - massive star are present at separations smaller than this limit value, and therefore, the defined profile remains valid for our calculations. The

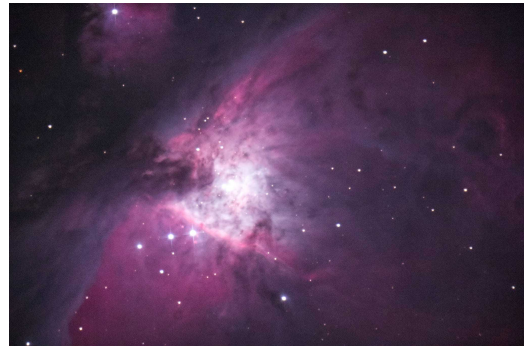
Table 8.3: Parameters of the volume density profiles used for the star-forming regions included in this work.

Region	α_1	β_1	r_{lim}	α_2	β_2	r_b	γ
UppSco	7.36×10^{-5}	1.83	1.5×10^{-1}	2.16×10^{-4}	6.04×10^{-1}	12	1.6
Lupus	1×10^{-6}	2.6	3×10^{-2}	2×10^{-5}	1.7	18	1.7
ρ Oph	3×10^{-5}	2.4	1×10^{-2}	1×10^{-2}	0.8	3	1.2
Taurus	2×10^{-4}	2.2	3×10^{-2}	1.5×10^{-3}	1.5	20	2.1
ChamI	1.5×10^{-4}	2.3	2×10^{-2}	6×10^{-3}	1.3	3.8	1.5
CrA	1×10^{-4}	2.2	5×10^{-2}	2×10^{-3}	1.22	10	1.8
ONC	-	-	-	3×10^{-2}	1.8	3	2.5
σ Ori	-	-	-	1×10^{-2}	2.2	8.0	1.8
λ Ori	-	-	-	1×10^{-2}	2.3	10	2
NGC 1977	-	-	5×10^{-3}	8×10^{-3}	1.0	4.0	1.8
25 Ori	-	-	-	3×10^{-2}	0.9	6	1.5
Serp Main	-	-	5×10^{-2}	3×10^{-3}	1.1	8	1.9
Serp NE	-	-	6×10^{-2}	2×10^{-3}	1.05	8	2
Serp South	-	-	1×10^{-2}	1×10^{-3}	1.2	10	1.3

Notes. Parameters adopted for the volume density profiles of the regions investigated in this work. ChamII is not included as no relevant massive OBA type star is found near the studied stars-hosting discs. While, NGC2024 presents a sample of confirmed members which is too small and spread to reliably define a density profile. The listed parameters refer to Eq. (8.10) for UppSco, Lupus, ρ Oph, Taurus, ChamI, and CrA. For the ONC, σ Ori, and λ Ori we used Eq. (8.11), while NGC1977 and Serp regions are described by Eq. (8.10) at distances larger than r_{lim} .

Cham II region does not contain any significant OBA stars, and therefore, we did not use the local density profile to evaluate the FUV flux in this region.

Mapping Orion in FUV flux



Stars forged in Orion's Sword - Image of M42, dense and bright star-forming region in the Orion complex, illuminated by the intense radiation emitted by the OB stars of the Trapezium.

The content of this Chapter, and part of the next Chapter, is taken from Anania et al. (2025, in prep.), a manuscript of which I am the first author and that will be soon submitted on the A&A Journal.

Typical star-forming environments in our Galaxy are illuminated by UV radiation emitted by OB-type stars, that significantly contribute in shaping the physical and chemical properties of protoplanetary discs and their planetary systems. In the solar neighbourhood ($\lesssim 2$ kpc from the Sun), stars are typically subject to FUV fluxes in the range $[10^2 - 10^4]$ G_0 . Since the star-forming regions located within $\lesssim 200$ pc from the Sun such as Lupus, Taurus, Upper Sco, are weakly FUV-irradiated (< 100 G_0), pursuing the goal of investigating typically FUV-irradiated discs requires observing more distant regions, which introduces complications due to sensitivity and resolution limits, and greater uncertainty in astrometric parameters. In this Chapter, we focus on Orion, the closest young stellar association (~ 360 -460 pc, Padovani & Galli 2018, Wright 2020) hosting late O-type and early B-type stars, and active star formation. Orion has previously been divided into several macro-areas based on position on the sky, stellar proper motion, and average age (Kounkel et al. 2018a, e.g.): Orion A, B, C, and D. The Orion A molecular cloud includes the Orion Nebula Cluster (ONC) and extends to the south-west part of the region. Orion B occupies the north-west part, Orion C corresponds to the central Orion belt, while Orion D spans the north-east region (see Chap. 3).

Accurately estimating the FUV radiation field is essential to select a representative stellar sample to address the role of the environment on disc properties. The FUV flux at the position of stars might be challenging to evaluate as it strongly depends on the three-dimensional (3D) geometry of a stellar cluster, which is highly influenced by parallax measurements. Anania et al. (2025b) proposed using the 2D geometry of a stellar cluster and the argument of isotropy to derive the best estimate 3D separation from OBA-type stars, and then compute the best estimate FUV flux with associated uncertainty. This procedure allowed for accuracy in the FUV flux calculation (see Chap. 8). We used the OPTICS clustering algorithm to divide a region of 6 degree radius around the ONC core in sub-clusters, and we applied the method of Anania et al. (2025b) to compute the FUV flux at the position of the stars members of the sub-clusters, with the aim of probing the range of FUV flux covered by Orion. We provide a publicly available table of FUV fluxes for a vast stellar population in Orion that will be fundamental to identify the most suitable targets for future studies investigating disc evolution in connection with the environment.

9.1 Method: Sub-clustering and FUV flux calculation

We adopted the method detailed in Anania et al. (2025b) (see Chap. 8) to infer the FUV flux at the position of a large sample of stars in the Orion region. This approach accounts for parallax uncertainties in the FUV flux calculation by using the 2D geometry of a stellar cluster to make arguments about the 3D separation between target stars and massive stars. We divided the Orion region into sub-clusters, as described in Sec. 9.1.1, and we evaluated the FUV flux as summarised in Sec. 9.1.2.

9.1.1 Sub-clustering of Orion

We selected from the Gaia DR3 catalogue all the stars contained in a 6D box defined on the basis of limits on right ascension (RA), declination (Dec), proper motion (μ_{RA} , μ_{Dec}), parallax ($\bar{\omega}$), and parallax-over-error ($\bar{\omega}/\sigma_{\bar{\omega}}$). Specifically, we selected stars in a circular region of radius six degrees centred at the $\theta^1\text{C}$ star, which is located in the core of the ONC, the region presenting the highest stellar density in Orion. Then, we limited the proper motion, parallax, and parallax-over-error, considering a large region of the parameter space around the typical ONC values (e.g., Da Rio et al. 2012; Kounkel et al. 2018a), which is extended to avoid missing potential stellar members. Specifically, we used $-4 < \mu_{\text{RA}}/(\text{mas s}^{-1}) < 4$, $-4 < \mu_{\text{Dec}}/(\text{mas s}^{-1}) < 4$, $2 < \bar{\omega}/\text{mas} < 3.5$, and $\bar{\omega}/\sigma_{\bar{\omega}} > 5$, to constrain the selection of the Orion members. Subsequently, we used the OPTICS (Ordering Points To Identify the Clustering Structure, Ankerst et al. 1999) clustering algorithm to identify sub-groups of stars, using as clustering criteria the position on the sky (RA, Dec), the proper motion, the parallax for each star, and a minimum number of stars per sub-cluster of 30 (Vioque et al. 2023). In Sec. 8.5, we show that requiring 30-40 stars per cluster is sufficient to reliably define the profile of the local density function (i.e., the distribution of pairwise separations or, in case of centrally concentric clusters, the separation of the stars from the centre of the cluster, see Sec. 9.1.2). For each star i , the algorithm computes a *core distance*, which is the distance to its nearest neighbourhood, and a *reachability distance* from every other star j , which is defined as

$$\text{reachability}(i, j) = \max(\text{core_distance}(i), \text{distance}(i, j)), \quad (9.1)$$

where $\text{core_distance}(i)$ and $\text{distance}(i, j)$ must be intended in 6 dimensions because they account for the clustering criteria we imposed (i.e., RA, Dec, μ_{RA} , μ_{Dec} , $\bar{\omega}$, $\sigma_{\bar{\omega}}$). The reach-

ability reflects the minimum density level at which i and j belong to the same cluster. By ordering the stars according to their reachability, the algorithm produces a hierarchic plot in which the transitions between clusters that contain at least the minimum number of stars that was initially set (i.e., 30 in our case) are indicated by peaks in the reachability. In total, we isolated 22 main sub-clusters in Orion, as discussed in Sec. 9.2.

The work by Kounkel et al. (2018a) has previously investigated the sub-division of Orion into clusters using APOGEE-2 and Gaia DR2, assigning sub-cluster memberships to ~ 5200 stars. In contrast, our current sample includes ~ 8600 stars across 22 sub-clusters. Of the sub-cluster members identified by Kounkel et al. (2018a), $\sim 70\%$ overlap with our sample, and of those, $\sim 96\%$ have sub-cluster memberships consistent with our assignments. Since most of the sub-clusters we found are consistent with those identified by Kounkel et al. (2018a), in this work we adopt a similar nomenclature for consistency. An important difference between our work and Kounkel et al. (2018a) is that they divided the main Orion sub-clusters into even smaller sub-groups containing a number of members ranging from 10 to ~ 300 , leading to a total of 190 sub-groups. Instead, we required a minimum of 30 stars per cluster in order to assign reliable density distribution functions and compute FUV fluxes (see Sec. 9.1.2 and 9.3.2), which resulted in 22 main sub-clusters spanning the Orion area. In general, the sub-clusters identified in this work contain a number of members between ~ 60 and ~ 3000 and have a diameter that spans a few degrees in the sky.

An important consideration on the division in sub-clusters presented in this work is that the stellar membership of the low-mass stars (i.e., types later than A) is not complete, as Gaia DR3 may miss some of the faintest stars (e.g., those embedded in dense cloud cores such as NGC 2024), and the low-mass stars (i.e., $G \gtrsim 21$ mag, which is the limit magnitude of Gaia and corresponds to $M_\star \lesssim 0.1$ at 400 pc). However, cross-matching our final sample with IR-based catalogues of YSOs (Megeath et al. 2012; Großschedl et al. 2019; Roquette et al. 2025), we verified that $\sim 72\%$ of the sample in the IR-based catalogues associated with the Orion region has Gaia DR3 counterpart, and therefore we can conclude that Gaia DR3 has good level of completeness in Orion. Moreover, we highlight that the main goal of this work is not to provide complete and definitive memberships to the entire Orion population. Instead, the level of completeness that we need for our calculation is related to the ability to define a reliable shape of the 2D density distribution function of the region, to compute the FUV flux. As will be shown in Sec. 9.3.2, setting the minimum number of stars per cluster to 30-40 allows us to recover the density distribution function profile, and thus evaluate the FUV flux. A future addition of stars to the sub-clusters will not significantly influence the shape of the density function, and therefore will negligibly impact our FUV flux calculation.

Our aim is computing the FUV flux of a large population of Orion members, and therefore it is essential to locate the OBA-type stars responsible for the UV radiation and define which of them are members of the identified Orion sub-clusters or field stars. We gained information on the massive stars in the region by combining multiple stellar catalogues in order to compensate for the limitations of each catalogue. In particular, following the approach in Anania et al. (2025b), we combined the ALSIII catalogue of massive stars (Pantaleoni González et al. 2025), from which we selected all the most massive stars in 2 kpc from the Sun, the Gaia DR3 Extended Stellar Parametrizer for Hot Stars (ESP-HS, Creevey et al. 2023; Fouesneau et al. 2023) catalogue, and Zari et al. (2021) catalogue of hot stars. Moreover, we included the OBA-type stars that are not present in Gaia, because of their high luminosity, but are instead listed in Hipparcos Perryman et al. (1997). Moreover, we cross-matched the selected sample of OBA stars with the

OBA stars in Orion listed in Quintana et al. (2025) to add the stars that may be missing from our previous selection. In order to derive the FUV luminosity of the OBA stars, we first derived their effective temperature from either spectral type classification Maíz Apellániz et al. (2017); Gray & Corbally (2009b) or the values of effective temperature contained in the ESP-HS Gaia DR3 pipeline. We used these effective temperatures, and the stellar parameters evaluated from MIST isochrones (Dotter 2016; Choi et al. 2016), in the stellar atmosphere models ATLAS9 (Castelli & Kurucz 2004) to compute the stellar luminosities and integrate them in the FUV range of wavelengths.

In conclusion, we selected 22 main sub-clusters located in correspondence to the Orion A and B molecular clouds and spanning Orion C (central belt), Orion D (north-east), and clusters in the extended population of Orion (see Sec. 9.2). We found ~ 115 OBA stars classified as members of the sub-clusters. In our calculation, we also included the contribution of the field OBA-type stars to the total FUV flux, following the approach detailed in the next Section.

9.1.2 Computing the FUV flux using the local density function

As described in Anania et al. (2025b), the FUV flux in a given location p is computed by summing up the individual FUV fluxes induced by the surrounding OBA stars: $F_{\text{FUV},p} = \sum_m (L_{\text{FUV},m}/4\pi|x_p - x_m|^2)$, where the separation from massive stars in 3D space, $|x_p - x_m|$, is the main source of uncertainty due to parallax measurements. Despite the fact that Orion is a relatively close star-forming association (~ 400 pc from the Sun), astrometric measurements of low-mass stars, which may appear faint, or very bright stars, which may saturate, might be challenging, resulting in a very uncertain (or not defined) parallax. Moreover, the FUV flux tends to be underestimated if the uncertainty in the radial distance of a certain star is larger than the relative separation between that star and the nearest OBA stars. We addressed this problem using the information on the 2D geometry of the stellar clusters in Orion to infer, for each star, the 3D separation from the OBA stars members of the clusters. Specifically, for the stellar clusters presenting a substructured 2D geometry, we define the probability of finding a pair of stars ij separated by a 3D distance r_{ij} , given their 2D projected separation in the sky plane R_{ij} :

$$dP(r_{ij}|R_{ij}) = dP(R_{ij}|r_{ij}) dP(r_{ij}) = \begin{cases} \frac{4\pi R_{ij} r_{ij}}{\sqrt{r_{ij}^2 - R_{ij}^2}} \hat{\rho}_{\text{pairs}}(r_{ij}) & R_{ij} < r_{ij} \\ 0 & \text{otherwise} \end{cases}, \quad (9.2)$$

where $\hat{\rho}_{\text{pairs}}(r)$ is the normalized 3D local density function of the cluster (i.e., the normalised 3D pairwise distribution). Assuming isotropy in the neighbourhoods of each star, $\hat{\rho}_{\text{pairs}}(r)$ is derived from the normalized 2D local density function $\hat{\Sigma}_{\text{pairs}}(R)$ using the Abel's inversion (Abel 1826):

$$\hat{\rho}_{\text{pairs}}(r) = -\frac{1}{\pi} \int_r^\infty \frac{d\hat{\Sigma}_{\text{pairs}}(R)}{dR} \frac{1}{\sqrt{r^2 - R^2}} dR, \quad (9.3)$$

which is valid if $\hat{\rho}_{\text{pairs}}(r) \rightarrow 0$ as $r \rightarrow \infty$, and where $\hat{\Sigma}_{\text{pairs}}(R)$. Instead, for the Orion sub-clusters showing a centrally concentric 2D geometry, we employed the assumption of spherical symmetry and the same expressions in Eq. (9.2) and Eq. (9.3) where R and r are the 2D and 3D distances of the stars from the centre of the cluster. Finally, we used the best estimate 3D separation to compute the FUV flux at the position of each star and its associated uncertainty. Using this approach, we build an FUV flux map of the Orion

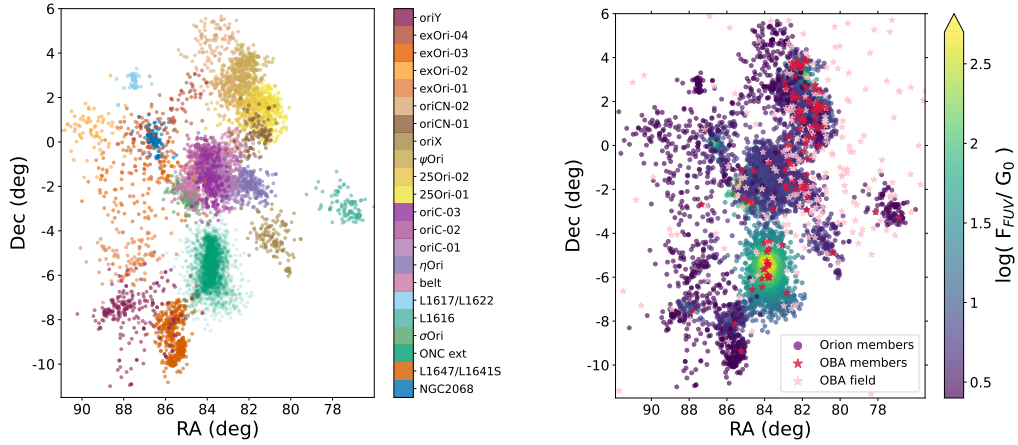


Figure 9.1: *Left panel:* The main 22 Orion sub-clusters obtained using the OPTICS clustering algorithm (Sec. 9.1.1). The sub-clusters are ordered in the colorbar from bottom to top by increasing average age, from < 1 Myr to > 6 Myr. *Right panel:* Median FUV flux at the position of the Orion members identified in this work. The full table of FUV flux values will be available at the CDS. Red and pink points mark the OBA members and field stars, respectively.

region (Sec. 9.2). In Appendix 9.5.1 we discuss the use of dust emission maps as an alternative method to compute G_0 and the advantages of using the method presented in this Section.

9.2 Results: Orion FUV flux map

In this Section, we show the FUV flux values resulting from applying the local density function method described in Sec. 9.1.2 at the position of the Orion members in each sub-cluster. The complete table of FUV fluxes and uncertainty will be publicly available at the CDS, while a code to compute the FUV flux at the position of additional stars in the Orion sub-clusters identified is available on Github¹.

Using the OPTICS algorithm, we identified 22 sub-clusters in Orion, as shown in the left panel of Fig. 9.1. The geometry of the clusters defines the shape of the local density function that we used to ultimately compute the FUV flux. The right panel in Fig. 9.1 shows the median FUV flux at each stellar object, resulting from applying the method described in Sec. 9.1.2. The locations of the OBA stars, members of the clusters and field stars, are marked in red and pink, respectively. The distribution of the median FUV fluxes for each sub-cluster is presented in Fig. 9.2, where we isolated the ONC inner core (i.e., the Trapezium, which hosts the most massive stars in the region) and NGC 1977 from the extended ONC region. This is done in order to extract these two small clusters from the full ONC identified by OPTICS (see below). We highlight here the main properties and the age range (for the age estimate see e.g., Warren & Hesser 1978; Briceño et al. 2005; Kounkel et al. 2018a) of the sub-clusters:

- The ONC region includes its innermost core (the Trapezium), and also NGC 1977. Despite of the astrometric accuracy of the Gaia DR3 data, the uncertainties in parallaxes and proper motions for stars in the densest stellar region like the ONC

¹The code is available on Github https://github.com/Rossella4712/Orion_FUV.git

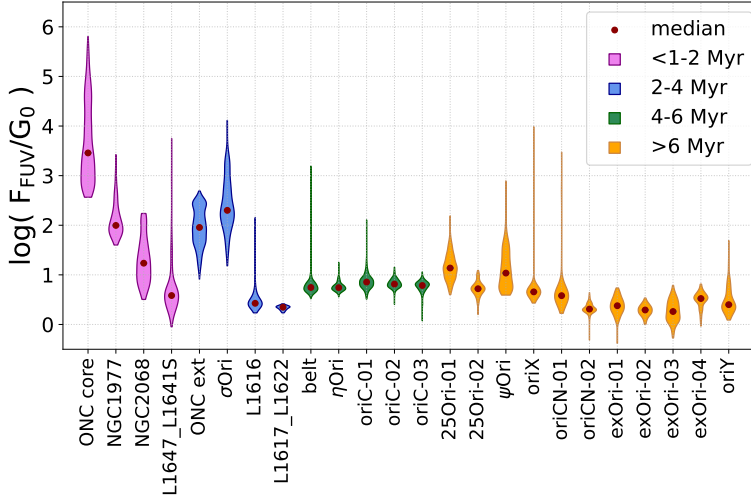


Figure 9.2: Distribution of the median FUV fluxes for the stellar sample identified in this work in the Orion sub-clusters. x-axis: Commonly known name of the sub-clusters identified in this work, divided in four age bins and ordered from the youngest (left) to the oldest (right). Notes: We isolated here the ONC core (~ 3 pc radius from θ^1 C) and we included the FUV fluxes for the NGC1977 sources investigated in Anania et al. (2025b).

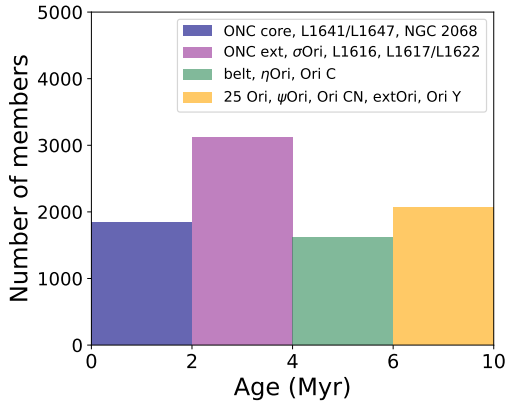


Figure 9.3: In this work, we divided the Orion stellar population in four age bins based on the average age of the sub-clusters. The histogram shows the number of selected members in each age bin. The sub-clusters included in each age bin are listed in the legend of the Figure, where we used different colours for the age bins.

are still too high to allow a solid separation of the ONC in kinematically distinct sub-populations. Therefore, we treated the ONC as a cluster containing multiple substructures and defined the corresponding local density function that describes its geometry (see Anania et al. 2025b for the detailed numerical approach related to this geometry). We only isolated the ONC core (~ 0.3 deg from θ^1 C) and NGC 1977, which are younger (< 2 Myr, Beccari et al. 2017) than the extended ONC stellar population ($\sim 2 - 4$ Myr, Beccari et al. 2017; Kounkel et al. 2018a), and therefore are treated with their own local density function. In Fig. 9.2, the FUV fluxes of the sources located in the innermost ONC (Ricci et al. 2008; Eisner et al. 2018a ONC core in Fig. 9.2), and in NGC 1977 (Kim et al., in prep.) are taken from Anania et al. (2025b), where a more detailed study on these regions was performed.

- The regions presenting young ages of $\lesssim 1-2$ Myr are: the ONC core and L1647/L1641-South, which are located in the Orion A cloud, NGC 2068 and NGC 2024, which are part of the Orion B cloud.
- The outer part of the ONC is known to be older than its core, with an age range $\sim 2 - 3$ Myr. A similar age is associated with L1616, which is located in the eastern part of Orion, and the cometary clouds L1617/L1622 which are suggested to be not physically associated with Orion B but more likely a foreground population (Reipurth et al. 2008).
- The σ Ori cluster is in projection closer to Orion OB1b. However, its age is more similar to that of the outer part of the ONC (i.e., Orion OB1c), which is $\sim 3 - 4$ Myr (Jeffries et al. 2006; Hernández et al. 2014). The kinematic association of this region to Orion OB1b is still debated (e.g., Caballero & Solano 2008; Jeffries et al. 2006). In our division of Orion, NGC 2024 and σ Ori fall within the same sub-cluster, as the former is still embedded and the stellar astrometry in this region remains highly uncertain, leaving only a few stars that can be reliably assigned as members (i.e., not enough to assign a density function of the region). Nevertheless, we emphasize that NGC 2024 is very young, with an age comparable to NGC 2068 and the ONC core.
- The sub-cluster named after *belt* in Fig. 9.1 and Fig. 9.2 spans the Orion belt in correspondence of the stars ζ Ori, ϵ Ori, and δ Ori. Together with the clusters named after η Ori and Ori C -[01 - 02 - 03], this region defines the central part of Orion corresponding to Orion OB1b. The average age of these clusters spans $\sim 4 - 6$ Myr (Briceño et al. 2005; Calvet et al. 2005; Bally 2008).
- The sub-clusters 25 Ori-[01 - 02], ψ Ori, and Ori CN-[01 - 02] are located in the north-eastern part of Orion, in the Orion OB1a region, where there is little or no gas left. This region hosts the oldest stellar population with average age of $\sim 7 - 10$ Myr (Briceño et al. 2005; Calvet et al. 2005).
- The remaining clusters form the extended population of Orion, which is of similar age to Orion OB1a, $\sim 7 - 10$ Myr (Briceño et al. 2005).

We emphasise that the upper limits of the FUV flux distributions presented in Fig. 9.2 may change in the future if following studies will identify candidate members closely located to the most massive stars, which may not be included in our list due to their uncertain astrometry because of the proximity to luminous stars. Since we are aware of the fact that future studies and different techniques may lead to improvements of the

stellar membership, and that we may need to compute the FUV flux at the position of additional stars, we released a `Python` script that, given the main astrometric parameter of a star (RA, Dec, μ_{RA} , μ_{Dec} , $\bar{\omega}$, $\sigma_{\bar{\omega}}$), provides the probability that the star is associated with any of the 22 identified clusters in Orion, and then returns the FUV flux and its uncertainty as the 16th and 84th percentiles of the distribution. The code is structured to accept as input a single star or multiple stars at the same time. Future improvements will allow updates to both cluster division and density functions. See Sec. 9.3.2 for a discussion on the cluster membership and the limitations of the method.

The FUV flux in young and more nearby regions (within ~ 200 pc) such as Lupus and Taurus does not exceed ~ 10 G₀ (it may reach up to ~ 100 G₀ in isolated cases in Upper Sco and ρ Oph, Anania et al. 2025b), while Orion is the closest region that also hosts a population of young ($\lesssim 6$ Myr), intermediately irradiated stars, with FUV flux in the $\sim 10^2 - 10^4$ G₀ range (see right panel of Fig. 9.1). Therefore, if we are interested in characterising typical protoplanetary disc evolution and planet formation in relation to the stellar formation environment, Orion hosts a population of key potential candidate targets in this FUV flux range. In contrast, our study demonstrates that Orion is not the ideal region for investigating the effect of extreme UV radiation on a population of protoplanetary discs. Indeed, the sample of stars illuminated by extreme FUV radiation fields (i.e., $\gtrsim 10^4$ G₀), including proplyds, is statistically insufficient to perform a population study ($\sim 5\%$ of the investigated stellar population). For this purpose, more distant regions such as Carina, M16, and M17 that host early O-type stars, are more suited.

Several Orion sub-clusters experience FUV fluxes in the range $1 - 10^2$ G₀ (see Fig. 9.2). Most of these sub-clusters are located in the north-east region of Orion (Orion OB1a). Although this region hosts numerous B-type stars (~ 55), it presents a lower stellar density due to its older age, resulting in greater average distances from the most massive stars compared to the ONC. As a result, even though the FUV radiation fields are comparable to those in more nearby regions (< 200 pc), the weakly irradiated areas of Orion are in general older (> 6 Myr) and located in an OBA-rich environment, where cluster dynamics may have significantly influenced the FUV radiation history and the evolution of protoplanetary discs. In Orion OB1a, 25 Orionis shows moderate FUV radiation fields of $\sim 10 - 100$ G₀ and evolved YSOs ($\sim 5 - 10$ Myr), making it comparable to Upper Scorpius. However, the stellar structure of the two regions is different: 25 Orionis is illuminated by a central B1V star (25 Orionis itself) and a sparse population of late-type B stars, whereas Upper Scorpius presents a sub-structured population around many late-type B stars (Anania et al. 2025a). A future comparison between these two stellar populations could offer insight into the role of the stellar environment on disc evolution.

9.3 Discussion

Accurately estimating the FUV flux at the position of stars is fundamental to investigate the connection between disc evolution and the external environment. We focused on Orion, the closest star-forming region that hosts numerous OBA-type stars and exhibits higher stellar densities compared to more nearby (< 200 pc) and surveyed regions (e.g. Lupus, Taurus). We evaluated the FUV flux and its uncertainty at the position of the stars identified as members of Orion sub-clusters in a region of a six-degree radius around the ONC core. We discuss the results and the application of the findings in Sec. 9.3.1. The limitations and future improvements of the method are discussed in Sec. 9.3.2.

9.3.1 Where to look next?

The FUV flux characterization of Orion that we presented in this work is primarily driven by the need to define the range of FUV fluxes covered by Orion, which is fundamental to identify suitable target regions and YSOs to characterise the impact of environmental FUV radiation on disc evolution. We showed that Orion is not the ideal target region for characterizing the effect of extreme environmental radiation (i.e. $> 10^4 G_0$) on disc evolution on a statistical basis, because most of the population is illuminated by weak and moderate FUV radiation fields $\lesssim 10^2 G_0$, as shown in Fig. 9.1 and 9.2. The extended and old population of Orion is subject to FUV fluxes in the range $1 - 10 G_0$, similar to those found in the more nearby star-forming regions (within ~ 200 pc from the Sun), while $\sim 35\%$ of the analysed sample is illuminated by FUV fluxes in the range $10^2 - 10^4 G_0$, which is the typical levels of FUV flux for star-forming regions in the solar neighbourhood (within 2 kpc from the Sun, see Chap. 3). Therefore, Orion holds potential for further observations particularly aimed at characterising protoplanetary disc evolution and planet formation at FUV radiation fields from low to intermediate levels. Candidate targets could be selected by combining our sub-cluster identification and H_α emission of Delfini et al. (2025), which is a tracer of ongoing accretion and, potentially, of a circumstellar disc, and with IR-based catalogues of YSOs. We comment on some suitable regions for future investigation as follows.

- In Orion A, the extended part of the ONC, outside the Trapezium area, is a suitable target region for studying intermediately FUV-irradiated objects. In this region, the FUV flux is lower than $10^4 G_0$ and proplyds are not detected.
- The South-West part of the ONC includes the L1641 cloud, which is located close to L1647 and is weakly FUV-irradiated ($\sim 5 - 100 G_0$). This part of Orion provides an ideal environment for studying the properties of young discs exposed to FUV fluxes more intense than in Taurus or Lupus, and more similar to those in Upper Sco. The Survey of Orion Disks with ALMA (SODA) observed 873 protoplanetary discs in mm-continuum emission in L1641 and L1647 (van Terwisga & Hacar 2023). However, $\sim 42\%$ of the total disc sample in this region did not present detectable flux emission at 1.3 mm. Therefore, further investigation on disc and stellar properties is needed to understand whether a deeper integration is needed, or whether discs are completely dispersed due to photoevaporation.
- In Orion B, NGC 2068 presents a young stellar population around the B-type star HD 38563 and FUV flux reaching $> 10^2 G_0$.
- Orion OB1a contains older clusters (> 6 Myr) such as 25 Orionis, and many B-type stars in a spread and low-density region. The fact that the B-type stars in this region are spread across a vast area is a potential consequence of cluster dynamics and expansion across time. Exploring YSOs in 25 Orionis and the surrounding clusters, such as Ori CN, would help characterise the advanced stages of moderately irradiated discs.

Our investigation suggests that the extended ONC is the best candidate region in Orion to investigate intermediately irradiated objects experiencing $\sim 10^2 - 10^4 G_0$. Moreover, we emphasise that, to investigate disc evolution across ages, it is crucial to compare young and old disc populations subjected to similar FUV radiation levels. The oldest and closest disc population that has been well-investigated by disc surveys is Upper Sco, which is moderately irradiated, with FUV fluxes in the range $\sim 5 - 100 G_0$ (Anania et al. 2025b), and hosts numerous B-type stars compared to younger regions such as

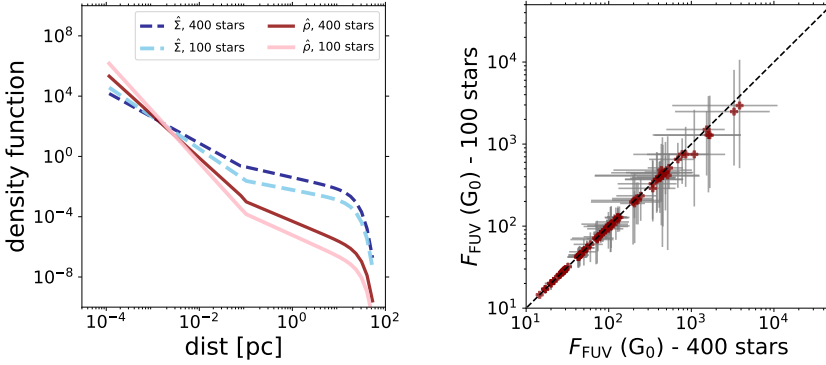


Figure 9.4: Left: Normalised surface (dashed lines) and volume (solid lines) local density function of a synthetic cluster hosting 400 stars (dark blue and red), and when removing 300 stars randomly (light blue and pink). Right: Relation between the FUV fluxes (and the uncertainties, shown as the 16th and the 84th percentiles) computed using the local density function of a cluster with 400 stars (x-axis) and with 100 stars (y-axis). The dashed lines indicate a perfect correlation (1:1).

Lupus and Taurus. In this context, the young and weakly irradiated regions of Orion would provide, to first order, a more appropriate representation of the early evolutionary stages of Upper Sco discs, enabling us to further characterize disc evolution.

9.3.2 Limitations on the cluster selection and future improvements

We computed the FUV flux at each star position using the best estimate 3D separation from massive stars, which is extracted from the posterior distribution in Eq. (9.2). Since this probability involves the use of the 2D geometry of stellar clusters to make arguments about their 3D structure, good knowledge of the membership of a certain cluster is needed. In this work, we used the OPTICS clustering algorithm on a stellar sample extracted from Gaia DR3. The main limitations and future improvements of the method are discussed as follows.

(i) *Membership of the sub-clusters.* The membership of the Orion sub-clusters identified in this work should not be regarded as complete, since the low-mass population and the stars with very uncertain astrometric parameters in the Orion region may be missing from our selection. However, we discussed in Sec. 9.1.1 that Gaia DR3 presents a good level of completeness in Orion and that for our purpose, membership completeness is required only to the extent that enables us to reliably determine the shape of the density function of the sub-clusters (i.e., the normalised distribution of the separation between pairs of stars, or from the cluster centre in case of a centrally symmetric cluster). Specifically, to compute the FUV flux, the density distribution function must be defined at the scales corresponding to the average separation between cluster members and the nearest massive stars. To achieve this, we require at least 5-6 stars per bin separation, considering an average bin widths of fractions of a parsec, which translates into a minimum of 30-40 stars per cluster (i.e., at least $\sim 7 - 8$ bins per cluster).

We used a synthetic stellar cluster to investigate the effect of membership incompleteness on the FUV flux calculation. We considered a substructured synthetic cluster that contains initially 400 stars (built as detailed in Chap. 8, Sec. 8.3.4). Then, we removed randomly 300 stars, without impacting the OBA stars that provide the irradiation, and we computed the local density function and the FUV flux at the position of the remaining

stars. In the left panel of Fig. 9.4, we show that the surface and volume normalised density distribution function profiles ($\hat{\Sigma}$ and $\hat{\rho}$, respectively) are not significantly impacted by the removal of members. Consequently, we obtain a similar FUV flux, as shown in the right panel of Fig. 9.4. However, we verified that with a sample of $\lesssim 40$ stars, deriving the shape of the density function is challenging and the final result may be not accurate, leading to FUV fluxes affected by large uncertainties.

(iii) *Unidentified sub-clusters.* Stellar groups sharing similar astrometric parameters but consisting of less than 30 Gaia DR3 stars cannot be distinguished as separated clusters in our calculation. An example is NGC 2024, where most of the stars in the dense and embedded core of the region are not detected by Gaia due to bad astrometry and the small number of Gaia sources in this region are associated by OPTICS to the close σ Ori cluster. We verified in Anania et al. (2025b) that even considering 2MASS members in the dense core of NGC2024, the shape of the density function cannot be reliably accessed at distances greater than 0.1 pc from the centre. Two potential solutions to this issue are: (i) considering σ Ori and NGC 2024 as part of a single cluster and evaluating the FUV fluxes using 3D separations, being aware of the large uncertainties, or (ii) using the 2D projected separation from the closest massive stars, providing estimate of the upper limit FUV flux. Since we show in Anania et al. (2025b) that the FUV flux upper limit can significantly differ from the true estimate, the first option is preferred, especially if we need to use the FUV flux estimate in a statistical analysis.

(iii) *Sub-clusters in the ONC.* Similar to what presented in the previous point, the inner (and denser) ONC core, corresponding to the Trapezium, and small clustered regions such as NGC 1977, and the north part of L1641 with NGC 1981 have astrometry not enough accurate to allow the clustering algorithm to reliably kinematically distinguish them. Considering the ONC as a unique large sub-cluster allows us to compute the FUV flux at stars in its external part, accounting for the 2D geometry of the contained substructures into the profile of the normalised density function. The innermost core can be modelled as a centrally concentric region around $\theta^1\text{C}$, and isolating the members of NGC1977 we can use a separate density function to compute the FUV flux in this cluster. Considering NGC 1977 as part of a big substructured cluster with the rest of the ONC, or isolating it, results in consistent FUV fluxes for its members.

This work does not investigate the λ Orionis region, which is located in the north-east part of Orion. λ Orionis is an old (~ 5 Myr) cluster, spanning ~ 10 degree in diameter on the sky, and where a supernova explosion happened ~ 1 Myr ago and has been proposed as a trigger for star formation (Dolan & Mathieu 2001; Cunha & Smith 1996). The decrease in mm-continuum disc emission with increasing separation from the most massive star was interpreted as indirect evidence of the effect of UV radiation on discs (?). Due to the peculiarities of the cluster and its large angular extent, we excluded this region from this work. However, a calculation of the FUV flux for the Class IIs in this region can be found in Anania et al. (2025b).

9.4 Conclusions

Connecting the evolution of YSOs and their protoplanetary discs to the stellar formation environment is essential to characterise the resulting planetary systems. We are interested in accurately determining the FUV flux range that is probed by the Orion stellar population. We selected from Gaia DR3 the stars in a region of radius of six degree around $\theta^1\text{C}$. Using the clustering algorithm OPTICS, we isolated 22 main sub-

clusters among the selected stellar population (Fig. 9.1). We used the 2D geometry of the sub-clusters to evaluate the best estimate 3D separation from OBA-type stars, and then compute the FUV flux and its uncertainty at the stars members of the sub-clusters. The results obtained are summarised as follows.

- We provided a publicly available table of FUV fluxes at the stars members of the 22 Orion sub-clusters, which includes ~ 8600 stars (the table will be available at the CDS). Moreover, we provided an accessible `Python` script to compute the FUV flux at additional stars in the sub-clusters identified in this work.
- Most of the Orion stellar population analysed is weakly irradiated, $\lesssim 10^2 G_0$ (Fig. 9.1). In particular, stars in Orion OB1a show FUV radiation levels comparable to those in more nearby regions such as Lupus and Taurus, but they are on average older (> 6 Myr). 25 Orionis presents a similar age and FUV radiation field of Upper Sco. Future comparison between disc populations experiencing similar FUV flux levels in Orion and in more nearby regions may enhance our understanding of disc evolution across ages and different star-forming environments.
- Orion is the closest region that, while containing numerous weakly irradiated stars, also hosts a substantial subset of intermediately irradiated stars exposed to FUV fluxes in the range $10^2 - 10^4 G_0$. This intermediately irradiated population accounts for $\sim 35\%$ of our sample. In contrast, only $\sim 5\%$ of the investigated stellar population is exposed to strong FUV radiation fields ($> 10^4 G_0$), which means that Orion is not the ideal region for investigating the effect of extreme UV radiation fields on discs on a population basis.

With this work, we provided some important tools and accessible tables that encourage further investigation of stellar and disc properties in the Orion region with the aim of characterising the role of environmental radiation on disc evolution and planet formation.

9.5 Appendix

9.5.1 Deriving the FUV flux from dust emission

Another method for computing the FUV flux in young star-forming regions has been previously introduced by Hollenbach et al. (1991) and consists of linking the dust temperature with the FUV radiation field. Hollenbach et al. (1991) used a theoretical 1D slab model that solves thermal and chemical balance considering incident FUV radiation on one side, to derive the relation between the dust temperature and the FUV flux in units of G_0 :

$$T_{\text{dust}} = \{8.9 \times 10^{-11} \nu_0 G_0 e^{-1.8A_v} + 2.7^5 + 3.4 \times 10^{-2} [0.42 - \ln(3.5 \times 10^{-2} \tau_{100} T_0)] \times \tau_{100} T_0^6\}^{0.2}, \quad (9.4)$$

where $\nu_0 = 3 \times 10^{15}$ Hz, T_0 is the equilibrium dust temperature at the slab surface due to the incident FUV photons, $T_0 = 12.2 G_0^{0.2}$, and $\tau_{100} = (2.7 \times 10^2 G_0)/T_0^5$ is the effective optical depth at $100\mu\text{m}$ that, accordingly to the model of Hollenbach et al. (1991), takes into account the attenuation of the FUV photons through the dust grains. The *Herschel*

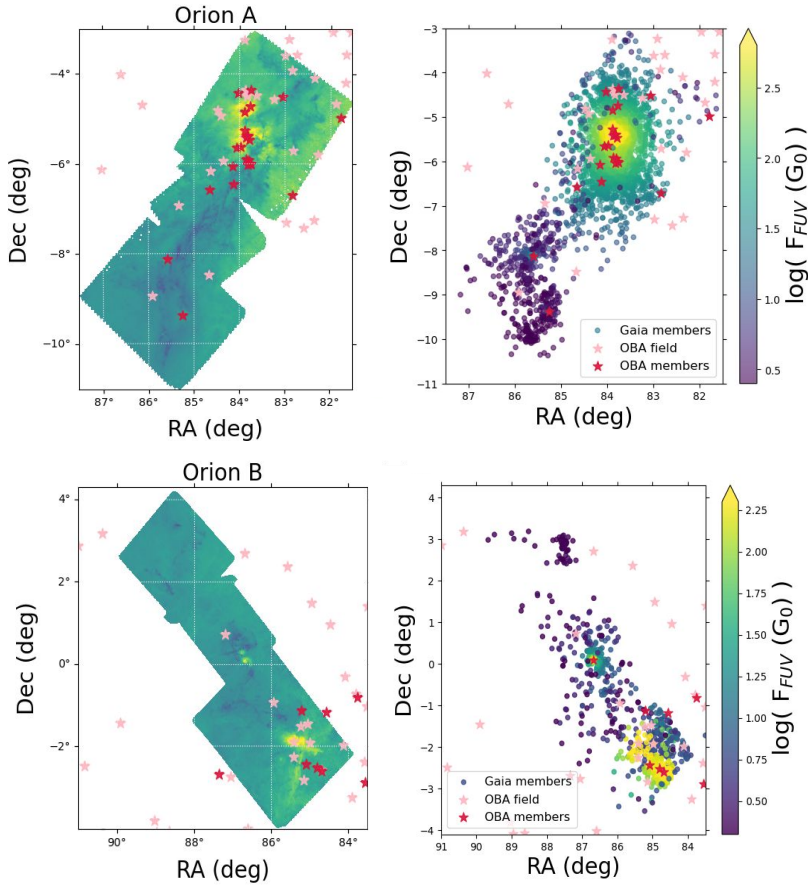


Figure 9.5: Top row: FUV flux map of Orion A as derived from the HGBS map of the dust temperature using Eq. (9.4) (left panels), and FUV flux at the positions of the stars included in that area, using the local density function method described in Sec. 9.1.2 (right panels). OBA-type stars are marked in red (members) and pink (field stars). Bottom row: Same as top row but for Orion B.

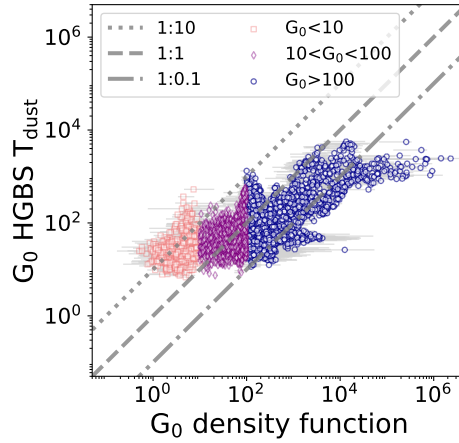


Figure 9.6: Comparison between the FUV flux values (in units of G_0) computed using the HGBS dust temperature maps of Orion A and Orion B, using Eq. (9.4), and those resulting from applying the local density function method (Sec. 9.1.2). Gray lines indicates where G_0 computed from dust temperature is [10, 1, 0.1] times the G_0 resulting from local density function.

Gould Belt survey (HGBS, André et al.2010; Könyves et al.2020) released the dust temperature maps for Orion A and B (Pezzuto et al. 2021)², that we used to compute G_0 from the expression in Eq. (9.4).

There is also an alternative method involving the use of dust emission for computing G_0 , which was proposed by Kramer et al. (2008) and is based on the assumption that the FUV radiation is completely absorbed by interstellar dust grains and re-emitted at FIR wavelengths. The FIR intensity, that will be converted into G_0 , can be computed by using the FIR maps at $70\mu\text{m}$ and $160\mu\text{m}$ provided by the HGBS (Ishii et al. 2019; Shimajiri et al. 2017). However, since the Herschel broadband filters do not provide full coverage of the entire FIR, significant part of the radiation may be missing from the calculation, resulting in significantly underestimated FUV fluxes. Therefore, we did not focus on this method, while we did compare our proposed method with the approach described by Eq. 9.4.

Herschel FIR observations do not cover the entire Orion region explored in this study. However, since the HGBS mapped Orion A and Orion B, providing the column density and dust temperature maps, we computed the FUV flux for these regions. In the panels of the left column of Fig. 9.5, we show the FUV flux maps of Orion A and Orion B resulting from using the relation between the dust temperature provided by the HGBS and G_0 (Eq. 9.4). The HGBS dust temperature maps cover Orion A and B with a resolution of $36.3''$. In the panels of the right column of Fig. 9.5, we present the FUV flux at each stellar position covered by the Orion A and Orion B dust maps, computed applying the local density function method.

We interpolated the FUV flux map derived from Herschel at the position of our stellar sample to make a comparison between the results of the two methods. The comparison is

²http://www.herschel.fr/cea/gouldbelt/en/Phoceia/Vie_des_labos/Ast/ast_visu.php?id_ast=66

shown in Fig. 9.6³. We found that the FUV flux values derived from dust temperature are consistent within 1σ error with those evaluated by applying the local density function method. The $\sim 30\%$ of the sample show consistent FUV fluxes within a factor of 2, while the remnant sample remains within an order of magnitude error. The main limitations in computing G_0 using dust emission maps are summarised as follows:

(i) G_0 resulting from dust maps depends on the amount of dust in the region. In areas with very low or no dust, the computed FUV flux is artificially low (i.e. not because the radiation is weak or there are no massive stars in the region, but because there is insufficient dust to trace it). As a result, low G_0 can be found near massive stars if they lie in dust-poor regions (e.g. very old regions). This is evident in the comparison made for Orion B, in Fig. 9.5, where the left plot shows low G_0 values in correspondence with the σ Orionis cluster, where only small amounts of dust are present, probably due to the older age (~ 3 Myr) with respect to NGC 2024 and the Orion A region (< 2 Myr). G_0 is predicted to be higher in NGC 2024, where the dust density is very high. In this last cluster, we evaluated G_0 only for a few Gaia stars which are not close enough to the most massive star to reach $> 10^4 G_0$, which is instead registered using the dust temperature map.

(ii) As discussed in Hollenbach et al. (1991), below $\sim 10 G_0$ the dust is mainly heated by visible and far-infrared photons, whereas FUV photons do not contribute to the majority of the emitted FIR radiation and the measured dust temperature. At higher, but moderate, FUV flux level between 10 and 100 G_0 , the dust heating can still be largely contaminated by visible and far-infrared. Therefore, using HGBS maps in moderate-to-low G_0 regions may lead to an overestimate of the actual FUV flux value. In Fig. 9.6, we marked in pink and purple the points at $G_0 < 10$ and $10 < G_0 < 100$, while colouring in blue the region at higher FUV flux.

(iii) When using the relation between FUV flux and dust temperature described in Eq. 9.4, the main source of uncertainty is the extinction term A_v , which is usually associated with large uncertainty in young and embedded regions.

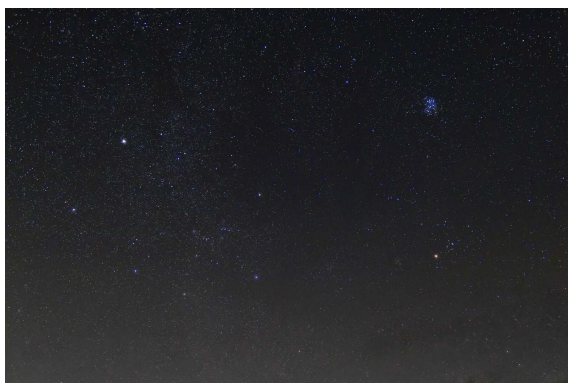
(iv) The relation between dust temperature and G_0 shown in Eq. (9.4) is based on 1D slab models, where UV radiation is incident on one side of the slab (Hollenbach et al. 1991). In the worst-case scenario, not including dust extinction when computing the FUV flux from the local density function can result in an order-of-magnitude difference in the FUV flux. However, in less extreme (and more probable) cases, ignoring the influence of 3D geometry and thus using a 1D slab model may contribute to a general trend for our method in predicting higher FUV fluxes, as shown by the blue region in Fig. 9.6.

Overall, although using dust emission maps to evaluate G_0 gives an indicative value of the average FUV radiation field in the region of interest, it has numerous limitations, from the number of regions in which it can be applied, which is limited to the regions mapped in dust emission and FIR with good resolution, to the application to older regions, where dust density is low and the result is not reliable, to the model-dependent limitations discussed in the list above. Conversely, our method relies on considerations on the 2D geometry of a stellar cluster. This can lead to intrinsic large error bars in the FUV flux depending on the shape of the local density function: the flatter the local density function is, the flatter the associated cumulative distribution function (CDF) will be, and therefore, larger will be the range of most probable 3D separations at which two stars can be. However, our method provides a way to consider the main source of uncertainty

³The comparison is made interpolating the 2D dust temperature map at the star positions where we evaluated the FUV flux. However, our FUV flux calculation considers the 3D geometry, and therefore the comparison may be influenced by projection effects

in the FUV flux calculation, which is the 3D structure of the stellar cluster, providing an estimate of the uncertainty in the FUV flux. Furthermore, if we are interested in the FUV flux at the position of a certain star, using dust emission maps is not accurate, as we must assume that there is dust in the exact position of the star that we are targeting. The FUV flux distribution at the position of each star can be used in statistical analysis to investigate, more accurately, the connection between protoplanetary disc properties and the external environment.

FUV flux and disc properties



It seems empty, but is rather full of young stars and discs - Taurus-Auriga association, as seen from Sampeyre (Italy). You can spot the bright, red Aldebaran on the bottom right and the Pleiades shining in the top right part.

This Chapter contains the investigation conducted in Anania et al. (2025b) about the relation between FUV flux and disc dust mass in nearby star-forming regions, and the study performed in Anania et al. (2025, in prep.) on the relation between FUV flux and accretion luminosity in Orion.

Evidence of external photoevaporation in action on protoplanetary discs is probed by direct observations of proplyds, which are protoplanetary discs seen in optical lines as ionised gas clouds with a characteristic comet-like shape, with the cusp pointing toward the nearest massive star (Sec. 3.1). These fascinating objects are found in highly irradiated environments ($\gtrsim 3 \times 10^3$ G₀, Kim et al. 2016) in close proximity to massive O-type or early-type B stars (within ~ 0.1 pc). However, proplyds are only representative of the most extreme environmental conditions, while the majority of discs form and evolve in moderately irradiated environments ($10^2 - 10^4$ G₀, Fatuzzo & Adams 2008), where external irradiation is driven by less massive stars, and the peculiar proplyd-like shape is not detectable. In the absence of proplyds, an indirect way to trace the effect of external photoevaporation on discs consists of comparing disc properties with the FUV radiation field experienced. In this Chapter, we first focus on the dust disc mass, which is the most

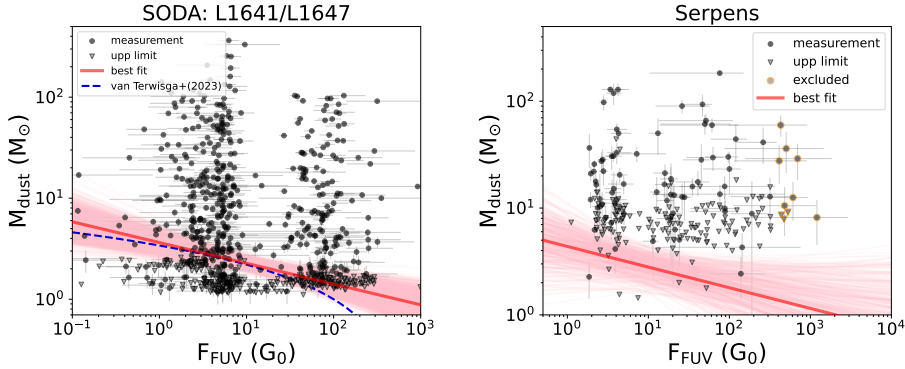


Figure 10.1: Dust disc mass measured from mm-flux observations, and FUV flux computed using the best estimate separation from massive stars in 3D space, for the SODA sample in Orion (left) and Serpens (right). The red line indicates the best log-log relation found, while the blue dashed line in the left panel indicates the lin-log relation in van Terwisga & Hacar (2023). The investigation on the Serpens sample will be presented in van Terwisga, Anania et al. (in prep.).

accessible disc quantity to determine, particularly in distant regions (Sec. 10.1). Subsequently, we investigate the relation between stellar accretion and FUV flux in Orion (Sec. 10.2).

10.1 Dust disc mass

10.1.1 Two young, moderately irradiated regions: L1641/L1647 and Serpens

In intermediately irradiated environments and where proplyds are not detected, the effect of photoevaporation is indirectly suggested by the dependence of the (sub-)millimetre continuum flux emission on the distance from the most massive star, such as in σ Orionis (Ansdell et al. 2017b, Maucó et al. 2023b) and λ Orionis (Ansdell et al. 2020). The relation between dust disc mass and FUV flux has been investigated by the SODA Survey in the extended population of Orion A, which includes the L1641 and L1647 clouds and presents FUV fluxes in the range ~ 1 -100 G_0 (van Terwisga & Hacar 2023). In that work, the FUV flux was evaluated using the 2D projected separation from the massive stars in the close proximity of each object. Following the method presented in the previous Chapter, we evaluated the FUV flux at the position of each SODA object accounting for the uncertainty in the 3D geometry of the stellar clusters, to provide a best estimate FUV flux and its uncertainty. The median FUV fluxes are shown in the left panel of Fig. 10.1 compared to the dust disc masses. The error bars on the FUV fluxes extend from the 16th to the 84th percentile of the distributions. We considered as *detected* the sources presenting 1.3 mm flux measurements $> 3\sigma$ significance, and we set the upper limits dust masses corresponding to the 3σ mm-flux limit. By fitting the single points and accounting for the upper limits dust masses and the uncertainties in FUV fluxes and dust disc masses, we found a relation of the kind $\log(M_{\text{dust}}/M_{\oplus}) = (-0.19 \pm 0.05) \log(F_{\text{FUV}}/G_0) + (0.55 \pm 0.06)$, shown by the red solid line in the left panel of Fig. 10.1. This relation is consistent, within the error bars, with the relation in lin-log space found by van Terwisga & Hacar (2023), which is shown by the blue dashed line in the left panel of Fig. 10.1. Overall, since upper limits in dust disc masses account

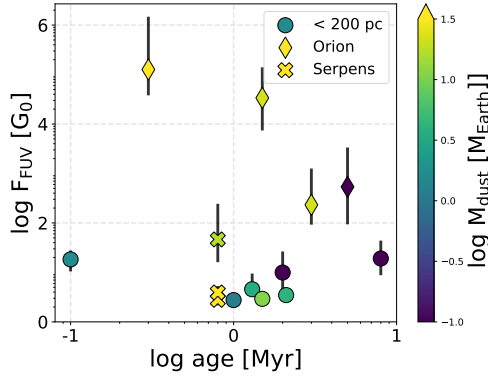


Figure 10.2: Median FUV flux as a function of the estimated age of the regions. Each dot represents a star-forming region in this work and is colour-coded based on the median dust disc mass available for the sample of discs. NGC 1977 and 25 Ori are not included because they lack dust disc mass measurements. The FUV flux presented for NGC 2024 is evaluated using Method 2 (Sect. 8.3.2).

for $\sim 42\%$ of the total observed sample, they play a crucial role in shaping the inferred relation with the FUV flux in this region.

We performed a similar investigation on FUV fluxes and dust disc masses in the Serpens star-forming region, considering the stellar sample presenting mm-flux measurements in Anderson et al. (2022) and the FUV fluxes evaluated as in Chap. 8. The results are shown in the right panel of Fig. 10.1 and will be part of a paper in preparation to which I have contributed as second author (van Terwisga, Anania et al. , 2025 in prep.). In the Serpens region, the upper limits of the dust disc masses account for $\sim 65\%$ of the total sample observed, and therefore the statistics are dominated by these values even more strongly than in the SODA survey (see the left panel of Fig. 10.1 for a comparison). As a consequence, the best-fit relation between dust disc mass and FUV flux is pushed to lower amplitudes and is affected by large uncertainties, as shown by the red line in the right panel of Fig. 10.1. The best-fit relation in Serpens was found excluding the sources experiencing $> 400 G_0$, marked in orange, which are located in correspondence to the centre of the H_2 region W4; their mm-flux measurements could be highly contaminated by the presence of the cloud.

L1641/L1647 and Serpens are extremely interesting regions to investigate the role of external photoevaporation in weakly-to-moderately irradiated discs. However, given the numerous upper limits and the large scatter in the obtained relations between dust disc mass and FUV flux, we conclude that the current observational limits prevent us from obtaining strong evidence of the effect of external photoevaporation in intermediately irradiated regions. However, the hints of a dependence of the disc mm-flux emission from the FUV flux, in addition to previous evidence in σ Ori and λ Ori, encourage future investigation and deeper observations that will provide measurements of disc properties (i.e., gas and dust disc masses and radii), and thus assess the role of external photoevaporation in moderately irradiated discs.

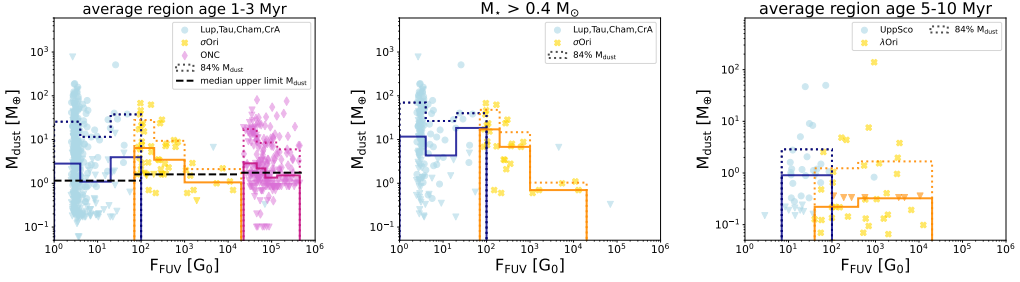


Figure 10.3: Dust disc masses and FUV fluxes for the regions with average age 1-3 Myr (top and central panel) and 5-10 Myr (bottom panel). The histograms show median dust disc masses in flux bins, including upper limits (shown as coloured triangles). The dotted histograms contain the 84% of the distributions. The top panel presents the median upper limit dust mass in each region as dashed black lines. The central panel only includes objects with a stellar mass $> 0.4 M_{\odot}$, and no upper limit in dust disc mass is present in σ Ori. The ONC is not included in this plot as the stellar mass is only defined for about one-third of the total disc sample.

10.1.2 Nearby star-forming regions

Externally irradiated discs can be significantly depleted in their gas and dust components (e.g. Owen et al. 2012, Sellek et al. 2020a). In order to understand the impact of external photoevaporation on the disc evolution and composition, it is crucial to investigate the relation between disc properties and the external radiation they experience. In this section, we focus on the dust disc mass because the gas disc mass and disc size are more challenging to determine and are only available for a few discs in regions at distances larger than ~ 200 pc.

The dust content in discs is a more accessible quantity and can be associated with the thermal emission at (sub-)millimeter wavelengths, assuming optically thin emission and setting a certain dust temperature and opacity (e.g., Miotello et al. 2023). The dust disc mass as retrieved from continuum observations is available for part of the investigated disc sample in the catalogue presented in Chap. 8. Specifically, 12 of the 14 regions investigated present dust mass measurements (the discs of 25 Ori and NGC 1977 currently lack dust mass measurements). We selected from each of these 12 regions the discs with a measured dust mass. In Fig. 10.2 we show the median FUV flux as a function of the average age of the region, colour-coded based on the median dust disc mass. The uncertainties on the FUV flux are the 16th and 84th percentiles of the distributions. The median FUV flux of NGC 2024 was obtained by using 2D projected separations from the OBA stars members of the region because the local density profile of the region, which is needed to apply the method proposed in this study, cannot be reliably derived here. However, since the NGC 2024 disc sample presents dust mass measurements, we decided to show the results, taking into account that the median flux value is an overestimate of the actual flux.

Figure 10.2 shows that less massive discs are generally observed in older regions, a trend that has been extensively investigated by disc surveys in the most nearby star-forming regions (e.g. Manara et al. 2023b). However, protoplanetary discs in more and less irradiated regions are expected to evolve differently, and therefore, the disc properties in different regions cannot be compared on the basis of the age alone, but the irradiation from the environment should also be considered. Theoretical and numerical models predict that discs exposed to stronger external irradiation deplete more quickly (e.g.

Winter & Haworth 2022), and therefore, we would expect to observe less massive discs in more highly irradiated environments. Observations within individual star-forming regions, such as σ Ori and the ONC, support this prediction, showing a clear dependence of disc millimeter flux emission on FUV flux, where the FUV flux values involved were obtained considering 2D projected separations from the closest massive stars (Ansdell et al. 2017b, Maucó et al. 2023b, Eisner et al. 2018b). This raises the question of whether a similar correlation might hold on a larger scale in multiple star-forming regions based on a more accurate estimate of the FUV fluxes. Firstly, we investigated this potential correlation from a region-averaged perspective, considering a median dust disc mass and the FUV fluxes (evaluated using Method 3, Sect. 8.3.3) in each region, as presented in Fig. 10.2. Subsequently, we focused on the distribution of dust disc masses and FUV fluxes in regions with a similar age, as shown in Fig. 10.3.

No clear correlation between the median dust mass and the median FUV flux was identified (Fig. 10.2). However, several considerations need to be addressed. First of all, the regions in Orion and Serpens presenting high median FUV fluxes in Fig. 10.2 are very young. This might explain the high dust mass observed with respect to what is measured in the most nearby regions, as external photoevaporation did not have enough time to significantly remove disc material. Secondly, interstellar dust can efficiently shield discs from high-energy external UV photons during the first half million years of the disc (Ali & Harries 2019), and therefore, the detected discs might have experienced intense UV radiation only very recently. Thirdly, in more distant regions, the observations are biased toward detecting more massive discs, leading to median dust disc masses that are not entirely representative of the regions. Additionally, the stellar mass distribution is not homogenous in Serpens, Orion, and the nearest regions. More distant regions present only a few objects that were identified as low-mass stars ($\lesssim 0.3 M_{\odot}$) and many more are lacking stellar mass measurements.

Last but not least, Fig. 10.2 shows a clear lack of information on the dust disc mass at intermediate- to high-irradiation levels ($> 10^2 G_0$) with respect to less irradiated regions. We suggest that future observations should aim to fill this gap.

We considered region-averaged values so far, and we now focus on individual values by dividing the disc sample into age bins. We considered regions with a similar average age as the discs disperse over time. Specifically, we considered the age range ~ 1 -3 Myr, where more than two regions present available dust masses. The selected regions are Lupus, Taurus, Cham I, Cham II, CrA, σ Ori, and the ONC. Due to the high concentration of dust mass measurements at low FUV fluxes compared to intermediate- and high-irradiation levels, we divided our dust disc mass sample into flux bins of varying width. We show the distribution of the median dust masses (in each bin) against the distribution of fluxes in the histograms in Fig. 10.3. The bin widths were chosen based on the number of objects and upper-limit dust masses in each region. Specifically, the sample of the most nearby discs, covering ~ 1 -100 G_0 , was divided into three parts, as was the σ Ori sample, which covered fluxes from $\sim 10^2$ to $\sim 2 \times 10^4 G_0$. The ONC disc sample, which consists of detections and non-detections in Eisner et al. (2018b), was divided into ~ 40 stars each, with at least four detections in each bin. Since in the ONC $\sim 60\%$ of the total disc sample is undetected, the median dust disc mass is dominated by upper-limit dust masses (shown as dashed black lines in the top panel of Fig. 10.3) and is comparable to the observed median in the other regions. Due to the known dependence of the dust mass on the stellar mass (e.g. Manara et al. 2023b), the medians can be highly influenced by the range of stellar masses covered. In order to reduce the effect produced by the dependence on stellar mass, we restricted the disc sample in these regions to the

relatively high-mass stars ($M_{\star} > 0.4 M_{\odot}$), which are the majority detected in σ Ori. The results are shown in the central panel of Fig. 10.3. We did not include the ONC in this panel because the stellar mass is defined for only one-third of the total disc sample. A decrease in dust disc mass is visible between $\sim 10^3$ and $\sim 10^4 G_0$, and it is related to σ Ori (Fig. 10.3 central panel), where this trend was found previously (Ansdell et al. 2017b, Maucó et al. 2023b). This is now supported by the use of our updated FUV flux estimate. The correlation in σ Ori is more pronounced when it is restricted to high stellar masses (Fig. 10.3). A weaker negative dependence of the dust mass on the external flux appears in the ONC, which is more clearly visible in the 84th percentile of the mass distribution, as the median is highly influenced by the undetected objects. The lack of stellar properties and disc millimeter continuum detections at intermediate- to high-irradiation levels limits our ability to draw more exhaustive conclusions for a wide range of distances, regions, and FUV fluxes. Therefore, we strongly support explorations of this parameter space in future observations.

When we applied the same study to the oldest regions of our sample, which are Upper Sco and λ Ori, we only detected a small decrease in the median dust mass from lower to higher FUV fluxes (from Upper Sco to λ Ori). The results are shown in the bottom panel of Fig. 10.3. As the disc sample is limited to only two regions, the data are insufficient for a deeper statistical investigation to determine whether the small decrease in the dust disc mass is due to external FUV irradiation or observational biases. An analysis of new mm-continuum ALMA observations for the disc population in 25 Ori is ongoing (project ID: 2023.1.01591.S), and the results will be crucial to further explore the relation.

10.2 Stellar accretion luminosity and FUV flux in Orion

We provide insight into the relation between stellar accretion and environmental FUV radiation in Orion. This investigation is motivated by the fact that it is still poorly understood how the properties and evolution of the innermost disc regions ($\lesssim 10$ au) are linked to the outer disc, which is affected by external photoevaporation and interactions with the environment. Observations show contradictory results on the influence of external UV radiation on the chemistry of the inner and outer disc regions (e.g., Berné et al. 2024; Frediani et al. 2025). Theoretical models show that viscous externally FUV-irradiated protoplanetary discs tend to evolve toward a constant ratio between the rate of mass lost in photoevaporative winds and that accreting onto the central protostar (Clarke 2007; Winter et al. 2020a; Sellek et al. 2020b). The timescale for reaching this equilibrium is around a couple of Myr for discs around stellar masses $\sim 0.6 M_{\odot}$ experiencing $\sim 1 - 10 G_0$ (Anania et al. 2025a). In this Section, we present a study of the stellar accretion luminosity as a function of the FUV flux in the Orion star-forming region. We complement our analysis with a population synthesis model of viscous externally irradiated protoplanetary discs, comparing the results of the model with the observations.

10.2.1 Method

We explored the link between the inner and outer protoplanetary disc regions by investigating the relation between stellar accretion luminosity and external FUV radiation field in Orion. Specifically, we investigated whether the accretion luminosity could be a useful parameter to test the impact of external photoevaporation on protoplanetary discs. We used the stellar accretion luminosities from Delfini et al. (2025), which were derived from Gaia DR3 XP spectra (De Angeli et al. 2023) of young stellar objects located within

500 pc of the Sun that show H_α line emission. Among the accretors included in Delfini et al. (2025), 1275 objects are candidate members of the Orion sub-clusters based on the membership obtained by using the OPTICS clustering algorithm (see Chap. 9) at which we computed the FUV flux. For 75 objects among these 1275, Delfini et al. (2025) reports H_α emission compatible with chromospheric activity, so their accretion luminosities are considered here as upper limits. For the 5073 objects in Orion that have Gaia XP spectra but no H_α emission is reported in Delfini et al. (2025), we defined representative upper limits to accretion luminosities using the following procedure. For each of these stars, we first evaluate the expected pre-main sequence bolometric luminosity as

$$\frac{L_\star}{L_\odot} = 10^{0.4(M_{\text{bol},\odot} - M_{\text{bol},\star})}, \quad (10.1)$$

where $M_{\text{bol},\odot} = 4.74$ mag is the bolometric magnitude of the Sun, and $M_{\text{bol},\star}$ is computed using the observed G magnitude, Bailer-Jones distance, and extinction magnitude in the G band, A_G , provided by Gaia DR3, and a representative bolometric correction in the G band of -0.8 mag, which corresponds to an M4 pre-main sequence star (Pecaut & Mamajek 2013b). For objects without defined A_G in Gaia, we adopted the average A_G among the closest neighbourhoods. Subsequently, we estimated the accretion luminosities corresponding to the computed stellar luminosities by considering that the detected sourced in Delfini et al. (2025) in Orion show the correlation:

$$\log\left(\frac{L_{\text{acc}}}{L_\odot}\right) = (-1.41 \pm 0.02) + (1.7 \pm 0.03) \log\left(\frac{L_\star}{L_\odot}\right), \quad (10.2)$$

with a spread of 0.4 dex. By employing this procedure, we provided limit accretion luminosities. For a detailed calculation, see also Appendix 10.5.1.

This list of objects provides us with a homogeneous sample of YSOs to study the relation between stellar accretion and external FUV radiation field. Although Delfini et al. (2025) reported both accretion luminosities and mass accretion rates of the stellar sample, we decided to compare FUV fluxes with accretion luminosities. This choice is motivated by the fact that accretion rate, $\dot{M}_{\text{acc}}$, is derived from accretion luminosity, L_{acc} , and stellar parameters via the standard relation

$$\dot{M}_{\text{acc}} = \frac{L_{\text{acc}} R_\star}{GM_\star} \left(1 - \frac{R_\star}{R_{\text{in}}}\right)^{-1}, \quad (10.3)$$

where $M_\star$ and $R_\star$ are the stellar mass and radius, respectively, and R_{in} is the inner disc radius. Since these stellar parameters are typically inferred from stellar evolutionary isochrones and are strongly affected by extinction, which is uncertain in Orion, we opted to base our analysis on L_{acc} , which is less affected by extinction uncertainties.

In this work, we used the sample of Gaia-based accretion luminosities because it is spread across the entire Orion region and allows us to explore a wide range of FUV fluxes. Conversely, there are only 50 stars in the σ Ori region (Maucó et al. 2023a), and 37 stars in the Orion Nebula (Piscarreta et al. 2025) that have stellar accretion luminosities and mass accretion rates derived from spectroscopic measurements, specifically using X-Shooter. This sample of spectroscopic observations contains a small number of objects that span only two of the 22 Orion sub-clusters identified in Sec. 9.2. Figure 10.4 shows that the accretion luminosities measured using H_α line emission from Gaia XP spectra correlate with those obtained from X-Shooter spectra in Orion, with a relation $\log(L_{\text{acc,Gaia}}/L_\odot) = (-0.9 \pm 0.3) + (0.66 \pm 0.21) \log(L_{\text{acc,XShooter}}/L_\odot)$ and a spread of 0.15

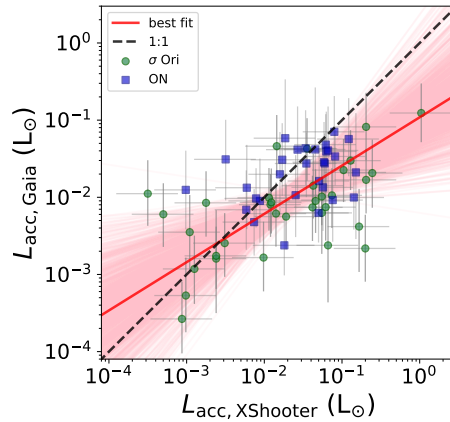


Figure 10.4: Comparison between accretion luminosities measured from X-Shooter spectra, in the σ Orionis (Maucó et al. 2023a) and the Orion Nebula (ON, Piscarreta et al. 2025) regions, and the corresponding value measured from H_α emission from Gaia XP spectra (Delfini et al. 2025). The best fit relation and its spread is shown in red and pink. The red line corresponds to $\log(L_{\text{acc,Gaia}}/L_\odot) = 0.66 \log(L_{\text{acc,XShooter}}/L_\odot) - 0.9$.

dex. The sample used for this comparison consists of objects having X-Shooter spectra in Orion (i.e., in the σ Ori, Maucó et al. 2023a, and ON Piscarreta et al. 2025, regions) and counterpart in Delfini et al. (2025). Since the two estimates of L_{acc} do not exhibit a perfect 1:1 correlation, we tested whether the observational errors on the Gaia-based accretion luminosities are sufficiently large to account for the intrinsic scatter observed in their relation with the X-Shooter measurements. To do this, we assumed that $L_{\text{acc,XShooter}}$ represents the true value. We then compared the distribution of deviations obtained when using $L_{\text{acc,Gaia}}$ instead (i.e., the difference between the Gaia and the X-Shooter values) with the distribution of the observational uncertainties associated with $L_{\text{acc,Gaia}}$. By evaluating the reduced χ^2 and the p-value, we found $\simeq 0.63$ and $\simeq 0.98$, respectively. These results indicate that the scatter between the two measurements is consistent with the observational errors in $L_{\text{acc,Gaia}}$, meaning that the additional scatter between Gaia and X-Shooter does not need to be explicitly included, as it is already accounted for by the uncertainties in $L_{\text{acc,Gaia}}$. As will be shown in the next Section, we perturbed the accretion luminosities derived from our model on the basis of the observational errors.

Disc evolution models: Parameter space explored

We complemented our analysis with a population synthesis model of protoplanetary discs subject to viscous evolution and external photoevaporation. We updated the population synthesis code Diskpop (Somigliana et al. 2024) to include the process of external photoevaporation. The Diskpop code allows us to simulate the evolution of a population of protoplanetary discs exploring a set of initial stellar and disc parameters at the same time. We updated the code by assuming that each disc of the population is subject to a constant FUV flux at the outer disc radius, and we defined the mass loss rate due to external photoevaporation by interpolating the FRIEDv2 grid of mass loss rates (Haworth et al. 2023a) based on the stellar and disc parameters. Disc material is removed outside-in following the numerical approach of Sellek et al. (2020b). We modelled a population of geometrically thin and axisymmetric discs undergoing viscous evolution

(Lynden-Bell & Pringle 1974) and external photoevaporation, where the time evolution of the disc surface density profile is described by

$$\frac{\partial}{\partial t} \Sigma_g = \frac{3}{R} \frac{\partial}{\partial R} \left(R^{1/2} \frac{\partial}{\partial R} (\nu \Sigma_g R^{1/2}) \right) - \dot{\Sigma}_{\text{ext}}, \quad (10.4)$$

where $\nu = \alpha c_s H_g$ is the kinematic viscosity, depending on the gas scale height H_g and the sound speed c_s , α is the Shakura & Sunyaev (1973) parameter describing the disc viscosity, and $\dot{\Sigma}_{\text{ext}}$ is the mass loss rate due to external photoevaporation. Assuming that the various sub-clusters in Orion have a similar initial mass function (IMF), we modelled a population of 7500 objects with stellar masses randomly selected from a Kroupa IMF (Kroupa 2001) in the range $[0.1 - 3] M_\odot$, which is the range of stellar masses supported by the FRIEDv2 grid. We applied a cut to the sample of stellar masses considering an average extinction based on that observed in Orion (Lallement et al. 2022), and a limit in G magnitude based on Gaia XP spectra, i.e. we excluded objects with G magnitude higher than 17.65, which corresponds approximately to stellar masses smaller than $\sim 0.3 M_\odot$ at ~ 400 pc. The final sample consists of ~ 7400 stars with disc that we modelled using the parameter space of the initial conditions presented in Table 10.1. Specifically, the FUV fluxes are randomly selected from the distribution of FUV fluxes of Orion derived in this work. The initial disc mass is set to 5% of the stellar mass. The initial disc extent is set to an indicative value of 50 au; however, we verified that increasing or decreasing the initial disc extent only leads to a shift in the accretion luminosity and mass accretion rate of less than one order of magnitude, without significant consequences on the relation with the FUV flux, which is the relation we are focusing on. The condition imposed on the α viscosity parameter, $\alpha = 10^{-3} (M_\star / M_\odot)$ ensures to achieve the commonly observed relation between accretion and stellar mass $\dot{M}_{\text{acc}} \propto M_\star^2$ (Alexander & Armitage 2006). We evolved the disc population up to 10 Myr, which is the average age of the old Orion stellar population (e.g., Kounkel et al. 2018a). The results of the population synthesis model are presented in the following Section.

Table 10.1: Parameter space used for population synthesis

Parameter	Values
$M_\star (M_\odot)$	Kroupa IMF $[0.1 - 3]$
$M_{d,0} (M_\odot)$	$0.05 M_\star$
$R_{c,0} (\text{au})$	50
α	$10^{-3} (M_\star / M_\odot)$
FUV flux (G_0)	sampling $[1-10^7]$

Notes. We modelled a population of ~ 7500 stars with disc, where the stellar masses are drawn from a Kroupa IMF in the range $[0.1 - 3] M_\odot$, and the FUV fluxes are randomly selected from the distribution of Orion FUV fluxes computed in this work. The initial disc mass ($M_{d,0}$) is set to 5% of the stellar mass, the initial disc characteristic radius ($R_{c,0}$) is fixed to 50 au, and the α viscosity parameter increases linearly with stellar mass.

10.2.2 Results

In this Section, we show the results of the investigation on stellar accretion luminosity and FUV radiation field in Orion and the comparison with numerical disc evolution models.

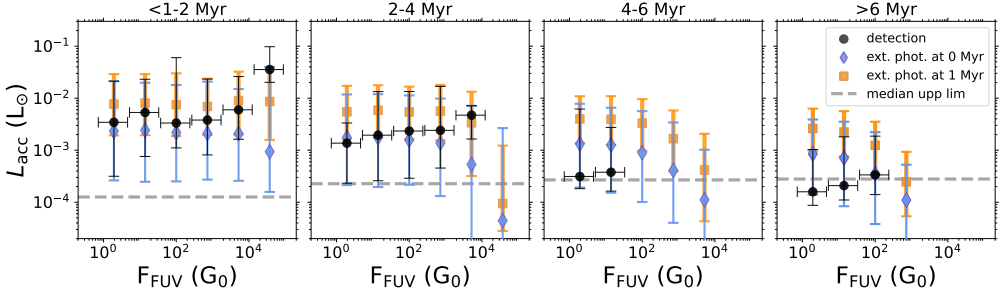


Figure 10.5: The markers indicate the median accretion luminosity L_{acc} in each bin of FUV flux as derived from the Orion objects in Delfini et al. (2025) (black dots) and predicted by the disc population synthesis model where external photoevaporation acts from the beginning (blue diamonds) and after 1 Myr to mimic the effect of shielding (orange squares). The vertical error bars extend from the 16th to the 84th percentile of the distributions, while the horizontal bars indicate the width of the FUV flux bins. The four panels refer to four increasing age bins from left to right: We divided the observed Orion sample by average cluster age and, for consistency, the results of the model are taken at 1 Myr, 3 Myr, 5 Myr, and 10 Myr, respectively. Dashed lines indicate the median upper limit L_{acc} observed in each age bin.

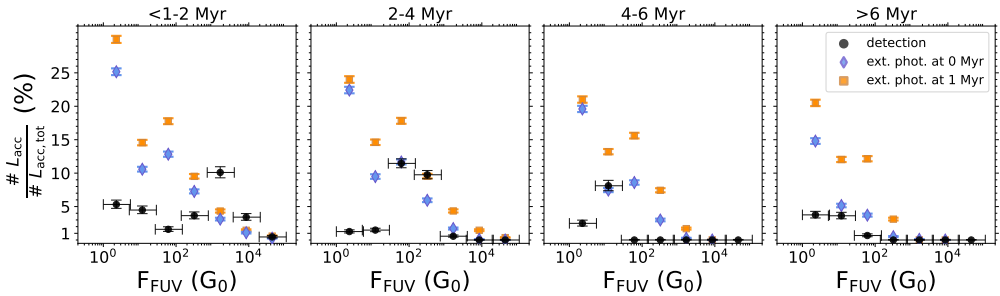


Figure 10.6: Detection fraction of L_{acc} per bin FUV flux and increasing age from the left to the right panel. Black dots refer to the observations, while blue diamonds and orange squares are the results of the population synthesis model where external photoevaporation is included from the beginning of the evolution and after 1 Myr (to mimic the effect of shielding), respectively. The vertical error bars are the uncertainties considering a binomial distribution, while the horizontal bars indicate the width of the FUV flux bins. The age of the observed sample refer to the average age of the Orion sub-clusters. For consistency with the Orion age bins selected, the results of the model are taken at 1 Myr, 3 Myr, 5 Myr, and 10 Myr, respectively from the left to the right panel.

Figure 10.5 presents the distribution of L_{acc} in each FUV flux bin for the Orion sample investigated in this work (black dots) and the results of the population synthesis models (blue diamonds and orange squares). Specifically, markers indicate medians and vertical error bars correspond to 16th and 84th percentiles of the distribution. The panels in Fig. 10.5 illustrate the results in four age bins from the youngest (left) to the oldest (right). Specifically, the Orion population is divided into the following age bins (see also Fig. 9.2, and for references on age range see Warren & Hesser 1978; Briceño et al. 2005; Kounkel et al. 2018a).

- $\lesssim 1 - 2$ Myr : stars in the ONC core (within 0.3 deg from $\theta^1\text{C}$), L1647/L1641-S, and NGC 2068;
- $2 - 4$ Myr : stellar population of σ Ori, L1616, and L1617/L1622;
- $4 - 6$ Myr : stellar sample in belt, η Ori, and Ori C-[01, 02, 03];
- > 6 Myr : the remaining clusters in Orion OB1a and the extended population of Orion.

The population synthesis consists of 1D viscous evolution models of gaseous protoplanetary discs where the dust component is neglected for simplicity and the effect of external photoevaporation is included, as described in Sec. 10.2.1. The initial conditions of the disc population are presented in Table 10.1. We performed two types of population synthesis model. In the first model, external photoevaporation acts from the beginning of disc evolution (blue markers in Fig. 10.5), while in the second model we introduce external photoevaporation after 1 Myr of evolution to mimic the effect of dust extinction (orange markers in Fig. 10.5). Indeed, dust extinction is capable of shielding protoplanetary discs in the first stages of their evolution. This scenario is supported by molecular cloud simulations, which show that dust shields efficiently protoplanetary discs against external FUV radiation during the first $\sim 0.5 - 1$ Myr, whereas the effectiveness of this shielding decreases sharply after that time (e.g., Ali & Harries 2019; Qiao et al. 2022). In our models, the accretion luminosity is evaluated, for each object at each timestep, using the relation in Eq. (10.3), where the mass accretion rate is an output of the model, and the stellar mass and radius are constant during disc evolution. The stellar mass is selected from a Kroupa IMF distribution (see Sec. 10.2.1) and the stellar radius is evaluated from the standard relation $(R_*/R_\odot) \simeq 1.5(M_*/M_\odot)^{0.6}$, which is valid for young stars. We perturbed the distribution of the resulting accretion luminosities with a spread based on the observed uncertainties in L_{acc} , which are of the order of 0.33 dex.

In Fig. 10.6, we show the detection fraction of accretors per FUV flux bin and age bin, as derived from observations and population synthesis models. In the observed sample, the detection fraction of accretors is the percentage of objects with measured L_{acc} in Delfini et al. (2025) over the total number of objects presenting Gaia XP spectra, in each age and FUV flux bin (shown in black in Fig. 10.6). To estimate the fraction of L_{acc} that would be detected in our model based on the observations, we applied to our synthetic accretion luminosities a probability distribution function that, based on the stellar luminosity, reproduces the distribution of limit accretion luminosities in the Orion sample. A detailed description of the function applied can be found in Appendix 10.5.1. The results of the models are shown in blue and orange in Fig. 10.6.

By examining Fig. 10.5 and 10.6, we derive the following main results.

- (i) Although the model does not perfectly reproduce the observed distribution of detected L_{acc} and the detection fraction of accretors, it captures the broad trend: both L_{acc} of the detected objects (Fig. 10.5) and the detection fraction (Fig. 10.6) decline with

age, and the fraction of detected objects drops more rapidly for stars exposed to stronger FUV fluxes than for those in weaker FUV-irradiated environments. The absence of old accretors at high FUV fluxes suggests that they may have dispersed their discs enough for accretion to fall below the H_α detection threshold, preventing us from detecting them. This may reflect a combined effect of age and strong external irradiation. We highlight that the distribution of L_{acc} is broad (see 1σ error bars in Fig. 10.5) and that the upper limits dominate the statistics of the sample. In Fig. 10.5 we show that the detected L_{acc} values of the oldest sources (4–6 Myr, and > 6 Myr) are consistent with or lower than the median upper limit value (dashed gray line). Furthermore, Fig. 10.6 shows that the model reproduces the trend with FUV flux more effectively than the trend with age: the observed decline in the detection fraction with age is steeper in the observations than in the model, suggesting that the model likely lacks a dependence of the accretion efficiency on age.

(ii) The model overestimates the detection fraction in all age bins considered and particularly for weakly irradiated sources, as shown in Fig. 10.6. This result is particularly interesting because it indicates that the accretion efficiency of the disc population may lack an important dependence on time in the framework used and may vary as a function of additional external and internal factors. In Sec. 10.3.2 we discuss the potential reasons that may lead to this result and the caveats of the presented analysis.

(iii) The leftmost panel of Fig. 10.5 shows that the young stellar population (< 2 Myr) presents a sample of strong accretors ($L_{\text{acc,median}} > 0.01 L_\odot$) that are externally irradiated by strong FUV radiation fields ($> 10^4 G_0$). Our models fail to reproduce this young population of strong accretors experiencing high FUV fluxes. Introducing shielding in the first 1 Myr helps to explain the median L_{acc} in the highest FUV flux bins (see orange squares), but it cannot still perfectly recover the observations, and L_{acc} remains underestimated. Moreover, a delayed photoevaporation leads to a larger discrepancy between the model and observations for the oldest population ($\gtrsim 4$ Myr). We discuss in Sec. 10.3.2 that additional factors may act on this young population in a strongly irradiated environment.

The lack of a tight correlation between accretion luminosity and FUV flux in each of the age bins and for both observations and models suggests that L_{acc} is not a preferred tracer of external photoevaporation. However, the results shown in Fig. 10.5 and 10.6 provide valuable insights into the observed population and the disc evolution framework. We provide a detailed discussion on the results in Sec. 10.3.2.

10.3 Discussion

10.3.1 On the relation between dust disc mass and FUV flux

Of the catalogue of nearby regions considered in Chap. 8, only σ Ori shows a decrease in the dust disc mass at higher G_0 , as presented in Maucó et al. (2023b). A similar decrease is presented at lower FUV fluxes in the SODA sample (van Terwisga & Hacar 2023). In the Serpens region, as well as in the sample in the ONC core, the statistics are dominated by upper limits. Figure 10.3 highlights key areas of the FUV flux-age parameter space to explore with future observations in order to investigate in detail how external irradiation contributes to shaping the dust disc content. Specifically, more stellar information (in particular stellar mass measurements) and deeper disc observations in intermediately and highly irradiated environments ($> 10^2 G_0$) in a range of ages of 1–10 Myr would be fundamental to complement our study. Therefore, we strongly encour-

age measurements of disc properties in other star-forming regions to explore the relation with the environmental irradiation.

In proplyds, direct information on the disc mass-loss rate due to external photoevaporation is provided by ionised gas emission lines (e.g. [NII], [Ne III], and [OIII]), which are bright in PDRs and seen at optical wavelengths. One of the most sensitive lines to external UV radiation and a good tracer of the inner region of the wind is [OI] 6300 (e.g. Bally et al. 1998, Störzer & Hollenbach 1998), whose luminosity is predicted to increase with the external FUV field strength (e.g. Ballabio et al. 2023). However, at low irradiation levels ($\lesssim 10^3 G_0$), the line luminosity is dominated by the irradiation from the central star, and it cannot be used as tracer of external winds. The observed line luminosity is therefore similar in nearby regions and σ Ori (Maucó et al. 2025). More irradiated regions included in this work, such as the ONC, have too few or missing detections for a statistically reliable study of the optical lines. The physical and chemical characteristics of PDRs are also provided by models and observations of IR lines (e.g., rovibrational lines of H_2 and CO, mid-IR features of polycyclic aromatic hydrocarbons Hollenbach & Tielens 1999), and are now investigated in detail by JWST in the solar neighbourhood (e.g., Van De Putte et al. 2024). However, the problem of using these lines to trace external photoevaporation in intermediately and moderately irradiated protoplanetary discs remains, as the radiation from the central star will obscure the contribution of an external UV source. This highlights the need for an independent method to evaluate the FUV radiation in PDR models to accurately compare with observations on the one hand, and alternative PDR tracers in the absence of a proplyd-like morphology on the other.

10.3.2 On the relation between accretion luminosity and FUV flux

We explored the relation between stellar accretion luminosity and FUV flux across the Orion region. The stellar luminosities were derived from H_α emission using Gaia XP spectra (Delfini et al. 2025), while the FUV fluxes were computed as detailed in Sec. 9.1. In this Section, we highlight some important caveats to consider when analysing the results that we presented.

First, the statistics of the accretion luminosities in Orion are largely dominated by upper limits, which are associated with the stars in Orion that present Gaia XP spectra but no H_α emission ($\sim 80\%$ of the total population). In Fig. 10.6 we show the detection fraction related to each age bin. Within the sample of upper limits, part of the stellar population may be non accreting, while another fraction may be weakly accreting and its L_{acc} falls below the detection limit. Since Gaia-based accretion luminosities are not retrieved directly from the H_α line flux, but using the H_α pseudo-equivalent width, as detailed in Delfini et al. (2025), distinguishing between a non-accreting and a weakly-accreting star is challenging. Therefore, we computed *limit accretion luminosities* for this sample of objects from their stellar luminosity. These are the accretion luminosities that we expect for PMS stars in Orion having the given stellar luminosity $L_\star$ (see Sec. 10.2.1). Then, to ensure a reliable comparison with the model, we applied to the simulated population a probability distribution function based on the distributions of stellar luminosities observed and the limit accretion luminosities evaluated (see Appendix 10.5.1 for a detailed calculation).

Second, we showed in Fig. 10.4 that Gaia-based accretion luminosities correlate with X-Shooter measurements and that the observational uncertainty in Gaia-based values is sufficiently large to account for the scatter from the X-Shooter values. However, Gaia-based accretion luminosities still have significantly large observational uncertainties because they are based on very low resolution spectra. Moreover, H_α background contam-

ination in Orion might complicate the derivation of accretion luminosities (De Marchi et al. 2010; Beccari et al. 2017). Therefore, we encourage additional and more precise spectroscopic measurements of stellar accretion in the Orion region.

We analysed the distribution of detected L_{acc} , and the detection fraction of L_{acc} as a function of age and FUV flux (Fig. 10.5 and 10.6, respectively), obtaining three main results that we discuss as follows.

Lack of strong accretors at high FUV fluxes in older regions

The lack of strong accretors in old and highly FUV-irradiated regions is shown in Fig. 10.5 and 10.6 and is broadly consistent with the results of the population synthesis model. In general, the median of the detected L_{acc} and the detection fraction decrease with age in both observations and models. However, the fact that in strongly FUV-irradiated environments we lose strong accretors faster than at weaker FUV fluxes, may suggest that external photoevaporation contributes to depleting discs, making L_{acc} fall below the detection limit and reducing the number of detections. Nevertheless, this result alone cannot be taken as strong evidence of the effect of external photoevaporation on the Orion protoplanetary disc population. A complementary investigation of disc properties (e.g., disc masses, radii) as a function of the FUV radiation field, covering a wide range and FUV fluxes, in the Orion region is needed.

Smaller detection fraction than predicted

The fact that the population synthesis model overestimates the fraction of detected accretors than is observed, in all age bins, suggests that the accretion process efficiency is not entirely captured by a viscous and external photoevaporation model. Moreover, we found that the model reproduces the decline in the detection fraction more accurately in FUV flux than age. This result suggests that the model may lack some dependence on the accretion efficiency on time. We discuss here some of the factors that can influence the accretion process.

First, our model assumes a constant level of disc turbulence with stellar mass, time, and across the disc extent by using the parametrisation $\alpha = 10^{-3}(M_{\star}/M_{\odot})$. However, the level of turbulence in a protoplanetary disc can vary locally and on different timescales due to several factors such as late infalls, close encounters, internal winds, and planet formation (e.g., for a review on disc turbulence see Rosotti 2023). However, it is not obvious that varying the accretion efficiency can fully solve the tension between models and observations. A lower accretion efficiency allows discs to evolve more slowly and survive longer, thus increasing the detection fraction. In contrast, a higher accretion rate leads to more rapid disc dispersal, resulting in a smaller detection fraction at older age but yet not resolving the discrepancy observed at younger ages. Another parameter that may play an important role in determining the efficiency of accretion is the disc magnetisation. Time-variation of this parameter can affect the efficiency of the change in angular momentum across the disc, which drives accretion (e.g., Tabone et al. 2025).

Additionally, the initial conditions adopted in our model may not be representative of the entire Orion population. Because Orion includes several sub-groups that differ in age, number of members, and OBA stars, the properties of the youngest population may not represent those of the oldest population at its earlier evolutionary stages. As a consequence, a single model is insufficient to capture the diversity in evolution frameworks

and the main processes driving disc evolution across different FUV radiation fields.

Young population of highly irradiated strong accretors

We detected a sample of accretors that are young ($\lesssim 1 - 2$ Myr), highly irradiated ($\gtrsim 10^4 G_0$), and have high accretion luminosities ($\gtrsim 10^{-2} L_\odot$), which cannot be reproduced by a population model of viscous externally photoevaporating discs. A potential explanation may involve the role of dust extinction in efficiently shielding protoplanetary discs. We showed that a population synthesis model in which the effect of external photoevaporation is delayed by 1 Myr supports in part this hypothesis, although it does not entirely solve the discrepancy because L_{acc} remains underestimated. We propose three scenarios to address this discrepancy. The first involves different initial conditions for our population synthesis model, implying that the parameter space that we explored is not fully representative of both the younger and older Orion population. As mentioned in the previous Section, and recently supported by (Weder et al. 2025), a single disc evolution model faces challenges in explaining properties in different star-forming regions. The second scenario involves the role of cluster dynamics in varying the FUV flux across time. Substantial temporal fluctuations in FUV flux can lead to significant variations in the external photoevaporation mass loss rate and, since the amount of material in the disc varies, it is not excluded that this will affect the accretion process. Considering, for example, an OB star that has been closer or farther by 1 pc from an accreting source at some point in the past, the corresponding FUV flux at the accreting source would have varied by a factor of four. However, since the models do not predict a strong dependence of L_{acc} on the FUV flux (in the wide range explored, $\sim 1 - 10^5 G_0$), such variation is unlikely to be the dominant factor required to explain the observed accretion luminosities in Orion. The last scenario involves infalling material from the external environment that is able to raise the rate of stellar accretion. To support this last scenario, we notice that the sample of young and highly irradiated sources is located in the ONC core, which is the part of Orion presenting the highest stellar and gas density. The correlation between accretion rate and stellar and gas density was investigated in the Lupus region, revealing that clustered sources present higher accretion rates than those that are non-clustered, and that accretion rates are consistent with the predictions of a Bondi-Hoyle-Lyttleton accretion model (Winter et al. 2024a). A similar effect may complement dust extinction shielding in explaining the high accretion luminosities detected for the strongest irradiated sources.

10.3.3 Accretion against depletion

The results of the population synthesis model presented in Sec. 10.2.2 show that the relation between L_{acc} and FUV flux does not provide strong constraints on the relation between accretion (affecting the inner disc) and depletion (truncating the outer disc). Comparing accretion rate and photoevaporative mass loss rate may serve this purpose more effectively. Indeed, the full information of the effectiveness of external photoevaporation on discs is not retained in the FUV flux alone, but instead depends also on the stellar mass, disc mass, and disc extent. The photoevaporation rate depends on all these factors and would allow us to better assess the connection between inner and outer disc dynamics. However, measuring the disc mass loss rate is extremely challenging. A direct measurement is only possible in proplyds, throughout measurements of the ionization front detectable in optical lines (e.g., Winter & Haworth 2022; Aru et al. 2024),

and using hydrogen recombination lines (Boyden et al. 2025). In the future, additional wind tracers will provide a direct way to assess the effect of external photoevaporation (e.g., CI, Haworth & Owen 2020) and estimates of the mass loss rate. Indirect measurements involve computing the mass loss rate from models (e.g., interpolating the FRIED grid Haworth et al. 2023a) and require knowledge of the stellar mass, disc mass, and disc radius. Observations to assess these properties are challenging to be performed in Orion due to the deep sensitivity and high resolution needed; however they are fundamental to characterise disc evolution in typical star-forming environments.

10.4 Conclusions

In this Chapter, we focused on the relation between disc properties and external FUV radiation. First, we investigated dust disc masses in the nearby star-forming regions (< 500 pc from the Sun) as a function of the FUV flux. The dust disc mass is the most accessible disc parameter that can be inferred, assuming optically thick emission, from mm-continuum observations. Subsequently, we focused on Orion and investigated the stellar accretion luminosities as a function of the FUV radiation field and age. Specifically, for this last analysis we used accretion luminosity evaluated from H_α line emission using Gaia XP spectra in Orion, and we compared with a population synthesis model of viscous discs experiencing external photoevaporation. The main results can be summarised as follows.

- We verified the relation between dust disc mass and FUV flux in the L1641 and L1647 regions targeted by the SODA survey, finding a linear correlation in log-log space (Fig. 10.1). An hint of a correlation between disc dust mass and FUV flux is also found in the Serpens region. However, we emphasise that the statistics is dominated by upper limits in both L1641/L1647 and, particularly, in Serpens. Future observations enabling measurements of disc parameters are essential to further investigate the role of external photoevaporation in these moderately irradiated regions.
- We showed, in Fig. 10.2 and 10.3, that we lack disc measurements at typical levels of FUV radiation field ($10^2 - 10^4 G_0$). A dependence of the disc dust mass on the FUV flux appears in σ Ori. However, the high concentration of disc dust measurements in weakly irradiated regions compared to regions experiencing $> 10^2 G_0$, and the large number of undetected sources in the ONC, prevent us from verifying whether a general correlation between disc dust content and FUV flux holds.
- We obtained that in Orion the accretion luminosity decreases with age, and that the fraction of detected accretors drops with age at higher FUV fluxes ($> 10^2 G_0$) faster than at lower FUV fluxes ($< 10^2 G_0$). This trend is broadly reproduced by the population synthesis model (Fig. 10.5 and 10.6).
- The population synthesis model overestimates the detection fraction of accretors with respect to observations (Fig. 10.6). This suggests that the current viscous external photoevaporation model may not fully capture the dependence of accretion on time and as a function of additional factors. However, we underline the need for more precise measurements of accretion luminosities in Orion, as Gaia-based measurements are affected by large uncertainties.

The results of this Chapter emphasise what was stated in Chap. 5 and Chap. 6 from the investigation of the Upper Sco disc sample: probing external photoevaporation in

moderately and weakly irradiated regions with the currently available dataset in the nearby star-forming regions is challenging, even considering the Orion population and its range of FUV fluxes. Our results suggest that protoplanetary discs in moderately irradiated regions undergo some degree of photoevaporation during their lifetime (e.g., disc evolution models, the dependence of dust disc mass on the FUV flux, the lack of old, strong accretors at high FUV flux), but differently from the highly irradiated proplyds, the evidence remains elusive. Moreover, the results presented in this Chapter provide important considerations on the disc evolution model used and encourage deeper observations of stellar and disc properties to characterise the evolution of externally irradiated discs.

10.5 Appendix

10.5.1 Inferring upper limits L_{acc} from the detection curve

We modelled the intrinsic relation between accretion luminosity and stellar luminosity using the following procedure. For each source i we define the parameters

$$x_i \equiv \log_{10} \left(\frac{L_{\star, i}}{L_{\odot}} \right), \quad \ell_{i, \text{true}} \equiv \log_{10} \left(\frac{L_{\text{acc}, i}}{L_{\odot}} \right). \quad (10.5)$$

The true stellar luminosities, $L_{\star}$, are not known exactly; instead we observe $x_{i, \text{obs}}$ with uncertainty $\sigma_{x, i}$. Therefore, we treated the true stellar luminosity as a latent variable, $x_{i, \text{true}} \sim \mathcal{N}(x_{i, \text{obs}}, \sigma_{x, i})$, and placed a linear model with intrinsic scatter on the true accretion luminosity,

$$\mu_L(x_{i, \text{true}}) = \alpha_0 + \alpha_1 (x_{i, \text{true}} - x_{\text{ref}}), \quad \ell_{i, \text{true}} \sim \mathcal{N}(\mu_L(x_{i, \text{true}}), \sigma_L), \quad (10.6)$$

where α_0 , α_1 , and the intrinsic scatter σ_L (in dex) are free parameters, and x_{ref} is a fixed reference luminosity (we take x_{ref} to be the mean of $x_{i, \text{obs}}$).

The detection of accretion emission is a function of stellar luminosity. Since it is not trivial to determine the shape of this function a priori, we modelled it by employing reasonable assumptions. At the low luminosity end, the accretion luminosity may be undetected because emission is weak, and drops below some threshold sensitivity. In the intermediate regime, the detection limit may scale roughly with the stellar luminosity. At the highest stellar luminosities, the stellar spectrum may peak at shorter wavelengths associated with accretion signatures, and this may further decrease sensitivity to these accretion signatures.

We therefore encoded completeness through a parametric detection curve in accretion luminosity. Specifically, for a given stellar luminosity $L_{\star}$ we defined the limiting accretion luminosity as

$$L_{\text{lim}}(L_{\star}) = L_0 + f L_{\star} \left[1 + \left(\frac{L_{\star}}{L_{\text{thr}}} \right)^2 \right], \quad (10.7)$$

with L_0 and L_{thr} in units of $L_{\odot}$ and f dimensionless. The square law of the second brackets is arbitrary, but corresponds to a steepening of the detection limit at high $L_{\star}$. The corresponding limit parameter in dex is

$$\ell_{\text{lim}}(L_{\star}) \equiv \log_{10} \left(\frac{L_{\text{lim}}(L_{\star})}{L_{\odot}} \right) = \log_{10} L_{\text{lim}}(L_{\star}). \quad (10.8)$$

Given the true accretion luminosity $\ell_{i,\text{true}}$ and $L_{\star,i}$, we assumed a “soft” transition between non-detections and detections. This softness is due to the fact that Gaia XP spectra quality varies between sources. The probability that a source i is detected is

$$z_i = \frac{\ell_{i,\text{true}} - \ell_{\text{lim},i}}{\sigma_{\text{det}}}, \quad p_i \equiv P(\text{det}_i = 1 \mid \ell_{i,\text{true}}, L_{\star,i}) = \Phi(z_i), \quad (10.9)$$

where Φ is the standard normal cumulative distribution function and σ_{det} (in dex) controls the width of the transition. The observed detection flag $y_i \in \{0, 1\}$ is then modelled as

$$y_i \sim \text{Bernoulli}(p_i). \quad (10.10)$$

For detected sources ($y_i = 1$) we also modelled the measured accretion luminosity (in dex) as

$$\ell_{i,\text{obs}} \sim \mathcal{N}(\ell_{i,\text{true}}, s_{\ell,i}), \quad (10.11)$$

where $s_{\ell,i}$ is the measurement uncertainty on $\ell_{i,\text{obs}}$ (if individual errors are not available, we adopted a fixed value $s_{\ell,i} \approx 0.05$ dex). For non-detections ($y_i = 0$) the numerical value of L_{acc} is not used in the likelihood; only the detection flag enters via the Bernoulli term above.

We inferred the parameters $\Theta = \{\alpha_0, \alpha_1, \sigma_L, L_0, f, L_{\text{thr}}, \sigma_{\text{det}}\}$ using the joint likelihood of detection flags and the measured accretion luminosities for the detected sources in a Bayesian framework. We adopted weakly informative priors listed in Table 10.2 and sampled from the posterior $P(\Theta \mid \{x_{i,\text{obs}}, \sigma_{x,i}, y_i, \ell_{i,\text{obs}}\})$ using the No-U-Turn Sampler as implemented in PYMC. Posterior summaries (medians and credible intervals) for all parameters are reported in Table 10.2.

In practice we adopted the posterior median values of the detection-curve parameters,

$$L_0 \approx 1.6 \times 10^{-4} L_{\odot}, \quad f \approx 4.4 \times 10^{-3}, \quad L_{\text{thr}} \approx 0.42 L_{\odot}, \quad (10.12)$$

and used Eq. (10.7) to define an effective upper limit on the accretion luminosity of each non-detected source:

$$L_{\text{acc},i}^{\text{UL}} \equiv L_{\text{lim}}(L_{\star,i}; L_0, f, L_{\text{thr}}), \quad \ell_i^{\text{UL}} = \log_{10} L_{\text{acc},i}^{\text{UL}}. \quad (10.13)$$

By construction, $L_{\text{acc},i}^{\text{UL}}$ is the accretion luminosity for which a source with stellar luminosity $L_{\star,i}$ has a 50% probability of being detected in our survey, given the jointly inferred intrinsic distribution and completeness model.

We verified on a synthetic sample that using this approach we reproduce the detection fraction, with a broadly similar distribution of detected luminosities. Therefore, for our purposes we adopt these upper limits to compare our model with observations. Refining the model for the upper limit provided by Gaia XP spectra will be topic of future work.

Table 10.2: Prior and Posterior of the parameters included

Parameter	Prior	Values
α_0	$\mathcal{N}(-3, 1^2)$	-3.58 ± 0.0521
α_1	$\mathcal{N}(2, 1^2)$	1.79 ± 0.0328
σ_L	HalfNormal(0.2)	0.482 ± 0.0228
$\log_{10}(L_0/L_\odot)$	$\mathcal{N}(-4, 0.3^2)$	-3.79 ± 0.145
$\log_{10} f$	$\mathcal{N}(-2, 0.2^2)$	-2.35 ± 0.0486
$\log_{10}(L_{\text{thr}}/L_\odot)$	$\mathcal{N}(\log_{10} 3, 0.15^2)$	-0.375 ± 0.0343
σ_{det}	HalfNormal(0.15)	0.439 ± 0.045

Notes. The parameters inferred by the Bayesian framework are listed in the left column. The central column includes the weakly informative prior used in the model, while the right column provides the posterior parameters as medians and credible intervals.

Conclusions and future directions

In this thesis, I explored the role of the process of external photoevaporation in protoplanetary discs, focusing on the disc populations in the nearby star-forming regions ($\lesssim 500$ pc from the Sun). The intense UV radiation emitted by OBA-type stars heats the uppermost layers of protoplanetary discs, producing thermal winds that are launched from the disc surface, and causing disc depletion outside-in. The concept of disc external photoevaporation is commonly associated with proplyds. However, these evaporating protoplanetary discs, which show the characteristic comet-like morphology in optical lines, are typically detected at FUV fluxes greater than $10^4 G_0$, while no direct observational evidence of the effect of external photoevaporation is currently available at weaker FUV fluxes. Moreover, previous works have investigated the level of FUV irradiation in the solar neighbourhood ($\lesssim 2$ kpc from the Sun), finding that typical environments for star (and planet) formation present intermediate FUV radiation fields in the range $3 \times 10^2 - 3 \times 10^4 G_0$. Therefore, to understand how disc evolution and planet formation typically proceeds in the near Galaxy, understanding the impact of external radiation on discs in this range of FUV fluxes is essential.

The investigation conducted in this thesis is motivated by the need to assess (i) whether external photoevaporation affects disc dynamics under moderate FUV radiation fields, and how disc properties, particularly mass and size, evolve across different irradiation environments; and (ii) whether the current dataset in the nearby star-forming regions can provide indirect evidence of the effect of external photoevaporation on discs or, alternatively, what disc properties and regions should be investigated in the future to characterise the evolution of typical planet birthplaces. Following a presentation of the theoretical framework on which the work is built (Chap. 4), I addressed these open topics in the second and third part of the thesis. Specifically, in Part II I compared disc evolution models with recent observations of protoplanetary discs in Upper Sco. In Part III, I provided a new method to evaluate the FUV flux and a catalogue for nearby dis-hosting stars and for the stellar population of the Orion complex. Moreover, I investigated the demographic of young stellar objects in the Orion region in connection with the FUV radiation field.

Most of the content of the second and third part of this thesis is based on three publications of which I am the first author (two refereed publications and one in preparation), and works in which I have contributed as a co-author. Additional results included to support the dissertation have appeared as refereed publications in scientific journals. The main results of this thesis are summarised as follows.

- **The impact of external photoevaporation on disc masses and radii in Upper Scorpius.**

Due to its proximity and the presence of a large population of old protoplanetary discs, Upper Sco is often used as a reference region for studying the latest stages of disc evolution in comparison to younger regions such as Lupus, Taurus, and Chamaeleon. However, this comparison requires caution. Upper Sco hosts numerous B-type stars and is exposed to FUV radiation fields that, even if moderate $\lesssim 100 G_0$, are capable of driving disc external photoevaporation. In contrast, the other young nearby regions contains only a couple of late-type B stars. Using disc evolution models of viscous externally irradiated discs, I demonstrated that discs can be truncated due to external photoevaporation and shrink outside-in even at low-to-moderate FUV fields of $1 - 100 G_0$. By including external photoevaporation in a pure viscous evolution framework, I achieved a closer agreement with the gas masses and radii of the Upper Sco disc sample investigated by the AGE-PRO ALMA Large Program, compared to a viscous model alone.

This was a fundamental result in the context of AGE-PRO. Indeed, the aim of this survey is to characterise gas disc evolution across ages, considering discs in Ophiuchus, Lupus, and Upper Sco, and I emphasised that the role of the environment is a fundamental factor to be considered when comparing average disc properties in different star-forming regions.

- **The extended Upper Sco disc sample**

Using the largest dataset of discs available in the Upper Sco region, I demonstrated that the inclusion of external photoevaporation to a population model of viscous discs is able to reproduce, on a statistical basis, the observed distribution of gas disc sizes. However, finding a less model-dependent proofs of the role of external photoevaporation in shaping disc evolution in Upper Sco, e.g., correlations between gas and dust disc properties and the FUV flux, remains a currently open problem.

- **A novel method to evaluate the FUV flux.**

Accurately measuring the FUV flux experienced by protoplanetary discs is crucial to assess the role of external photoevaporation on them. I illustrated a new method to evaluate the FUV flux at the position of stars: I used a statistical framework and the 2D geometry of a stellar cluster to account for the uncertainty in the separation from massive stars in 3D space. I provided a publicly available catalogue of the FUV flux at numerous disc-hosting stars in the nearby regions. Discs that are located close to BA-type stars can experience a significant level of FUV radiation even in weakly irradiated regions.

- **Mapping Orion in FUV flux.**

The Orion complex contains the closest star-forming region hosting active star formation and numerous OB stars. We mapped Orion in FUV flux, providing a catalogue and a script to compute the FUV flux, which will be soon available. These tools are extremely useful for identifying targets for future observations and for studying disc and stellar properties in connection with the environment. I showed that Orion hosts a large population of weakly-to-intermediately irradiated stars ($1 - 10^4 G_0$). Many of those objects show sign of the presence of an accretion disc from the detection of the H_α line, and have not been investigated yet.

- **Using disc dust mass and accretion luminosity to provide indirect evidence of the effect of external photoevaporation**

In Chap. 10, I compared the dust disc masses retrieved from observations in the nearby star-forming regions with the FUV fluxes experienced, showing that the current gap of knowledge of disc and

stellar parameters at $> 10^2 G_0$ prevents us from determining whether a general relation between dust mm-flux emission and FUV flux exists. Furthermore, I compared the accretion luminosity of the objects showing H_α emission in Gaia XP spectra with the FUV flux experienced, showing a dependence on age and FUV flux, which is, in part, reproduced by the models. However, I showed that the FUV flux does not hold full information on the effect of photoevaporation on discs, which instead depends also on the stellar mass, disc mass, and disc extent. A stronger indicator of the effectiveness of external photoevaporation would be provided by the mass loss rate; however, measuring it when the proplyd-like shape is not observed is challenging.

Future directions

Following the results and the discussion performed in this work, I indicate some major future steps that would enable us to characterise disc evolution in connection with the environment.

- **Direct wind tracers in intermediately irradiated discs.**

Although the FUV flow alone may drive the wind (Haworth et al. 2023a), the EUV flow is required to produce the bright ionisation front. Consequently, the comet-like shape is less apparent in FUV-dominated flows, the detection is more challenging and different tracers are required. I showed that there is indirect evidence and multiple hints indicating the effect of external photoevaporation on disc properties (e.g., disc dust mass, accretion luminosity). However, gathering direct evidence is the only way to prove whether these features are due to external photoevaporation. Theoretical models show that atomic carbon (CI) is the most promising tracer of external photoevaporation in action, in absence of a clear proplyd-like morphology (Haworth & Owen 2020). Indeed, CO, which is the most widely detected molecule in discs, is expected to photodissociate toward the low density gas launched by external photoevaporation, forming CI. Because the CI layer is located just beyond the CO layer, it serves as an ideal candidate to trace the inner part of the PDR. We expect to detect an excess in CI emission above the disc level in strongly UV-irradiated objects.

Another possibility consists of detecting CO in the wind. In moderately and intermediately irradiated discs, a thin layer of CO below the CI layer could lie in the wind, and Facchini et al. (2016) pointed out that, since the specific angular momentum in the wind scales as R^{-2} (assuming a spherical wind), a non-Keplerian rotation of the wind would provide a signature of external photoevaporation. Candidate targets for these observations have been proposed (e.g., Law et al. 2023; Teague 2019). However, additional studies on the disc properties and observational parameters (e.g., sensitivity and resolution) in connection with the external FUV radiation field are needed to determine whether and in what targets these observations are feasible. Detecting CO and CI in the photoevaporative wind will allow us to directly study wind kinematics.

- **Improvement of the models**

In this work, I used 1D evolution models of viscous protoplanetary discs where I included the effect of external photoevaporation. However, the alternative scenario to the viscous model driving disc accretion are MHD winds, that allow accretion onto the star by removing material and angular momentum from the disc through magnetic field lines (e.g., Lesur et al. 2014). The combined effect of MHD

wind-driven accretion and external photoevaporation still needs to be investigated, on the perspective of both gas and dust disc components.

- **Measurements of stellar and disc properties in Orion**

The easiest disc property to estimate from observations is the dust disc mass, which is usually derived from mm- and sub-mm observations, assuming optically thick emission from the dust grains. Comparing dust disc masses for objects in the nearby star-forming regions with the FUV flux experienced, I showed in Chap. 10 that there is a gap of knowledge in the disc and stellar parameters at FUV fluxes $> 10^2 G_0$. Therefore, in order to assess the role of photoevaporation, it is imperative to observe protoplanetary discs at higher FUV fluxes. The closest candidate region where we could populate this gap is Orion. There is a large population of accreting stars showing H_α emission, and potentially a circumstellar disc, that has not been observed with ALMA and may serve as future target.

Moreover, in Chap. 5 I showed that we expect that the gas disc size (e.g., computed from ^{12}CO line) should be a more efficient tracer of the effect of external photoevaporation, with respect to gas disc mass, which is extremely challenging to measure, and dust disc mass and radius, which are highly influenced by traps and substructures. However, CO observations at $> 10^2 G_0$ are only available for a few discs in σ Orionis. Deeper CO observations aimed to measure gas disc properties are needed to assess the role of photoevaporation.

- **Exploring higher-mass star-forming regions**

In this work, we considered as *typical* FUV flux experienced by young stars the range $\sim 10^2 - 10^4 G_0$, which refers to the solar neighbourhood (< 2 kpc from the Sun). Expanding the perspective to our Galaxy, most stars form in regions even more massive than Orion, where the numerous O-type stars contribute to FUV radiation fields $> 10^4 G_0$. These environments are better representative of the birth habitat for the exoplanet-hosting stars that formed 4-10 Gyr ago (e.g., Winter et al. 2018, 2020b; Hallatt & Lee 2025) and the cluster environment where the Sun formed (Adams 2010). This thesis emphasises that all our knowledge of protoplanetary discs is based on an extremely biased sample, which consists of weakly irradiated star-forming regions ($< 10^2 G_0$), which are only the tip of the iceberg. I showed that Orion is a good candidate region where to look for higher FUV-irradiated discs; however, even Orion is not representative of the typical conditions for planet formation in our Galaxy. More suitable regions to investigate for this purpose are located at distances > 700 pc from the Sun, such as Tr14 in the Carina Nebula (e.g., Itrich et al. 2024; Zeidler et al. 2016) and M16 (e.g., Feigelson et al. 2013; Lim et al. 2025).

To summarise, protoplanetary disc evolution can significantly be affected by external photoevaporation across a wide range of FUV fluxes. Given the FUV flux distribution that we provided in this work, it is reasonable to state that typical protoplanetary discs in the solar neighbourhood undergo some degree of external photoevaporation during their lifetime. At the same time, the results of this thesis emphasise the difficulty in finding model-independent proofs of the effect of external photoevaporation in action on discs with the current dataset, even when exploring the Orion region. The dust disc mass and the accretion luminosity show some degree of dependence on the FUV flux; however, measurements of disc and stellar properties at $> 10^2 G_0$, as well as estimates of the mass loss rate due to photoevaporation are crucial for future investigation and

stronger evidence.

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List of Publications

As November 2025

Refereed publications

First author publications

- *A novel method for estimating the far-ultraviolet flux, and a catalogue for disc-hosting stars in nearby star-forming regions*, **Anania Rossella**, Winter Andrew J., Rosotti Giovanni P., Vioque Miguel, Zari Eleonora, Pantaleoni González Michelangelo, Testi Leonardo, *Astronomy & Astrophysics*, Volume 695, id.A74, 20 pp., (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). VIII. The Impact of External Photoevaporation on Disk Masses and Radii in Upper Scorpius*, **Anania Rossella**, Rosotti Giovanni P., Gárate Matías, Pinilla Paola, Vioque Miguel, Trapman Leon, Carpenter John, Zhang Ke, Pascucci Ilaria, Cieza Lucas A., Sierra Anibal, Kurtovic Nicolas T., Miley James, Pérez Laura M., Tabone Benoît, Hogerheijde Michiel, Deng Dingshan, Agurto-Gangas Carolina, Ruiz-Rodriguez Dary A., González-Ruilova Camilo, TorresVillanueva Estephani E., *The Astrophysical Journal*, Volume 989, Issue 1, id.8, 20 pp. (2025)

Co-author publications

- *The evolution of the M_d - M_* and $-M_*$ correlations traces protoplanetary disc dispersal*, Somigliana Alice, Testi Leonardo, Rosotti Giovanni, Toci Claudia, Lodato Giuseppe, **Anania Rossella**, Tabone Benot, Tazzari Marco, Klessen Ralf, Lebreuilly Ugo, Hennebelle Patrick, Molinari Sergio, *Astronomy & Astrophysics*, Volume 689, id.A285, 17 pp., (2024)
- *Running with the bulls: The frequency of star-disc encounters in the Taurus star-forming region*, Winter Andrew J, Benisty Myriam, Shuai Linling, Dûchene Gaspard, Cuello Nicolás, **Anania Rossella**, Cadiou Corentin, Joncour Isabelle, *Astronomy & Astrophysics*, Volume 691, id.A43, 19 pp., (2024)
- *The past, present and future of observations of externally irradiated disks*, Planet formation environments collaboration, *The Open Journal of Astrophysics*, vol. 8, id. 54, (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). I. Program Overview and First Results*, Zhang Ke, and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)

- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). II. Gas and Dust in the Ophiuchus Star-forming Region*, Ruiz-Rodriguez Dary A., and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). III. Dust and Gas Disk Properties in the Lupus Star-forming Region*, Deng Dingshan, and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
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- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). V. Protoplanetary gas disk masses*, Trapman Leon, and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). VI. Comparison of Dust Evolution Models to AGE-PRO Observations*, Kurtovic Nicolas T., and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). VII. Testing Accretion Mechanisms from Disk Population Synthesis*, Tabone Benoît, and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). IX. Hints of Planet Formation Signatures in a Large-cavity Disk Studied in the AGE-PRO ALMA Large Program*, Sierra Anibal, and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). X. Dust Substructures, Disk Geometries, and Dust-disk Radii*, Vioque Miguel, and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). XI. Beam-corrected Gas Disk Sizes from Fitting ^{12}CO Moment Zero Maps*, Trapman Leon, and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
- *The ALMA Survey of Gas Evolution of PROtoplanetary Disks (AGE-PRO). XII. Extreme millimetre variability detected in a Class II disk*, Miley James M., and AGE-PRO collaboration, *The Astrophysical Journal*, Volume 989, Issue 1, (2025)
- *Observational Constraints on Evolution of Dust Disc Properties in Upper Scorpius*, Pinilla Paola, Sierra Anibal, Kurtovic Nicolas T., **Anania Rossella**, Andrews Sean, Carpenter John, Guerra-Alvarado Osmar, Long Feng, Marino Sebastian, Vioque Miguel, Zhang Ke, accepted for publication on MNRAS (2025)

Publications under review

Co-author publications

- *A tension between dust and gas radii: the role of substructures and external photoevaporation in protoplanetary disks*, Delussu Luca, **Anania Rossella**, Birnstiel Tilman, Toci Claudia, Rosotti Giovanni P., Stammer Sebastian Markus, Lau Tommy Chi Ho, Miotello Anna, *Astronomy & Astrophysics* (2025)

Submitted publications

Co-author publications

- *Survival of protoplanetary disks in Upper Scorpius from population synthesis models with external photoevaporation*, Ping Jingyi, **Anania Rossella**, Pinilla Paola, Vioque Miguel, *Astronomy & Astrophysics*
- *Timescales diagnostics for saving viscously- and MHD-driven dusty discs from external photoevaporation*, Pichierri Gabriele, Rosotti Giovanni P., **Anania Rossella**, Lodato Giuseppe, *Astronomy & Astrophysics*

Publications in preparation

- *Far-ultraviolet flux distribution in Orion and its relation with stellar accretion*, **Anania Rossella**, Winter Andrew J., Vioque Miguel, Rosotti Giovanni P., Beccari Giacomo, Lodato Giuseppe, Malanga Lorenzo, Piscarreta Lara, Somigliana Alice, Testi Leonardo, Toci Claudia, *Astronomy & Astrophysics*
- *Compact CO emission in planet-forming discs in Upper Sco*, Zagaria Francesco, Leon Trapman, **Anania Rossella**, Empey Aaron, Rosotti Giovanni P., Vioque Miguel, Toci Claudia, Andrews Sean, Carpenter John M., Manara Carlo F., Mamajek Eric E., Zhang Ke, *Astronomy & Astrophysics*

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I provide here, for completeness, the acknowledgment section of the first-author refereed publications

- **Chapter 5.**

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- **Chapter 8 and 10.**

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